complementation test (see Sections 2.2 and 4.8), and indeed, two units of function, the *rIIA* cistron and the *rIIB* cistron, were defined by complementation. The limits of *rIIA* and *rIIB* are shown in Figures 8.21 and 8.23. The complementation between *rIIA* and *rIIB* is observed when two types of T4 phage, one with a mutation in *rIIA* and the other with a mutation in *rIIB*, are used simultaneously to infect *E. coli* strain K12(λ). The multiply infected cells produce normal numbers of phage progeny, most of which carry either the parental *rIIA* mutation or the parental *rIIB* mutation. However, rare recombination between the mutant sites results in recombinant progeny phage that are rII^{+}. In contrast, when K12(λ) is simultaneously infected with two phage having different *rIIA* mutations, or two phage having different *rIIB* mutations, no progeny phage are produced.

Besides the meaning of function, clarified by use of the term *cistron*, the term *gene* has two other distinct meanings: (1) the unit of genetic transmission that participates in recombination and (2) the unit of genetic change or mutation. Physically, both the recombinational and the mutational units correspond to the individual nucleotides in a gene. Despite potential ambiguity, the term *gene* is still the most important word in genetics, and in most cases, the shade of meaning intended is clear from the context.

8.7— Genetic Recombination in Temperate Bacteriophages

Temperate bacteriophages have two alternative life cycles—a lytic cycle and a lysogenic cycle. The lytic cycle is depicted in Figure 8.3. A temperate phage, such as *E. coli* phage λ, when reproducing in its lytic cycle, undergoes general recombination much as phage T4 does. A map of the phage λ genome is depicted in Figure 8.24; it is linear rather than circular. The DNA molecule in the λ phage particle is also linear. Unlike the DNA molecules in T4 phage, however, every phage λ DNA molecule has identical ends. Indeed, the ends are single-stranded and complementary in sequence so that they can pair, forming a circular molecule.

The single-stranded ends are called **cohesive ends** to indicate their ability to undergo base pairing. The packaging of DNA in phage λ does not follow a headful mechanism, like T4. Rather, the λ packaging process recognizes specific sequences that are cleaved to produce the cohesive ends.

The complete DNA sequence of the genome of λ phage has been determined, and many of the genes and gene products and their functions have been identified. The map of genes in Figure 8.24 shows where each gene is located along the DNA molecule, scaled in kilobase pairs (kb), rather than in terms of its position in the genetic map. However, a genetic map with coordinates in map units (centimorgans, or percent recombination) has been placed directly below the molecular scale. Comparing the scales reveals that the frequency of recombination is not uniform along the molecule. For example, the 5 map units between genes *H* and *I* span about 3.1 kb, whereas the 5 map units between the genes *int* and *cIII* span about 4.8 kb.

The λ map is also interesting for what it implies about the evolution of the phage genome. The genes in λ show extensive clustering by function. The left half of the map consists entirely of genes whose products (head and tail proteins) are required for assembly of the phage structure, and within this region, the head genes and the tail genes themselves form subclusters. The right half of the λ genome also shows several gene clusters, which include genes for DNA replication, recombination, and lysis. The genes are clustered not only by function but also according to the time at which their products are synthesized. For example, the N gene acts early; genes *O* and *P* are active later; and genes *Q*, *S*, *R*, and the head-tail cluster are expressed last. The transcription patterns for mRNA synthesis are thus very simple and efficient. There are only two rightward transcripts, and all late genes except for *Q* are transcribed into the same mRNA.

Lysogeny

The lysogenic cycle comes about when the λ DNA molecule becomes integrated into the bacterial chromosomes, where it is passively replicated as part of the bacterial

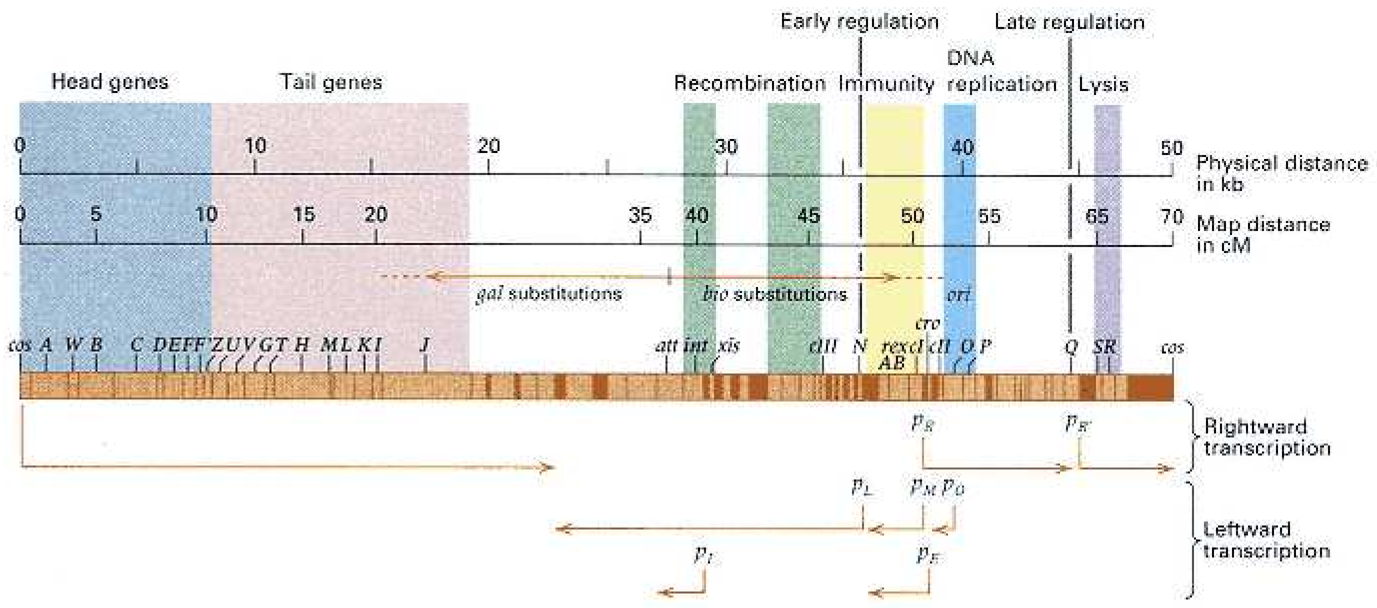


Figure 8.24
Molecular and genetic maps of bacteriophage λ. The scale of the molecular map is length in kilobase pairs (kb); genetic distances are given in centimorgans (cM), equivalent in this case to percent recombination. Clusters of genes with related functions are indicated by the pastel colored boxes. The positions of the regulatory genes *N* and *Q* are indicated by vertical lines at about 35 and 45 kb, respectively. Promoters are indicated by the letter *p*. The direction of transcription and length of transcript are indicated by arrows. Many of the genes known through mutant phenotypes are identified. Coding regions of unidentified function are represented as unlabeled, light-colored rectangles; dark-colored regions indicate sequences unlikely to code for proteins. The λ *att* (attachment site) is the site of recombination for integration of the λ prophage. The origin of replication (*ori)* is the region denoted O. The extents of possible substitutions of λ DNA with *E. coli* DNA in the specialized transducing phages *λdgal* and *λdbio* are indicated by arrows just above the gene map; such transducing phages are discussed in the section on specialized transduction.

chromosome; phage particles are not produced (Figure 8.25). The inserted DNA is called a **prophage,** and the surviving cell is called a **lysogen.** A strain lysogenic for λ has the symbol (λ) appended to its name. For example, the strain *E. coli* K12(λ) that made possible the fine-structure analysis of the *rII* region of phage T4 was a K12 strain that had become lysogenic for λ.

As noted, the cohesive ends of the λ DNA molecule are single-stranded, with 12 unpaired bases at either end. The ends are complementary, however. Upon entering the cell, the complementary ends anneal to form a nicked circle, and ligation seals the nicks (Figure 8.26). Circularization, which takes place early in both the lytic and lysogenic cycles, is a necessary event in both cycles: for DNA replication in the lytic mode, for prophage integration in the lysogenic cycle. In about 75 percent of infected cells, the circular molecule replicates, and the lytic cycle ensues. However, in about 25 percent of infected cells, the circular λ molecule and the circular *E. coli* DNA molecule interact and undergo a single, site-specific recombination event in which the phage DNA becomes incorporated into the bacterial chromosome. Because λ can exist either as an autonomous genetic element (in the lytic cycle) or as an integrated element in the chromosome (in lysogeny), λ, like the F factor, is classified as an episome.

The positions of the site-specific recombination in the bacterial and phage DNA are called the **bacterial** and **phage attachment sites,** respectively. Each attachment site consists of three segments. The central

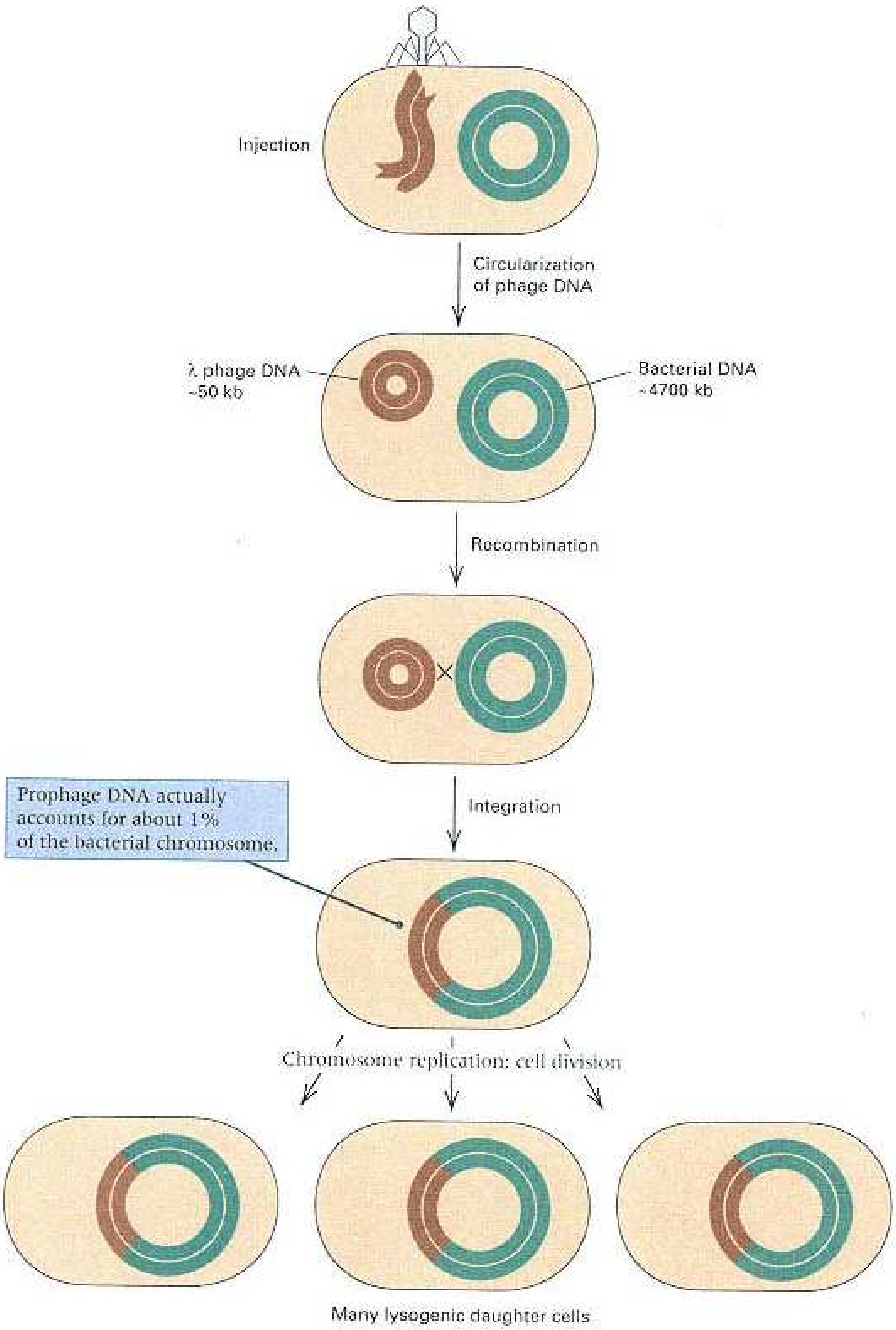


Figure 8.25
The general mode of lysogenization by integration of phage DNA into the bacterial chromosome. Some genes (those needed to establish lysogeny) are expressed shortly after infection and are then turned off. The inserted brown DNA is the prophage. For clarity, the phage DNA is drawn much larger than to scale; the size of phage λ DNA is actually about 1 percent of the size of the *E. coli* genome.

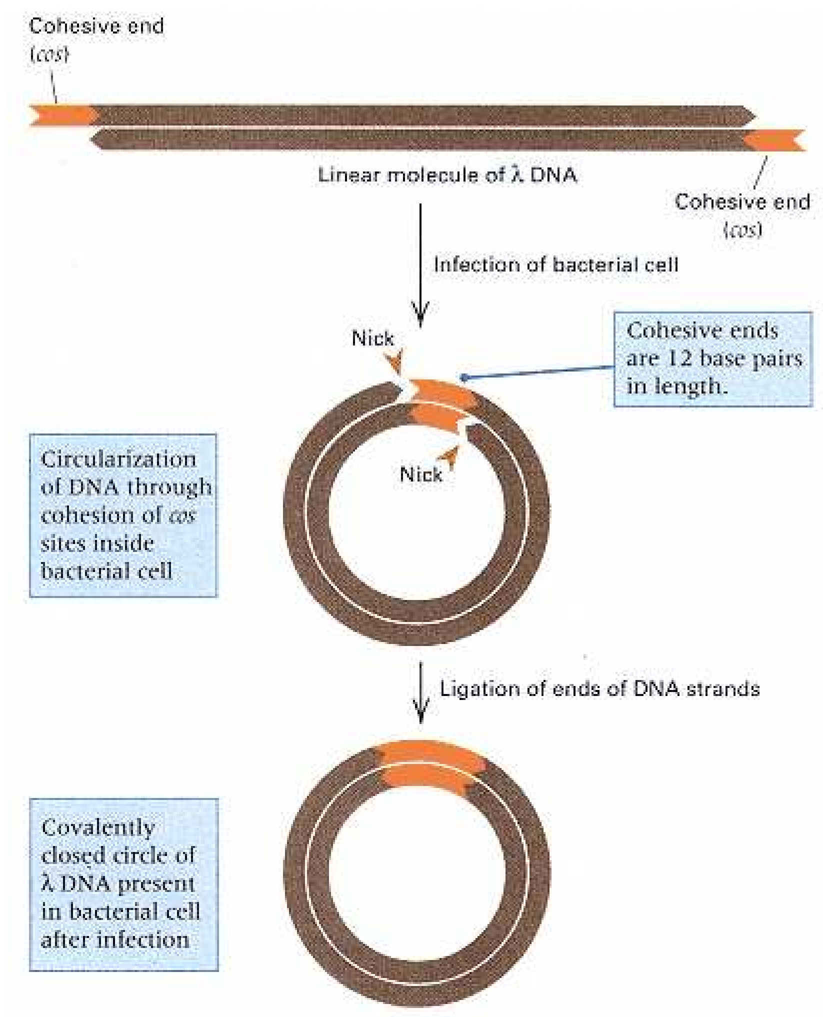


Figure 8.26
A diagram of a linear λ DNA molecule showing the cohesive ends (complementary single-stranded ends). Circularization by means of base pairing between the cohesive ends forms an open (nicked) circle, which is converted into a covalently closed (uninterrupted) circle by sealing (ligation) of the single-strand breaks. The length of the cohesive ends is 12 base pairs in a total molecule of approximately 50 kb.

segment has the same nucleotide sequence in both attachment sites and is the region in which the recombination actually takes place. The phage attachment site is denoted *POP'* (*P* for phage), and the bacterial attachment site is denoted *BOB'* (*B* for bacteria). A comparison of the genetic maps of the phage and the prophage indicates that *POP'* is located near the middle of the linear form of the phage DNA molecule (see Figure 8.24). A phage protein, **integrase,** catalyzes a site-specific recombination event; the integrase recognizes the phage and bacterial attachment sites and causes the physical exchange that results in integration of the λ DNA molecule into the bacterial DNA. The geometry of the exchange is shown in Figure 8.27. As a result of the recombination event, the genetic map of the prophage is not the same as the map of the phage; it is a circular permutation of it that arises from the central location of λ *att* (the *POP'* site) and the circularization of the DNA.

The correct model for prophage integration was first suggested by Allan Campbell in 1962. The model was confirmed by bacterial crosses of lysogens with nonlysogens, as well as by transduction by phage P1. Campbell found that the order of genes in the integrated prophage was

where *m6* is a mutation affecting head formation and *mi* is one affecting lysis. In contrast, the gene order of genes in the free phage as determined by general recombination is

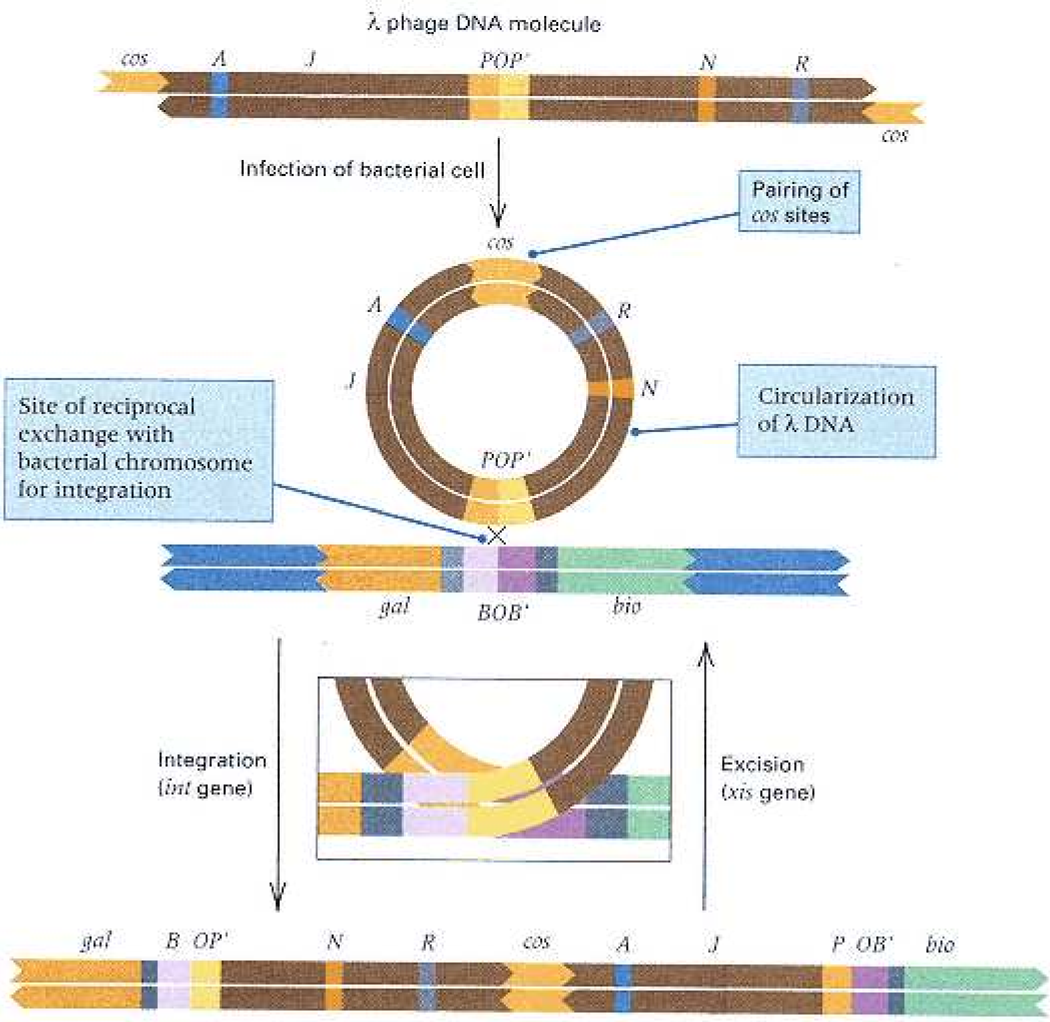


Figure 8.27
The geometry of integration and excision of phage λ. The phage attachment site is *POP*'. The bacterial attachment site is *BOB*'. The prophage is flanked by two hybrid attachment sites denoted *BOP*' and *POB'*.

The prophage map is thus a circular permutation of the map of the free phage. The prophage is inserted into the *E. coli* chromosome between the genes *gal* and *bio,* as indicated in Figure 8.28. An additional finding that confirmed the physical insertion of λ was that the integrated prophage increased the distance between *gal* and *bio* so that these markers could no longer be cotransduced by phage P1. The distance between *gal* and *bio* in a λ lysogen is about 2 minutes, compared to 1 minute in a nonlysogen.

When a cell is lysogenized, the phage genes become part of the bacterial chromosome, so it might be expected that the phenotype of the bacterium would change. But most phage genes in a prophage are kept in an inactive state by a **repressor** protein, the product of one of the phage genes. The repressor protein is synthesized initially by the infecting phage and then continually by the prophage. The gene that codes for the repressor is frequently the only prophage gene that is expressed in lysogens. If a lysogen is infected with a phage of the same type as the prophage—for example, λ infecting a λ lysogen—then the repressor present within the cell from the prophage prevents expression of the genes of the infecting phage. This resistance to infection by a phage identical with the prophage, which is called **immunity,** is the usual criterion for determining whether a bacterial cell contains a particular prophage. For example, λ will not form

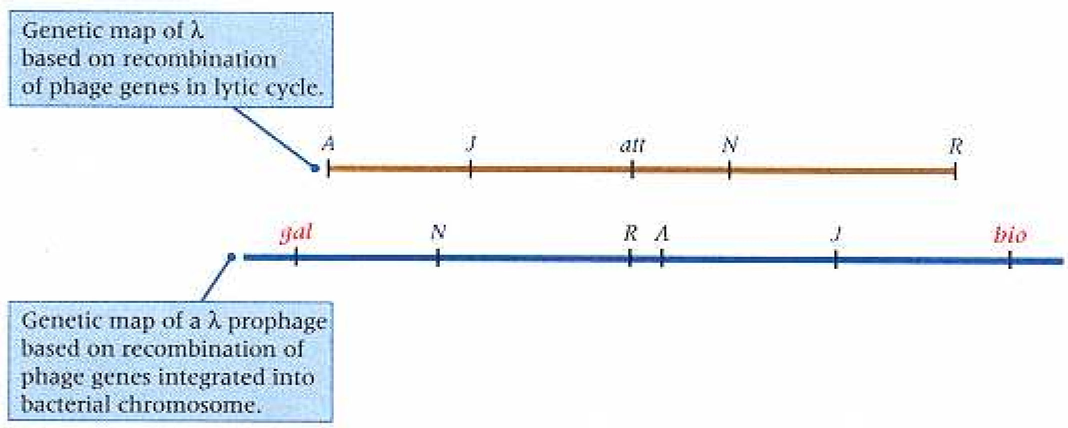


Figure 8.28
The map order of genes in phage λ as determined by phage recombination (lytic cycle) and in the prophage (prophage order). The genes have been selected arbitrarily to provide reference points. Bacterial DNA is labeled in red letters.

plaques on bacteria that contain a λ prophage.

A lysogenic cell can replicate nearly indefinitely without the release of phage progeny. However, the prophage can sometimes become activated to undergo a lytic cycle in which the usual number of phage progeny are produced. This phenomenon is called **prophage induction,** and it is initiated by damage to the bacterial DNA. The damage sometimes happens spontaneously but is more often caused by some environmental agent, such as chemicals or radiation. The ability to be induced is advantageous for the phage because the phage DNA can escape from a damaged cell. The biochemical mechanism of induction is complex and will not be discussed, but the excision of the phage is straightforward.

Excision is another site-specific recombination event that reverses the integration process. Excision requires the phage enzyme integrase plus an additional phage protein called **excisionase.** Genetic evidence and studies of physical binding of purified excisionase, integrase, and λ DNA indicate that excisionase binds to integrase and thereby enables the latter to recognize the prophage attachment sites *BOP'* and *POB';* once bound to these sites, integrase makes cuts in the *O* sequence and recreates the *BOB'* and *POP'* sites. This reverses the integration reaction, causing excision of the prophage (Figure 8.27).

Specialized Transducing Phage

When a bacterium lysogenic for phage λ is subjected to DNA damage that leads to induction, the prophage is usually excised from the chromosome precisely. However, in about 1 cell per 10^6 to 10^7 cells, an excision error is made (Figure 8.29), and a chance breakage in two nonhomologous sequences takes place—one break within the prophage and the other in the bacterial DNA. The free ends of the excised DNA are then joined to produce a DNA circle capable of replication. The sites of breakage are not always located so as to produce a length of DNA that can fit in a λ phage head, and the DNA may be too large or too small. Sometimes, however, a molecule forms that can replicate and be packaged. In λ lysogens, the prophage lies between the *gal* and *bio* genes, and because the aberrant cut in the host DNA can be either to the right or to the left of the prophage, particles can arise that carry either the *bio* genes (cut at the right) or the *gal* genes (cut at the left). The resulting phage are called λ*bio* and λ*gal* transducing particles. These are **specialized transducing phages** because they can transduce only certain bacterial genes (*gal* or *bio),* in contrast with the P1-type generalized transducing particles, which can transduce any gene.

Often the specialized transducing phages are defective, because essential λ

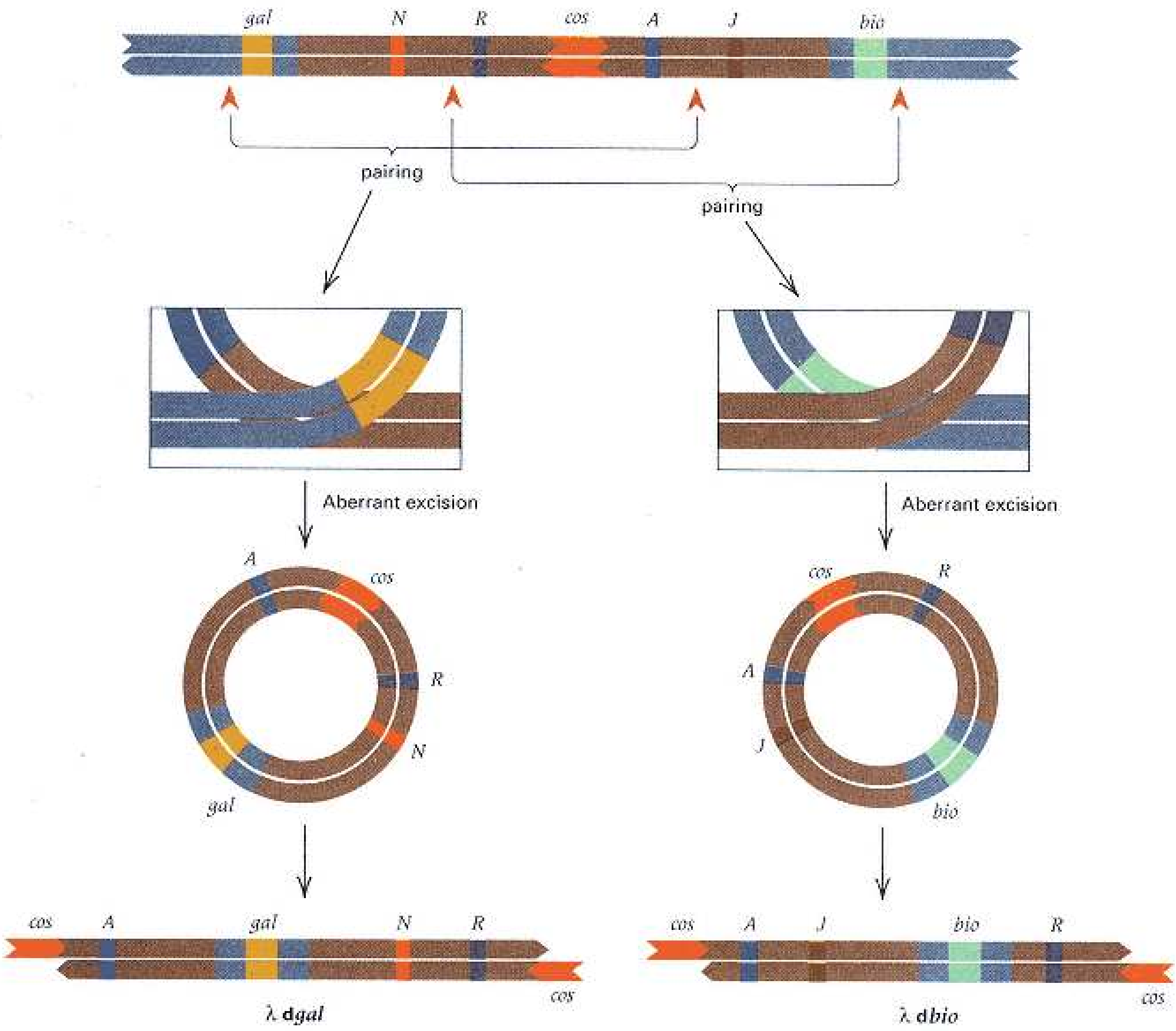


Figure 8.29
Aberrant excision leading to the production of specialized λ transducing phages. (A) Formation of a *gal* transducing phage (*λdgal*). (B) Formation of a bio transducing phage (*λdbio*).

genes are replaced by bacterial genes during formation of the *gal*$^+$ -bearing or *bio*$^+$-bearing λ molecules. The phages are thus called λ*dgal* (defective, *gal*$^+$-transducing) or λ*dbio* (defective, *bio*$^+$ -transducing). They are unable by themselves to infect *E. coli* productively. Addition of a wildtype helper phage both allows production of a lysate rich in λ*dgal* or λ*dbio* transducing phage and formation of a double lysogen. Presumably, integration of the wildtype λ provides hybrid attachment sites POB' and BOP' that allow integration of λ*dbio* or λ*dgal* by homologous recombination between identical hybrid attachment sites.

8.8— Transposable Elements

DNA sequences that are present in a genome in multiple copies and that have the capability of occasional movement, or **transposition,** to new locations in the genome were described in Section 6.8. These *transposable elements* are widespread in living organisms.

Many transposable elements in bacteria have been extensively studied. The bacterial elements first discovered—called **insertion sequences** or **IS elements**—are small and do not contain any known host genes. Like many transposable elements in eukaryotes, these bacterial elements possess inverted-repeat sequences at their termini, and most code for a **transposase** protein required for transposition and one or more additional proteins that regulate the rate of transposition. The DNA organization of the insertion sequence IS*50* is diagrammed in Figure 8.30A. Insertion sequences are instrumental in the origin of Hfr bacteria from F^+ cells (Section 8.4), because the F plasmid normally integrates through genetic exchange between insertion sequences present in F and homologous copies present at various sites in the bacterial chromosome.

Other transposable elements in bacteria contain one or more bacterial genes that can be shuttled between different bacterial hosts by transposing into bacterial plasmids (such as the F plasmid), which are capable of conjugational transfer. These gene-containing elements are called **transposons,** and their length is typically several kilobases of DNA, but a few are much longer. Some transposons have composite structures with the bacterial genes sandwiched between insertion sequences, as is the case with the Tn*5* element illustrated in Figure 8.30B, which terminates in two IS*50* elements in inverted orientation. Transposons are usually designated by the abbreviation Tn followed by an italicized number (for example, Tn*5). When it is necessary to refer to genes carried in such an element, the usual designations for the genes are used. For example, Tn5 *(neo-r ble-r str-r)* contains genes for resistance to three different antibiotics: neomycin, bleomycin, and streptomycin. Such genes provide markers, making it easy to detect transposition of the composite element, as is shown in Figure 8.31. An F' *lac*$^+$ plasmid is transferred by conjugation into a bacterial

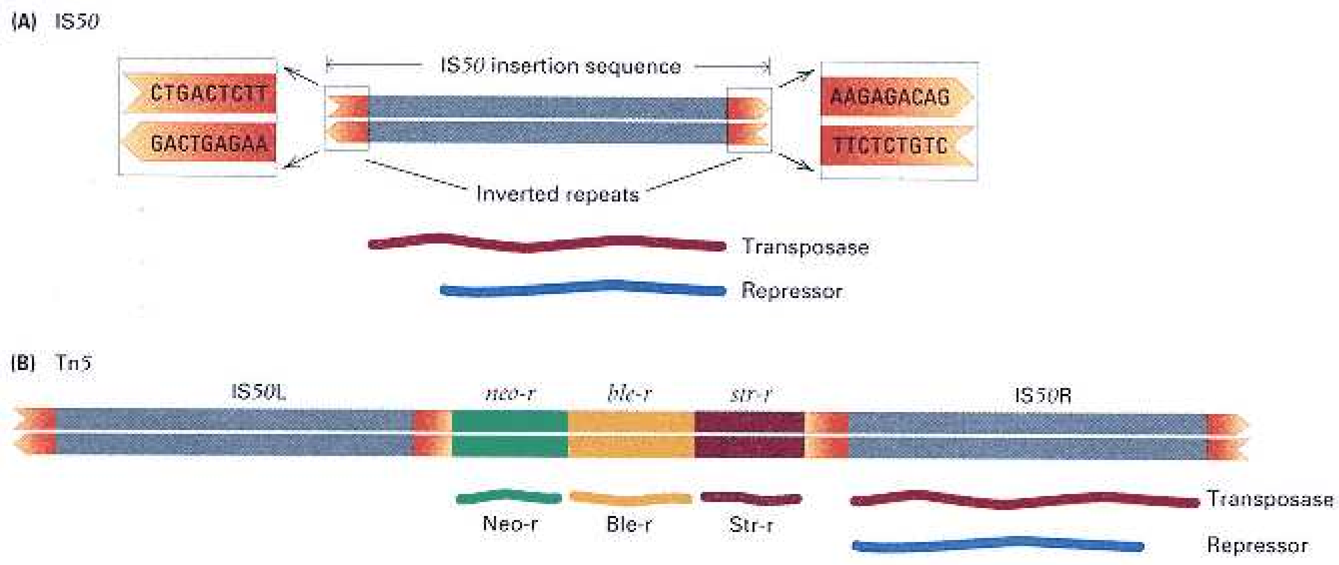


Figure 8.30
Transposable elements in bacteria. (A) Insertion sequence IS*50.* The element is terminated by short, nearly perfect inverted-repeat sequences, the terminal nine base pairs of which are indicated. IS*50* contains a region that codes for the transposase and for a repressor of transposition. The coding regions are identical in the region of overlap, but the repressor is somewhat shorter because it begins at a different place. (B) Composite transposon Tn*5.* The central sequence contains genes for resistance to neomycin, *neo-r*; bleomycin, *ble-r*, and streptomycin, *str-r;* it is flanked by two copies of IS*50* in inverted orientation. The left-hand element (IS*50L*) contains mutations and is nonfunctional, so the transposase and repressor are made by the right-hand element (IS*50R*).

cell containing a transposable element that carries the *neo-r* (neomycin-resistance) gene. The bacterial cell is allowed to grow, and in the course of multiplication, transposition of the transposon into the F' plasmid occasionally takes place in a progeny cell. Transposition yields an F' plasmid containing both the *lac*$^+$ and *neo-r* genes. In a subsequent mating to a Neo-s Lac$^-$ cell, the *lac*$^+$ and *neo-r* markers are transferred together and so are genetically linked.

In nature, sequential transposition of transposons containing *different* antibiotic-resistance genes into the same plasmid results in the evolution of plasmids that confer resistance to multiple antibiotics. These multiple-resistance plasmids are called **R plasmids.** The evolution of R plasmids is promoted by the use (and regrettable overuse) of antibiotics, which selects for resistant cells because, in the presence of antibiotics, resistant cells have a growth advantage over sensitive cells. The presence of multiple antibiotics in the environment selects for multiple-drug resistance. Serious clinical complications result when plasmids resistant to multiple drugs are transferred to bacterial **pathogens,** or agents of disease. Infections with some pathogens containing R factors are extremely difficult to treat because the pathogen is resistant to all known antibiotics.

When transposition takes place, the transposable element can be inserted in any one of a large number of positions. The existence of multiple insertion sites can be shown when a wildtype lysogenic *E. coli* culture is infected with a temperate phage

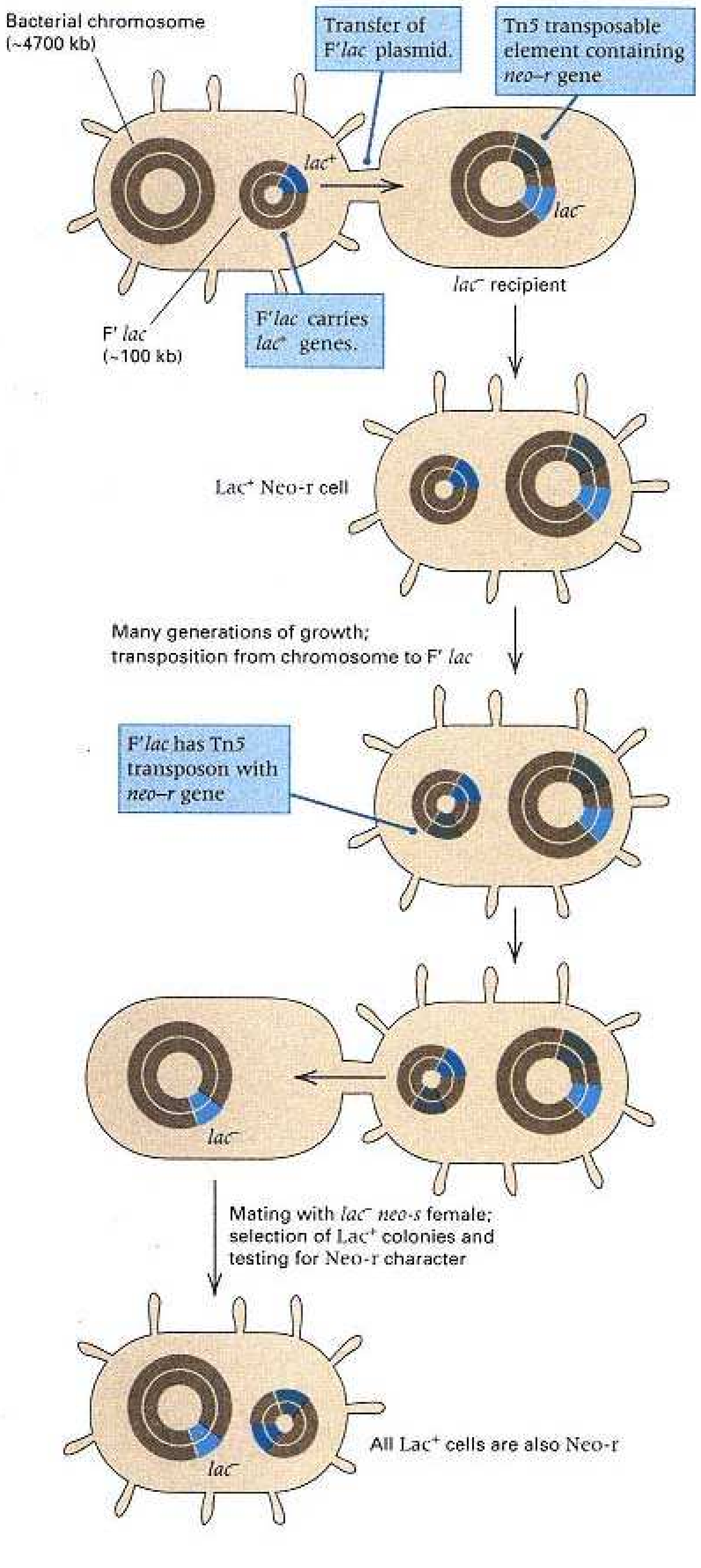


Figure 8.31

An experiment demonstrating the transposition of transposon Tn5, which contains a neomycin-resistance gene, from the chromosome to an F' plasmid containing the bacterial gene for lactose utilization (F' *lac*). The bacterial chromosome is *lac*$^{-}$ and the cells are unable to grow on lactose unless the F' *lac* is present. After the transposition, the F' *lac* plasmid also contains Tn*5*, indicated by the linkage of the *neo-r* gene to the F' factor. Note that transposition to the F' plasmid does not eliminate the copy of the transposon in the chromosome.

that is identical to the prophage (immunity prevents the phage from killing the cell) but carries a transposable element with an antibiotic-resistance marker. Transposition events can be detected by the production of antibiotic-resistant bacteria that contain new mutations resulting from insertion of the transposon into a gene in the bacterial chromosome. For example, a lac^{+} leu^{+} *neo-s* culture of *E. coli* infected with a *neo-r* transposon can yield both lac^{-} *neo-r* and leu^{-} *neo-r* mutants. If many hundreds of Neo-r bacterial colonies are examined and tested for a variety of nutritional requirements and for the ability to utilize different sugars as a carbon source, then colonies can usually be found bearing a mutation in almost any gene that is examined. This observation indicates that potential insertion sites for transposons are scattered throughout the chromosomes of *E. coli* and other bacterial species.

The end result of the transposition process is the insertion of a transposable element between two base pairs in a recipient DNA molecule. During the insertion process, most transposition events also create a duplication of a short (2–12 nucleotide pairs) sequence of host DNA, which after transposition flanks the insertion site (Section 6.8). The insertion of a transposable element does not require DNA sequence homology or the use of most of the enzymes of homologous recombination, because transposition is not inhibited even in cells in which the major enzyme systems for homologous recombination have been eliminated. Direct nucleotide sequence analysis of many transposable elements and their insertion sites confirms the absence of DNA sequence homology in the recipient with any sequence in the transposable element.

Transposons in Genetic Analysis

Transposons can be employed in a variety of ways in bacterial genetic analysis. Three features make them especially useful for this purpose.

1. Transposons can insert at a large number of potential target sites that are essentially random in their distribution throughout the genome.

2. Many transposons code for their own transposase and require only a small number of host genes for mobility.

3. Transposons contain one or more genes for antibiotic resistance that serve as genetic markers for selection.

Genetic analysis using transposons is particularly important in bacterial species that do not have readily exploited systems for genetic manipulation or large numbers of identified and mapped genes. Many of these species are bacterial pathogens, and transposons can be used to identify and manipulate the disease factors.

The use of a *neo-r* transposable element to identify a particular disease gene in a pathogenic bacterial species is diagrammed in Figure 8.32. In this example, the transposon is introduced into the pathogenic bacterium by a mutant bacteriophage that cannot replicate in the pathogenic host. A variety of other methods of introduction also are possible. After introduction of the transposon and selection on medium containing the antibiotic, the only resistant cells are those in which the transposon inserted into the bacterial chromosome, because the phage DNA in which it was introduced is incapable of replication. Any of a number of screening methods are then used to identify cells in which a particular disease gene became inactivated (nonfunctional) as a result of the transposon inserting into the gene. This method is known as **transposon tagging,** because the gene with the insertion is tagged (marked) with the antibiotic-resistance phenotype of the transposon.

Once the disease gene has been tagged by transposon insertion, it can be used in many ways (for example, in genetic mapping) because the phenotype resulting from the presence of the tagged gene is antibiotic resistance, which is easily identified and selected. The lower part of Figure 8.32 shows how the transposon tag is used to transfer the disease gene into *E. coli*. In the first step, DNA from the pathogenic strain that contains the tagged gene is purified, cut into fragments of suitable size, and inserted into a small plasmid capable of replication in *E. coli*. (Details of these genetic engineering methods are discussed

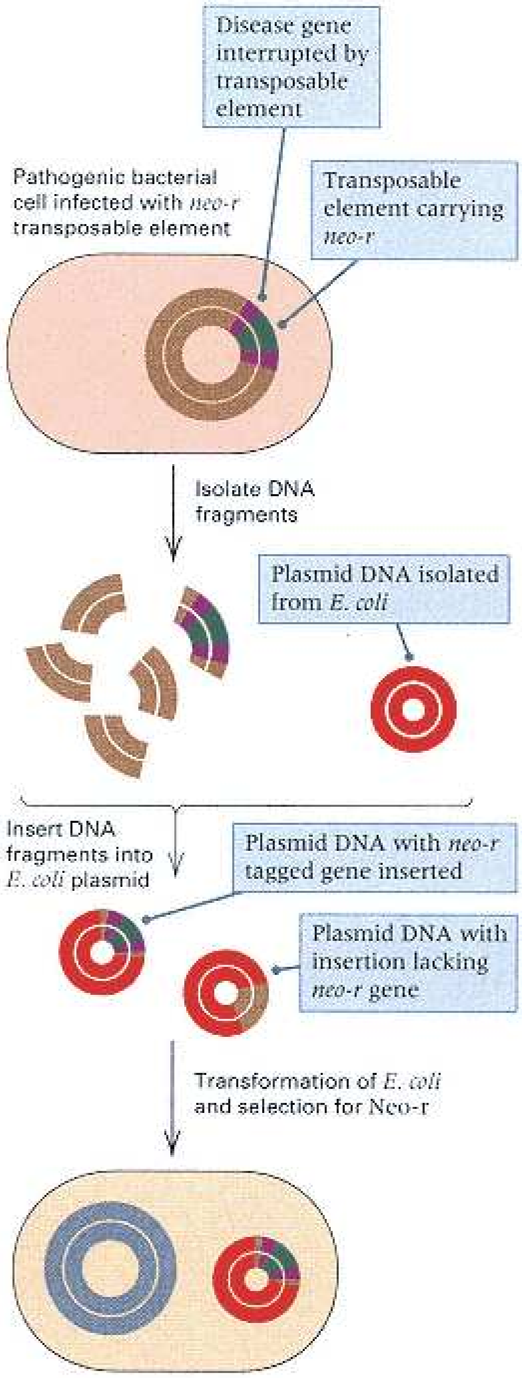


Figure 8.32
Transposon tagging making use of a *neo-r* transposon to mutate a disease gene by inserting into it. The *neo-r* resistance marker is then used to select plasmids containing the disease gene after transfer into *E. coli*, where the gene can be analyzed more safely and easily.

in Chapter 9.) The result is a heterogeneous collection of plasmids, most containing DNA fragments other than the one of interest. However, plasmids that do contain the fragment of interest also contain the transposon tag and so are able to confer antibiotic resistance on the host. The heterogeneous collection of plasmids is introduced into *E. coli* cells by transformation, and the cells are grown on solid medium containing the antibiotic. The only cells that survive contain the desired plasmid, which contains the transposon along with DNA sequences from the desired gene. The transposon-tagged gene is said to be **cloned** in *E. coli*.

The cloning of genes in *E. coli* greatly facilitates further genetic and molecular studies, such as DNA sequencing. The disease gene usually poses no hazard when it is present in *E. coli* for several reasons: (1) More than one gene in the pathogen is usually required to cause disease; and (2) the cloned transposon-tagged gene is inactivated by the insertion; and (3) the disease gene cloned in *E. coli* is usually incomplete or, if complete, is often inactive in this host. In cases in which some danger might result from cloning, potential problems are minimized by using special kinds of *E. coli* strains and following procedures that ensure containment in the laboratory.

Chapter Summary

DNA can be transferred between bacteria in three ways: transformation, transduction, and conjugation. In transformation, free DNA molecules, obtained from donor cells, are taken up by recipient cells; by a recombinational mechanism, a single-stranded segment becomes integrated into the recipient chromosome, replacing a homologous segment.

In conjugation, donor and recipient cells pair and a single strand of DNA is transferred by rolling-circle replication from the donor cell to the recipient. Transfer is mediated by the transfer genes of the F plasmid. When F is present as a free plasmid, it becomes established in the recipient as an autonomously replicating plasmid; if F is integrated into the donor chromosome—that is, if the donor is an Hfr cell—then only part of the donor chromosome is usually transferred, and it can be maintained in the recipient only after an exchange event. About 100 minutes is required to transfer the entire *E. coli* chromosome, but the mating cells usually break apart before transfer is complete. DNA transfer starts at a particular point in the Hfr chromosome (the site of integration of F) and proceeds linearly. The times at which donor markers first enter recipient cells—the times of entry—can be arranged in order, yielding a map of the bacterial genome. This map is circular because of the multiple sites at which an F plasmid integrates into the bacterial chromosome. Occasionally, F is excised from the chromosome in an Hfr cell; aberrant excision, in which one cut is made at the end of F and the other cut is made in the chromosome, gives rise to an F' plasmid, which can transfer bacterial genes. The term *episome* applies to DNA elements, such as the F factor, that can be maintained within the cell either as an autonomously replicating molecule or by integration into the chromosome.

In transduction, a generalized transducing phage infects a donor cell, fragments the host DNA, and packages fragments of host DNA into phage particles. The transducing particles contain no phage DNA but can inject donor DNA into a recipient bacterium. By genetic exchange, the transferred DNA can replace homologous DNA of the recipient, producing a recombinant bacterium called a transductant.

When bacteria are infected with several phages, genetic exchange can take place between phage DNA molecules, generating recombinant phage progeny. Measurement of recombination frequency yields a genetic map of the phage. A common feature of these maps is clustering of phage genes with related function. The genetic map of bacteriophage T4, which infects cells of *E. coli,* is circular, but the genome consists of a linear DNA molecule. The circularity of the genetic map results from a packaging mechanism in which successive "headfuls" of DNA, each slightly longer than the minimum size required to contain all genes, are cleaved from a long concatemer of phage genomes. In a population of phage, the DNA molecules are related as circular permutations, and every DNA molecule has a short duplication at each end. Bacteriophage T4 undergoes efficient recombination. The fine-structure genetic analysis of the *rII* region led to the discovery of hotspots of mutation and to clarification of the term *gene* as meaning (1) the unit of function (cistron) defined by a complementation test, (2) the unit of genetic transmission that participates in recombination, or (3) the unit of genetic change or mutation.

In contrast to virulent phage, such as T4, temperate phages (including phage λ) possess mechanisms for recombining phage and bacterial DNA. A bacterium that contains integrated phage DNA is called a lysogen, the integrated phage DNA is a prophage, and the overall phenomenon is lysogeny. Integration results from site-specific recombination between particular sequences, called attachment sites, in the bacterial and phage DNA. The bacterial and phage attachment sites are not identical but have a short common sequence in which the recombination takes place. Temperate phages circularize their DNA after infection. Because the phage has a single attachment site, which is not terminally located, the order of the prophage genes is a circular permutation of the order of the genes in the phage particle. The integrated prophage DNA is stable, but if the bacterial DNA is damaged, then prophage induction is initiated, the phage DNA is excised, and the lytic cycle of the phage ensues. Defective excision of a prophage yields a specialized transducing phage capable of transducing bacterial genes on either side of the phage attachment site.

Most bacteria possess transposable elements, which are capable of moving from one part of the DNA to another. Transposition does not require sequence homology. Bacterial insertion sequences (IS) are small transposable elements that code for their own transposase and that have inverted-repeat sequences at the ends. Some larger transposons have a composite structure consisting of a central region, often containing one or more antibiotic-resistance genes, flanked by two IS sequences in inverted orientation. Accumulation of such transposons in plasmids gives rise to multiple-drug-resistant R plasmids, which confer resistance to several chemically unrelated antibiotics. Transposable elements are useful in genetic analysis because they can create mutations by inserting within a host gene and interrupting its continuity. Such mutations make possible transposon tagging for genetic analysis of bacterial pathogens and other species.

Key Terms

antibiotic-resistant mutant	episome	lytic cycle
attachment site	excisionase	merodiploid
auxotroph	F plasmid	minimal medium

bacterial attachment site	F' plasmid	nonselective medium
bacteriophage	generalized transduction	nucleoid
carbon-source mutant	Hfr strain	nutritional mutant
circular permutation	hotspot	partial diploid
cistron	immunity	pathogen
clone	insertion sequence	phage
cloned gene	integrase	phage attachment site
cohesive end	interrupted-mating technique	phage repressor
conjugation	IS element	plaque
cotransduction	lysis	plasmid
cotransformation	lysogen	prophage
counterselected marker	lysogenic cycle	prophage induction

prototroph	terminal redundancy	transposon
R plasmid	time of entry	transposon tagging
selected marker	transducing phage	virulent phage
selective medium	transduction	
site-specific recombination	transformation	
specialized transduction	transposase	
temperate phage	transposition	

Review the Basics

• What are three phenotypes of mutations that are particularly useful for genetic analysis in bacteria.

• What is the difference between an F plasmid and an F' plasmid?

• What is the difference between a selected marker and a counterselected marker? Why are both necessary in analyzing the progeny of an Hfr × F^- mating?

• Distinguish between generalized and specialized transduction.

• Transfer of DNA in conjugation is always accompanied by DNA replication. What is the mode of replication and in which cell does it take place?

• In a mating between Hfr and F^- cells, why does the recipient cell usually remain F^-?

• What prevents a bacterial cell that is lysogenic for some temperate bacteriophage from being reinfected with a phage of the same type?

Guide to Problem Solving

Problem 1: In an Hfr × F^- cross in *E. coli*, the Hfr genotype is a^+ b^+ *str-s* and the F^- genotype is a^- b^- *str-r*. Recombinants of genotype a^+ *str-r* are selected, and almost all of them prove to be b^+. How can this result be explained?

Answer: Because most colonies that are a^+ are also b^+, it is clear that a and b are close together in the bacterial chromosome.

Problem 2: Consider the following Hfr × F^- cross.

Hfr genotype: a^+ b^+ c^+ *str-s*

F^- genotype: a^- b^- c^- *str-r*

The order of gene transfer is *a b c,* with a transferred at 9 minutes, *b* at 11 minutes, *c* at 30 minutes, and *str-s* at 40 minutes. Recombinants are selected by plating on a medium that lacks particular nutrients and contains streptomycin. Which of the following statements are true, and why? (*Note:* Genetic markers introduced before a selected marker, and not closely linked to it, are incorporated into the recipient genome about 50 percent of the time.)

(a) a^+ *str-r* colonies≈ b^+ *str-r* colonies.

(b) a^+ *str-r* colonies > c^+ *str-r* colonies.

(c) b^+ *str-r* colonies < c^+ *str-r* colonies.

(d) Among colonies selected for

b^+ *str-r*, a^+ b^+ *str-r* ≈ a^- b^+ *str-r.*

(e) Among colonies selected for

c^+ *str-r*, a^+ c^+ *str-r* ≈ a^- c^+ *str-r.*

(f) Among colonies selected for

a^+ c^+ *str-r*, a^+ b^+ c^+

str-r < a^+ b^- c^+ *str-r.*

Answer:

(a) True, because *a* and *b* are only 2 minutes apart, and if chromosome transfer has gone far enough to include *a*, it will usually also include *b.*

(b) True, because *a* enters long before *c*.

(c) False, because *b* enters long before *c*.

(d) False, because *a* and *b* are only 2 minutes apart, and strains selected for b^+ will usually also contain a^+.

(e) True, because *a* is transferred long before *c*, so the probability of incorporation of a^+ is about 50 percent.

(f) False, because *a* and *b* are only 2 minutes apart, and strains selected for a^+ will usually also contain b^+.

Problem 3: A temperate bacteriophage has the gene order *a b c d e f g h*, whereas the order of genes in the prophage present in the bacterial chromosome is *g h a b c d e f*. What information does this give you about the location of the attachment site in the phage?

Answer: The gene order in the prophage is circularly permuted with respect to the gene order in the phage, and the attachment site defines the site of breakage and rejoining with the bacterial chromosome. Because adjacent genes *f* and *g* are separated by prophage integration, the attachment site must be located between *f* and *g*.

Problem 4: An Hfr *str-s* strain that transfers genes in alphabetical order

$$a^+b^+c^+ \ . \ . \ .x^+y^+z^+$$

with *a* transferred early and *z* a terminal marker, is mated with F^- z^- *str-r* cells. The mating mixture is agitated violently 15 minutes after mixing to break apart conjugating cells and then plated on a medium that lacks nutrient *Z* and contains streptomycin. The *z* gene is far from the *str* gene. The yield of z^+ *str-r* colonies is about one per 10^7 Hfr cells. What are two possible modes of origin and genotypes of such a colony? How could the two possibilities be distinguished?

Answer: The time of 15 minutes is too short to allow the transfer of a terminal gene. Thus, one explanation for a z^+ *str-r* colony is that aberrant excision of the F factor occurred in an Hfr cell, yielding an F' plasmid carrying terminal markers. When the F' z^+ plasmid is transferred into an F^- recipient, the genotype of the resulting cell would be F' z^+/z^- *str-r.* Alternatively, a z^+ *str-r* colony can originate by reverse mutation of z^- (ρ)z^+ in the F^- recipient. The possibilities can be distinguished because the F' z^+/z^- *str-r* cell is able to transfer the z^+ marker to other cells.

Analysis and Applications

8.1 In the specialized transduction of a gal^- bio^- strain of *E. coli* using bacteriophage λ from a gal^+ bio^+ lysogen, what medium would select for gal^+ transductants without selecting for bio^+?

8.2 Given that bacteriophage λ has a genome size of 50 kilobase pairs, what is the approximate genetic length of λ prophage in minutes? (*Hint:* There are 100 minutes in the entire *E. coli* genetic map.)

8.3 How could you obtain a lysate of bacteriophage P1 in which some of the transducing particles contained bacteriophage λ?

8.4 A temperate bacteriophage has gene order *A B C att D E F.* What is the gene order in the prophage?

8.5 Why are λ specialized transducing particles generated only by inducing a lysogen to produce phages rather than by lytic infection?

8.6 If leu^+ *str-r* recombinants are desired from the cross Hfr leu^+ *str-s* × F^- leu^- *str-r*, on what kind of medium should the matings pairs be plated? Which are the selected and which the counterselected markers?

8.7 How many plaques can be formed by a single bacteriophage particle? A bacteriophage adsorbs to a bacterium in a liquid growth medium. Before lysis, the infected cell is added to a suspension of cells that are plated on solid medium to form a lawn. How many plaques will result?

8.8 Phage T2 (a relative of T4) normally forms small, clear plaques on a lawn of *E. coli* strain B. Another strain of *E. coli*, called B/2, is unable to adsorb T2 phage particles, and no plaques are formed. T2*h* is a host-range mutant capable of adsorbing to *E. coli* B and to B/2, and it forms normal-looking plaques. If *E. coli* B and the mutant B/2 are mixed in equal proportions and used to generate a lawn, how can plaques made by T2 and T2*h* be distinguished by their appearance?

8.9 If 10^6 phages are mixed with 10^6 bacteria and all phages adsorb, what fraction of the bacteria remain uninfected?

8.10 An Hfr strain transfers genes in alphabetical order. When tetracycline sensitivity is used for counterselection, the number of h^+ *tet-r* colonies is 1000-fold lower than the number of h^+ *str-r* colonies found when streptomycin sensitivity is used for counterselection. Suggest an explanation for the difference.

8.11 An Hfr strain transfers genes in the order *a b c.* In an Hfr a^+ b^+ c^+ *str-s* × F^- a^- b^- c^- *str-r* mating, do all b^+ *str-r* recombinants receive the a^+ allele? Are all b^+ *str-r* recombinants also a^+? Why or why not?

8.12 If the genes in a bacterial chromosome are in alphabetical order and an Hfr cell transfers genes in the order *g h* *i* . . . *d e f*, what types of F' plasmids could be derived from the Hfr?

8.13 In an Hfr *lac*$^+$ *met*$^+$ *str-s* $\times$ F$^-$ *lac*$^-$ *met*$^-$ *str-r* mating, *met* is transferred much later than *lac*. If cells are plated on minimal medium containing glucose and streptomycin, what

Chapter 8 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at
http://www.jbpub.com/genetics
Select the link to ***Genetics: Principles and Analysis*** and then choose the link to ***GeNETics on the web.*** You will be presented with a chapter-by-chapter list of highlighted keywords.
GeNETics EXERCISES
To view a list of keywords corresponding to the bold terms in the four GeNETics Exercises below, visit the *GeNETics on the Web* home page and choose the link for Chapter 8, then select "GeNETics Exercises." Clicking on a keyword will transfer you to a web site that contains information necessary to complete the exercise. Each exercise suggests a specific, written eport that makes use of the information available at that site.
1. Strains of *E. coli* as well as an updated **genetic map** are available at this keyword site. Browse the Working Map and find the *gal* genes, the bacterial lambda attachment site (called *attLAM),* and the *bio* genes around 17 minutes. If assigned to do so, use the WebServer at the site to find out how many bacterial attachment sites *(att)* have been identified; make a list of the phage that have attachment sites in the *E. coli* chromosome.
2. A listing of available **Hfr** strains can be found by searching at this site. How many entries are retrieved? Clicking on any entry—say, *3310 Pairs Hfr*—will open a link to more detailed information. The *PO#* will give you the origin of transfer of the Hfr (in this case, 96.9 minutes) and the direction of transfer (1 = clockwise, 0 = counterclockwise). Check out the "classic" Hfr strains, Hfr Hayes and Hfr Cavalli. If assigned to do so, choose any other 10 Hfr strains, and draw a diagram of the *E. coli* genetic map showing the name, origin, and direction of transfer of each of the strains you selected.
3. The keyword **genes and metabolism** will link you with an encyclopedia of many of the metabolic pathways present in this model organism. You can choose any of an extensive list of pathways and also view the molecular structures of the intermediates. If assigned to do so, outline the metabolic pathway for the

(text box continued on next page)

fraction of the cells are expected to be *lac*$^+$? If cells are plated on minimal medium containing lactose, methionine, and streptomycin, what fraction of the cells are expected to be *met*$^+$?

8.14 In the mating

Hfr *met*$^-$ *his*$^+$ *leu*$^+$ *trp*$^+$ × F$^-$ *met*$^+$ *his*$^-$ *leu*$^-$ *trp*$^-$

the *met* marker is known to be transferred very late. After a short time, the mating cells are interrupted and the cell suspensions plated on four different growth media. The amino acids in the growth media and the number of colonies observed on each are as follows:

histidine + tryptophan	250 colonies

histidine + leucine	50 colonies
leucine + tryptophan	500 colonies
histidine	10 colonies

What is the purpose of the *met*⁻ mutation in the Hfr strain? What is the order of transfer of the genes? Why is the number of colonies so small for the medium containing only histidine?

8.15 Bacterial cells of genotype pur^- pro^+ his^+ were transduced with P1 bacteriophage grown on bacteria of genotype pur^+ pro^- his^-. Transductants containing pur^+ were selected and tested for the unselected markers *pro* and *his.* The numbers of pur^+ colonies with each of four genotypes are as follows:

pro^+ his^+	102
pro^- his^+	25
pro^+ his^-	160
pro^- his^-	1

What is the gene order?

8.16 For generalized transduction, the theoretical relationship between map distance in minutes and frequency of cotransduction is

$$\text{map distance} = 2 - 2 \times (\text{cotransduction frequency})^{1/3}$$

(a) What map distance in minutes corresponds to 75 percent cotransduction? To 50 percent cotransduction? To 25 percent cotransduction?

(b) What is the cotransduction frequency between genetic markers separated by half a minute? One minute? Two minutes? Greater than two minutes?

(c) If genes *a*, *b,* and *c* are in alphabetical order and the cotransduction frequencies are 30 percent for *a–b* and

(text box continued from previous page)

synthesis of the vitamin biotin, and sketch the molecular structure of biotin.

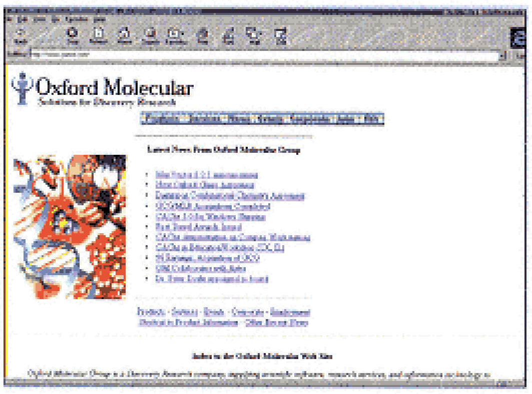
4. A highly annotated genetic map of **bacteriophage** λ can be found at this keyword site. You can get detailed information about any of the phage genes. To find the location of, for example, *int* and *xis*, select *GeneList* and then the genes. From the main menu, choose DNA and then GenBank report to access the entire annotated lambda DNA sequence. Using this report, find the nucleotide sequences corresponding to the site, *attP*, at which the phage recombines with the bacterial chromosome. Where is this site in relation to that of *int* and *xis?* If assigned to do so, draw a diagram of a portion of phage DNA, to the correct scale, showing the locations of the open reading frames for *int* and *xis* and the location of *attP*.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 8, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 8.



10 percent for *b–c*, what is the expected cotransduction frequency between *a* and *c?*

8.17 An ampicillin-resistant (Amp-r) strain of *E. coli* isolated from nature is infected with λ, and a suspension of phage progeny is obtained. This phage suspension is used to infect a culture of laboratory *E. coli* that is lysogenic for λ, sensitive to the antibiotic, and unable to undergo homologous recombination. Because the λ repressor is present in the lysogen, infecting λ molecules cannot replicate and are gradually diluted out of the culture by growth and continued division of the cells. However, rare Amp-r cells are found among the lysogens. Explain how they might have arisen.

8.18 On continued growth of a culture of a λ lysogen, rare cells become nonlysogenic by spontaneous loss of the prophage. Such loss is a result of prophage excision without subsequent development of the phage. The lysogen is said to have been "cured." A λ phage that carries a gene for tetracycline resistance is used to lysogenize a Tet-s cell. At a later time, a cell is isolated that is Tet-r but no longer contains a λ prophage. Explain how the cured cell could remain Tet-r.

8.19 A time-of-entry experiment is carried out in the mating

Hfr a^+ b^+ c^+ d^+ *str-s* × F^- a^- b^- c^- d^- *str-r*

where the spacing between the genes is equal. The data in the accompanying table are obtained. What are the times of entry of each gene? Suggest one possible reason for the low frequency of d^+ *str-r* recombinants.

Time of mating in minutes	**Number of recombinants of indicated genotype per 100 Hfr**			
	a^+ ***str-r***	b^+ ***str-r***	c^+ ***str-r***	d^+ ***str-r***
0	0.01	0.006	0.008	0.0001
10	5	0.1	0.01	0.0004
15	50	3	0.1	0.001
20	100	35	2	0.001
25	105	80	20	0.1
30	110	82	43	0.2
40	105	80	40	0.3
50	105	80	40	0.4
60	105	81	42	0.4
70	103	80	41	0.4

Challenge Problems

8.20 You have reason to believe that a number of *rII* mutants of bacteriophage T4 are deletion mutations. You cross them in all possible combinations in *E. coli* strain B and plate them on *E. coli* strain K12(λ) to determine whether r^+ recombinants are formed. The formation of r^+ recombinants indicates that the mutations can recombine and so, if they are deletions, they must be nonoverlapping. The results are given in the accompanying table, in which *a* through *f* indicate an *rII* mutation and + indicates the formation of r^+ recombinant progeny in the cross.

	a	*b*	*c*	*d*	*e*	*f*
a	−	−	−	−	−	−
b		−	−	+	+	−
c			−	+	−	+
d				−	−	+
e					−	+
f						−

Assemble a deletion map for these mutations, using a line to indicate the DNA segment that is deleted in each mutant.

8.21 Seven *rII* mutants, *t* through *z*, thought to carry point mutations, are crossed to the six deletion mutants in Challenge Problem 8.20 and scored for the ability to produce r^+ recombinants. The results are given in the accompanying table.

	a	*b*	*c*	*d*	*e*	*f*
t	−	−	−	+	+	+
u	−	+	−	+	+	+
v	+	+	+	−	+	+
w	+	+	+	+	+	−
x	−	+	+	+	+	−
y	−	−	+	+	+	−
z	−	+	+	−	−	+

Using the deletion end points to define genetic intervals along the *rII* gene, position each point mutation within an interval. Refine the deletion end points if required by the data.

8.22 You cross the *rII* point mutations *t* through *z* in Challenge Problem 8.21 in all possible combinations in K12 (λ) to assess complementation. The results are shown in the accompanying complementation matrix.

	t	*u*	*v*	*w*	*x*	*y*	*z*
t	−	−	+	−	−	−	+
u		−	+	−	−	−	+
v			−	+	+	+	−
w				−	−	−	+
x					−	−	+
y						−	+
z							−

The mutations *v* and *z* complement deletion *c* but not deletion *e*. The mutations *v* and *z* also fail to complement a mutant known to be defective for *rIIB* function but not for *rIIA* function. Which mutants are in the *rIIA* cistron and which in the *rIIB* cistron? Where does the boundary between the two genes lie? Assemble a complementation matrix for the deletion mutants, filling in all the squares if possible.

8.23 Bacteriophage 363 can be used for generalized transduction of genetic markers in *E. coli.* Francois Jacob used phage 363 to demonstrate that the lysogenic state of bacteriophage λ can be genetically transferred from one strain to another so long as the experiment is carried out at 20°C, a temperature at which λ phage do not multiply. Jacob studied four strains of bacteria:

A: $thr^- \; lac^- \; gal^-$

B: $thr^- \; lac^- \; gal^-$ (λ)

C: $thr^+ \; lac^+ \; gal^+$

D: $thr^+ \; lac^+ \; gal^+$ (λ)

In these designations, the symbol (λ) means that the strain is lysogenic for phage λ. Jacob made two kinds of crosses. Phage grown on strain D were used to transduce strain A, and phage grown on strain C were used to transduce strain B. The results are given in the table below.

Give a genetic explanation of these results. What genetic element is being transduced to give or to remove lysogeny?

Problem 8.23:

		Thr^+		**Lac^+**		Gal^+	
Donor strain	**Recipient strain**	**Colonies tested**	**Colonies lysogenic**	**Colonies tested**	**Colonies lysogenic**	**Colonies tested**	**Colonies lysogenic**
D	A	400	0	400	0	400	24
C	B	400	400	400	400	400	368

Further Reading

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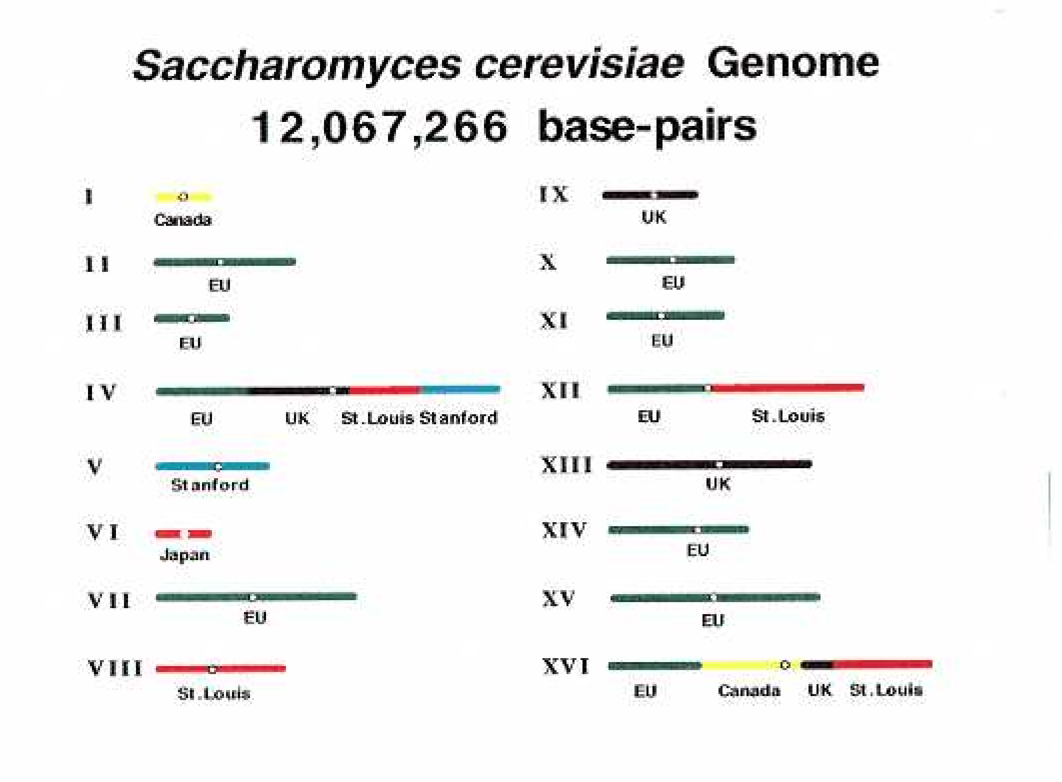


Who sequenced what? Shown are the locations of the research groups whose collaborative effort resulted in the complete sequence of the genome of the yeast, Saccharomyces cerevisiae—the first eukaryotic genome sequenced in its entirety. EU stands for a consortium of laboratories in Europe, and UK is the United Kingdom.
[Courtesy of H. Mark Johnston.]

Chapter 9— Genetic Engineering and Genome Analysis

CHAPTER OUTLINE

PRINCIPLES

• In recombinant DNA (gene cloning), DNA fragments are isolated, inserted into suitable vector molecules, and introduced into bacteria or yeast cells where they are replicated.

• Recombinant DNA is widely used in research, medical diagnostics, and the manufacture of drugs and other commercial products.

• In reverse genetics, gene function is analyzed by introducing an altered gene carrying a designed mutation into the germ line of an organism.

• Specialized methods for manipulating and cloning large fragments of DNA—up to one million base pairs—have made it possible to develop physical maps of the DNA in complex genomes.

• Large-scale automated DNA sequencing has resulted in the complete sequence of the genomes of several species of bacteria, the yeast *Saccharomyces cerevisiae* (the first eukaryote completely sequenced), and most of the genome of the nematode *Caenorhabditis elegans*. Large-scale sequencing projects are underway in many other organisms of genetic or commercial interest.

CONNECTIONS

CONNECTION: Hello, Dolly!

Ian Wilmut, Anagelika E. Schnieke, Jim McWhir, Alex J. Kind, and Keith H. S. Campbell 1997
Viable offspring derived from fetal and adult mammalian cells

CONNECTION: YAC-ity YAC

David T. Burke, Georges F. Carle and Maynard V. Olson 1987
Cloning of large segments of exogenous DNA into yeast by means of artificial chromosome vectors

Genetic analysis is a powerful experimental approach for understanding the mechanisms and evolution of biological processes. Genetics has played an equally important role in the development of techniques for the deliberate manipulation of biological systems to create organisms with novel genotypes and phenotypes. These organisms may be created for experimental studies or for economic reasons, such as to obtain superior varieties of domesticated animals and crop plants or to create organisms that produce molecules used in the treatment of human disease. The traditional foundations of genetic manipulation have been mutation and recombination. These processes are essentially random; selective procedures, often quite complex, are required to identify organisms with the desired characteristics among the many other types of organisms produced. Since the 1970s, techniques have been developed in which the genotype of an organism can instead be modified in a directed and predetermined way. This approach is called **recombinant DNA** technology, **genetic engineering,** or **gene cloning.** It entails isolating DNA fragments, joining them in new combinations, and introducing the recombined molecules back into a living organism. Selection of the desired genotype is still necessary, but the probability of success is usually many orders of magnitude greater than with earlier procedures. The basic technique is quite simple: Two DNA molecules are isolated and cut into fragments by one or more specialized enzymes, and then the fragments are joined together *in any desired combination* and introduced back into a cell for replication and reproduction. Recombinant DNA procedures complement the traditional methods of plant and animal breeding in supplying new types of genetic variation for improvement of the organisms.

Much of the current interest in genetic engineering is motivated by its many practical applications. Among these are: (1) the isolation of a particular gene, part of a gene, or region of a genome; (2) the production of a particular RNA or protein molecule in quantities formerly unobtainable; (3) improved efficiency in the production of biochemicals (such as enzymes and drugs) and commercially important organic chemicals; (4) the creation of organisms with particular desirable characteristics (for example, plants that require less fertilizer and animals that exhibit faster growth rates or increased resistance to disease); and (5) potentially, the correction of genetic defects in higher organisms, including human beings. Some specific examples will be considered later in this chapter.

9.1—
Restriction Enzymes and Vectors

The technology of recombinant DNA relies heavily on restriction enzymes, which are discussed in Section 5.7 with reference to their site-specific DNA cleavage in the generation of DNA fragments with defined ends that can be separated by gel electrophoresis, isolated, sequenced, or otherwise manipulated. In recombinant DNA, after the DNA has been cleaved by a restriction enzyme at specific sites, the restriction fragments are ligated into a vector molecule capable of replication in a living host organism, typically *E. coli* or yeast. In this section, we take a closer look at the special features of restriction enzymes that make them so useful in DNA cloning. Some of the most popular types of vectors used in recombinant DNA are also examined.

Production of Defined DNA Fragments

DNA fragments are usually obtained by the treatment of DNA samples with restriction enzymes. A **restriction enzyme** is a type of nuclease that cleaves a DNA duplex wherever the DNA molecule contains a particular short sequence of nucleotides matching the **restriction site** of the enzyme (see Section 5.7). Most restriction sites consist of four or six nucleotides, within which the restriction enzyme makes a break in each DNA strand. Each break creates a free 3' hydroxyl (3'-OH) and a free 5' phosphate (5'-P) at the site of breakage. Several hundred restriction enzymes, each with a different restriction site, have been isolated from microorganisms. Most restriction sites are symmetrical in the sense that the recognition sequence is identical in both strands of the DNA duplex. For example, the restriction enzyme *Eco*RI, isolated from *Escherichia coli*, has the restriction site 5'-GAATTC-3' and cuts the strand between the G and the A. The sequence of the other strand is 3'-CTTAAG-5', which is identical but is written with the 3' end at the left. The term **palindrome** is used to denote this type of symmetry.

Soon after restriction enzymes were discovered, observations with the electron microscope indicated that the fragments produced by many restriction enzymes could spontaneously form circles. The circles could be made linear again by heating; however, if after circularization they were treated with *E. coli* DNA ligase, which joins 3'-OH and 5'-P groups (Section 5.6), then the ends became covalently joined. This observation was the first evidence for three important features of restriction enzymes:

• Restriction enzymes make breaks in palindromic sequences.

• The breaks need not be directly opposite one another in the two DNA strands.

• Enzymes that cleave the DNA strands asymmetrically generate DNA fragments with complementary ends.

These properties are illustrated for *Eco*RI in Figure 9.1.

Most restriction enzymes are like *Eco*RI in that the cuts in the DNA strands are staggered, producing single-stranded ends called **sticky ends** that can adhere to each other because they contain complementary nucleotide sequences. Some restriction

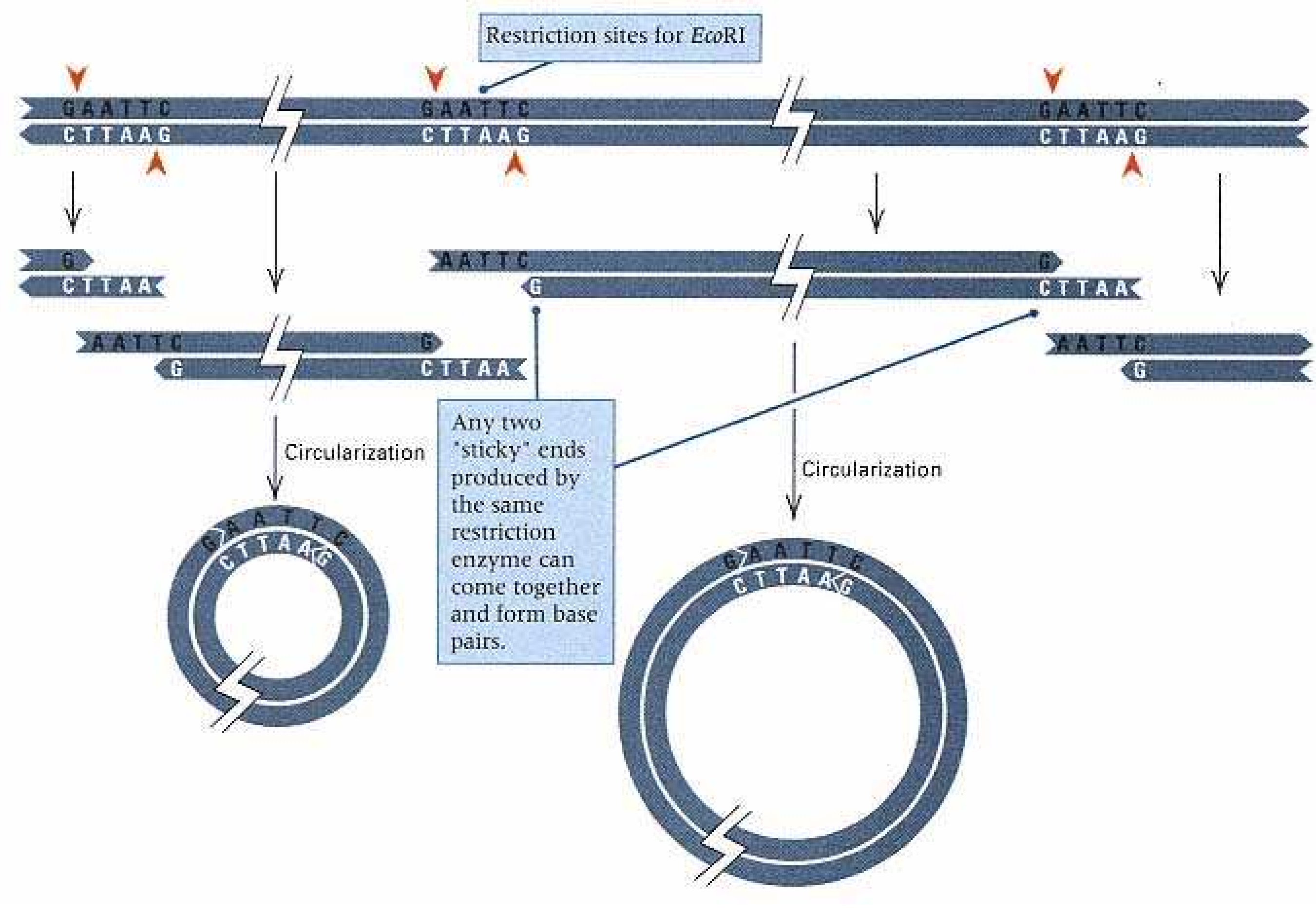


Figure 9.1
Circularization of DNA fragments produced by a restriction enzyme. The red arrowheads indicate the *Eco*RI cleavage sites.

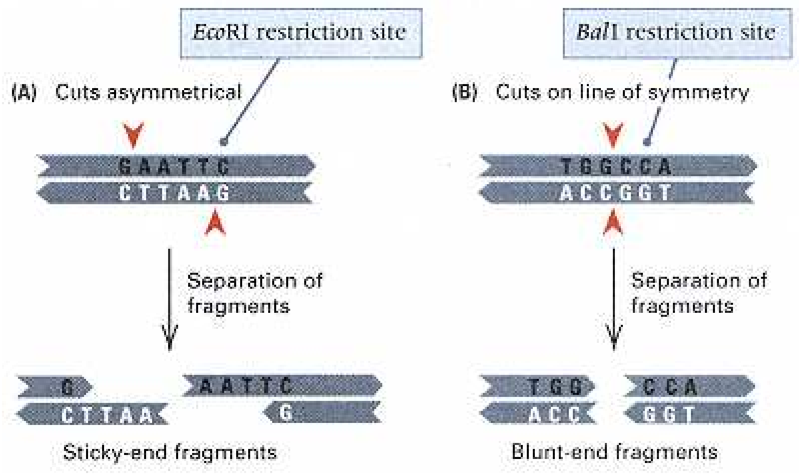


Figure 9.2
Two types of cuts made by restriction enzymes. The red arrowheads indicate the cleavage sites. (A) Cuts made in each strand at an equal distance from the center of symmetry of the restriction site. In this example with the enzyme *Eco*RI, the resulting molecules have complementary 5' overhangs. Other restriction enzymes produce fragments with complementary 3' overhangs. (B) Cuts made in each strand at the center of symmetry of the restriction site. The products have blunt ends. The specific enzyme in this example is *Bal*I.

enzymes, including *Eco*RI, leave a single-stranded overhang at the 5' end (Figure 9.2A); others leave a 3' overhang. A number of restriction enzymes cleave both DNA strands at the center of symmetry, forming **blunt ends.** Figure 9.2B shows the blunt ends produced by the enzyme *Bal*I. Blunt ends also can be ligated by DNA ligase. However, whereas ligation of sticky ends recreates the original restriction site, any blunt end can join with any other blunt end and not necessarily create a restriction site.

Most restriction enzymes recognize their restriction sequence without regard to the source of the DNA:

> DNA fragments obtained from one organism will have the same sticky ends as the fragments from another organism if they have been produced by the same restriction enzyme.

This principle is one of the foundations of recombinant DNA technology.

Because most restriction enzymes recognize a unique sequence, the number of cuts made in the DNA of an organism by a particular enzyme is limited. For example, an *E. coli* DNA molecule contains 4.7×10^6 base pairs, and any enzyme that cleaves a six-base restriction site will cut the molecule into about a thousand fragments because, with equal and random frequencies of each of the four nucleotides, a six-base restriction site is expected to occur, on average, every $4^6 = 4096$ base pairs. Mammalian nuclear DNA, because the genome is much larger than in *E. coli,* would be cut into about a million fragments. These large numbers still are small compared with the number that would be produced by random cuts. Of special interest are cleavage products from smaller DNA molecules, such as viral or plasmid DNA, which may have from only one to ten sites of cutting (or possibly no sites) for any particular enzyme. Plasmids containing a single site for a particular enzyme are especially valuable, as we will see shortly.

Recombinant DNA Molecules

A **vector** is a DNA molecule into which a DNA fragment can be cloned and which can replicate in a suitable host organism. In genetic engineering, a particular DNA

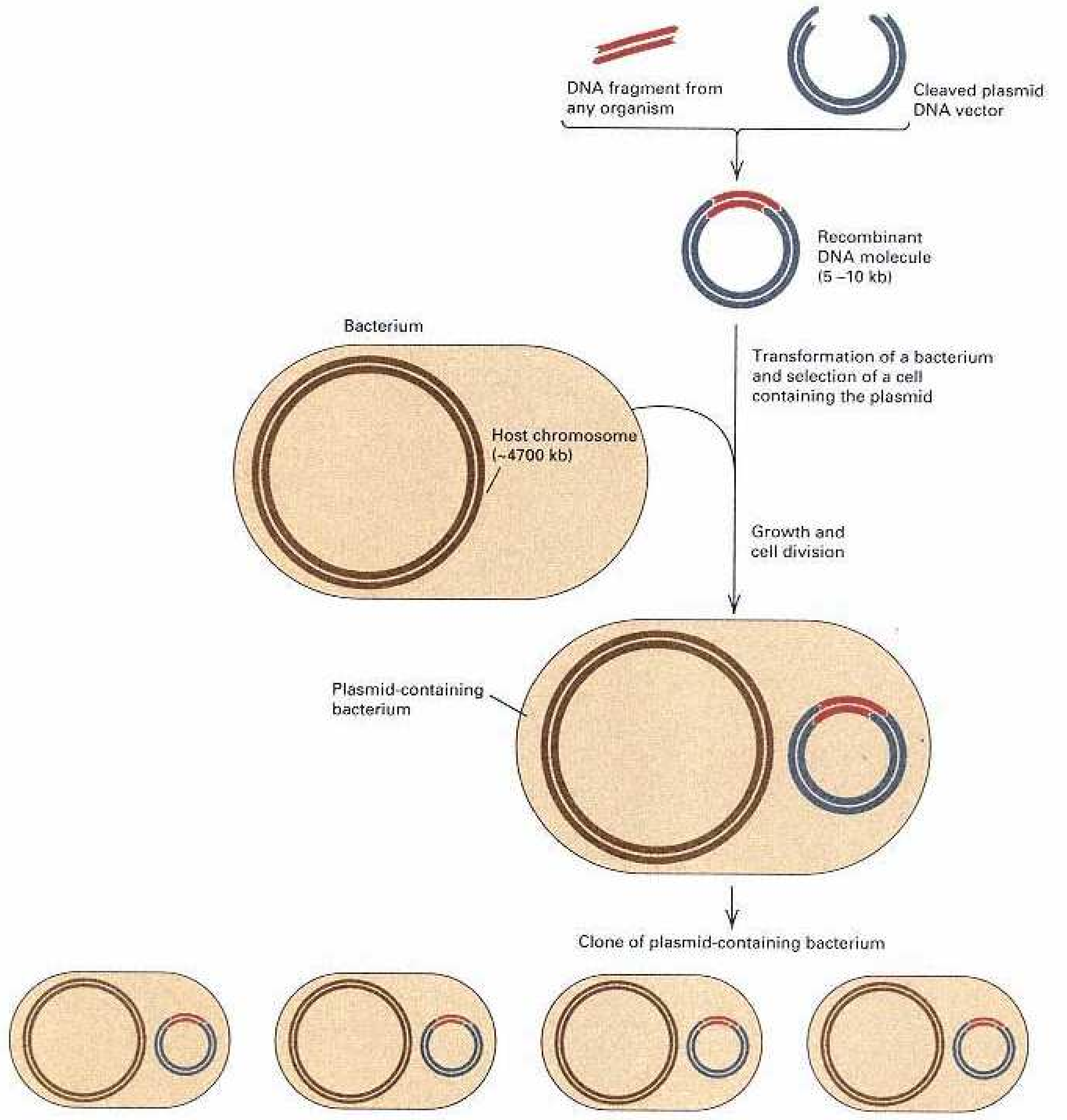


Figure 9.3
An example of cloning. A fragment of DNA from any organism is joined to a cleaved plasmid. The recombinant plasmid is then used to transform a bacterial cell, and thereafter, the foreign DNA is present in all progeny bacteria. The bacterial host chromosome is not drawn to scale. It is typically about 1000 times larger than the plasmid.

segment of interest is joined to a vector DNA molecule. By a transformation procedure very similar to that used by Avery, MacLeod, and McCarty in proving that DNA is the genetic material (Chapter 1), this recombinant molecule is placed in a cell in which replication can take place (Figure 9.3). When a stable transformant has been isolated, the genes or DNA sequences linked to the vector are said to be **cloned.** In the following section, several types of vectors are described.

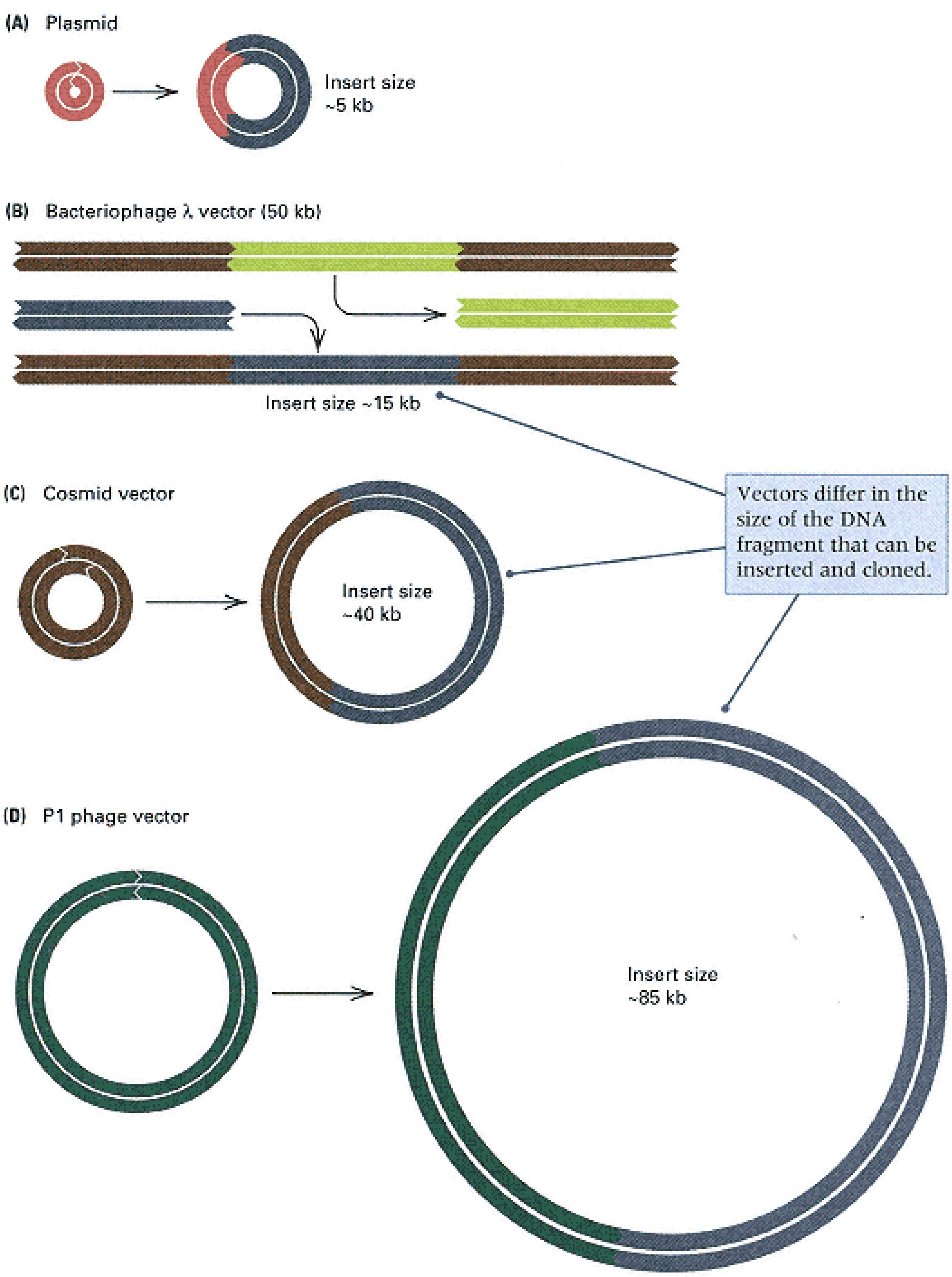


Figure 9.4
Common cloning vectors for use with *E. coli*, drawn approximately to scale. (A) Plasmid vectors are ideal for cloning relatively small fragments of DNA. (B) Bacteriophage λ vectors contain convenient restriction sites for removing the middle section of the phage and replacing it with the DNA of interest. (C) Cosmid vectors are useful for cloning DNA fragments up to about 40 kb; they can replicate as plasmids but contain the cohesive ends of phage λ and so can be packaged in phage particles. (D) Vectors based on the bacteriophage Pl are used for cloning DNA fragments of about 85 kb.

Plasmid, Lambda, Cosmid, and P1 Vectors

The most generally useful vectors have three properties:

• The vector DNA can be introduced into a host cell.

• The vector contains a replication origin so that it can replicate inside the host cell.

• Cells containing the vector can usually be selected in a straightforward manner, most conveniently through a selectable phenotype, such as antibiotic resistance, conferred on the host by genes present in the vector.

At present, the most commonly used vectors are *E. coli* plasmids and derivatives of the bacteriophages λ and M13. Other plasmids and viruses also have been developed for cloning into cells of animals, plants, and bacteria. Recombinant DNA can be detected in host cells by means of genetic features or particular markers made evident in the formation of colonies or plaques. Plasmid and phage DNA can be introduced into cells by a transformation procedure in which cells gain the ability to take up free DNA by exposure to a $CaCl_2$ solution. Recombinant DNA can also be introduced into cells by a kind of electrophoretic procedure called **electroporation.** After introduction of the DNA, the cells containing the recombinant DNA are plated on a solid medium. If the added DNA is a plasmid, then colonies that consist of bacterial cells containing the recombinant plasmid are formed, and the transformants can usually be detected by the phenotype that the plasmid confers on the host cell. For example, plasmid vectors typically include one or more genes for resistance to antibiotics, and plating the transformed cells on a selective medium with antibiotic prevents all but the plasmid-containing cells from growing (Section 8.1). Alternatively, if the vector is phage DNA, the infected cells are plated in the usual way to yield plaques. Variants of these procedures are used to transform animal or plant cells with suitable vectors, but the technical details may differ considerably.

Four types of vectors commonly used for cloning into *E. coli* are illustrated in Figure 9.4. Plasmids (Figure 9.4A) are most convenient for cloning relatively small DNA fragments (5–10 kb). Somewhat larger fragments can be cloned with bacteriophage λ (Figure 9.4B). The wildtype phage is approximately 50 kb in length, but the central portion of the genome is not essential for lytic growth and can be removed and replaced with donor DNA. After the donor DNA has been ligated in place, the recombinant DNA is packaged into mature phage *in vitro,* and the phage is used to infect bacterial cells. However, to be packaged into a phage head, the recombinant DNA must be neither too large nor too small, which means that the donor DNA must be roughly the same size as the portion of the λ genome that was removed. Most λ cloning vectors accept inserts ranging in size from 12 to 20 kb. Still larger DNA fragments can be inserted into cosmid vectors (Figure 9.4C). These vectors can exist as plasmids, but they also contain the cohesive ends of phage λ (Section 8.6), which enables them to be packaged into mature phages. The size limitation on cosmid inserts usually ranges from 40 to 45 kb. A vector for cloning even larger fragments of DNA is a vector derived from the bacteriophage P1 (Figure 9.4D). The genome of the wildtype P1 phage is approximately 100 kb, but only about 15 kb are required for the genome to replicate and persist as a plasmid inside a suitable host bacterium. Therefore, P1 vectors that contain only the minimal 15 kb of essential DNA can be used to clone DNA fragments averaging approximately 85 kb.

9.2— Cloning Strategies

In genetic engineering, the immediate goal of an experiment is usually to insert a *particular* fragment of chromosomal DNA into a plasmid or a viral DNA molecule. One strategy for isolating DNA from a known gene is *transposon tagging,* which was discussed in Chapter 8. In transposon tagging, the target gene to be cloned is first inactivated by means of a transposon insertion. Because the insertion inactivates the gene, the mutant phenotype and the position of the mutation in the genetic map indicate that the correct gene has been hit. Because the transposon contains an antibiotic-resistance gene, the resistance affords a direct selection for any vector molecules that contain the transposon. The DNA flanking the transposon in the recombinant clone must necessarily derive from the target gene.

Although transposon tagging is an important approach in gene cloning, the method is feasible only in organisms whose genetics is already highly developed, such as maize and *Drosophila.* Transposon tagging is not possible in many organisms of interest, including human beings. Even in organisms in which the technique is available, particular genes may not be clonable with the approach because the gene contains no target sites for the transposon. It was therefore necessary to develop a set of alternative strategies for cloning genes. These are considered next.

Joining DNA Fragments

The circularization of restriction fragments having terminal single-stranded, "sticky" ends that have complementary bases was described in Section 9.1. Because a particular restriction enzyme produces fragments with *identical* sticky ends, without regard for the source of the DNA, fragments from DNA molecules isolated from two different organisms can be joined, as shown in Figure 9.5. In this example, the restriction enzyme *Eco*RI is used to digest DNA from any organism of interest and to cleave a bacterial plasmid that contains only one *Eco*RI restriction site. The donor DNA is digested into many fragments (one of which is shown) and the plasmid into a single linear fragment. When the donor fragment and the linearized plasmid are mixed, recombinant molecules can form by base pairing between the complementary single-stranded ends. At this point, the DNA is treated with DNA ligase to seal the joints, and the donor fragment becomes permanently joined in a combination that may never have existed before. The ability to join a donor DNA fragment of interest to a vector is the basis of the recombinant DNA technology.

Joining sticky ends does not always produce a DNA sequence that has functional genes. For example, consider a linear DNA molecule that is cleaved into four fragments—A, B, C, and D—whose sequence in the original molecule was A B C D. Reassembly of the fragments can occasionally yield the original molecule, but if B and C have the same pair of sticky ends, then molecules with different arrangements of the fragments are also formed (Figure 9.6), including arrangements in which one or more of the restriction fragments are inverted in orientation (indicated in the figure by the reversed letters). Restriction fragments from the vector can join together in the wrong order, but this potential problem can be eliminated by the use of a vector that has only one cleavage site for a particular restriction enzyme. When a circular molecule has only one cleavage site for a restriction enzyme, cleavage with the restriction enzyme opens the circle at this single point, and any other DNA fragment of interest, if it has complementary ends, can be inserted at the point of cleavage. Many plasmids with single cleavage sites are available (most have been created by genetic engineering). Many vectors contain unique sites for several different restriction enzymes, but generally only one enzyme is used at a time.

DNA molecules that lack sticky ends also can be joined. A direct method uses the DNA ligase made by *E. coli* phage T4. This enzyme differs from other DNA ligases in that it not only heals single-stranded breaks in double-stranded DNA but can also join molecules with blunt ends.

Insertion of a Particular DNA Molecule into a Vector

In the cloning procedure described so far, a collection of fragments obtained by digestion with a restriction enzyme can be made to anneal with a cleaved vector molecule, yielding a large number of recombinant molecules containing different fragments of donor DNA. However, if a particular gene or segment of DNA is to be cloned, then the

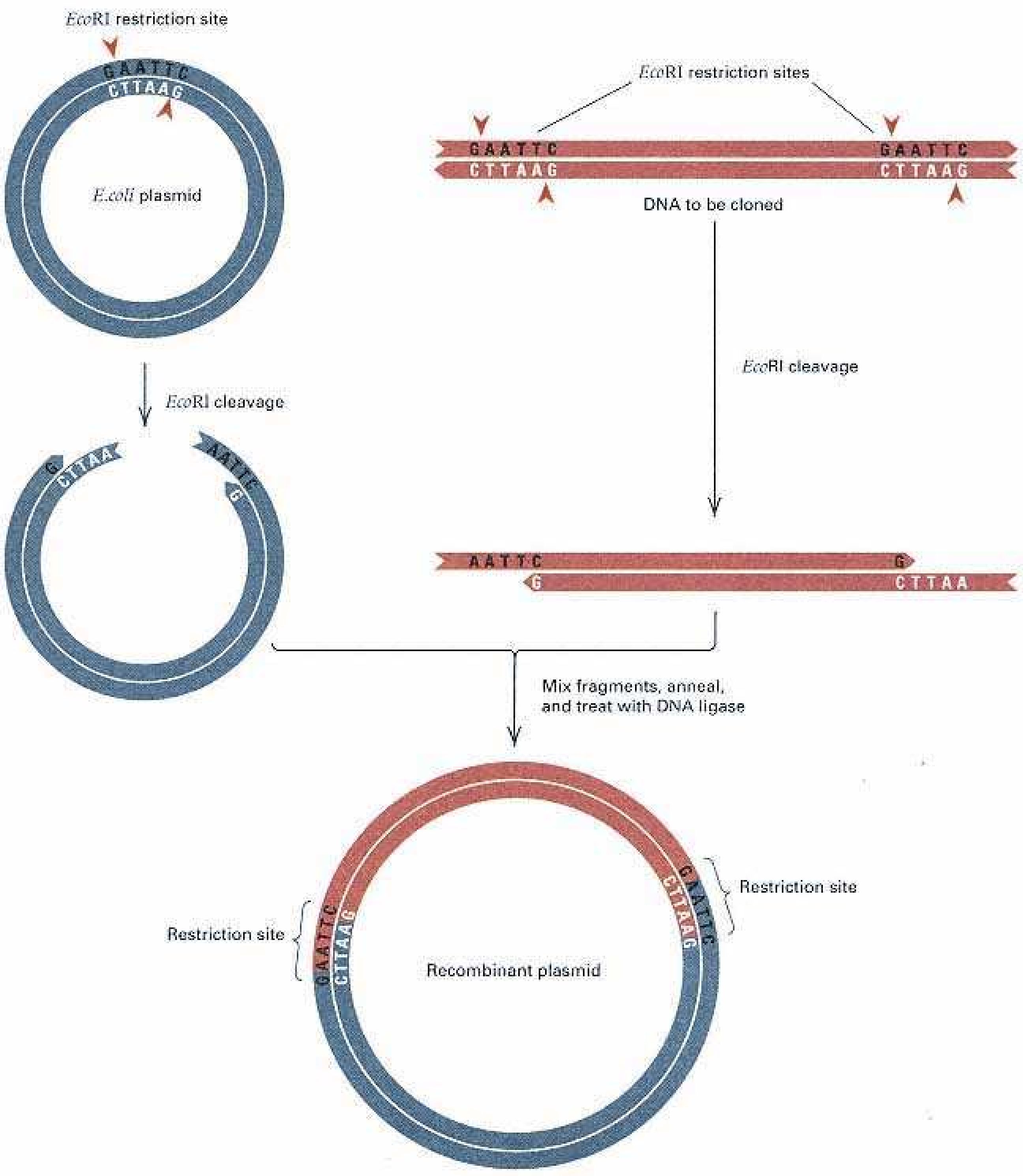


Figure 9.5
Construction of recombinant DNA plasmids containing fragments derived from a donor organism, by the use of a restriction enzyme (in this example *Eco*RI) and cohesive-end joining. Short red arrowheads indicate cleavage sites.

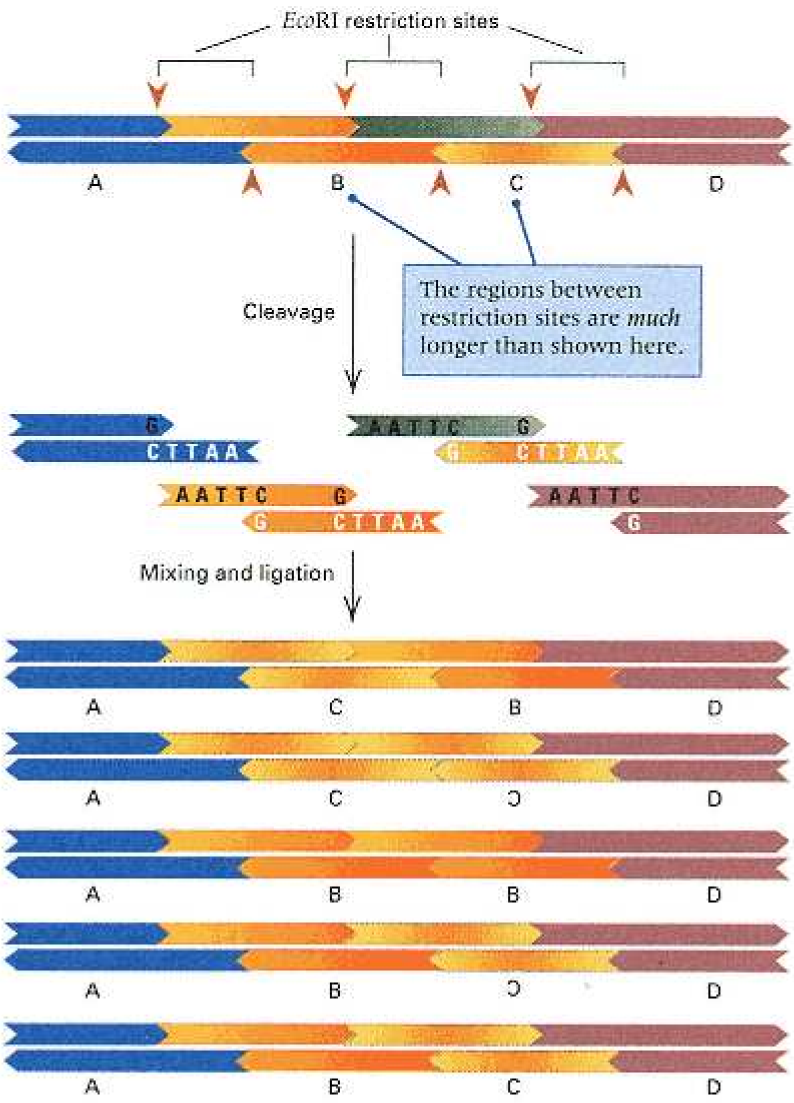


Figure 9.6
Fragments produced by a restriction enzyme can be rejoined in arbitrary ways because the ends of the cleavage sites are compatible. In this example, a DNA molecule is cut into four different fragments (A, B, C, and D) by a restriction enzyme. A few examples of how these can be rejoined are shown. Even among the rejoined molecules that have A and D at the ends and two fragments in the middle, many have one fragment represented twice and the other not at all, or they have a fragment present in the reverse orientation. Only occasionally does a rejoining recreate the original order A B C D.

recombinant molecule possessing that particular segment must be isolated from among all the recombinant molecules that contain donor DNA. The simplest method is direct selection of the desired clone because of a phenotypic attribute it conveys. Indirect kinds of selection will be described shortly.

As an example of direct selection for the recovery of recombinant molecules containing a particular gene, we may consider the cloning of a *leu*$^+$ gene from a bacterial cell. In this case, one would use a Leu$^-$ host, which cannot grow in medium lacking leucine. After transformation with recombinant vectors, a few host cells will contain the *leu*$^+$ gene. These transformants will each give rise to a Leu$^+$ colony on a medium lacking leucine.

Direct selection for recombinant clones, such as in the Leu$^+$ example, is not always possible. If the clone of interest is either very rare or very difficult to detect, it is often preferable to purify the DNA fragment that contains the gene of interest before joining it with the vector. Two methods of fragment isolation have already been discussed:

• Amplification by means of the **polymerase chain reaction (PCR),** discussed in Section 5.8, in which short oligonucleotide primers homologous to the ends of the fragment to be isolated are used repeatedly to replicate the fragment of interest. After a sufficient number of rounds of replication, the resulting solution contains predominantly replicas of the targeted DNA fragment.

• Purification of a restriction fragment known to contain the gene of interest (Section 5.7). For example, if a gene in a virus is known to be contained in a particular restriction fragment, then this fragment can be isolated from a gel after electrophoresis and joined to an appropriate vector.

Some genes in higher eukaryotes, such as human beings, are exceedingly long, as we shall see in Chapter 10. Any DNA fragment containing the complete gene is likely to contain multiple cleavage sites for any restriction enzyme and is also likely to be too long to be amplified by PCR. However, such large genes usually produce messenger RNA (mRNA) molecules of modest size. If the purpose of cloning the gene is to produce the protein product of the gene, then all the necessary information is contained in the mRNA. How to clone a DNA molecule whose sequence corresponds to a particular mRNA is described next.

The Use of Reverse Transcriptase: cDNA and RT-PCR

Some specialized animal cells make only one or a very small number of proteins in large amounts. In these cells, the cytoplasm contains a very high abundance of specific mRNA molecules, which constitute a large fraction of the total mRNA synthesized. Consequently, mRNA samples can usually be obtained that consist predominantly of a single mRNA species. An example is the chicken gene for the egg white protein ovalbumin, which is highly expressed in the oviduct of adult hens. In the cloning of genes whose products are major cellular proteins, the purified mRNA can serve as a starting point for creating a collection of recombinant plasmids, many of which contain coding sequences of the gene of interest.

Cloning from mRNA molecules depends on an unusual polymerase, **reverse transcriptase,** which can use a single-stranded RNA molecule as a template and synthesize a complementary strand of DNA called **complementary DNA,** or **cDNA.** Like other DNA polymerases, reverse transcriptase requires a primer. The stretch of A nucleotides usually found at the 3' end of eukaryotic mRNA serves as a convenient priming site, because the primer can be an oligonucleotide consisting of poly-T (Figure 9.7). Like any other single-stranded DNA molecule, the single strand of DNA produced from the RNA template can fold back upon itself at the extreme 3' end to form a "hairpin" structure that includes a very short double-stranded region consisting of a few base pairs. The 3' end of the hairpin serves as a primer for second-strand synthesis. The second strand can be synthesized either by DNA polymerase or by reverse transcriptase itself. Reverse transcriptase is the source of the second strand in RNA-based viruses that use reverse transcriptase, such as the human immunodeficiency virus (HIV). Conversion into a conventional double-stranded DNA molecule is achieved by cleavage of the hairpin by a nuclease.

In the reverse transcription of an mRNA molecule, the resulting full-length cDNA contains an uninterrupted coding sequence for the protein of interest. As we will see in Chapter 10, eukaryotic genes often contain DNA sequences, called *introns,* that are initially transcribed into RNA but are removed in the production of the mature mRNA. Because the introns are absent from the mRNA, the cDNA sequence is not identical with that in the genome of the original donor organism. However, if the purpose of forming the recombinant DNA molecule is to identify the coding sequence or to synthesize the gene product in a bacterial cell, then cDNA formed from

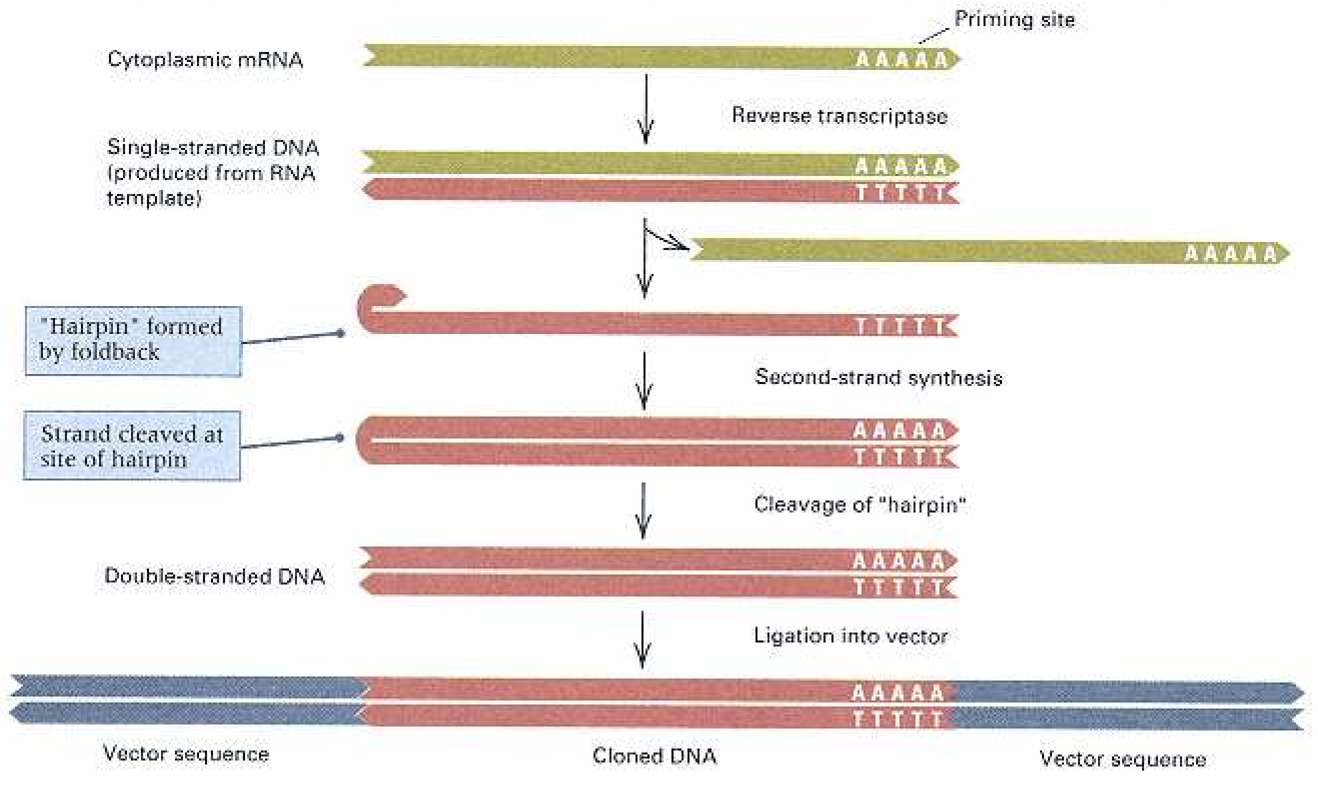


Figure 9.7
Reverse transcriptase produces a single-stranded DNA complementary in sequence to a template RNA. In this example, a cytoplasmic mRNA is copied. As indicated here, most eukaryotic mRNA molecules have a tract of consecutive A nucleotides at the 3' end, which serves as a convenient priming site. After the single-stranded DNA is produced, a foldback at the 3' end forms a hairpin that serves as a primer for second-strand synthesis. After the hairpin is cleaved, the resulting double-stranded DNA can be ligated into an appropriate vector either immediately or after PCR amplification. The resulting clone contains the entire coding region for the protein product of the gene.

processed mRNA is the material of choice for cloning. The joining of cDNA to a vector can be accomplished by available procedures for joining blunt-ended molecules (Figure 9.7).

Genes that are not highly expressed are represented by mRNA molecules whose abundance ranges from low to exceedingly rare. The cDNA molecules produced from such rare RNAs will also be rare. The efficiency of cloning rare cDNA molecules can be markedly increased by PCR amplification prior to ligation into the vector. The only limitation on the procedure is the requirement that enough DNA sequence be known at both ends of the cDNA for appropriate oligonucleotide primers to be designed. PCR amplification of the cDNA produced by reverse transcriptase is called **reverse transcriptase PCR (RT-PCR).** The resulting amplified molecules contain the coding sequence of the gene of interest with very little contaminating DNA.

Detection of Recombinant Molecules

When a vector is cleaved by a restriction enzyme and renatured in the presence of many different restriction fragments from a particular organism, many types of molecules result, including such examples as a self-joined circular vector that has not acquired any fragments, a vector containing one or more fragments, and a molecule consisting only of many joined fragments. To facilitate the isolation of a vector containing a particular gene, some means is needed to ensure (1) that the vector does indeed possess an inserted DNA fragment, and (2) that the fragment is in fact the DNA segment of interest. This section describes several useful procedures for detecting chimeric vectors.

In the use of transformation to introduce recombinant plasmids into bacterial cells, the initial goal is to isolate bacteria that contain the plasmid from a mixture of plasmid-free and plasmid-containing cells. A common procedure is to use a plasmid possessing an antibiotic-resistance marker and to grow the transformed bacteria on a medium that contains the antibiotic: Only cells that contain plasmid can form a colony. An example of a state-of-the-art cloning vector is the pBluescript plasmid illustrated in Figure 9.8A. The entire plasmid is 2961 base pairs. Different regions contribute to its utility as a cloning vector:

• The plasmid origin of replication is derived from the *E. coli* plasmid ColE1. The ColE1 is a high-copy-number plasmid, and its origin of replication enables pBluescript and its recombinant derivatives to exist in approximately 300 copies per cell.

• The ampicillin-resistance gene allows for selection of transformed cells in medium containing ampicillin.

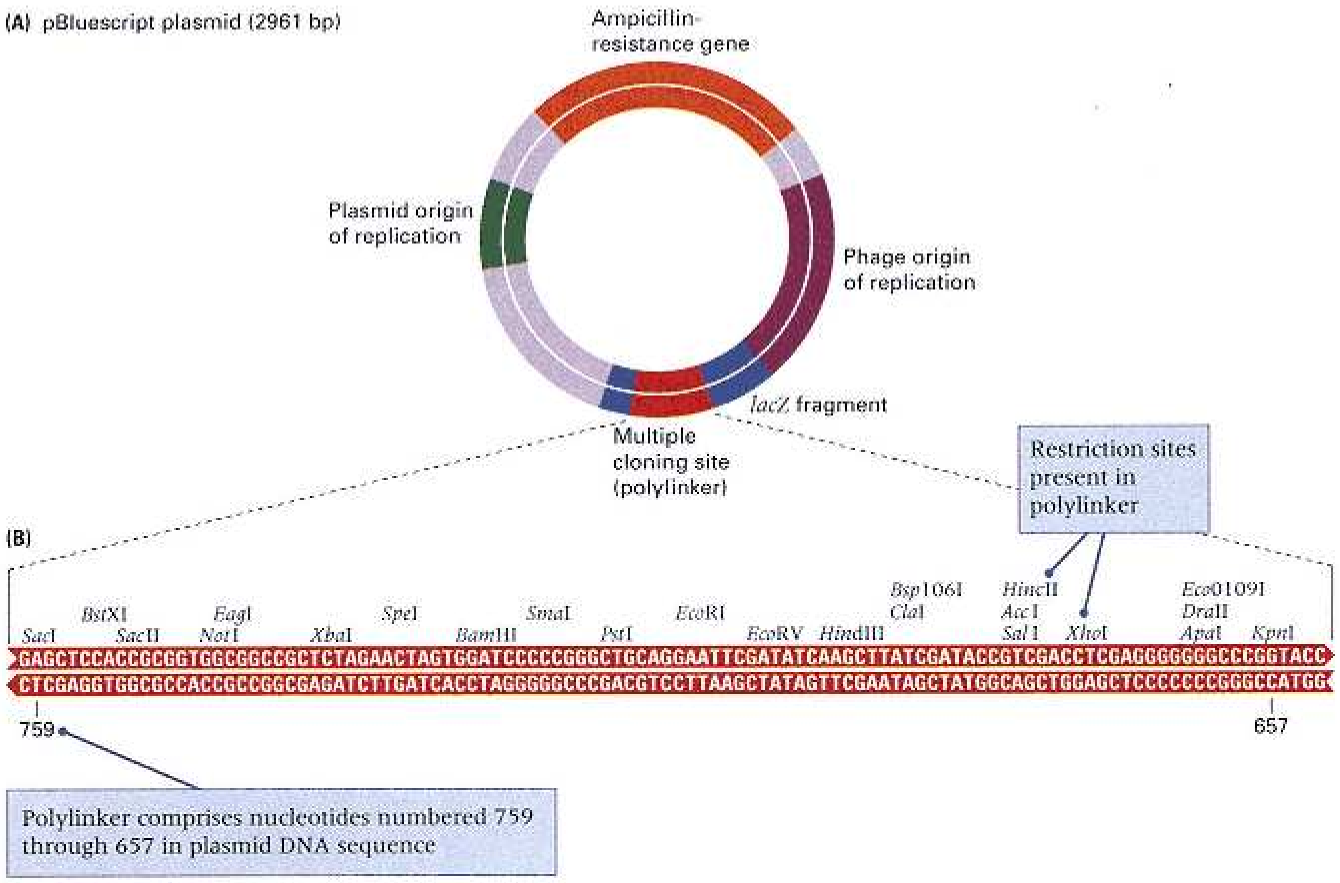


Figure 9.8
(A) Diagram of the cloning vector pBluescript II. It contains a plasmid origin of replication, an ampicillin-resistance gene, a multiple cloning site (polylinker) within a fragment of the *lacZ* gene from *E. coli*, and a bacteriophage origin of replication. (B) Sequence of the multiple cloning site showing the unique restriction sites at which the vector can be opened for the insertion of DNA fragments. The numbers 657 and 759 refer to the position of the base pairs in the complete sequence of pBluescript.
[Courtesy of Stratagene Cloning Systems, La Jolla, CA.]

• The cloning site is called a **multiple cloning site (MCS)** or **polylinker** because it contains unique cleavage sites for many different restriction enzymes and enables many types of restriction fragments to be inserted. In pBluescript, the MCS is a 108-bp sequence that contains cloning sites for 23 different restriction enzymes (Figure 9.8B).

• The detection of recombinant plasmids is by means of a region containing the *lacZ* gene from *E. coli*. When the *lacZ* region is interrupted by a fragment of DNA inserted into the MCS, the recombinant plasmid yields Lac^- cells. Nonrecombinant plasmids do not contain a DNA fragment in the MCS and yield Lac^+ colonies. The Lac^+ or Lac^- phenotypes can be distinguished by color when the cells are grown on a special β-galactoside compound called X-gal, which releases a deep blue dye when cleaved. On medium containing X-gal, Lac^+ colonies contain nonrecombinant plasmids and are a deep blue, whereas Lac^- colonies contain recombinant plasmids and are white.

• The bacteriophage origin of replication is from the single-stranded DNA phage f1. When cells containing a recombinant plasmid are infected with an fl helper phage, the fl origin enables a single strand of the inserted fragment, starting with *lacZ,* to be packaged in progeny phage. This feature is very convenient because it yields single stranded DNA for sequencing. The plasmid shown in Figure 9.8A is the SK(+) variety. There is also an SK(-) variety in which the fl origin is in the opposite orientation and packages the complementary DNA strand.

All good cloning vectors have an efficient origin of replication, at least one unique cloning site for the insertion of DNA fragments, and a second gene whose interruption by inserted DNA yields a phenotype indicative of a recombinant plasmid. Once a **library,** or large set of clones, has been obtained in a particular vector, the next problem is how to identify the particular recombinant clones that contain the gene of interest.

Screening for Particular Recombinants

The procedure of **colony hybridization** makes it possible to detect the presence of any gene for which DNA or RNA labeled with radioactivity or some other means is available (Figure 9.9). Colonies to be tested are transferred *(lifted)* from a solid medium onto a nitrocellulose or nylon filter by gently pressing the filter onto the surface. A part of each colony remains on the agar medium, which constitutes the reference plate. The filter is treated with sodium hydroxide (NaOH), which simultaneously breaks open the cells and denatures the DNA. The filter is then saturated with labeled DNA or RNA, complementary to the gene being sought, and the cellular DNA is renatured. The labeled nucleic acid used in the hybridization is called the **probe.** After washing to remove unbound probe, the positions of the bound probe identify the desired colonies. For example, with radioactively labeled probe, the desired colonies are located by means of autoradiography. A similar assay is done with phage vectors, but in this case plaques are lifted onto the filters.

If transformed cells can synthesize the protein product of a cloned gene or cDNA, then immunological techniques may allow the protein-producing colony to be identified. In one method, the colonies are transferred as in colony hybridization, and the transferred copies are exposed to a labeled antibody directed against the particular protein. Colonies to which the antibody adheres are those that contain the gene of interest.

Positional Cloning

Because of the large number of polymorphisms in DNA sequence that are found among members of natural populations of virtually every species, the genetic map of many species is covered almost completely with a dense concentration of molecular markers. The most common type of molecular marker is a *simple tandem repeat polymor-*

phism, or *STRP*, in which the polymorphism consists of variation in the number of copies of a simple repeating sequence. STRPs and similar types of molecular markers were examined in Section 4.4 in the context of the human genetic map. In the molecular map of the human genome, there is an average of approximately 700 kb between adjacent markers.

In a species in which there is a high density of molecular markers in the genetic map, the genetic map itself often supplies the information needed for cloning a gene. The procedure for cloning a mutation in any gene of interest has four steps.

1. Determine the genetic map position of the mutant gene as precisely as possible using molecular markers linked to it.

2. For molecular markers flanking the mutant gene, isolate clones that contain the marker sequences.

3. Verify that the DNA inserts in the isolated clones are mutually overlapping in such a way as to cover the entire region between the flanking markers without any gaps. Such a mutually overlapping set of clones is called a **contig** because adjacent clones are contiguous ("touching"). If the initial set of clones does include one or more gaps, complete the contig by screening the clone library with additional probes derived from the ends of the clones flanking each gap.

4. Identify the clone or clones in the contig that contain the gene of interest. This usually entails determining the nucleotide sequence of all or part of the contig in wildtype and mutant chromosomes to identify the site of the mutation.

Note that the cloning procedure outlined in these steps requires no knowledge of what the gene of interest does. All the method requires is that a mutation in the gene be mapped rather precisely with respect to molecular markers, which, in turn, are used to isolate a contig of clones covering the relevant region. Use of the genetic map to clone genes in this manner

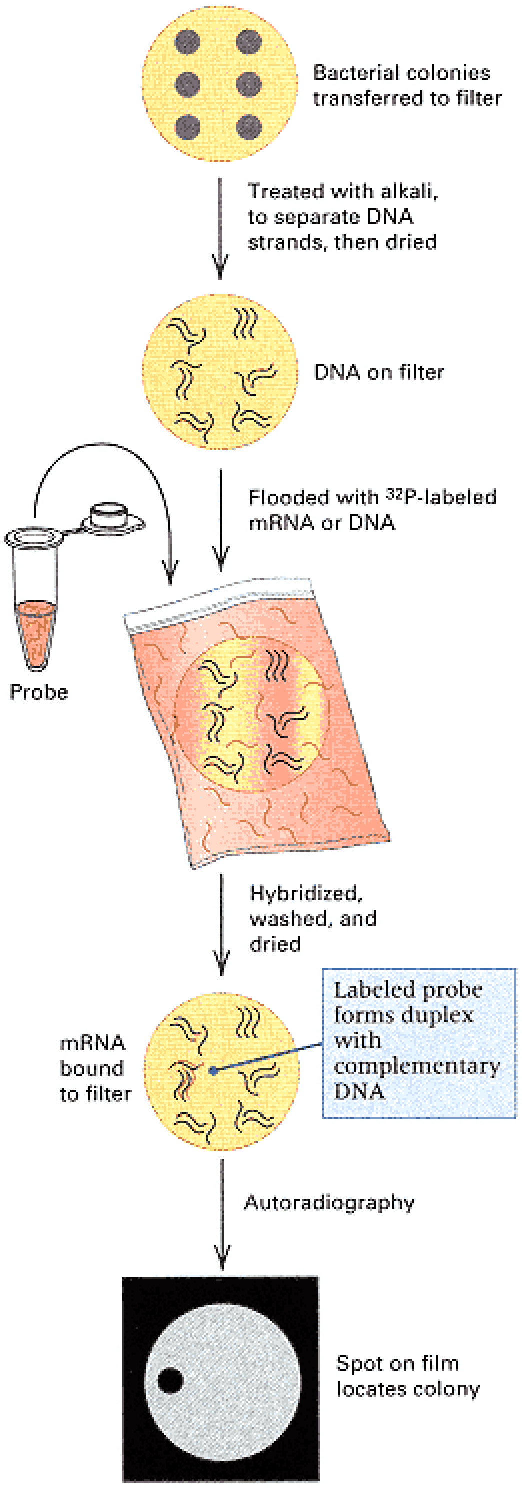


Figure 9.9
Colony hybridization. The reference plate, from which the colonies were obtained, is not shown.

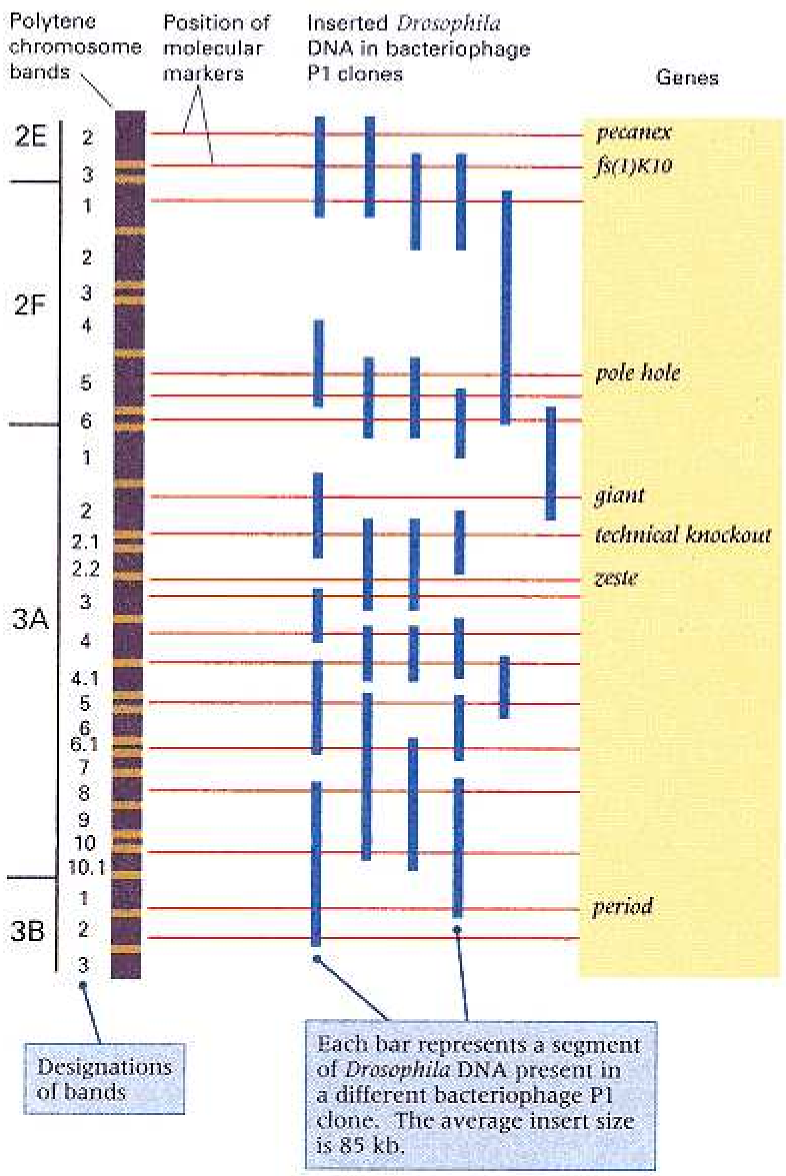


Figure 9.10
Positional cloning in *Drosophila* is made possible by a set of P1 clones that covers virtually the entire genome. At the right is illustrated a set of P1 clones covering a small region of the X chromosome. (Most of the X chromosome is also covered with cosmid contigs.) The P1 clones contain sequences for a set of molecular markers (red lines) that have known locations on the genetic map as well as in the polytene chromosomes of the salivary glands (left). The locations of several cloned genes in the region are shown. Any new mutation in the region is cloned by first determining its location with respect to flanking molecular markers and then identifying the gene within the P1 clones that contain the flanking markers.
[After D.L. Hartl, D.I . Nurminsky, R.W. Jones, and E.R. Lozovskaya, *Proc . Natl. Acad. Sci. USA* 1994. 91: 6824.]

is called **positional cloning** or **mapbased cloning.** It is one of the most important approaches to cloning genes in all species that have a dense genetic map, and it is particularly important in human molecular genetics.

In many well-studied organisms, contigs of clones are already available that cover almost the entire genome. These organisms include bacteria, such as *E. coli;* fungi, including *S. cerevisiae;* the nematode *C. elegans;* the fruit fly *D. melanogaster;* plants, such as *A. thaliana;* the mouse *Mus musculus;* human beings *Homo sapiens;* and many others. The utility of such contigs is illustrated for a small part of the *Drosophila* genome in Figure 9.10. On the left is a diagram of the banding pattern in the polytene salivary gland chromosomes across the region 2E2 through 3B3. (The conventions for naming the bands are discussed in Section 7.6.) These 26 bands comprise approximately 650 kb of DNA. To the right of the bands, the red lines represent the positions of molecular markers in the region, spaced at an average distance of about 30 kb. The blue bars depict P1 clones known to contain sequences of the molecular markers. Each clone contains an insert of approximately 85 kb. The red lines coincide with the positions of the molecular markers, and the points of intersection with the clones show the presence of each sequence within each clone. Seven cloned genes in the region are indicated. In the cloning of any new mutant gene that maps within the region, the location of the mutation with respect to the markers immediately identifies the clone or clones in which the gene is located.

9.3— Site-Directed Mutagenesis

Once a gene has been identified and cloned, the next step is usually to sequence the gene and ascertain whether similar sequences are already present in databases from other organisms. Similarity in sequence with a known gene sometimes gives an important clue to the function of a previously unknown gene. Other clues to function can be determined by deliberately introducing mutations into the gene and putting the gene back into the organism to observe the resulting phenotype. Methods by which a gene can be introduced into the germ line of a living organism are examined in Section 9.4. In this section we deal with the method of mutagenesis itself.

Virtually any desired mutation can be introduced into a cloned gene by direct manipulation of the DNA. An important method for producing changes in one or a few nucleotides is called **oligonucleotide site-directed mutagenesis.** Application of this method for the substitution of a single nucleotide in a gene is outlined in Figure 9.11. The first step is to isolate a single-stranded DNA molecule from a plasmid that contains the target gene and two antibiotic-resistance markers—in this example, kanamycin resistance *(kan-r)* and ampicillin resistance *(amp-r).* One of the genes is deliberately chosen to have a single-nucleotide substitution that inactivates the gene, which in this example is the *amp-r* gene. Hence, the double-stranded form of the plasmid would have the phenotype Kan-r Amp-s because of the mutation in *amp.* The single-stranded molecule is annealed with two short oligonucleotides (usually from 20 to 50 nucleotides), as shown in Figure 9.11A. One oligonucleotide is complementary to the target gene except for a single-nucleotide mismatch. The other oligonucleotide is complementary to a region of the *amp-s* allele except for a single-nucleotide mismatch that reverses the *amp* mutation. After annealing of the oligonucleotides, which act as primers, the complementary strand of the DNA is synthesized by the addition of DNA polymerase and DNA ligase. The oligonucleotide primers are incorporated into the new strand (Figure 9.11B).

The next step is transformation, in which the DNA is introduced into a bacterial cell. The circular plasmid DNA contains two mismatches, one to correct the antibiotic-resistance mutation and the other in the target gene, each of which is theoretically susceptible to repair by enzymes present in the cell. The mismatch repair enzymes work by cleaving one of the mismatched nucleotides out of the molecule, widening the resulting gap somewhat, and resynthesizing across the gap from the remaining template strand. In *E. coli,* the

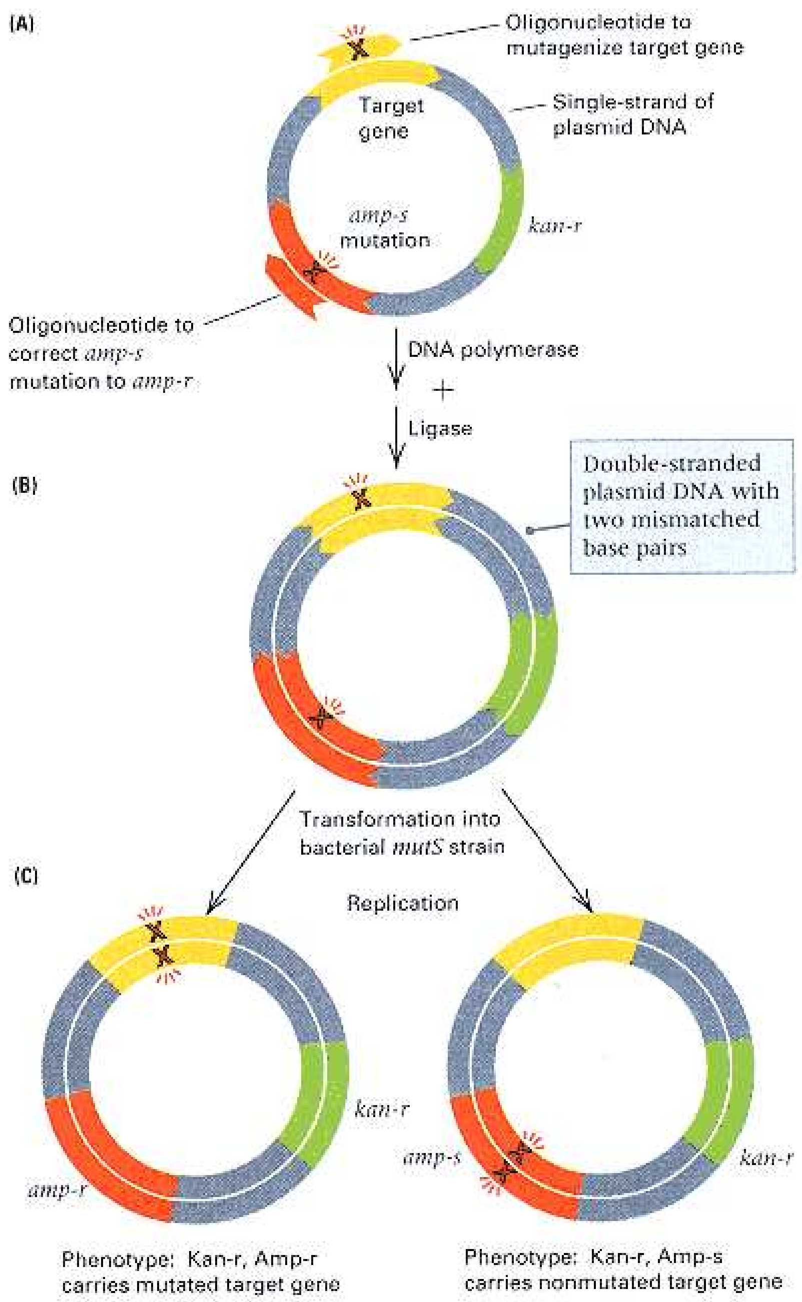


Figure 9.11
Method for introducing specific nucleotide substitutions into a DNA molecule. (A) A single strand of a plasmid is annealed with two oligonucleotides, one containing a mismatch that creates the desired mutation in the target gene and the other containing a mismatch that corrects a pre-existing mutation in an ampicillin-resistance gene. (B) The second DNA strand, which incorporates the oligonucleotides, is synthesized by DNA polymerase and the gaps are closed by ligase. (C) After transformation and one round of DNA replication, the mutated strand yields a plasmid that confers ampicillin resistance and that also contains the desired mutation in the target gene. The *mutS* mutation in the host knocks out the mismatch repair system.

possibility of repair is eliminated by using a *mutS* strain for transformation, because *mutS* mutants lack one of the key enzymes necessary for mismatch repair (Chapter 13). After one round of replication, the original molecule segregates into two daughter molecules, each without mismatches (Figure 9.11C). One daughter molecule contains the mutations introduced by the oligonucleotides, which create the desired mutation in the target gene and reverse the *amp* mutation. The other daughter molecule contains neither mutation and so has the unmutated target and the *amp-s* mutation. Therefore, if the transformed cells are plated on medium containing both kanamycin and ampicillin, only those cells that contain descendants of the mutagenized strand can survive, because only these cells are Kan-r as well as Amp-r. Furthermore, in a typical experiment of this kind, more than 90 percent of the Kan-r Amp-r cells also contain the desired mutation in the target gene.

Many genes are too large to be contained completely in a plasmid of the size used for site-directed mutagenesis. In such cases, only a fragment of the target gene is mutated, and then the mutated fragment is isolated from the plasmid and used to replace the nonmutant homologous fragment from the wildtype gene. Site-directed mutagenesis of part of the wildtype gene for the *E. coli* enzyme alkaline phosphatase is shown in Figure 9.12. The nucleotide substitution in the oligonucleotide is shown in red. The amino acid sequence coded by the wildtype and the mutant differ in a single amino acid, with serine replacing cysteine. This particular cysteine residue was thought to be critical in protein folding. As expected, the cysteine is critical, and the mutant enzyme was found to be completely nonfunctional.

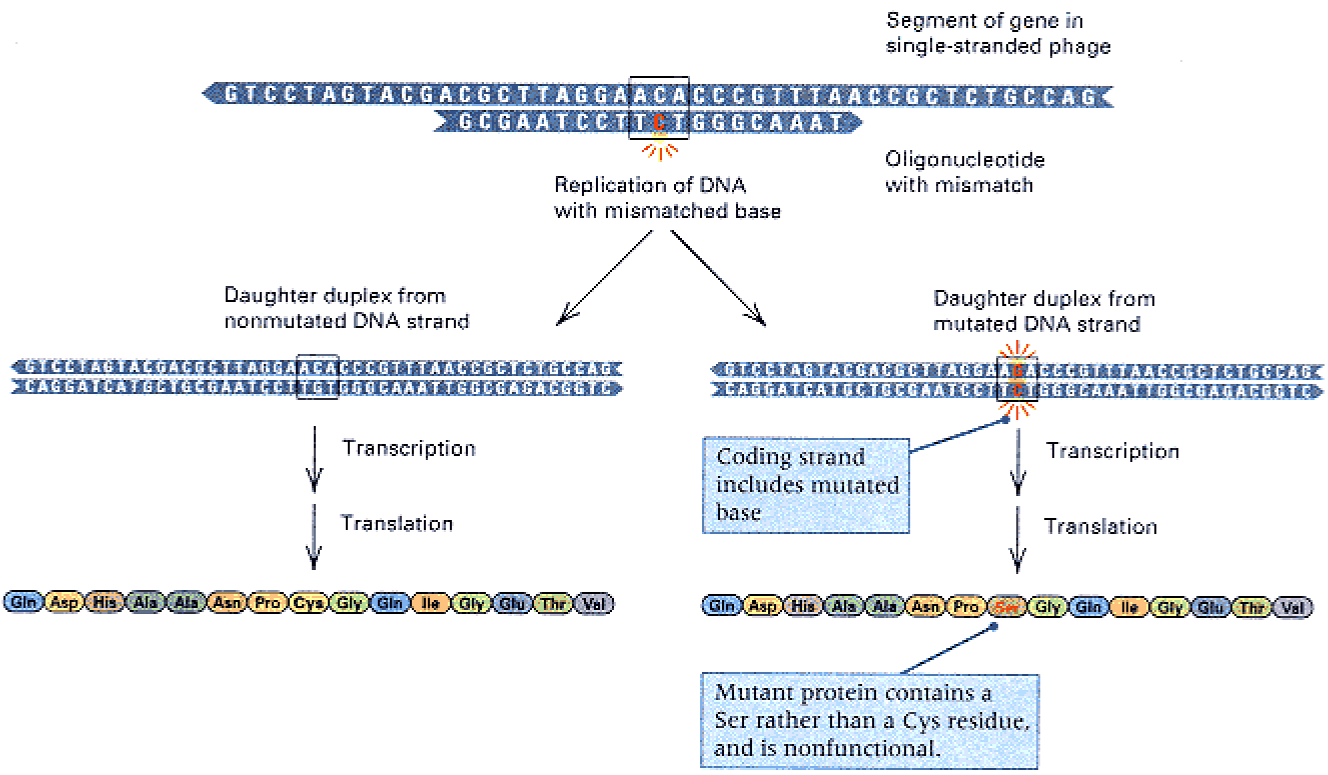


Figure 9.12
Production of a specific mutation in the *E. coli* alkaline phosphatase by the use of oligonucleotide site-directed mutagenesis. The mismatch in the oligonucleotide is in red. The mutant protein contains a cysteine → serine replacement, which renders the enzyme nonfunctional.

9.4— Reverse Genetics

Genetics has traditionally relied on mutation to provide the raw material needed for analysis. The customary procedure has been to use a mutant phenotype to recognize a mutant gene and then to identify the wildtype allele and its normal function. This approach has proved highly successful, as evidenced by numerous examples throughout this book. But the approach also has its limitations. For example, it may prove difficult or impossible to isolate mutations in genes that duplicate the functions of other genes or that are essential for the viability of the organism. As we saw in the previous section, site-directed mutagenesis opens up a quite different approach in which virtually any type of mutation can be created in almost any gene. Because the mutations are predetermined, a very fine level of resolution is possible in defining promoter and enhancer sequences that are necessary for transcription, the sequences necessary for messenger RNA production, and particular amino acids that are essential for protein function. This approach to genetic analysis is often called **reverse genetics** because it reverses the usual flow of study: Instead of starting with a mutant phenotype and trying to identify the nature of the mutation, reverse genetics starts with a known mutation and determines its effect on the phenotype. Of course, once a mutation has been created in the laboratory, there remains the problem of how to put it back into a living organism. The following sections describe some of the relevant techniques.

Germ-Line Transformation in Animals

Reverse genetics can be carried out in most organisms that have been extensively studied genetically, including the nematode *Caenorhabditis elegans*, *Drosophila*, the mouse, and many domesticated animals and plants. In nematodes, the basic procedure is to manipulate the DNA of interest in a plasmid that also contains a selectable genetic marker that will alter the phenotype of the transformed animal. The DNA is injected directly into the reproductive organs and sometimes spontaneously becomes incorporated into the chromosomes in the germ line. The result of transformation is observed and can be selected in the progeny of the injected animals because of the phenotype conferred by the selectable marker.

A somewhat more elaborate procedure is necessary for germ-line transformation in *Drosophila*. The usual method makes use of a 2.9-kb transposable element (Section 6.8) called the ***P* element,** which consists of a central region coding for transposase flanked by 31 base-pair inverted repeats (Figure 9.13A). A genetically engineered derivative of this *P* element, called *wings clipped*, can make functional transposase but cannot itself transpose because of deletions introduced at the ends of the inverted repeats (Figure 9.13B). For germ-line transformation, the vector is a plasmid containing a *P* element that includes, within the inverted repeats, a selectable genetic marker (usually one affecting eye color), as well as a large internal deletion that removes much of the transposase-coding region. By itself, this *P* element cannot transpose because it makes no transposase, but it can be mobilized by the transposase produced by the wings-clipped or other intact *P* elements.

In *Drosophila* transformation, any DNA fragment of interest is introduced between the ends of the deleted *P* element. The resulting plasmid and a different plasmid containing the wings-clipped element are injected into the region of the early embryo containing the germ cells. The DNA is taken up by the germ cells, and the wings-clipped element produces functional transposase (Figure 9.13B). This mobilizes the engineered *P* vector and results in its transposition into an essentially random location in the genome. Transformants are detected among the progeny of the injected flies because of the eye color or other genetic marker included in the *P* vector. Integration into the germ line is typically very efficient: From 10 to 20 percent of the injected embryos that survive and are fertile yield one or more transformed progeny. However, the efficiency decreases with the size of the DNA fragment in the *P* element, and the effective upper limit is approximately 20 kb.

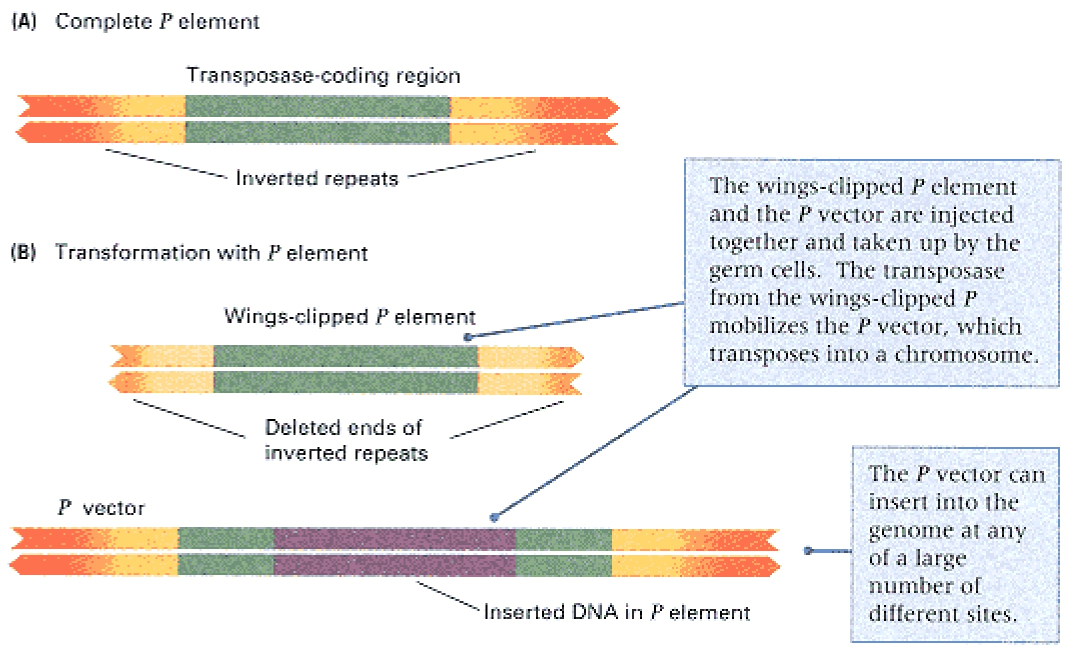


Figure 9.13
Transformation in *Drosophila* mediated by the transposable element *P*. (A) Complete P element containing inverted repeats at the ends and an internal transposase-coding region. (B) Two-component transformation system. The vector component contains the DNA of interest flanked by the recognition sequences needed for transposition. The wings-clipped component is a modified *P* element that codes for transposase but cannot transpose itself because critical recognition sequences are deleted.

Transformation of the germ line in mammals can be carried out in several ways. The most direct is the injection of vector DNA into the nucleus of fertilized eggs, which are then transferred to the uterus of foster mothers for development. The vector is usually a modified retrovirus. **Retroviruses** have RNA as their genetic material and code for a reverse transcriptase that converts the retrovirusus genome into double-stranded DNA that becomes inserted into the genome in infected cells (Section 7.7). Genetically engineered retroviruses containing inserted genes undergo the same process. Animals that have had new genes inserted into the germ line in this or any other manner are called **transgenic** animals.

Another method of transforming mammals uses **embryonic stem cells** obtained from embryos a few days after fertilization (Figure 9.14). Although embryonic stem cells are not very hardy, they can be isolated and then grown and manipulated in culture; mutations in the stem cells can be selected or introduced using recombinant DNA vectors. The mutant stem cells are introduced into another developing embryo and transferred into the uterus of a foster mother (Figure 9.14A), where they become incorporated into various tissues of the embryo and often participate in forming the germ line. If the embryonic stem cells carry a genetic marker, such as a gene for black coat color, then mosaic animals can be identified by their spotted coats (Figure 9.14B). Some of these animals, when mated, produce black offspring (Figure 9.14C), which indicates that the embryonic stem cells had become incorpo-

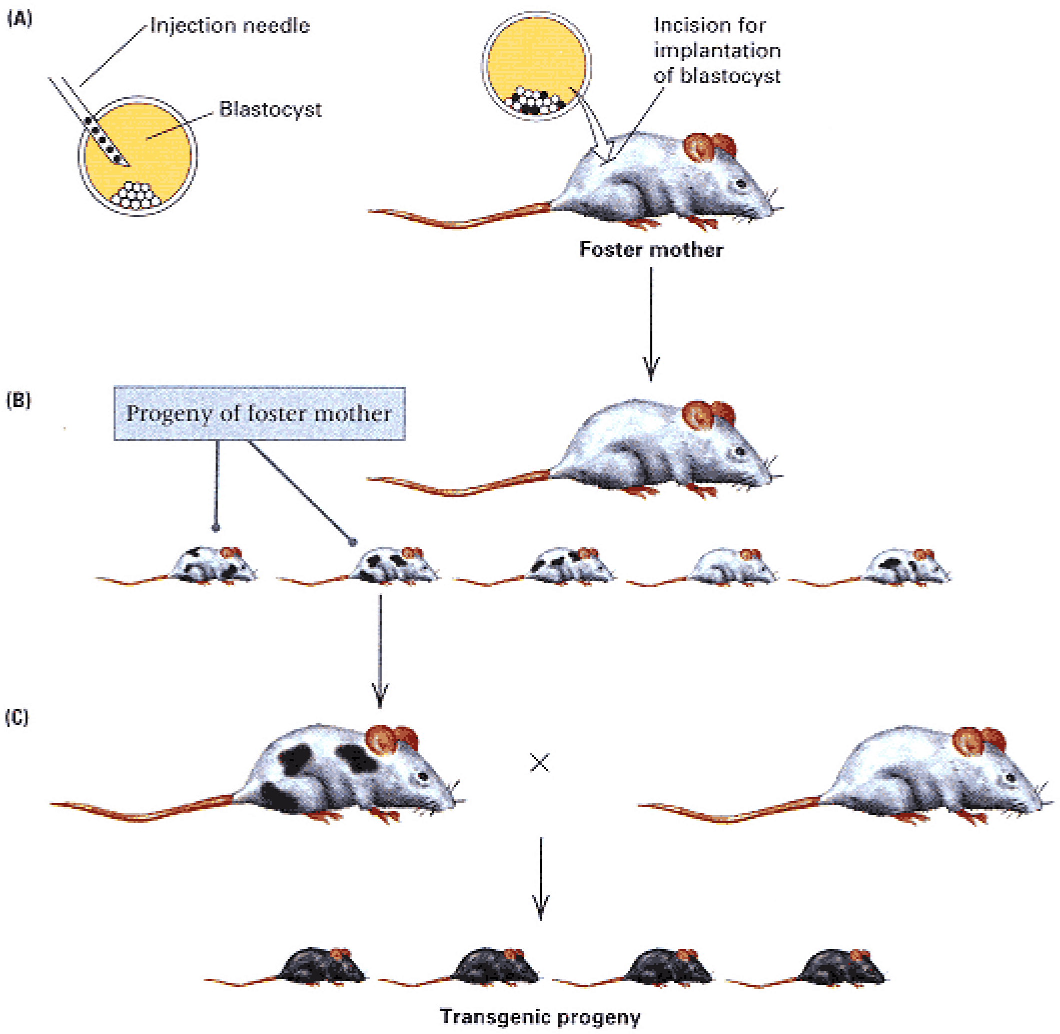


Figure 9.14
Transformation of the germ line in the mouse using embryonic stem cells. (A) Stem cells obtained from an embryo of a black strain are isolated and, after genetic manipulation in culture, injected into the embryo of a white strain, which is then introduced into the uterus of a foster mother. (B) The resulting offspring are often mosaics that contain cells from both the black and the white strains. (C) If cells from the black strain colonize the germ line, then the offspring of the mosaic animal will be black.
[After M. R. Capecchi. 1989. *Trends Genet*., 5: 70.]

rated into the germ line. In this way, mutations introduced into the embryonic stem cells while they were in culture may become incorporated into the germ line of living animals. The method in Figure 9.14 has been used to create strains of mice with mutations in genes associated with such human genetic diseases as cystic fibrosis. These strains serve as mouse models for studying the disease and for testing new drugs and therapeutic methods.

The procedure for introducing mutations into specific genes is called **gene targeting.** The specificity of gene targeting comes from the DNA sequence homology needed for homologous recombination. Two examples are illustrated in Figure 9.15, where the DNA sequences present in

Connection Hello, Dolly!

Ian Wilmut, Anagelika E. Schnieke, Jim McWhir, Alex J. Kind, and Keith H. S. Campbell 1997

Roslin Institute, Roslin, Midlothian, Scotland
Viable Offspring Derived from Fetal and Adult Mammalian Cells

The Scottish Finn Dorset ewe known as "Dolly" is the first mammal to have been cloned from the nucleus of a cell taken from an adult mammal. The experiment created a press sensation. The President of the United States said that he would be against cloning people. A professor of public health said, "I don't think reasonable, rational people would want to clone themselves, but an eccentric millionaire might want to leave his money to a clone. There's no way of stopping the super-rich from going offshore to clone themselves." Some people were concerned for the animals. A spokesperson for People for the Ethical Treatment of Animals was quoted as saying, "It is time that society learned to respect our fellows, not exploit them for every fool thing." (Note in the excerpt that follows that the experiment was carried out with the prior approval of the appropriate Animal Welfare Committee.) Many important ethical issues are raised by the possibility of cloning humans. There are also important issues concerning the status of cloned animals. Should the clone of a racehorse that won the Triple Crown compete under the same rules as other race-horses? Should the clones of champion show dogs and cats compete under the same rules? Who should decide?

Transfer of a single nucleus at a specific stage of development, to an enucleated unfertilized egg, provided an opportunity to investigate whether cellular differentiation to that stage involved irreversible genetic modification. The first offspring to develop

The fact that a lamb was derived from an adult cell confirms that differentiation of that cell did not involve the irreversible modification of genetic material required for development to term.

from a differentiated cell were born after nuclear transfer from an embryo-derived cell that had been induced to become quiescent. Using the same procedure, we now report the birth of live lambs from adult mammary gland, fetus, and embryo. The fact that a lamb was derived from an adult cell confirms that differentiation of that cell did not involve the irreversible modification of genetic material required for development to term. . . . If the recipient cytoplasm is prepared by enucleation of an ocyte at metaphase II, it is only possible to avoid chromosomal damage and maintain normal ploidy by transfer of diploid nuclei. Our studies with cultured cells suggest that there is an advantage if cells are quiescent . . .. Together our results indicate that nuclei from a wide range of cell types should prove to be totipotent after enhancing opportunities for reprogramming by using appropriate combinations of these cell-cycle stages. In turn, the dissemination of the genetic improvement obtained within elite selection herds will be enhanced by limited replication of animals with proven performance by nuclear transfer of cells derived from adult animals. . . . The lamb born after nuclear transfer from a mammary gland cell is, to our knowledge, the first mammal to develop from a cell derived from an adult tissue. . . . This is consistent with the generally accepted view that mammalian differentiation is almost all achieved by systematic, sequential changes in gene expression brought about by interactions between the nucleus and the changing cytoplasmic environment. . . . These experiments were conducted under the Animals (Scientific Procedure) Act 1986 [United Kingdom] and with the approval of the Roslin Institute Animal Welfare Experiments Committee.

Source: Nature 385:810–813

gene-targeting vectors are shown as looped configurations paired with homologous regions in the chromosome prior to recombination. The targeted gene is shown in pink. In Figure 9.15A, the vector contains the targeted gene interrupted by an insertion of a novel DNA sequence, and homologous recombination results in the novel sequence becoming inserted into the targeted gene in the genome. In Figure 9.15B, the vector contains only flanking sequences, not the targeted gene, so homologous recombination results in replacement of the targeted gene with an unrelated DNA sequence. In both cases, cells with targeted gene mutations can be selected by including an antibiotic resistance gene, or other selectable genetic marker, in the sequences that are incorporated into the genome through homologous recombination.

Genetic Engineering in Plants

A procedure for the transformation of plant cells makes use of a plasmid found in the soil bacterium *Agrobacterium tumefaciens* and related species. Infection of susceptible

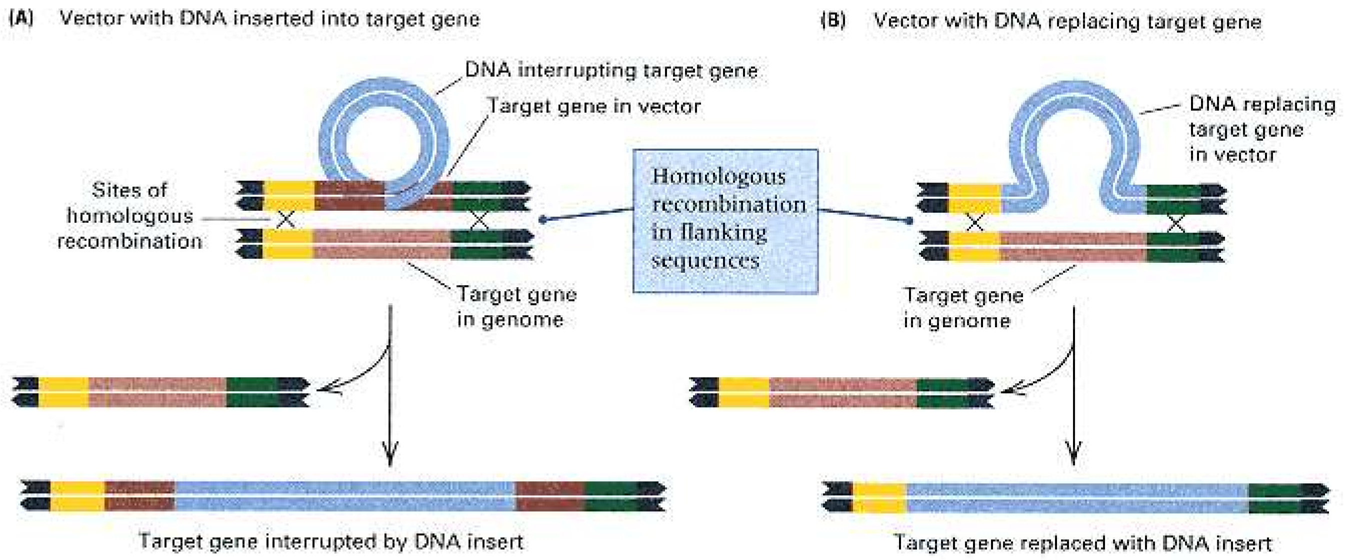


Figure 9.15
Gene targeting in embryonic stem cells. (A) The vector (top) contains the target sequence (red) interrupted by an insertion. Homologous recombination introduces the insertion into the genome. (B) The vector contains DNA sequences flanking the targeted gene. Homologous recombination results in replacement of the targeted gene with an unrelated DNA sequence .
[After M. R. Capecchi. 1989. *Trends Genet.*, 5: 70.]

plants with this bacterium results in the growth of tumors known as **crown gall tumors** at the entry site, usually a wound. Susceptible plants comprise about 160,000 species of flowering plants, known as the dicots, and include the great majority of the most common flowering plants.

The *Agrobacterium* contains a large plasmid of approximately 200 kb called the ***Ti* plasmid,** which includes a smaller (~25 kb) region known as the **T DNA** flanked by 25 base-pair direct repeats (Figure 9.16A). The *Agrobacterium* causes a profound change in the metabolism of infected cells because of transfer of the T DNA into the plant genome. The T DNA contains genes coding for proteins that stimulate division of infected cells, hence causing the tumor, and also coding for enzymes that convert the amino acid arginine into an unusual derivative, generally **nopaline** or **octopine** (depending on the particular type of *Ti* plasmid), that the bacterium needs in order to grow. The transfer functions are present not in the T DNA itself but in another region of the plasmid called the *vir* (stands for *virulence)* region of about 40 kb that includes six genes necessary for transfer.

Transfer of T DNA into the host genome is similar in some key respects to bacterial conjugation, which we examined in Chapter 8. In infected cells, transfer begins with the formation of a nick that frees one end of the T DNA (Figure 9.16A), which peels off the plasmid and is replaced by rolling-circle replication (Figure 9.16B). The region of the plasmid that is transferred is delimited by a second nick at the other end of the T DNA, but the position of this nick is variable. The resulting single stranded T DNA is bound with molecules of a single-stranded binding protein (SSBP) and is transferred into the plant cell and incorporated into the nucleus. There it is integrated into the chromosomal DNA by a mechanism that is still unclear (Figure 9.16C). Although the SSBP has certain similarities in amino acid sequence to the *recA* protein from *E. coli*, which plays a key role in homologous recombination, it is clear that integration of the T DNA does not require homology.

Use of T DNA in plant transformation is made possible by engineered plasmids in which the sequences normally present in T DNA are removed and replaced with those to be incorporated into the plant genome

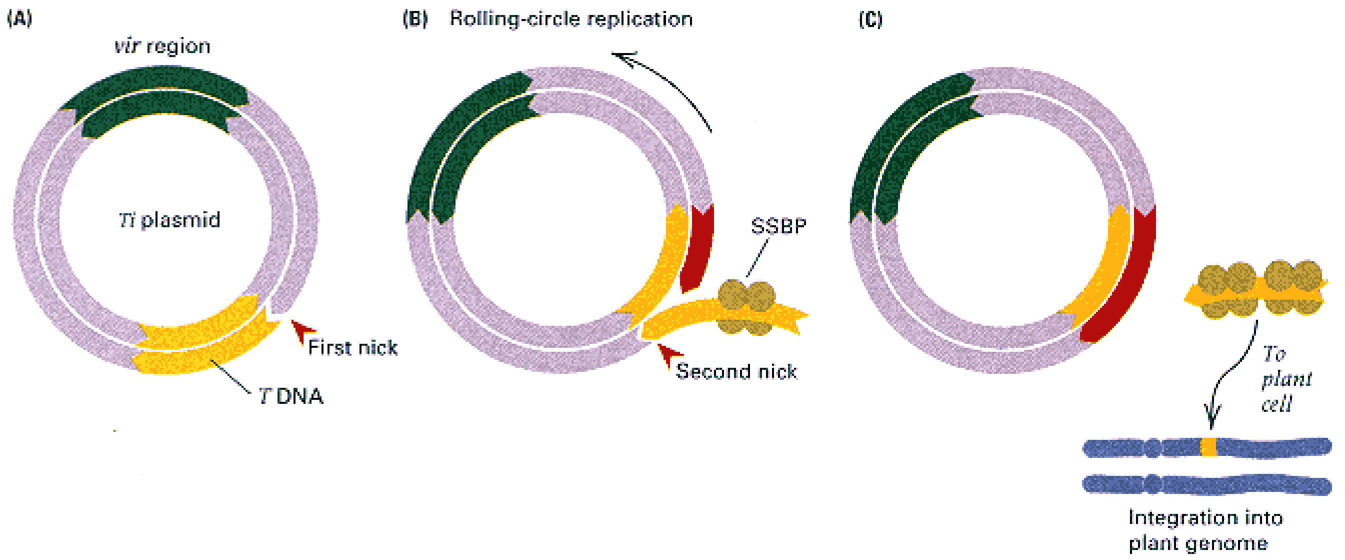


Figure 9.16
Transformation of a plant genome by *T* DNA from the *Ti* plasmid. (A) A nick forms at the 5' end of the *T* DNA. (B) Rolling-circle replication elongates the 3' end and displaces the 5' end, which is stabilized by single-stranded binding protein (SSBP). A second nick terminates replication. (C) The SSBP-bound *T* DNA is transferred to a plant cell and inserts into the genome.

along with a selectable marker. A second plasmid contains the *vir* genes and permits mobilization of the engineered T DNA. In infected tissues, the *vir* functions mobilize the T DNA for transfer into the host cells and integration into the chromosome. Transformed cells are selected in culture by the use of the selectable marker and then grown into mature plants in accordance with the methods described in Section 7.3.

9.5— Applications of Genetic Engineering

Engineered male sterility in plants is just one example of how recombinant DNA technology has revolutionized modern biology. At the present time, recombinant DNA technology is used mainly in (1) the efficient production of useful proteins, (2) the creation of cells capable of synthesizing economically important molecules, (3) the generation of DNA and RNA sequences as research tools or in medical diagnosis, (4) the manipulation of the genotype of organisms such as plants, and (5) the potential correction of genetic defects in animals, including human patients (gene therapy). Some examples of these applications follow.

Giant Salmon with Engineered Growth Hormone

In many animals, the rate of growth is controlled by the amount of growth hormone produced. Transgenic animals with a growth-hormone gene under the control of a highly active regulatory sequence are often larger than their normal counterparts. An example of a highly active sequence that can drive gene expression is the regulatory region of a gene for **metallothionein.** The metallothioneins are proteins that bind heavy metals. They are ubiquitous in eukaryotic organisms and are often encoded in a family of related genes. Human beings, for example, have more than ten metallothionein genes that can be separated into two major groups according to their sequences. The regulatory region of a metallothionein gene accelerates transcription of any genes to which it is attached in response to heavy metals or steroid hormones. When DNA constructs consisting of a rat growth-hormone gene under metallothionein control are used to produce transgenic mice, the resulting animals grow about twice as large as normal mice.

The effect of another growth-hormone construct is shown in Figure 9.17. The fish are coho salmon at 14 months of age. Those on the left are normal, whereas those on the right are transgenic animals that contain a salmon growth-hormone gene driven by a metallothionein regulatory region. Both the growth-hormone gene and the metallothionein gene were cloned from the sockeye salmon. As an index of scale, the largest transgenic fish on the right has a length of about 42 cm. On average, the transgenic fish are 11 times heavier than their normal counterparts; the largest transgenic fish was 37 times the average weight of the nontransgenic animals. Not only do the transgenic salmon grow faster and become larger than normal salmon, they also mature faster.

Engineered Male Sterility with Suicide Genes

Another practical application of genetic engineering is in the production of male-sterile plants. Compared with their inbred parents, hybrid plants are usually superior in numerous respects, including higher yield and increased disease resistance (Chapter 16). Male sterility is important in hybrid seed production, because it promotes efficient hybridization between the inbred lines. Cytoplasmic male sterility is widely used in corn breeding (Chapter 14), but analogous mutations are not available in many crop plants. To be optimally useful, male sterility should also be reversible by fertility restorers so that the inbred lines can be propagated.

A genetically engineered system of male sterility and fertility restoration has been introduced into a number of plant species, including the oilseed rape, *Brassica napus,* which is a major source of vegetable oil. (Canola oil comes from the oilseed rape.) The genetic engineering of male sterility and fertility restoration makes use of two engineered genes. The basis of the sterility is an extracellular RNA nuclease called **barnase,** which is produced by the bacterium *Bacillus amyloliquefaciens.* The barnase nuclease is an extremely potent

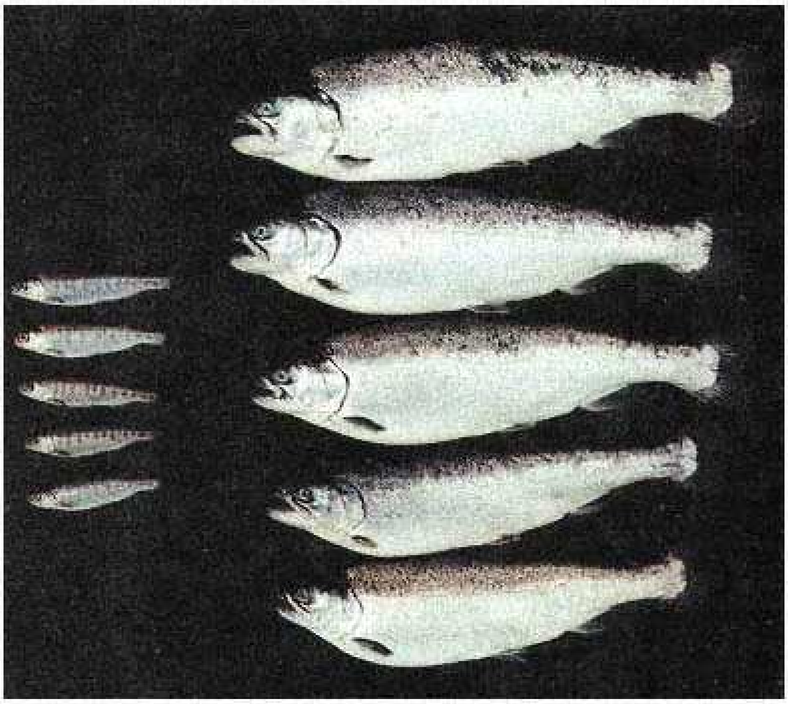


Figure 9.17
Normal coho salmon (left) and genetically engineered coho salmon (right) containing a sockeye salmon growth-hormone gene driven by the regulatory region from a metallothionein gene. The transgenic salmon average 11 times the weight of the nontransgenic fish. The smallest fish on the left is about 4 inches long.
[Courtesy of R. H. Devlin; see R. H. Devlin, T. Y. Yesaki, C. A. Biagi, E. M. Donaldson, P. Swanson, and W.-K. Chan. 1994. *Nature* 371: 209.]

cellular toxin. In the use of barnase to produce male sterility, the coding sequence of the bacterial gene is fused with the regulatory sequence of a gene, *TA29,* that has a tissue-specific expression in the tapetal cell layer that surrounds the pollen sacs in the anther (Figure 9.18A). When the artificial *TA29-barnase* gene is transformed into the genome by the use of T DNA from *Agrobacterium,* the resulting plants are male-sterile because of the destruction of tapetal cell RNA by the barnase enzyme and the resulting lethality of cells in the tapetum (Figure 9.18B).

Fertility restoration makes use of another protein, called **barstar,** also produced by *B. amyloliquefaciens.* Barstar is an intracellular protein that protects against the lethal effects of barnase by forming a stable, enzymatically inactive complex in the cytoplasm. Hence, barstar is the bacterial cell's self-defense against its own barnase. Plants transformed with an artificial *TA29-barstar* gene are healthy and fertile; they merely produce barstar in the tapetum because of the tapetum-specific expression of the *TA29* regulatory region. However, when the male-sterile *TA29-barnase* plants are crossed with those carrying *TA29-barstar,* the resulting genotype that combines *TA29-barnase* with *TA29-barstar* is male-fertile (Figure 9.18C). The reason for the restoration of fertility is that the barstar protein combines with the barnase nuclease and renders it ineffective.

The macroscopic appearance and the microscopic appearance of oilseed rape plants of various genotypes are shown in Figure 9.19. The wildtype phenotype is shown in parts A and B, and numerous pollen grains (PG) are apparent in the cross section of the anther. Parts C and D show the flower and anthers of plants carrying the *TA29-barnase;* the anthers are small. shriveled, and devoid of pollen. Parts E and F show the flower and anthers of plants whose genotypes include *TA29-barnase* plus *TA29-barstar;* the phenotype is virtually indistinguishable from wildtype.

Other Commercial Opportunities

As we have seen, organisms with novel phenotypes can be produced by genetic engineering, sometimes by combining the features of different organisms. Additional examples include a species of marine bacte-

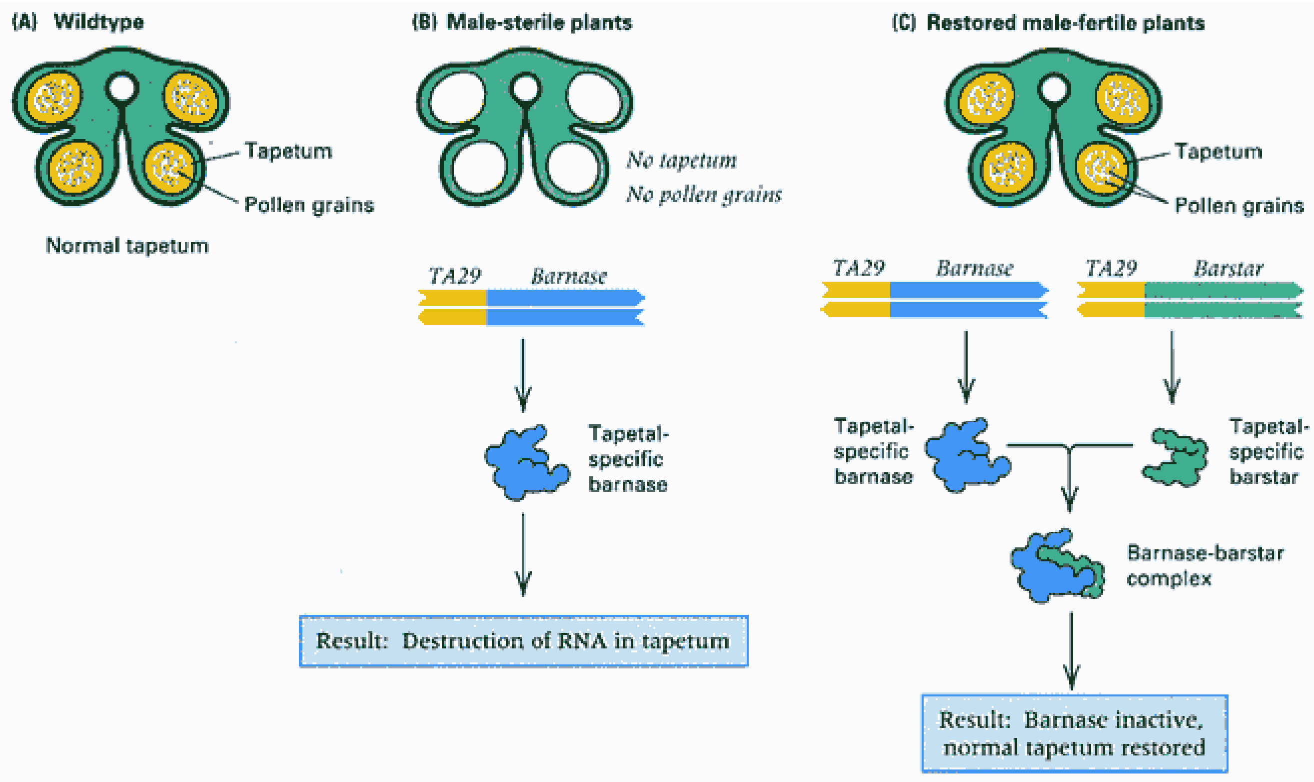


Figure 9.18
Engineered genetic male sterility and fertility restoration. (A) Pollen grains (PG) develop within a thin layer of tapetal cells (T). (B) Expression of the barnase RNA nuclease under the control of the tapetum-specific regulatory region of gene *TA29* destroys the tapetum and renders the plants male-sterile. (C) Expression of barstar (a barnase inhibitor) in tapetal cells inactivates barnase and restores fertility.
[Courtesy of Robert B. Goldberg.]

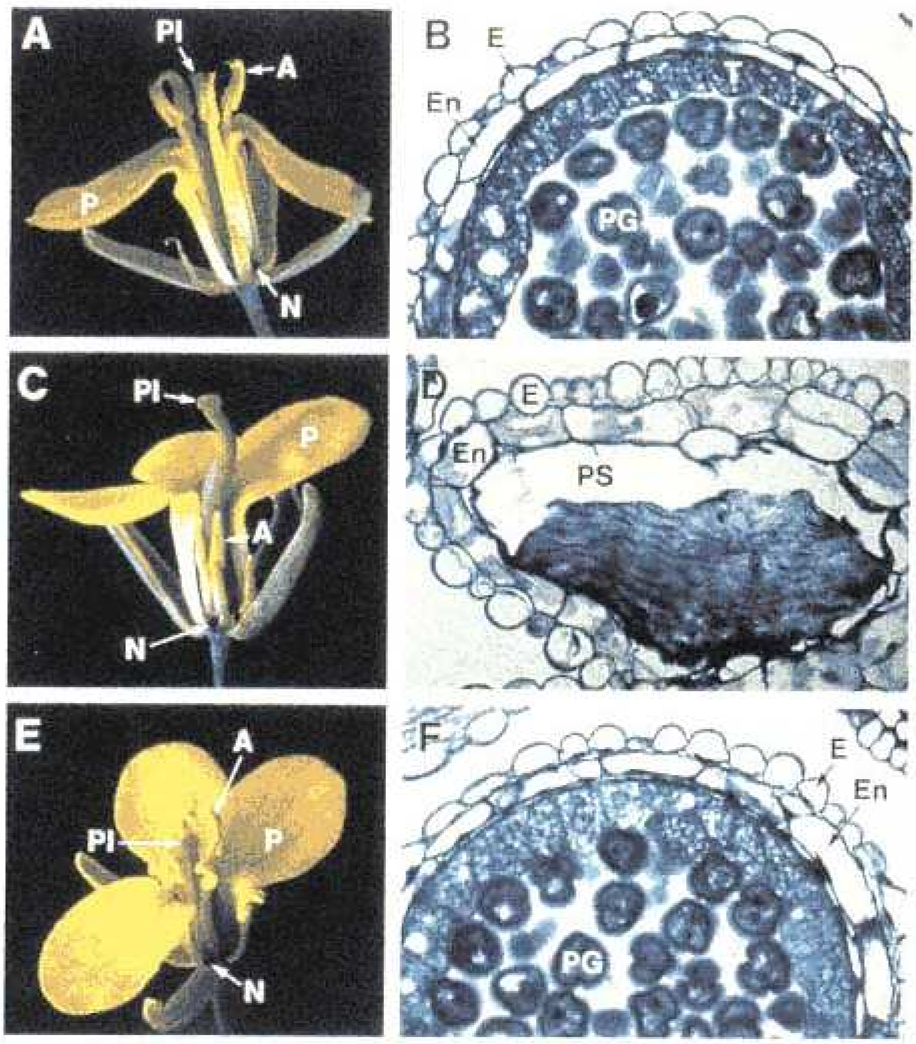


Figure 9.19
Flowers (A, C, and E) and cross sections of anthers (B, D, and F) of oilseed rape plants. The genotypes are as follows: A and B, wildtype; C and D, *TA29-barnase*; E and F, *TA29-barnase* + *TA29-barstar*. Note that the fertility-restored plants (E and F) are virtually indistinguishable from wildtype (A and B). The labeled parts of the flowers are A, anther; P, petal; P1, pistil; and N, nectary. The labeled parts of the cross sections are E, epidermis; En, endothecium; PG, pollen grain; PS, pollen sac; and T, tapetum.
[From C. Mariani, V. Gossele, M. De Beuckeleer, M. De Block, R. B. Goldberg, W. De Greef and J. Leemans. 1992. *Nature* 357: 384.]

ria that has been equipped with genes from other bacterial species for the metabolism of some components of petroleum, yielding an organism used in cleaning up oil spills in the oceans. Many biotechnology companies are at work designing bacteria that can synthesize industrial chemicals or degrade industrial wastes. Bacteria have also been created that are able to compost waste more efficiently and to fix nitrogen to improve the fertility of soil. A great deal of effort is currently being expended to create organisms that can convert biological waste into alcohol. A number of medicinally or commercially important molecules are routinely produced in genetically engineered cells, including human insulin and growth hormones produced in bacteria.

Altering the genotypes of plants (Figure 9.16) is an important application of recombinant DNA technology. It is possible to transfer genes from one plant species to another. The transferred genes affect yield, hardiness, or disease resistance. There are also attempts to alter the surface structure of the roots of grains, such as wheat, by introducing genes from legumes (peas, beans) to give the grains the ability that legumes possess to establish root nodules of nitrogen-fixing bacteria. If successful, this would eliminate the need to add nitrogenous fertilizers to soils where grains are grown.

The first engineered recombinant plant of commercial value was developed in 1985. An economically important herbicide is **glyphosate,** a weed killer that inhibits an enzyme in a metabolic pathway found only in plants and microorganisms. The pathway is the *shikimic acid pathway,* and glyphosate is a competitive inhibitor of the enzyme *EPSP synthase* (stands for *5-enolpyruvylshikimic acid-3-phosphate synthase).* Glyphosate is the active ingredient in the world's largest-selling herbicide, Roundup, which is used on more than 100 crops. Treatment requires that the compound be applied directly to the green foliage of the weeds because it is also toxic to the crop plant itself. However, the target gene of glyphosate is also present in the bacterium *Salmonella typhimurium.* A resistant form of the gene was obtained by mutagenesis and growth of *Salmonella* in the presence of glyphosate. Then the gene was cloned in *E. coli* and recloned in the T DNA of *Agrobacterium.* Transformation of plants with the glyphosate-resistance gene has yielded varieties of maize, cotton, tobacco, and other plants that are resistant to the herbicide. Thus fields of these crops can be sprayed with glyphosate at any stage of growth of the crop. The weeds are killed but the crop is unharmed.

Genetic engineering can also be used to control insect pests. For example, the black cutworm causes extensive crop damage and is usually combated with noxious insecticides. The bacterium *Bacillus thuringiensis* produces a protein that is lethal to the black cutworm, but the bacterium does not normally grow in association with the plants that are damaged by the worm. However, the gene coding for the lethal protein has been introduced into the soil bacterium *Pseudomonas fluorescens,* which lives in association with maize and soybean roots. Inoculation of soil with the engineered *P. fluorescens* helps control the black cutworm and reduces crop damage.

Uses in Research

Modern molecular genetics could not exist without recombinant DNA technology. It is an essential research tool. Reverse genetics makes it possible to isolate and alter a gene at will and introduce it back into a living cell or even the germ line. Besides saving time and labor, reverse genetics enables mutants to be constructed that cannot be formed in any other way. An example is the formation of double mutants of animal viruses, which naturally undergo recombination at such a low frequency that mutations can rarely be recombined by genetic crosses. The greatest impact of recombinant DNA on basic research has been in the study of eukaryotic gene regulation and development, and many of the principles of regulation and development summarized in Chapters 11 and 12 were determined by using these methods.

Production of Useful Proteins

Among the most important applications of genetic engineering is the production of large quantities of particular proteins that are otherwise difficult to obtain (for example, proteins that are present in only a few molecules per cell or that are produced in only a small number of cells or only in human cells). The method is simple in principle. A DNA sequence coding for the desired protein is cloned in a vector adjacent to an appropriate regulatory sequence. This step is usually done with cDNA because cDNA has all the coding sequences spliced together in the right order. Using a vector with a high copy number ensures that many copies of the coding sequence will be present in each bacterial cell, which can result in synthesis of the gene product at concentrations ranging from 1 to 5 percent of the total cellular protein. In practice, the production of large quantities of a protein in bacterial cells is straightforward, but there are often problems that must be overcome because, in the bacterial cell, which is a prokaryote, the eukaryotic protein may be unstable, may not fold properly, or may fail to undergo necessary chemical modification. Many important proteins are currently produced in bacterial cells, including human growth hormone, blood-clotting factors, and insulin. Patent offices in Europe and the United States have already issued well over 1000 patents for the clinical use of the products of genetically engineered human genes. Figure 9.20 gives a breakdown of the number of patents issued relative to clinical application.

Genetic Engineering with Animal Viruses

The genetic engineering of animal cells often makes use of retroviruses because their reverse transcriptase makes a double-stranded DNA copy of the RNA genome, which then becomes inserted into the chromosomes of the cell. DNA-to-RNA transcription occurs only after the DNA copy is inserted. The infected host cell survives the infection, retaining the retroviral DNA in its genome. These features of retroviruses make them convenient vectors for the genetic manipulation of animal cells, including those of birds, rodents, monkeys, and human beings.

Genetic engineering with retroviruses allows the possibility of altering the genotypes of animal cells. Because a wide variety of retroviruses are known, including many that infect human cells, genetic defects may be corrected by these procedures in the future. However, many retroviruses contain a gene that results in uncontrolled growth of the infected cell, thereby causing a tumor. When retrovirus vectors are used for genetic engineering, the tumor-causing gene is first deleted. The deletion also provides the space needed for the incorporation of foreign DNA sequences. The recombinant DNA procedure employed with retroviruses consists of synthesis in the laboratory of double-stranded DNA from the viral RNA, through use of reverse transcriptase. The DNA is then cleaved with a restriction enzyme and, by means of the techniques already described, foreign DNA is inserted. Trans-

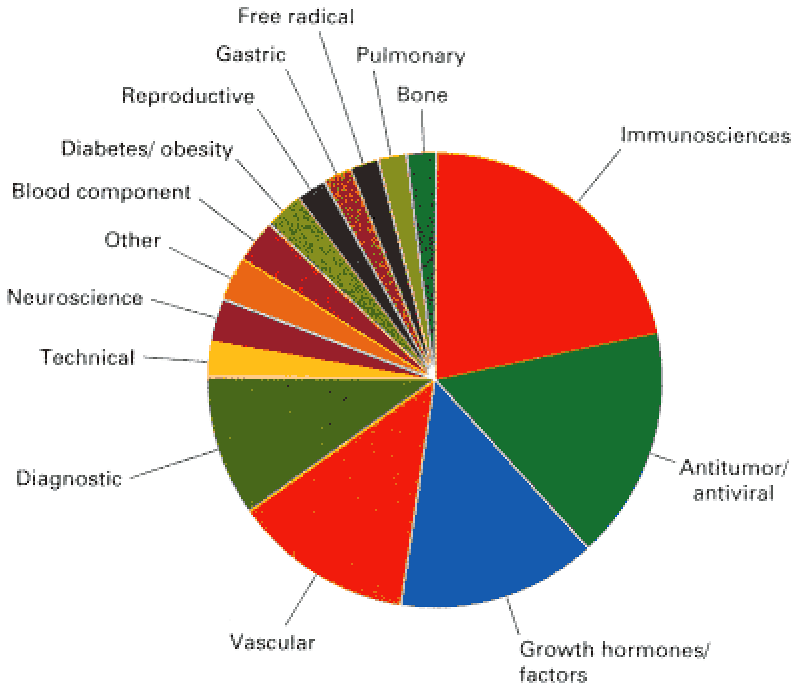


Figure 9.20
Relative number of patents issued for various clinical applications of the products of genetically engineered human genes
[Data from *Nature* 1996, 380: 388.]

formation yields cells with the recombinant retroviral DNA permanently inserted into the genome. In this way, the genotype of the cells can be altered.

Attempts are currently underway to assess the potential of genetic engineering in the treatment of cystic fibrosis. This disorder results from a mutation in a gene for chloride transport in the lungs, pancreas, and other glands, leading to abnormal secretions and the accumulation of a thick, sticky, honey-like mucus in the lungs. Among its chief symptoms are recurrent respiratory infections. A genetically engineered form of the common cold virus is being used in clinical trials in an attempt to introduce the nonmutant form of the cystic fibrosis gene directly into the lungs of affected patients. This is an example of **gene therapy.** The exciting potential of gene therapy lies in the possibility of correcting genetic defects—for example, restoring the ability to make insulin persons with diabetes or correcting immunological deficiencies in patients with defective immunity. However, a number of major problems stand in the way of gene therapy becoming widely used. At this time, there is no completely reliable way to ensure that a gene will be inserted only into the appropriate target cell or target tissue, and there is no completely reliable means of regulating the expression of the inserted genes.

A major breakthrough in disease prevention has been the development of synthetic vaccines. Production of certain vaccines is difficult because of the extreme hazards of working with large quantities of the active virus—for example, the human immunodeficiency virus (HIVI) that causes acquired immune deficiency syndrome (AIDS). The danger is minimized by cloning and producing viral antigens in a nonpathogenic organism. Vaccinia virus, the agent used in smallpox vaccination, has been very useful for this purpose. Viral antigens are often on the surface of virus particles, and some of these antigens can be engineered into the coat of vaccinia. For example, engineered vaccinia with surface antigens of hepatitis B, influenza virus, and vesicular stomatitis virus (which kills cattle, horses, and pigs) have produced useful vaccines in animal tests. A surface antigen of *Plasmodium falciparum*, the parasite that causes malaria, has also been placed in the vaccinia coat, a development that may ultimately lead to an antimalaria vaccine.

Diagnosis of Hereditary Diseases

Probes derived by recombinant DNA methods are widely used in prenatal detection of disease, including cystic fibrosis, Huntington disease, sickle-cell anemia, and hundreds of other genetic disorders. In some cases, probes derived from the gene itself are used; in other cases, molecular markers, such as STRPs, that are genetically linked to the disease gene are employed (Section 4.4). If the disease gene itself, or a region close to it in the chromosome, differs from the normal chromosome in the positions of one or more cleavage sites for restriction enzymes, then these differences can be detected with Southern blots using cloned DNA from the region as the probe (Section 5.7) or in DNA amplified by the polymerase chain reaction (Section 5.8).

The genotype of the fetus can be determined directly. These techniques are very sensitive and can be carried out as soon as tissue from the fetus (or even from the embryonic membranes) can be obtained.

9.6— Analysis of Complex Genomes

Gene structure and function are understood in considerable detail because of the manipulative capability of recombinant DNA technology combined with genetic analysis. Most recombinant DNA techniques are limited to the manipulation of DNA fragments smaller than about 40 kb. Although this range includes most cDNAs and many genes, some genes are much larger than 50 kb and so must be analyzed in smaller pieces. However, recent advances in recombinant DNA technology have made it possible for very large DNA molecules to be cloned and analyzed. These methods are suitable not only for the study of very large genes, but also for the analysis of entire genomes. This section discusses complex genomes and the recombinant DNA technology that has made their analysis possible.

Sizes of Complex Genomes

The simplest genomes are those of small bacterial viruses whose DNA is smaller than 10 kb—for example, the *E. coli* phage ϕX174 whose genome contains 5386 base pairs. Toward the larger end of the spectrum is the human genome, consisting of 3×10^9 base pairs. The range is so large that comparisons are often made in terms of information content, with the base pairs of DNA analogous to the letters in a book. The analogy is apt, because the range of sizes is from a few pages of text to an entire library.

The book analogy is used to compare genome sizes in Figure 9.21. The volumes are about the size of a big-city telephone book: 1500 pages per volume with 25,000 characters (nucleotides) per page. In these terms, the 50-kb genome of bacteriophage λ fills two pages, and the 4.7-Mb genome of *E. coli* needs about 200 pages. Among eukaryotes, the size range for organisms of

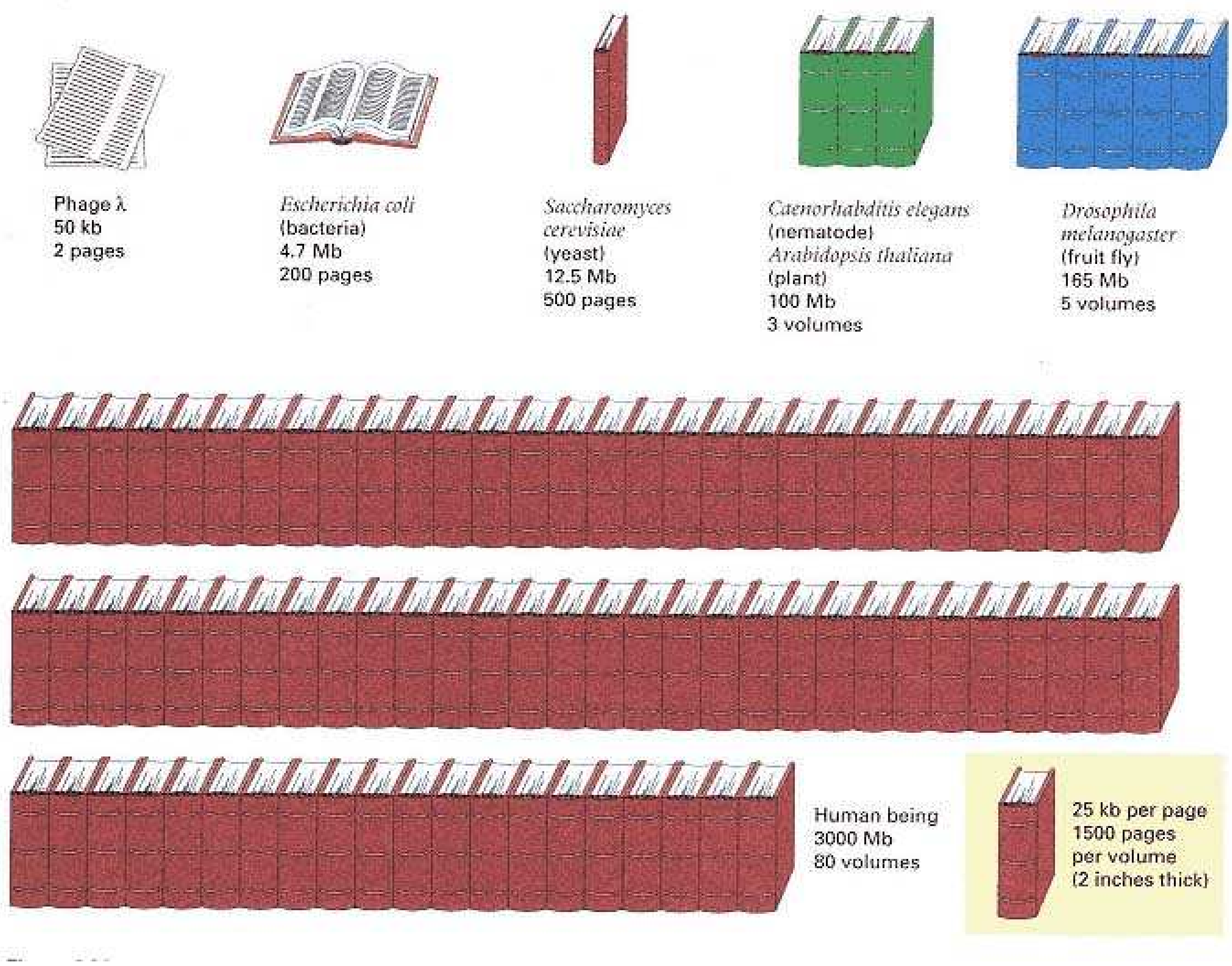


Figure 9.21
Relative sizes of genomes if they were printed at 25,000 characters per page and bound in 1500 page volumes. One volume would contain about as many characters as a telephone book 2.5 inches thick. The *E. coli* genome would require about 200 pages, yeast 500 pages, and so forth.

genetic interest varies considerably. Yeast would take up one-third of a volume (genome size 12.5 Mb), the nematode *Caenorhabditis elegans* and the flowering plant *Arabidopsis thaliana* 3 volumes (100 Mb) each, *Drosophila melanogaster* 5 volumes (165 Mb), and human beings 80 volumes (3000 Mb). To put the matter in practical terms, let us calculate how many cosmids would be necessary to clone an entire genome in a cosmid library in which the average insert size is 40 kb. To have a good chance of containing most sequences of interest, a genomic library must contain several copies of the genome. For example, for a genomic library to have a 95 percent chance of containing any given single-copy sequence, the library should consist of enough clones that the DNA fragments in the clones, added together, would equal 3 haploid genomes (3 **genome equivalents).**

The rationale for the 3 genome equivalents is that in a library of n clones, each containing a small fraction f of the haploid genome, the probability that any single-copy DNA fragment will be included in a particular clone is f, assuming that the DNA fragment is small in relation to the insert size of the clones and that there is no bias toward or against cloning the fragment of interest. The probability that none of the n

clones contains the fragment is therefore $(1 - f)^n$, so the probability that at least one clone contains the fragment is $1 - (1 - f)^n$, which we want to be greater than 95 percent. Hence,

$$1 - (1 - f)^n \geq 0.95 \quad \text{or} \quad (1 - f)^n < 0.05$$

Taking natural logarithms of both sides of the right-hand inequality and simplifying yields

$$n \geq \ln(0.05)/\ln(1 - f).$$

Now, In(0.05) ≈ -3, and when f is small, In$(1 - f) = -f$ to a very close approximation, as can be seen by substituting in any small number for f (for example, $f = 0.001$). Hence the inequality becomes $n \geq -3/-f$ or $nf \geq 3$. But what is nf? It is the fraction of the genome present in each clone multiplied by the total number of clones or, in other words, the number of haploid genome equivalents present in the library. (To verify that you understand this argument, you should try to show that 4.6 genome equivalents are needed to ensure with 99 percent confidence that any fragment will be present.)

Taking 3 genome equivalents as needed for 95 percent coverage of the single-copy sequences, this coverage of yeast would require a little less than 1000 clones in a cosmid vector averaging 40-kb inserts. Three genome equivalents of nematode or *A. thaliana* DNA would require 7500 clones; that of *Drosophila* DNA, 12,000 clones; and that of human DNA, 225,000 clones. The larger genomes require a formidable number of clones. An alternative to larger numbers of clones is larger fragments within the clones. Clones that contain large fragments of DNA can be made and analyzed by the procedures described in the following sections.

Manipulation of Large DNA Fragments

Section 6.4 included an examination of pulsed-field gel electrophoresis, a procedure by which DNA fragments exceeding several megabases can be separated. In only a few organisms are any chromosomes as small as several megabases, however. In *Drosophila*, the smallest wildtype chromosome is about 6 Mb, and even the smallest rearranged and deleted chromosome is 1 Mb. Most chromosomes in higher eukaryotes are very much larger. The smallest human chromosome is chromosome 21, and its long arm alone is approximately 42 Mb. Therefore, even with the ability to separate large DNA fragments by electrophoresis, it is necessary to be able to cut the DNA in the genome into fragments of manageable size. This can be done with a class of restriction enzymes, each of which cleaves at a restriction site consisting of eight bases rather than the usual six or four. For example, the restriction sites of the eight-cutter enzymes *Not*I and *Sfi*I are

```
NotI 5'-GC*GGCCGC-3'

SfiI 5'-GGCCNNNN*NGGCC-3'
```

Both enzymes cleave double-stranded DNA at the positions of the restriction sites, and the asterisks denote the position at which the backbone is cut in each DNA strand. (The N's in the *Sfi*I restriction site mean that any nucleotide can be present at this site.) In a genome with equal proportions of the four nucleotides and random nucleotide sequences, the average size of both *Not*I and *Sfi*I fragments is $4^8 = 65{,}536$ nucleotide pairs, or about 66 kb. Many genomes are relatively A + T-rich, and the average *Not*I and *Sfi*I fragment size is larger than 66 kb. In vertebrate genomes, there is a bias against long runs of G's and C's, so many *Not*I and *Sfi*I fragments are considerably larger than 66 kb. In any case, the use of eight-cutter restriction enzymes allows complex DNA molecules to be cleaved into a relatively small number of large fragments that can be separated, cloned, and analyzed individually.

Cloning of Large DNA Fragments

Large DNA molecules can be cloned intact in bacterial cells with the use of specialized vectors that can accept large inserts. An example is a vector derived from the bacteriophage P1 (Figure 9.4D), which is used to clone DNA fragments averaging approximately 85 kb. DNA fragments in the appropriate size range can be produced by breaking larger molecules into fragments of the desired size by physical means, by treatment with restriction enzymes that have infrequent cleavage sites (for example, *NotI* or *SfiI)*, or by treatment with ordinary restriction enzymes under conditions in which only a fraction of the restriction sites are cleaved (**partial digestion).** Cloning the large molecules consists of mixing the large fragments of source DNA with the vector, ligation with DNA ligase, introduction of the recombinant molecules into bacterial cells, and selection for the clones of interest. These methods are generally similar to those described in Section 9.2 for the production of recombinant molecules containing small inserts of cloned DNA.

DNA fragments as large as 1 Mb can be cloned intact in yeast cells with the use of special vectors for creating **yeast artificial chromosomes,** or **YACs.** The general structure of a YAC vector is diagrammed at the upper left in Figure 9.22. The YAC vector contains four types of genetic elements: (1) a cloning site, (2) a yeast centromere

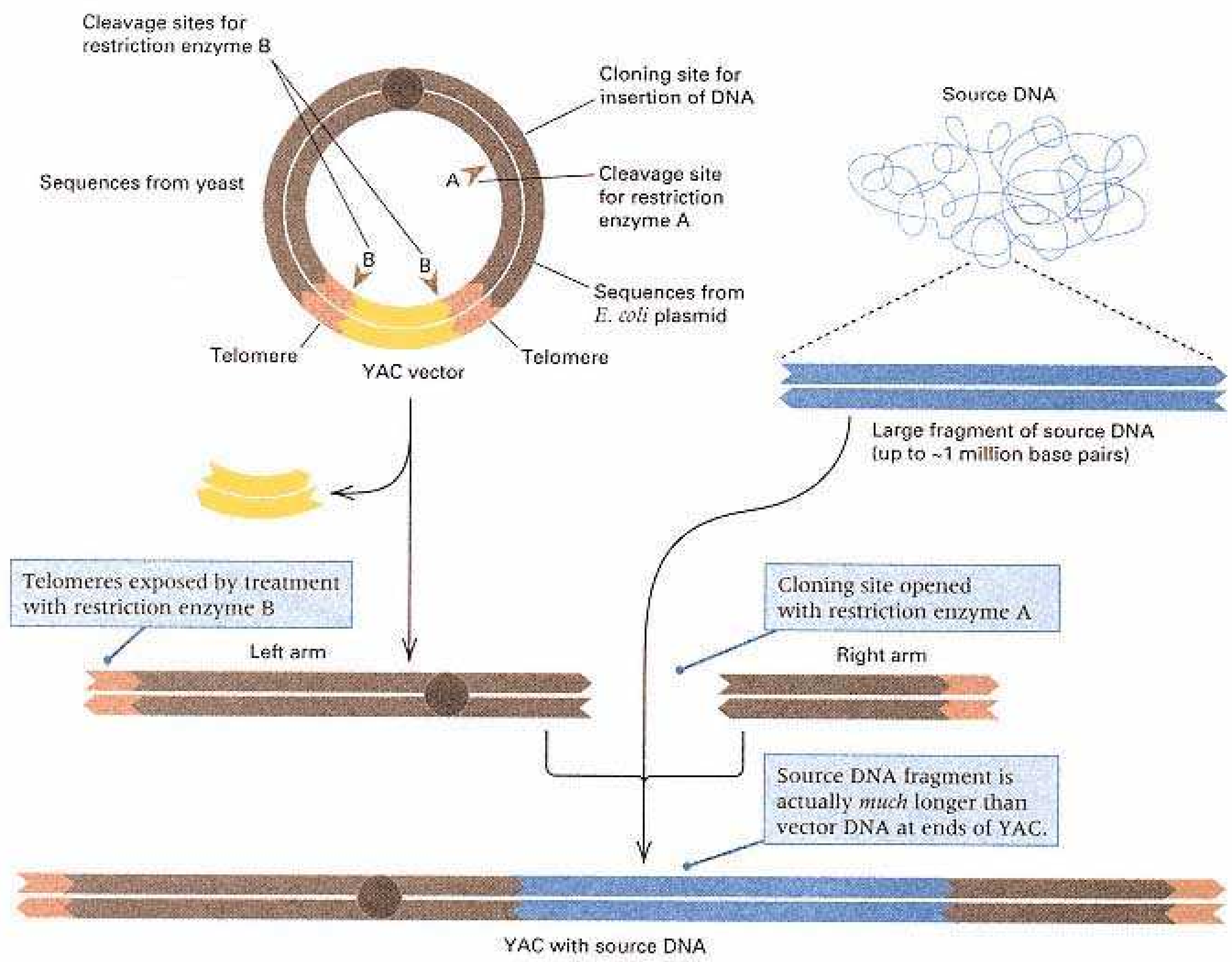


Figure 9.22
Cloning large DNA fragments in yeast artificial chromosomes (YACs). The vector (upper left) contains sequences that allow replication and selection in both *E. coli* and yeast, a yeast centromere, and telomeres from *Tetrahymena*. In producing the YAC clones, the vector is cut by two restriction enzymes (A and B) to free the chromosome arms. These arms are ligated to the ends of large fragments of source DNA, and yeast cells are transformed. Many ligation products are possible, but only those that consist of source DNA flanked by the left and right vector arms form stable artificial chromosomes in yeast.

and genetic markers that are selectable in yeast, (3) an E. coli origin of replication and genetic markers that are selectable in *E. coli,* and (4) a pair of telomere sequences from *Tetrahymena.* Therefore, a YAC vector is a **shuttle vector** that can replicate and be selected in both *E. coli* and yeast.

Use of the YAC vector in cloning is also illustrated in Figure 9.22. The circular YAC vector is isolated after growth in E. coli and cleaved with two different restriction enzymes—one that cuts only at the cloning site (denoted A) and one that cuts near the tip of each of the telomeres (denoted B). Discarding the segment between the telomeres results in the two telomere-bearing fragments shown, which form the arms of the yeast artificial chromosomes. Ligation of a mixture containing source DNA and YAC vector arms results in a number of possible products. However, transformation of yeast cells and selection for the genetic markers in the YAC arms yields only the ligation product shown in Figure 9.22, in which a fragment of source DNA is inserted at a site within the right arm of the YAC vector. This product is recovered because it is the only true chromosome possessing a single centromere and two telomeres. Products that contain two YAC left arms are dicentric, those with two YAC right arms are acentric, and both of these types of products are genetically unstable (Section 7.1). YACs that have donor DNA inserted at the cloning site can be identified because the inserted DNA interrupts a yeast gene present at the cloning site and renders it nonfunctional. Yeast cells containing YACs with inserts of particular sequences of donor DNA can be identified in a number of ways, including colony hybridization (Figure 9.9), and these cells can be grown and manipulated in order to isolate and study the YAC insert. Figure 9.23 shows a region of the *Drosophila* salivary gland chromosomes that hybridizes with a YAC containing an insert of 300 kb. The entire genome of *Drosophila* could be contained in only 550 YAC clones of this size.

Physical Mapping

The development of methods for isolating and cloning large DNA fragments has stimulated major efforts to map and sequence the human genome, which is the principal goal of an effort termed the **human genome project.** The project also aims to map and sequence the genomes of a number of model genetic organisms, including *E. coli,* yeast, *Arabidopsis thaliana,* the nematode *C. elegans, Drosophila,* and the laboratory mouse. The first stage in the analysis of complex genomes is usually the production of a **physical map,** which is a diagram of the genome depicting the physical loca-

Figure 9.23
Hybridization *in situ* between *Drosophila* DNA cloned into a yeast artificial chromosome and the giant salivary gland chromosomes. The yeast artificial chromosome contains DNA sequences derived from numerous adjacent salivary bands—in this example, the bands in regions 52B through 52E in the right arm of chromosome 2. On average, each salivary band contains about 20 kb of DNA, but the bands vary widely in DNA content.

Connection YAC-ity YAC

David T. Burke, Georges F. Carle, and Maynard V. Olson 1987

Washington University,
St. Louis, Missouri

Cloning of Large Segments of Exogenous DNA into Yeast by Means of Artificial Chromosome Vectors

Technological innovations often open up whole new areas of investigation by making novel types of experiments possible. A case in point is the development of methods for cloning large fragments of DNA, pioneered by the development of yeast artificial chromosomes (YACs). About the origin of the method, Olson recalls: "David Burke [then a graduate student] was in his third year of a molecular genetics project that was going well, when he did what all students are told not to do—start another project." The new approach was a great stimulus for genome analysis—the study of the organization of DNA in complex genomes. Large-fragment DNA cloning made it possible to obtain a physical map of the DNA in the entire human genome and to correlate the genetic map based on homologous recombination with the physical map based on the analysis of cloned DNA fragments. The result has been great activity in human gene mapping, including genetic mapping of mutations that cause inherited diseases, and also rapid progress in isolating the DNA of the mutant genes and identifying their normal functions.

Standard recombinant DNA techniques, . . .whose capacities for exogenous DNA range up to 50 kilobase pairs (kb), are well suited to the analysis and manipulation of genes from organisms in which the genetic information is tightly packed. It is increasingly apparent, however, that many of the functional genetic units in higher organisms span enormous tracts of DNA. For example, . . .recent estimates of the size of the gene that is defective in Duchenne's muscular dystrophy suggest that this single genetic locus, whose protein-coding func-

> ***We report here the development of a highcapacity cloning system that is based on the in vitro construction of linear DNA molecules that can be transformed into yeast, where they are maintained as artificial chromosomes.***

tion could be fulfilled by as little as 15 kb of DNA, actually covers more than a million base pairs. . . . We report here the development of a high-capacity cloning system that is based on the in vitro construction of linear DNA molecules that can be transformed into yeast, where they are maintained as artificial chromosomes. . . . The vector incorporates all necessary functions into a single plasmid that can replicate in *Escherichia coli.* This plasmid, called a "yeast artificial chromosome" (YAC) vector, supplies a cloning site within a gene whose interruption is phenotypically visible, an autonomous-replication sequence with properties expected of a replication origin, a yeast centromere, selectable markers on both sides of the centromere, and two sequences that seed telomere formation in yeast. . . . An initial test of the vector system involved cloning human DNA into the YAC vector. . . . A number of clones were analyzed to determine whether or not the artificial chromosomes that had been produced had the expected structure. . . . The test cases appear to be propagated as faithful copies of the source DNA. . . . Further experience with the YAC cloning system will be required to assess such issues as the stability of the clones, the extent to which the source DNA is randomly sampled, and the biological activity of the cloned DNA. Nevertheless, there are grounds for optimism that YAC vectors could even offer important advantages over standard cloning systems in these areas. . . . The demonstration of the basic feasibility of generating large recombinant DNA's in vitro and transforming them into easily manipulated host cells may stimulate experimentation with other combinations of vectors and hosts. There is a strong incentive to develop such systems since they are directed toward the major remaining gap in our ability to dissect the genomes of higher organisms.

Science 236: 806–812

tions of various landmarks along the DNA. The landmarks in a physical map usually consist of the locations of particular DNA sequences, such as coding regions or sequences present in particular cloned DNA fragments. If the landmarks are the locations of the cleavage sites for restriction enzymes, then the physical map is also a restriction map (Section 5.7). More useful landmarks are the positions of molecular markers, such as STRPs (simple tandem repeat polymorphisms), that have also been located in the genetic map (Section 4.4). Molecular markers serve to unify the genetic map and the physical map of an organism. The utility of a physical map is that it affords a single framework for organizing and integrating diverse types of genetic information, including the positions of chromosome bands, chromosome breakpoints, mutant genes, transcribed regions, and DNA sequences.

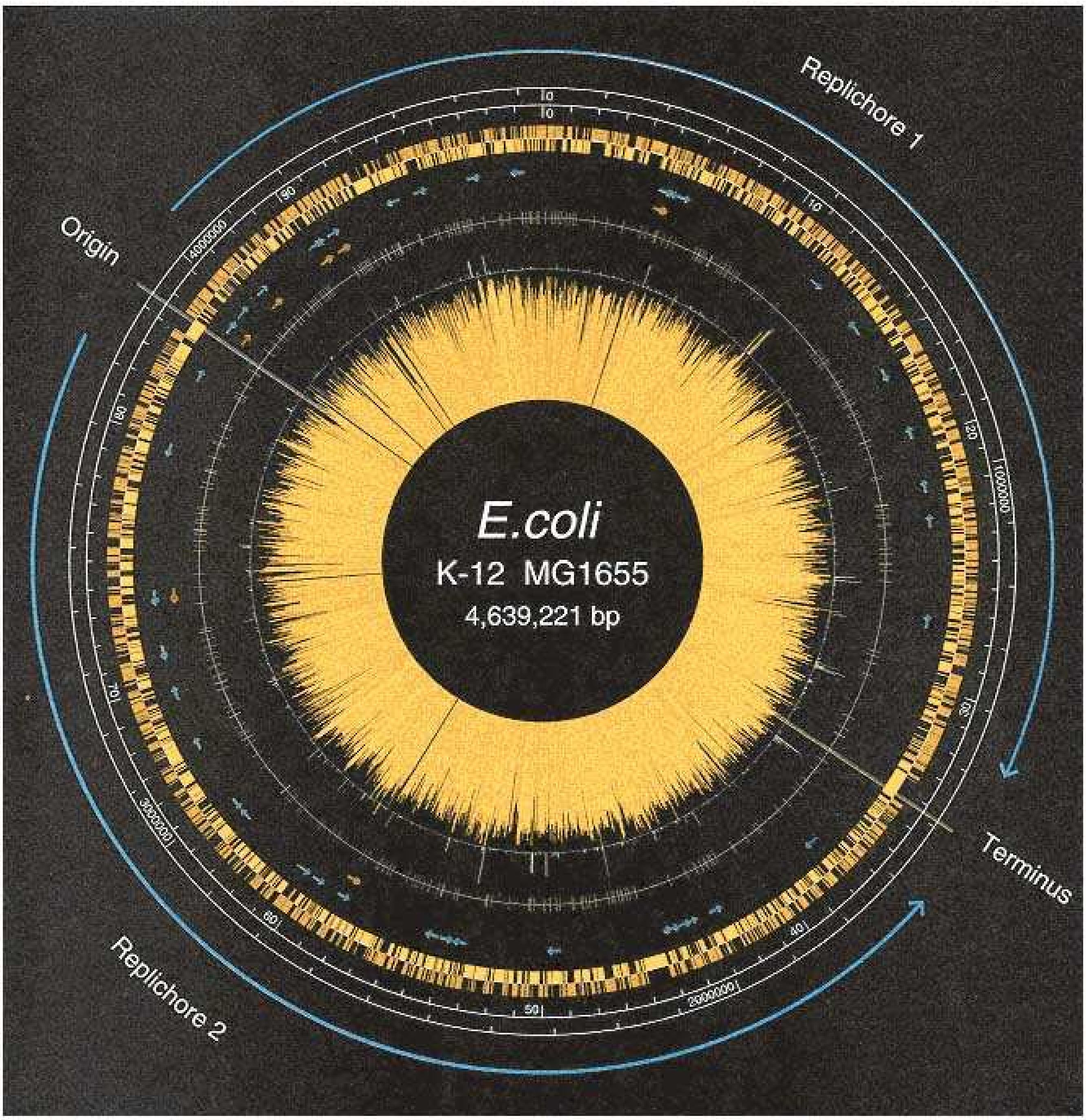


Figure 9.24
Diagram of the DNA sequence organization of *Escherichia coli* strain K-12. The coordinates are given in base pairs as well as in minutes on the genetic map. The coding sequences are shown as gold and yellow bars, which are transcribed in a clockwise (gold) or counterclockwise (yellow) direction. Green and red arrows denote genes for transfer RNAs or for ribosomal RNAs, respectively. The gold rays of the "sunburst" are proportional to the degree of randomness of codon usage in the coding sequences. Genes with the longest rays use the codons in the genetic code almost randomly. The origin and terminus of DNA replication are indicated. Bidirectional replication creates two "replichores." The peaks on the circle immediately outside the sunburst indicate coding sequences with high similarity to previously described bacteriophage proteins.
[Courtesy of Frederick R. Blattner and Guy Plunkett III. From F. R. Blattner et al. 1997. *Science* 277: 1453.]

The Genome of E. coli

A diagram of the genome of *E. coli*, based on the complete DNA sequence, is illustrated in Figure 9.24. The coordinates of the circle are given in minutes on the genetic map (0–100) as well as in base pairs. The "replichores" are the two halves of the circle replicated bidirectionally starting from the origin. The gold bars on the outside denote genes whose transcription is from left to right, the yellow bars on the inside denote genes transcribed from right to left. The green arrows show the positions of tRNA genes, red arrows rRNA genes. The circle just inside the red arrows shows the positions of a 40-base-pair repetitive sequence of unknown function that is present 581 times. The rays of the yellow "sunburst" in the middle show the usage of codons among all of the coding sequences. The length of each ray is proportional to the degree to which codons are used randomly. Short rays indicate genes with a highly biased usage of codons, which is usually associated with a high level of gene expression. This strain has a genome of 4.6 megabases.

The Human Genome

A more complex type of physical map is illustrated in Figure 9.25. The map covers a small part of human chromosome 16, and it illustrates how the physical map is used to organize and integrate several different levels of genetic information. A map of the metaphase banding pattern of chromosome 16 is shown across the top. (The entire chromosome contains about 95 Mb of DNA.) Beneath the cytogenetic map, the somatic cell hybrid map shows the locations of chromosome breakpoints observed in cultured hybrid cells that contain only a part of chromosome 16. The genetic linkage map depicts the locations of various genetic markers studied in pedigrees; most of the genetic markers are molecular markers, such as STRPs (Section 4.4). One region of the genetic linkage map shows the location of a YAC clone, which has been assigned a physical location in the cytogenetic map by *in situ* hybridization. The large DNA insert in the YAC clone is also represented in a set of overlapping cosmid or P1 clones; these clones define a contig covering a contiguous region of the genome without any gaps.

The various levels of the physical map in Figure 9.25 are connected by a special type of genetic marker shown at the bottom: a **sequence-tagged site,** or **STS.** An STS marker is a DNA sequence, present once per haploid genome, that can be amplified with a suitable pair of oligonucleotide primers by means of the polymerase chain reaction (PCR), described in Section 5.8. Hence an STS marker defines a unique site in the genome whose presence in a cloned DNA fragment can be detected by PCR amplification. In Figure 9.25, for example, the STS marker is present in two clones of the cosmid (or P1) contig, as well as in the YAC clone. Furthermore, the STS can be positioned on the somatic cell hybrid map by carrying out the PCR reaction with DNA from hybrid cells that contain rearranged or deleted chromosomes, and the amplified PCR product (or a clone containing the STS) can be used in *in situ* chromosome hybridization to localize the STS on the cytogenetic map. Therefore, STS markers are a type of genetic marker that can be used to integrate different types of information in a physical map. At present, 94 percent of the human genome is covered by a set of 16,494 YAC clones interconnected by 10,850 STS markers. The YAC clones define 377 contigs, averaging 8 Mb in size, and there is an average spacing between adjacent STS markers of 276 kb. About half of the STS markers have also been located on the human genetic linkage map.

Although an STS may consist of any type of sequence present once per haploid genome, some genome projects rely extensively on STS markers derived from cDNAs, because these markers represent coding regions that are likely to be of greater long-term interest than single-copy sequences that are noncoding. Intensive study of human cDNAs has identified a substantial proportion of the estimated 80,000 or so genes in the human genome. The most ambitious cDNA project to date included sequencing about 300,000 partial cDNAs, among them cDNAs obtained from cDNA libraries prepared from 37 distinct

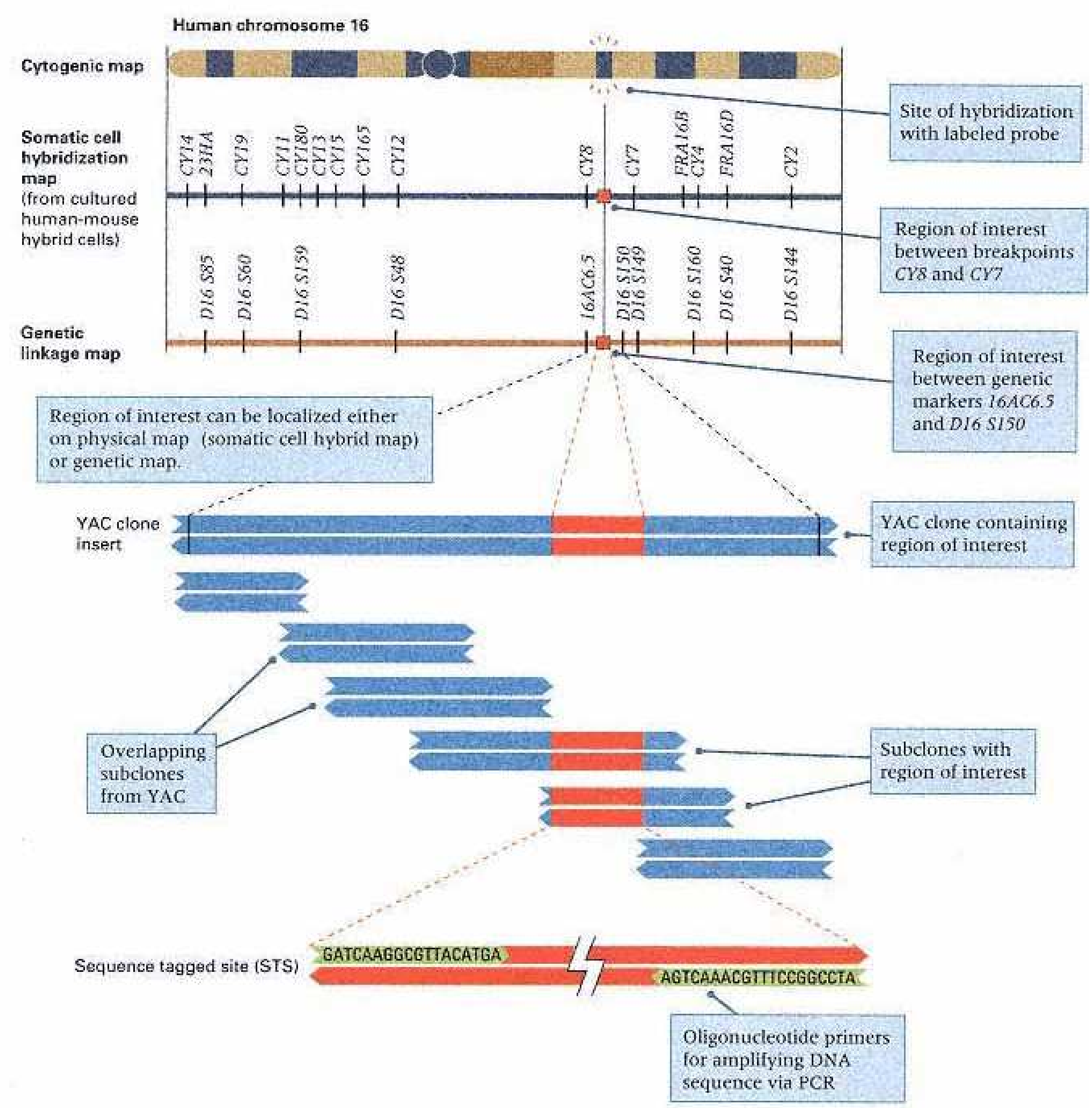


Figure 9.25
Integrated physical map of a small part of human chromosome 16. The map contains information about (1) the banding pattern of the metaphase chromosome (cytogenetic map), (2) the position of a particular sequence in the chromosome derived from *in situ* hybridization, (3) the locations of chromosome breakpoints in cultured cells (somatic cell hybrid map), (4) the positions of genetic markers analyzed by means of recombination in pedigrees (genetic linkage map), (5) the inserted DNA (blue) present in a YAC clone, and (6) a set of overlapping cosmid or P1 clones forming a contig (coverage of a contiguous region of the genome without any gaps). The various levels of the map are integrated by sequence-tagged sites (STSs), sequences present once per haploid genome that can be amplified with the polymerase chain reaction.
[After an illustration in *Human Genome: 1991–92 Program Report*, United States Department of Energy.]

human organs and tissues. The total DNA sequence obtained was 83 million base pairs. Computer matching among the cDNA sequences revealed 87,983 distinct sequences, many of which could be assigned a function on the basis of similarity with already known genes from human beings or from other organisms. Figure 9.26 gives a breakdown of the cDNA sequences by type of function. Approximately 40 percent of human genes are implicated in basic energy metabolism, cell structure, homeostasis, or cell division; a further 22 percent are concerned with RNA and protein synthesis and processing; and 12 percent are associated with signaling and communication between cells. Figure 9.27 summarizes the results of examining the tissue-specific cDNA libraries. For each organ or tissue type, the first number is the total number of cDNA clones sequenced, and the number in parentheses is the number of distinct cDNA sequences found among the total from that organ or tissue type.

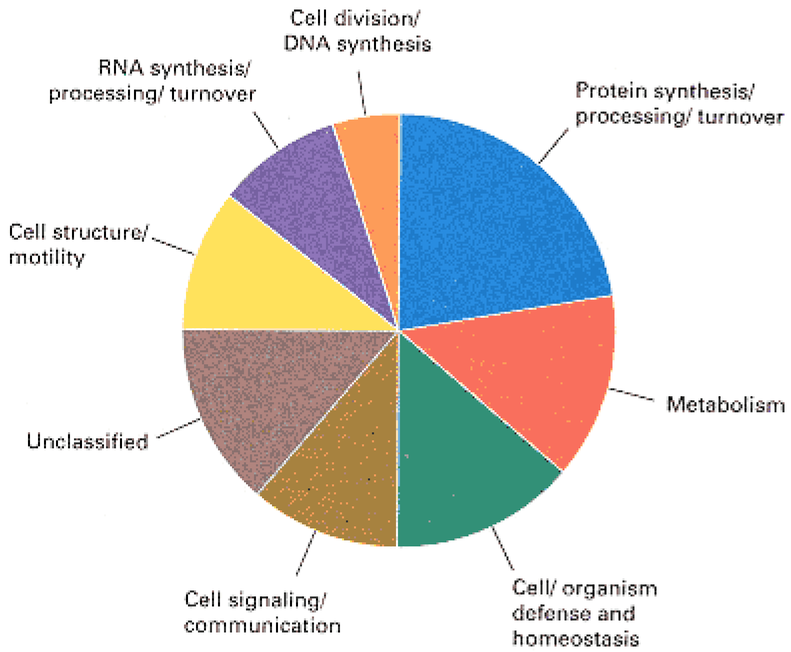
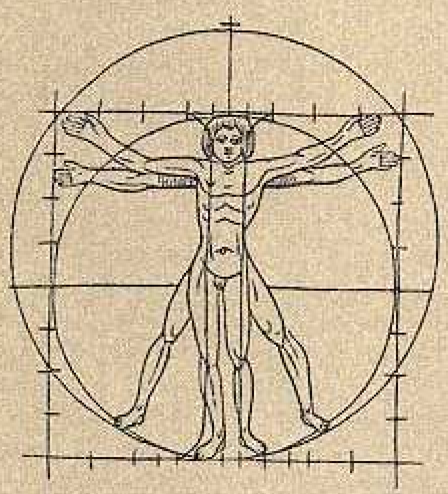


Figure 9.26
Classification of cDNA sequences by function. The chart is based on over 13,000 distinct, randomly selected human cDNA sequences.
[Data courtesy of Craig Venter and the Institute for Genomic Research.]



Source of RNA	Sequences	Source of RNA	Sequences
Skin	3,043 (629)	Brain	67,679 (3,195)
Blood cells	23,505 (2,194)	Eye	1,932 (574)
Bone	5,736 (904)	Salivary gland	186 (17)
Thyroid	2,381 (584)	Esophagus	194 (76)
Parathyroid	197 (46)	Adipose tissue	2,412 (581)
Endothelial cells	5,736 (1,031)	Heart	9,400 (1,195)
Liver	37,807 (2,091)	Peritoneum	884 (163)
Gall bladder	3,754 (768)	Spleen	7,924 (924)
Colon	4,832 (879)	Adrenal gland	3,427 (658)
Smooth muscle	297 (127)	Pancreas	5,534 (1,094)
Small intestine	1,009 (297)	Kidney	3,213 (712)
Prostate	7,971 (1,203)	Epididymus	1,716 (370)
Testis	7,117 (1,232)	Skeletal muscle	4,693 (735)
Ovary	3,848 (504)	Synovial membrane	3,889 (813)
Uterus	6,392 (1,059)	Embryo	19,291 (1,989)
Placenta	12,148 (1,290)	Thymus	2,412 (261)

Figure 9.27
Classification of cDNA sequences by organ or tissue type. In each category, the initial number is the total number of cDNA clones examined. The number in parentheses is the number of distinct sequences found per organ or tissue type.

[Data from M. D. Adams and 84 other authors. 1995. *Nature* 377 (Suppl.): 3.]

Genome Evolution in the Grass Family

The discovery of unexpectedly regular relationships among the genomes of several cereal grasses in the Family Gramineae must be numbered among the extraordinary results that have come from applications of genome analysis. The cereal grasses are among our most important crop plants. They include rice, wheat, maize, millet, sugar cane, sorghum, and other cereals. The genomes of grass species vary enormously in size. The smallest, at 400 Mb, is found in rice; the largest, at 17,000 Mb, is found in wheat. Although some of the difference in genome size results from the fact that wheat is an allohexaploid (Chapter 7) whereas rice is a diploid, a far more important factor is the large variation from one species to the next in types and amount of repetitive DNA sequences present. Each chromosome in wheat contains approximately 25 times as much DNA as each chromosome in rice. For comparison, maize has a genome size of 2500 Mb; it is intermediate in size among the grasses and approximately the same size as the human genome.

In spite of the large variation in chromosome number and genome size in the grass family, there are a number of genetic and physical linkages between single-copy genes that are remarkably conserved amid a background of very rapidly evolving repetitive DNA sequences. In particular, each of the conserved regions can be identified in all the grasses and referred to a similar region in the rice genome. The situation is as depicted in Figure 9.28. The rice chromosome pairs are numbered R1 through R12, and the conserved regions within each chromosome are indicated by lowercase letters, for example, R1a and R1b. In each of the other species, each chromosome pair is diagrammed according to the arrangement of segments of the rice genome that contain single-copy DNA sequences homologous to those in the corresponding region of the chromosome of the species in question. For example, the wheat monoploid chromosome set is designated W1 through W7. One region of W1 contains single-copy sequences that are homologous to those in rice segment R5a, another contains single-copy sequences that are homologous to those in rice segment R10, and still another contains single-copy sequences that are homologous to those in rice segment R5b. The genomes of the other grass species can be aligned with those of rice as shown. Each of such conserved genetic and physical linkages is called a **synteny group.** Synteny groups are found in other species comparisons as well. For example, many synteny groups are shared between the human and the mouse genomes. The human-mouse synteny groups are often useful in identifying the mouse homolog of a human gene.

Relative to the synteny groups in Figure 9.28, note that the maize genome has a repetition of segments, indicated by the connecting lines. The relationships confirm what some maize geneticists had long suspected, that maize is a complete, very ancient tetraploid with the complication that the two complete genomes are rearranged relative to each other. Further-more, most of the larger chromosomes (1, 2, 3, 4 and 6) comprise one of the genomes, and most of the short chromosomes (5, 7, 8, 9, and 10) comprise the other.

The synteny groups among the grass genomes are shown in a different format in Figure 9.29. In this case, the segments are formed into a circle in the same order in which they are aligned in the hypothetical ancestral chromosome (Figure 9.28G). There is no evidence that the ancestral cereal chromosome was actually a circle. It seems highly unlikely that it was anything other than a normal linear chromosome. However, the value of the circular diagram is that it shows the arrangement of the synteny groups in all of the grass genomes simultaneously, which makes comparisons much easier. Because of the synteny groups in the genomes, homologous genes can often be identified by location alone. For example, both wheat and maize have dwarfing mutations in which the mutant plants are insensitive to the plant hormone gibberellin. A line through the positions of these mutations in the wheat (Triticeae)

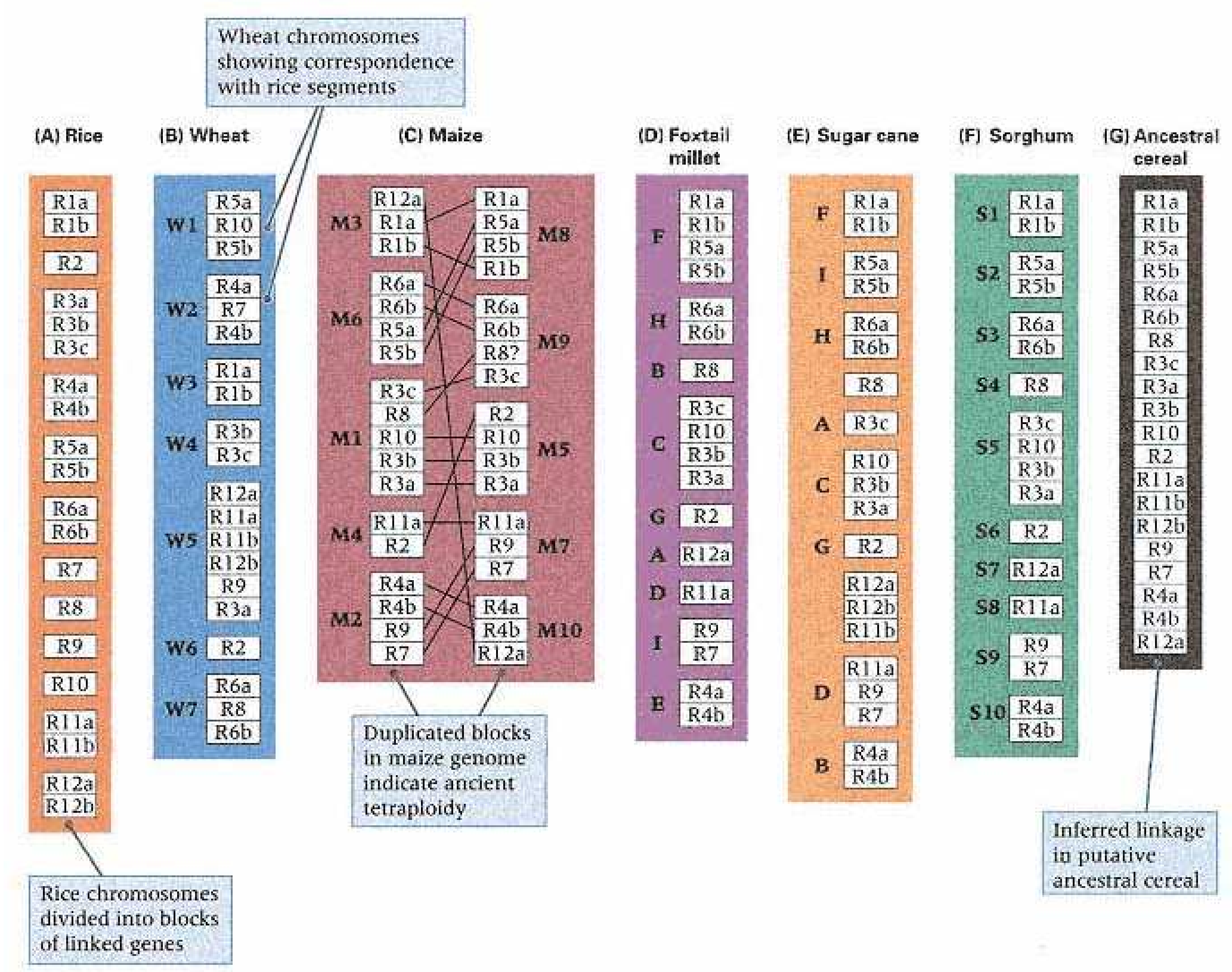


Figure 9.28
Conserved linkages (synteny groups) between the rice genome (A) and that of other grass species: wheat (B), maize (C), foxtail millet (D), sugar cane (E), and sorghum (F). Part (G) depicts the inferred or "reconstructed" order of segments in a hypothetical ancestral cereal genome consisting of a single chromosome pair. For each extant species, the hypothetical ancestral chromosome can be cleaved at different points to yield groups of blocks corresponding to the arrangement of the segments in the chromosomes of the species.
[Courtesy of Graham Moore. From G. Moore, K. M. Devos, Z. Wang, and M. D. Gale. 1995. *Current Opinion Genet. Devel.* 5: 737.]

and maize chromosomes passes through rice block 3b, which probably contains the homologous gene. Similarly, rice block 4 contains a gene for *liguless* that aligns with similar mutations in the chromosomes of barley (again Triticeae) and maize, which indicates that the mutations are almost certainly in homologous genes in all three species. Such relationships between genes based on position in the physical maps affords an important method of positional cloning in all species in the grass family.

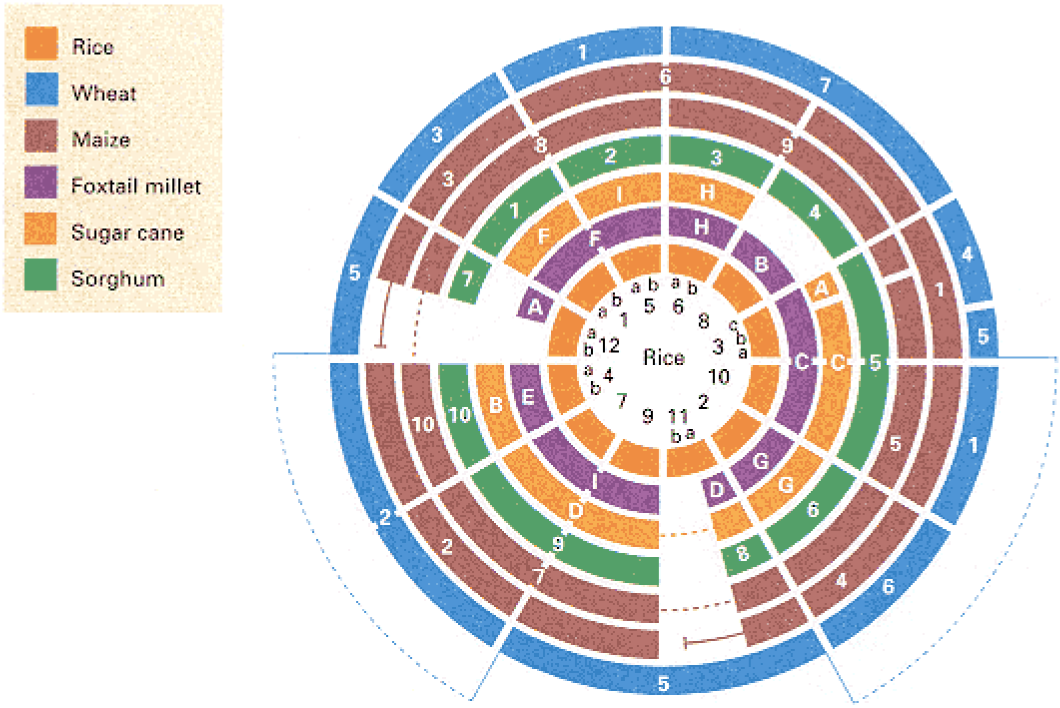


Figure 9.29
Circular arrangement of synteny groups in the cereal grasses makes simultaneous comparisons possible. The thin dashed lines indicate connections between blocks of genes. A number of transpositions of genetic segments are not shown. In some cases, a region within a synteny group is inverted; these inversions are not shown. The circular diagram is for convenience only; there is no indication that the ancestral grass chromosome was actually circular.
[Courtesy of Graham Moore. From G. Moore, K. M. Devos, Z. Wang, and M. D. Gale. 1995. *Current Opinion Genet. Devel.* 5: 737.]

9.7— Large-Scale DNA Sequencing

Large-scale DNA sequencing is well under way in a number of model organisms. The 12.5-Mb genome of the yeast *Saccharomyces cerevisiae* is the first eukaryotic genome to have been sequenced in its entirety. Some of the important conclusions are summarized next.

Complete Sequence of the Yeast Genome

A summary of the analysis of the 666,448 nucleotide pairs in the sequence of yeast chromosome XI is illustrated in Figure 9.30. Like other yeast chromosomes, chromosome XI has a high density of coding regions. A coding region includes an **open reading frame (ORF),** which is a region of sequence containing an uninterrupted run of amino-acid-coding triplets (codons) with no "stop" codons that would terminate translation. An ORF of greater than random length is likely to code for a protein of some kind. The coding region associated with an ORF also includes the flanking regulatory sequences. In chromosome XI, approximately 72 percent of the sequence is present within 331 coding regions averaging 2 kb. The average length of ORF codes for a sequence of 488 amino acids, but the longest ORF codes for the protein dynein with 4092 amino acids. Worthy of note in the chromosome XI sequence is the low number of introns (sequences that are transcribed but removed from the RNA in producing the mRNA); only about 2 percent of the genes have introns. The chromosome also includes sequences for 16 transfer-RNA genes used in protein translation as well as representatives of the transposable elements δ and σ.

The yeast sequence contains evidence of an ancient duplication of the entire genome, although only a small fraction of the genes are retained in duplicate. Protein pairs derived from the ancient duplication make up 13 percent of all yeast proteins. These include pairs of cytoskeletal proteins, ribosomal proteins, transcription factors, glycolytic enzymes, cyclins, proteins of the secretory pathway, and protein kinases. When yeast protein sequences are compared with mammalian protein sequences in GenBank (the international repository of protein sequences of all organisms), a mammalian homolog is found for about 31 percent of yeast proteins. This is a minimum estimate of homology, because the mammalian sequences available for comparison are only a small fraction of those present in mammalian genomes. The homologous proteins are of many types. Examples include proteins that catalyze metabolic reactions, subunits of RNA polymerase, transcription factors, translation initiation and elongation factors, enzymes of DNA synthesis and repair, nuclear-pore proteins, and structural proteins and enzymes of mitochondria and peroxisomes. In some instances the similarities are known to reflect conservation of function, because in more than 70 cases, a human amino-acid coding sequence will substitute for a yeast sequence. These include coding sequences for cyclins, DNA ligase, the *RAS* proto-oncogenes, translation initiation factors, and proteins involved in signal transduction.

The most common method of cloning human disease genes is positional cloning, which means cloning by map position (Figure 9.25). Usually nothing is known about the gene except that, when defective, it results in disease. The first clue to function often comes by recognizing homology to a yeast gene. Striking examples are the human genes that cause hereditary nonpolyposis colon cancer and Werner's syndrome, a disease associated with premature aging. In cells of patients with hereditary nonpolyposis colon cancer, short repeated DNA sequences are unstable. These findings stimulated studies of stability of repeated DNA sequences in yeast mutants, which revealed that repeated DNA sequences are unstable in yeast cells that are deficient in the repair of mismatched nucleotides in DNA, including *msh*2 and *mlh*1 mutants (Chapter 13). The prediction that the cancer genes might also encode proteins for mismatch repair was later verified when the cancer genes were cloned. Cells of patients with Werner's syndrome of premature aging show a limited life span in culture. The human gene encodes a protein highly similar to a DNA helicase encoded by *SGS1* in yeast. The *sgs1* mutant yeast cells show accelerated aging and a reduced lifespan, as well as other cellular phenotypes, including relocation of proteins from telomeres to the nucleolus and nucleolar fragmentation. Examples like these demonstrate how research on model organisms can have direct applicability to human health and disease.

Automated DNA Sequencing

Large-scale sequencing studies of other eukaryotic genomes are also well advanced and, judging by the insights gained from yeast sequences, can be expected to be of considerable value. Most eukaryotic chromosomes are much larger than those of yeast, so complete DNA sequencing is not feasible without the use of instruments that partly automate the process. The principle behind automated DNA sequencing is shown in Figure 9.31. Figure 9.31A illustrates a conventional sequencing gel obtained from the dideoxy procedure described in Section 5.9; each lane contains the products of DNA synthesis carried out in the presence of a small amount of a dideoxy nucleotide (dideoxy-G, -A, -T or -C), which, when incorporated into a growing DNA strand, terminates further elongation. The products of each reaction are separated by electrophoresis in individual lanes, and the gel is placed in contact with photographic film so that radioactive atoms present in one of the normal nucleotides will darken the

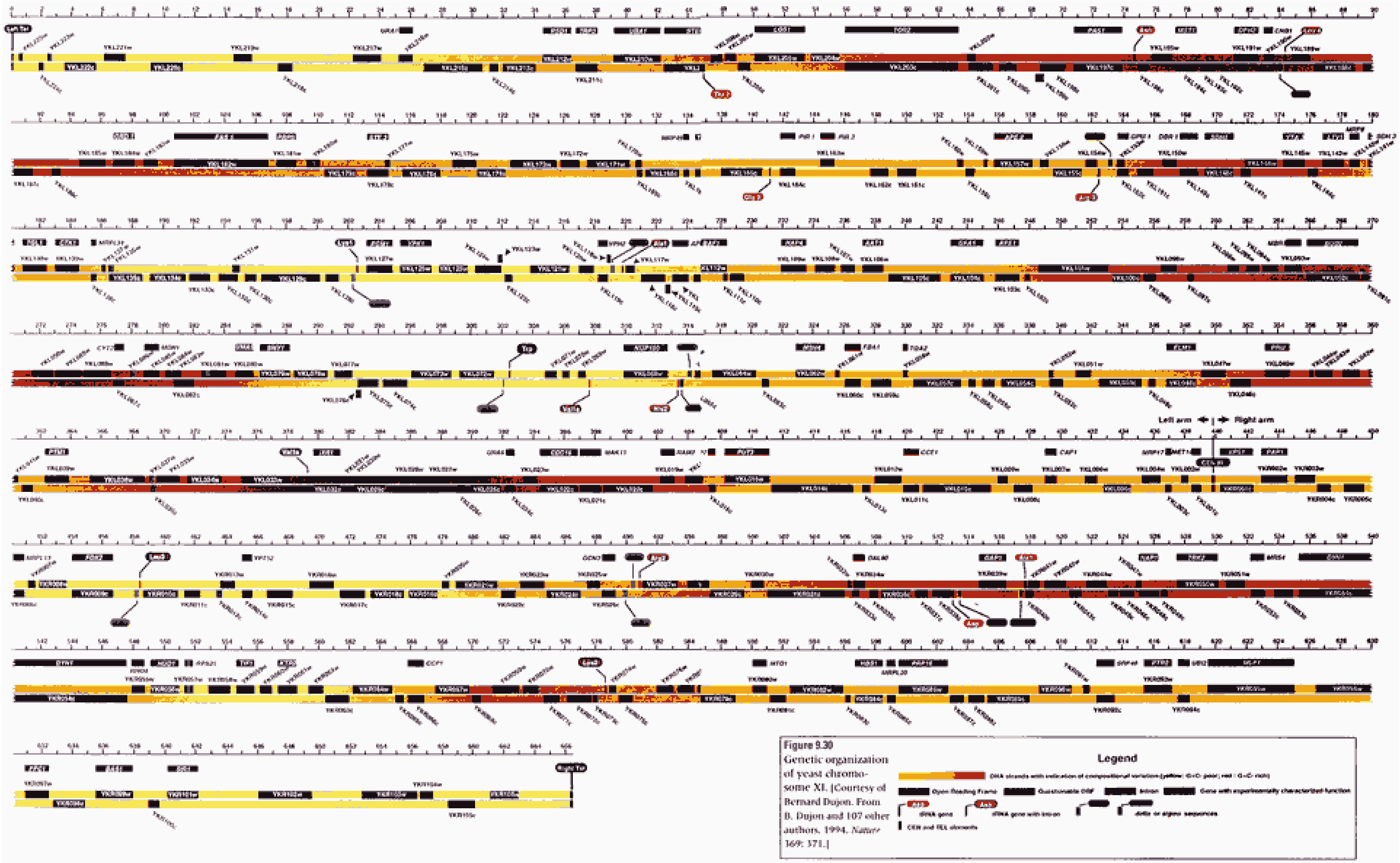


Figure 9.30
Genetic organization of yeast chromosome XI.
[Courtesy of Bernard Dujon. From B. Dujon and 107 other authors. 1994. *Nature* 369: 371.]

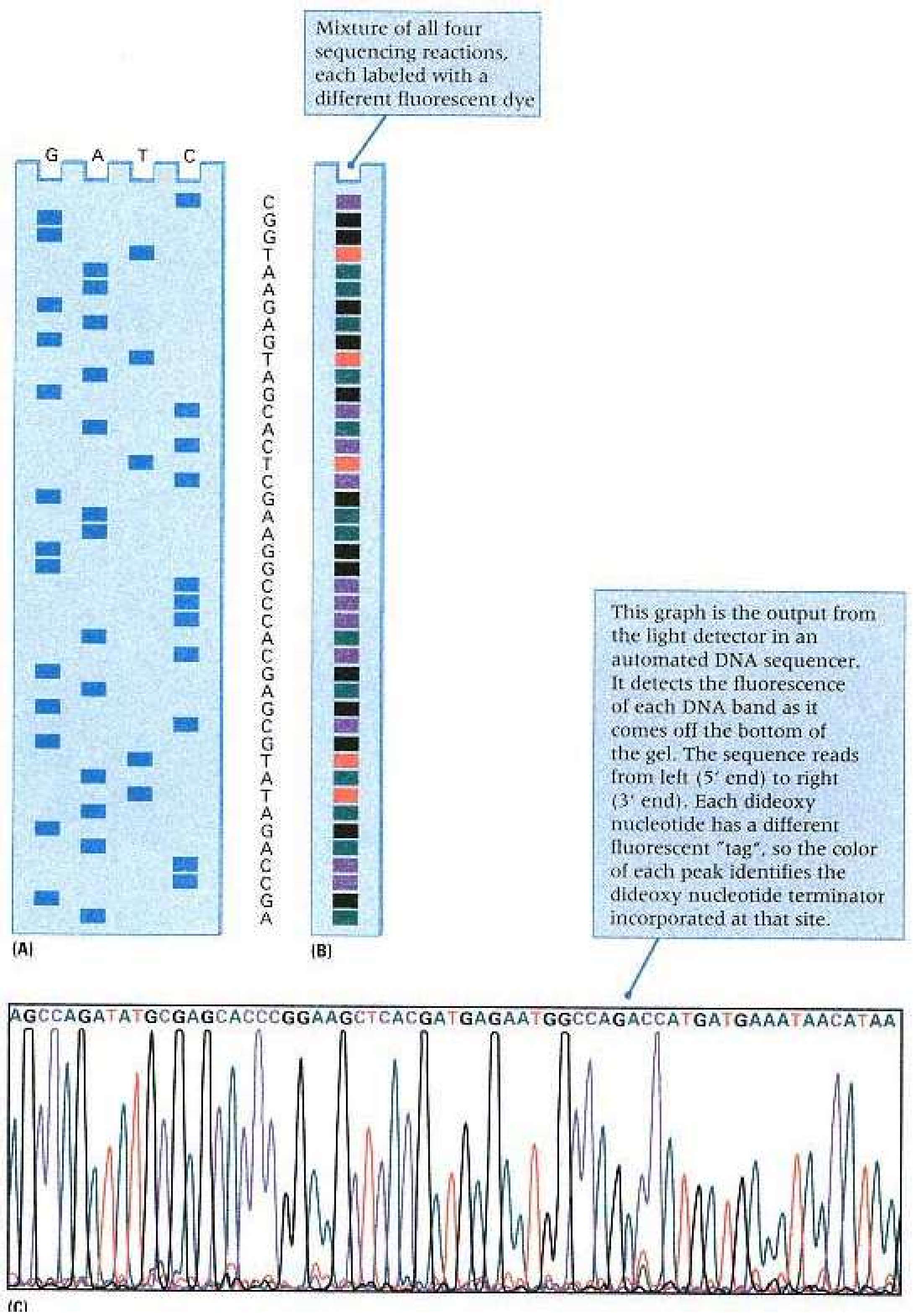


Figure 9.31
Automated DNA sequencing. (A) Conventional sequencing gel obtained from the dideoxy procedure (Section 5.9). The DNA sequence can be determined directly from photographic film according to the positions of the bands. (B) Banding pattern obtained when each of the terminating nucleotides is labeled with a different fluorescent dye and the bands are separated in the same lane of the gel. (C) Trace of the fluorescence pattern obtained from the gel in part B by automated detection of the fluorescence of each band as it comes off the bottom of the gel during continued electrophoresis.

film and reveal a band at the position to which each incomplete DNA strand migrated in electrophoresis. After the film is developed, the DNA sequence is read from the pattern of bands, as shown by the sequence at the right of the gel.

In automated DNA sequencing, illustrated in Figure 9.31B, the nucleotides that terminate synthesis are labeled with different fluorescent dyes (G, black; A, green; T, red; C, blue). Because the colors distinguish the products of DNA synthesis that terminate with each nucleotide, the products of all the synthesis reactions can be put together in the same tube and separated by electrophoresis in a single lane. In principle, the sequence could again be read directly from the gel, as shown in letters at the left of Figure 9.31B. However, a substantial improvement in efficiency is accomplished by continuing the electrophoresis until each band, in turn, drops off the bottom of the gel. As each band comes off the bottom of the gel, the fluorescent dye that it contains is excited by laser light, and the color of the fluorescence is read automatically by a photocell and recorded in a computer. Figure 9.31C is a trace of the fluorescence pattern that would emerge at the bottom of the gel in Figure 9.31B after continued electrophoresis. The nucleotide sequence is read directly from the colors of the alternating peaks along the trace. When used to maximum capacity, an automated sequencing instrument can ideally generate as much as 20 Mb of nucleotide sequence per year. The actual amount of finished sequence is considerably smaller, because in a sequencing project, each DNA strand needs to be sequenced completely for the sake of minimizing sequencing errors, and some troublesome regions need to be sequenced several times.

Chapter Summary

Recombinant DNA technology makes it possible to modify the genotype of an organism in a directed, predetermined way by enabling different DNA molecules to be joined into novel genetic units, altered as desired, and reintroduced into the organism. Restriction enzymes play a key role in the technique because they can cleave DNA molecules within particular base sequences. Many restriction enzymes generate DNA fragments with complementary single-stranded ends, which can anneal and be ligated together with similar fragments from other DNA molecules. The carrier DNA molecule used to propagate a desired DNA fragment is called a vector. The most common vectors are plasmids, phages, viruses, and yeast artificial chromosomes (YACs). Transformation is an essential step in the propagation of recombinant molecules because it enables the recombinant DNA molecules to enter host cells, such as those of bacteria, yeast, or mammals. If the recombinant molecule has its own replication system or can use the host replication system, then it can replicate. Plasmid vectors become permanently established in the host cell; phages can multiply and produce a stable population of phages carrying source DNA; retroviruses can be used to establish a gene in an animal cell; and YACs include source DNA within an artificial chromosome that contains a functional centromere and telomeres.

Recombinant DNA can also be used to transform the germ line of animals or to genetically engineer plants. These techniques form the basis of reverse genetics, in which genes are deliberately mutated in specified ways and introduced back into the organism to determine the effects on pheno-type. Reverse genetics is routine in genetic analysis in bacteria, yeast, nematodes, *Drosophila,* the mouse, and other organisms. In *Drosophila,* transformation employs a system of two vectors based on the transposable *P* element. One vector contains sequences that produce the *P* transposase; the other contains the DNA of interest between the inverted repeats of *P* and other sequences needed for mobilization by transposase and insertion into the genome. Germ-line transformation in the mouse makes use of retrovirus vectors or embryonic stem cells. Dicot plants are transformed with T DNA derived from the *Ti* plasmid found in species of *Agrobacterium,* whose virulence genes promote a conjugation-like transfer of T DNA into the host plant cell, where it is integrated into the chromosomal DNA.

Practical applications of recombinant DNA technology include the efficient production of useful proteins, the creation of novel genotypes for the synthesis of economically important molecules, the generation of DNA and RNA sequences for use in medical diagnosis, the manipulation of the genotype of domesticated animals and plants, the development of new types of vaccines, and the potential correction of genetic defects (gene therapy). Production of eukaryotic proteins in bacterial cells is sometimes hampered by protein instability, inability to fold properly, or failure to undergo necessary chemical modification. These problems are often eliminated by production of the protein in yeast or mammalian cells.

Cloning in bacteriophage P1 vectors or yeast artificial chromosomes (YACs) allows very large DNA molecules to be isolated and manipulated and has stimulated a major effort to analyze complex genomes, such as those in nematodes, *Drosophila,* the mouse, and human beings. These efforts include the development of detailed physical maps that integrate many levels of genetic information, such as the positions of sequence-tagged sites or the positions and lengths of contigs. Among the important discoveries of genome analysis is that of the relationships among cereal grass genomes. Although the genomes differ enormously in size from one species to the next, largely because of the abundance and types of repetitive DNA sequences, cross-hybridization of single-copy sequences demonstrates that the genomes can be brought into register by postulating rearrangements of 20 segments, or synteny groups, found in rice, the smallest of the genomes. Large-scale DNA sequencing of the yeast genome has revealed an unexpectedly high density of genetic information. Many coding regions have previously unknown functions and yield no recognizable phenotype when mutated. Large-scale genomic sequencing is also underway in other organisms through the application of automated DNA-sequencing machines.

Key Terms

blunt end
cDNA
cloning
cohesive end
colony hybridization assay
complementary DNA
contig
crown gall tumor
electroporation
embryonic stem cell
gene cloning
gene targeting
gene therapy
genetic engineering
genome equivalent
human genome project
insertional inactivation
library
map-based cloning
multiple cloning site
oligonucleotide site-directed mutagenesis
open reading frame
P1 bacteriophage
palindrome
partial digestion
physical map
polylinker
polymerase chain reaction (PCR)
positional cloning
primer oligonucleotides
probe
P transposable element
recombinant DNA technology
restriction enzyme
restriction site
retrovirus
reverse genetics
reverse transcriptase
reverse transcriptase PCR
sequence-tagged site (STS)
shuttle vector
sticky end
synteny group
T DNA
Ti plasmid
transformation
transgenic animal
vector
yeast artificial chromosome (YAC)

Review the Basics

• What is recombinant DNA?

• What features are essential in a bacterial cloning vector?

• What is the reaction catalyzed by the enzyme reverse transcriptase? How is this enzyme used in recombinant DNA technology?

• What is a transgenic organism?

• In the context of genome analysis, what is a YAC? What feature makes YACs useful in the analysis of complex genomes?

• What is a physical map? How is a physical map related to a genetic map?

Guide to Problem Solving

Problem 1: In genetic engineering, when we are expressing eukaryotic gene products in bacterial cells, why is it necessary.

(a) to use cDNA instead of genomic DNA?

(b) to fuse the cDNA with a bacterial promoter?

Answer:

(a) Most eukaryotic genes have introns, which cannot be removed by bacterial cells.

(b) Eukaryotic promoters have a different sequence than prokaryotic promoters and are not normally recognized in bacterial cells.

Problem 2: What is the average distance between restriction sites for each of the following restriction enzymes? Assume that the DNA substrate has a random sequence with equal amounts of each base. The symbol N stands for any nucleotide, R for any purine (A or G), and Y for any pyrimidine (T or C).

(a) TCGA *Taq*I

(b) GGTACC *Kpn*I

(c) GTNAC *Mae*III

(d) GGNNCC *Nla*IV

(e) GRCGYC *Acy*I

Answer: The average distance between restriction sites equals the reciprocal of the probability of occurrence of the restriction site. You must therefore calculate the probability of occurrence of each restriction site in a random DNA sequence.

(a) The probability of the sequence TCGA is $1/4 \times 1/4 \times 1/4 \times 1/4 = (1/4)^4 = 1/256$, so 256 bases is the average distance between *Taq*I sites.

(b) By the same reasoning, the probability of a *Kpn*I site is $(1/4)^6 = 4096$, so 4096 bases is the average distance between *Kpn*I sites.

(c) The probability of N (any nucleotide at a site) is 1, so the probability of the sequence GTNAC equals $1/4 \times 1/4 \times 1 \times 1/4 \times 1/4 = (1/4)^4 = 1/256$; therefore, 256 is the average distance between *Mae*III sites.

(d) The same reasoning yields the average distance between *Nla*IV sites as $(1/4 \times 1/4 \times 1 \times 1 \times 1/4 \times 1/4)^{-1} = 256$ bases.

(e) The probability of an R (A or G) at a site is 1/2, and the probability of a Y (T or C) at a site is 1/2. Hence the probability of the sequence GRCGYC is $1/4 \times 1/2 \times 1/4 \times 1/4 \times 1/2 \times 1/4 = 1/1024$, so there is an average of 1024 bases between *Acy*I sites.

Problem 3: What fundamental structural components are necessary for yeast artificial chromosomes to be stable in yeast cells?

Answer: Yeast artificial chromosomes need a centromere for segregation in cell division and a telomere at each end to stabilize the tips.

Analysis and Applications

9.1 The euchromatic part of the *Drosophila* genome that is highly replicated in the banded salivary gland chromosomes is approximately 110 Mb (million base pairs) in size. The salivary gland chromosomes contain approximately 5000 bands. For ease of reference, the salivary chromosomes are divided into about 100 approximately equal, numbered sections (1–100), each of which consists of six lettered subdivisions (A–F). On average, how much DNA is in a salivary gland band? In a lettered subdivision? In a numbered section? How do these compare with the size of the DNA insert in a 200-kb (kilobase pair) YAC? With the size of the DNA insert in an 80-kb P1 clone?

9.2 Restriction enzymes generate one of three possible types of ends on the DNA molecules that they cleave. What are the three possibilities?

9.3 Are the ends of different restriction fragments produced by a particular restriction enzyme always the same? Must opposite ends of each restriction fragment be the same? Why?

9.4 Will the sequences 5'-GGCC-3' and 3'-GGCC-5' in a double-stranded DNA molecule be cut by the same restriction enzyme?

9.5 In cloning into bacterial vectors, why is it useful to insert DNA fragments to be cloned into a restriction site inside an antibiotic-resistance gene? Why is another gene for resistance to a second antibiotic also required?

9.6 How frequently would the restriction enzymes *Taq*I (restriction site TCGA) and *Mae*III (restriction site GTNAC, in which N is any nucleotide) cleave double-stranded DNA molecules containing random sequences of

(a) 1/6 A, 1/6 T, 1/3 G, and 1/3 C?

(b) 1/3 A, 1/3 T, 1/6 G, and 1/6 C?

9.7 If the genomic and cDNA sequences of a gene are compared, what information does the cDNA sequence give you that is not obvious from the genomic sequence? What information does the genomic sequence contain that is not in the cDNA?

9.8 What might prevent a cloned eukaryotic gene from yielding a functional mRNA in a bacterial host? Assuming that these problems are overcome, why might the desired protein still not be produced?

9.9 When DNA isolated from phage J2 is treated with the enzyme *Sal*I, eight fragments are produced with sizes of 1.3, 2.8, 3.6, 5.3, 7.4, 7.6, 8.1, and 11.4 kilobase pairs. However, if J2 DNA is isolated from infected cells, only seven fragments are found, with sizes of 1.3, 2.8, 7.4, 7.6, 8.1, 8.9, and 11.4 kb. What form of the intracellular DNA can account for these results?

9.10 Phage X82 DNA is cleaved into six fragments by the enzyme *Bgl*I. A mutant is isolated with plaques that look quite different from the wildtype plaque. DNA isolated from

Chapter 9 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

http://www.jbpub.com/genetics

Select the link to ***Genetics: Principles and Analysis*** and then choose the link to ***GeNETics on the web.*** You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. The keyword **genome analysis** will lead you to an informative introduction to some of the methods used in geno research. Read the discussion of top-down and bottom-up physical mapping. If assigned to do so, write a brief summary of each type of mapping, and list some of the advantages and limitations of each.

2. Most people are very surprised to learn how many organisms have had their genomes sequenced either complete or in large part. An extensive list of **genome sequencing projects** is maintained at this keyword site. If assigned to so, make a list of microbial genomes whose sequences are completely known and which are available in public databases.

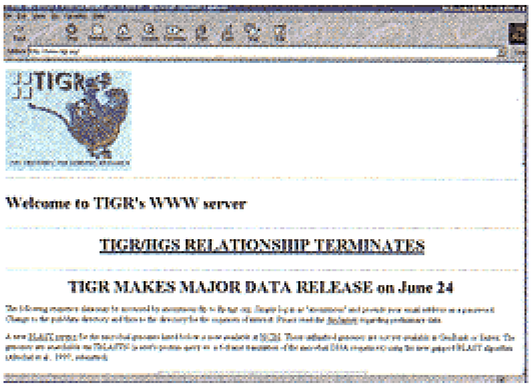
3. What are the **social implications** of modern genetics? Some groups are worried because the application of genet technologies poses ethical and legal issues of the foreknowledge of one's health as well as issues of genetic privacy and insurability. Others are optimistic that the technologies will yield great benefits for medicine and society. The debate continues. This keyword site will connect you to resources that will enable you to learn more about these issues. If assigned to do so, choose one controversial ethical or legal issue related to modern genetic technologies, a write a 250-word paper defining the issue and the opposing views.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 9, and you will be linke the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 9.



the mutant is cleaved into only five fragments. What possible genetic changes can account for the difference in the number of restriction fragments?

9.11 The wildtype allele of a bacterial gene is easily selected by growth on a special medium. Repeated attempts to clone the gene by digestion of cellular DNA with the enzyme *Eco*RI are unsuccessful. However, if the enzyme *Hind* is used, then clones are easily found. Explain.

9.12 A *kan-r tet-r* plasmid is treated with the restriction enzyme *Bgl*I, which cleaves the *kan* (kanamycin) gene. The DNA is annealed with a *Bgl*I digest of *Neurospora* DNA and then used to transform *E. coli*.

(a) What antibiotic would you put in the growth medium to ensure that each colony has the plasmid?

(b) What antibiotic-resistance phenotypes will be found among the resulting colonies?

(c) Which phenotype will contain *Neurospora* DNA inserts?

9.13 A *lac*$^+$ *tet-r* plasmid is cleaved in the *lac* gene with a restriction enzyme. The enzyme has a four-base restrictio and generates fragments with a two-base single-stranded overhang. The cutting site in the *lac* gene is in the codon f the second amino acid in the chain, a site that can tolerate any amino acid without loss of function. After cleavage, t single-stranded ends are converted to blunt ends with DNA polymerase I, and then the ends are joined by blunt-end ligation to recreate a circle. A *lac*$^-$ *tet-s* bacterial strain is transformed with the DNA, and tetracycline-resistant bacteria are selected. What is the Lac phenotype of the colonies?

9.14 You want to introduce the human insulin gene into a bacterial host in hopes of producing a large amount of human insulin. Should you use the genomic DNA or the cDNA? Explain your reasoning.

Challenge Problem

9.15 Plasmid pBR607 DNA is a double-stranded circle of 4 kilobase pairs. This plasmid carries two genes whose protein products confer resistance to tetracycline (Tet-r) and ampicillin (Amp-r) in host bacteria. The DNA has a single site for each of the following restriction enzymes: *Eco*RI, *Bam*HI, *Hin*dIII, *Pst*I, and *Sal*I. Cloning DNA into the *Eco*RI site does not affect resistance to either drug. Cloning DNA into the *Bam*HI, *Hin*dIII, and *Sal*I sites abolishes tetracycline resistance. Cloning into the *Pst*I site abolishes ampicillin resistance. Digestion with the following mixtures of restriction enzymes yields fragments with the sizes listed below. Indicate the positions of the *Pst*I, *Bam*HI, *Hin*dIII, and *Sal*I cleavage sites on a restriction map, relative to the *Eco*RI cleavage site.

Enzyme mixture	**Fragment size (kb)**
*Eco*RI + *Pst*I	0.70, 3.30
*Eco*RI + *Bam*HI	0.30, 3.70
*Eco*RI + *Hind*III	0.08, 3.92
*Eco*RI + *Sal*I	0.85, 3.15
*Eco*RI + *Bam*HI + *Pst*I	0.30, 0.70, 3.00

Further Reading

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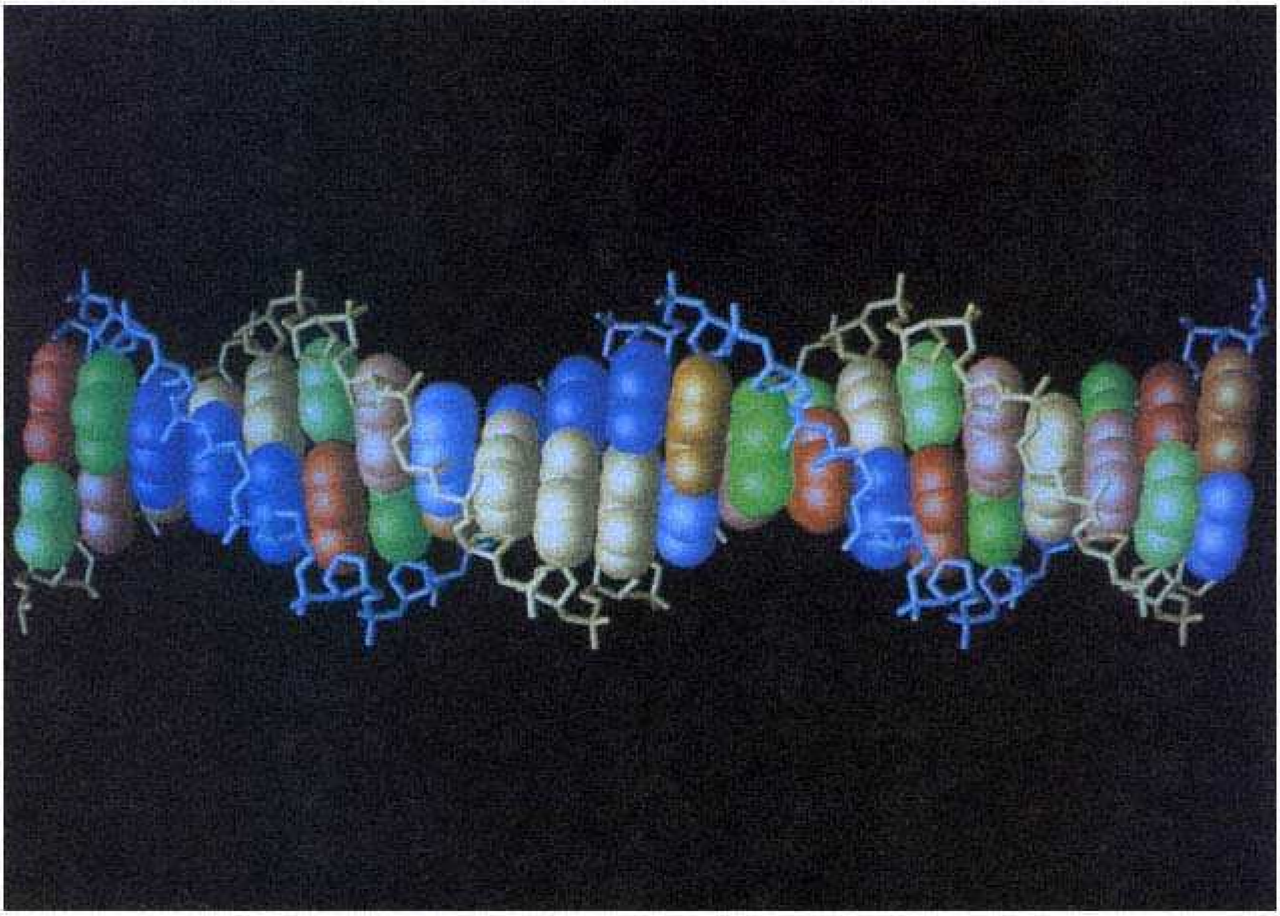


Structure of the DNA double helix showing the major (wide) and minor (narrow) grooves. Each base is shown in a dark or light color according to whether it is present on the template strand transcribed in RNA synthesis (light brown backbone) or on the nontemplate strand (blue backbone). The deoxyribose sugars are the pentagonal shapes, each adjacent pair connected by a phosphodiester bond (P–O–P). The base colors are brown (G), blue (C), red (A), and green (T). [Courtesy of Antony M. Dean.]

Chapter 10— Gene Expression

CHAPTER OUTLINE

Challenge Problems

Further Reading

GeNETics on the web

PRINCIPLES

• In gene expression, information in the base sequence of DNA is used to dictate the linear order of amino acids in a polypeptide by means of an RNA intermediate.

• Transcription of an RNA from one strand of the DNA is the first step in gene expression.

• In eukaryotes, the RNA transcript is modified and may undergo splicing to make the messenger RNA.

• The messenger RNA is translated on ribosomes in groups of three bases (codons), each specifying an amino acid through an interaction with molecules of transfer RNA, each "charged" (chemically bonded) with one amino acid.

• Ribosomes are particles consisting of special types of RNA (ribosomal RNA) and numerous proteins. Each transfer RNA molecule contains a region of three bases that recognizes one (in some cases more than one) codon by base pairing. Each transfer RNA also has a particular amino acid attached at one end that corresponds to the amino acid encoded by the codon (or codons) with which the transfer RNA binds.

• Almost all organisms use the same genetic code, but exceptions are found in certain protozoa and in the genetic codes of mitochondrial and other organelle DNA.

CONNECTIONS

CONNECTION: One Gene, One Enzyme

George W. Beadle and Edward L. Tatum 1941
Genetic control of biochemical reactions in Neurospora

CONNECTION: Messenger Light

Sydney Brenner, François Jacob and Matthew Meselson 1961
An unstable intermediate carrying information from genes to ribosomes for protein synthesis

CONNECTION: Uncles and Aunts

Francis H. C. Crick, Leslie Barnett, Sydney Brenner, and R. J. Watts-Tobin, 1961
General nature of the genetic code for proteins

Earlier chapters have been concerned with genetic analysis—with genes as units of genetic information, their relation to chromosomes, and the chemical structure and replication of the genetic material. In this chapter, we shift our perspective and consider the processes by which the information contained in genes is converted into molecules that determine the properties of cells and viruses. The transfer of genetic information from DNA into protein constitutes **gene expression.** The information transfer is accomplished by a series of events in which the sequence of bases in DNA is first copied into an RNA molecule and then the RNA is used, either directly or after some chemical modification, to determine the amino acid sequence of a protein molecule. The principle steps in gene expression can be summarized as follows:

1. RNA molecules are synthesized enzymatically by *RNA polymerase*, which uses the base sequence of a segment of a single strand of DNA as a template in a polymerization reaction similar to that used in replicating DNA. The overall process by which the segment corresponding to a particular gene is selected and an RNA molecule is made is called **transcription.**

2. In eukaryotes, the RNA usually undergoes chemical modification in the nucleus called **processing.**

3. Protein molecules are then synthesized by the use of the base sequence of a processed RNA molecule to direct the sequential joining of amino acids in a particular order, and so the amino acid sequence is a direct consequence of the base sequence. The production of an amino acid sequence from an RNA base sequence is called **translation,** and the protein made is called the **gene product.**

10.1— Proteins and Amino Acids

Proteins are the molecules responsible for catalyzing most intracellular chemical reactions (enzymes), for regulating gene expression (regulatory proteins), and for determining many features of the structures of cells, tissues, and viruses (structural proteins). A protein is composed of one or more chains of amino acids. Each of these chains is a series of covalently joined amino acids that constitute a **polypeptide.** The 20 different amino acids commonly found in polypeptides can be joined in any number and in any order. Because the number of amino acids in a polypeptide usually ranges from 100 to 1000, an enormous number of different protein molecules can be formed from the 20 common amino acids.

Each amino acid contains a carbon atom (the α carbon) to which is attached one carboxyl group (-COOH), one amino group ($-NH_2$), and a side chain commonly called an **R group** (Figure 10.1). The R groups are generally chains or rings of carbon atoms bearing various chemical groups. The simplest side chains are those of glycine (-H) and of alanine ($-CH_3$). For reference, the chemical structures of all 20 amino acids are shown in Figure 10.2. For each amino acid, the R group is indicated by a gold rectangle.

Polypeptide chains are formed when the carboxyl group of each amino acid becomes joined with the amino group of the next amino acid in line; the resulting chemical bond is an ordinary covalent bond called a **peptide bond** (Figure 10.3A). Thus the basic unit of a protein is a polypeptide chain in which α-carbon atoms alternate with peptide groups to form a backbone that has an ordered array of side chains (Figure 10.3B).

The two ends of every polypeptide molecule are distinct. One end has a free $-NH_2$ group and is called the **amino terminus;** the other end has a free -COOH group and is the **carboxyl terminus.** Polypeptides

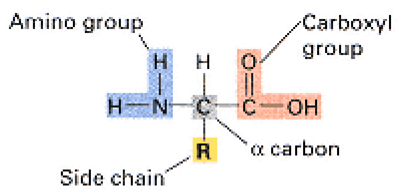


Figure 10.1
The general structure of an amino acid.

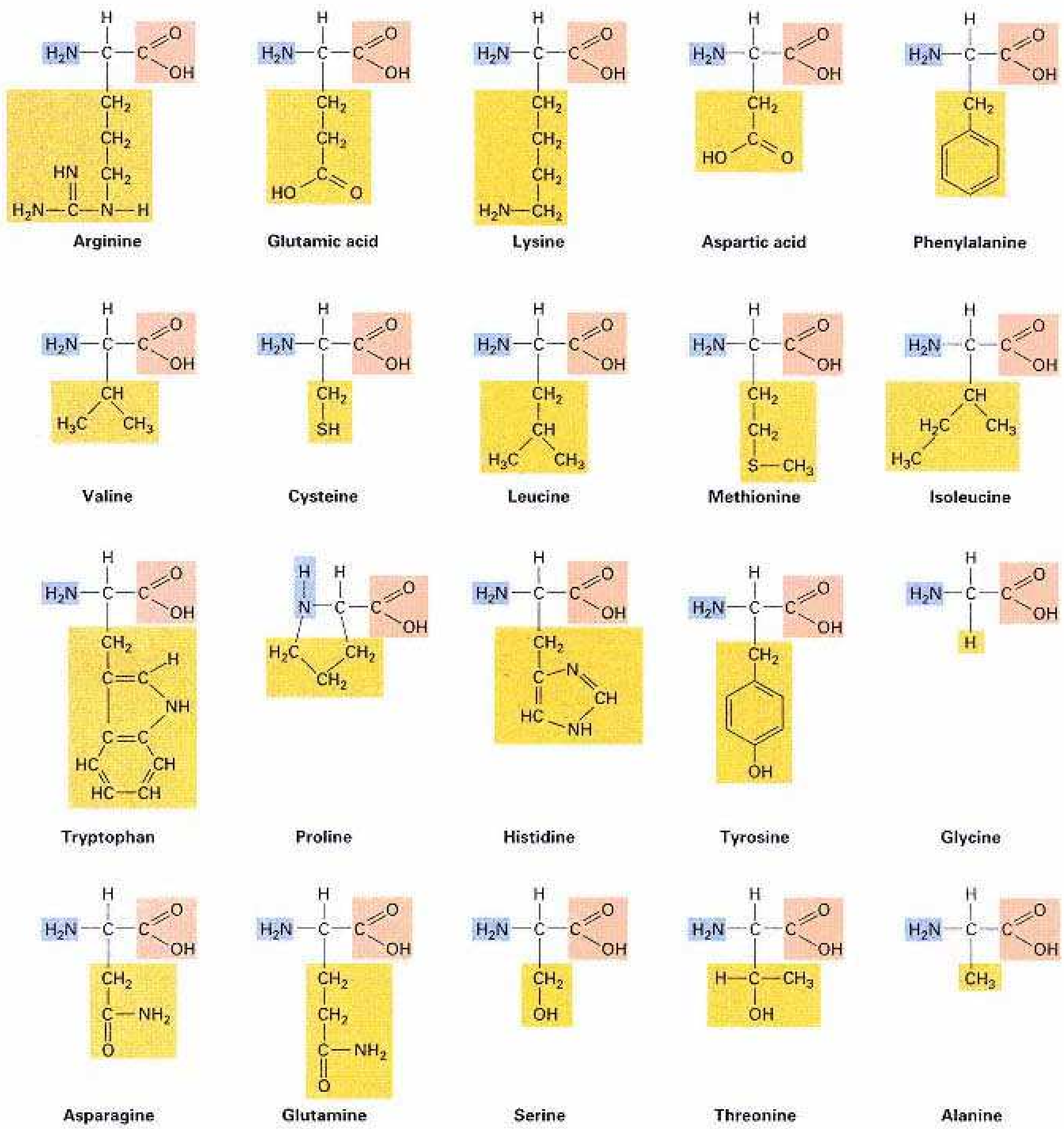


Figure 10.2
Chemical structures of the amino acids. Note that proline does not have the general structure shown in Figure 10.1 because it lacks a free amino group.

are synthesized by adding successive amino acids to the carboxyl end of the growing chain. Conventionally, the amino acids of a polypeptide chain are numbered starting at the amino terminus.

Most polypeptide chains are highly folded, and a variety of three-dimensional shapes have been observed. The manner of folding is determined primarily by the sequence of amino acids—in particular, by

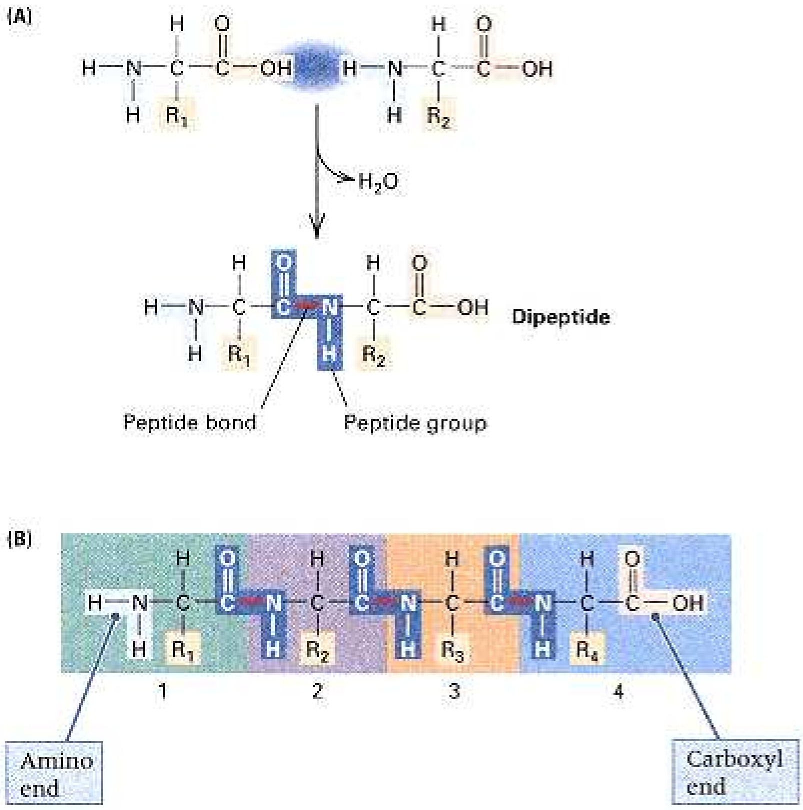


Figure 10.3
Properties of a polypeptide chain. (A) Formation of a dipeptide by reaction of the carboxyl group of one amino acid (left) with the amino group of a second amino acid (right). A molecule of water (HOH) is eliminated to form a peptide bond (red line). (B) A tetrapeptide showing the alternation of α-carbon atoms (black) and peptide groups (blue). The four amino acids are numbered below.

noncovalent interactions between the side chains—so each polypeptide chain tends to fold into a unique three-dimensional shape as it is being synthesized. In some cases, protein folding is assisted by interactions with other proteins in the cell called **chaperones.** Generally speaking, the molecules fold so that amino acids with charged side chains tend to be on the surface of the protein (in contact with water) and those with uncharged side chains tend to be internal. Specific folded configurations also result from hydrogen bonding between peptide groups. Two fundamental polypeptide structures are the α helix and the β sheet (Figure 10.4). The α helix, represented as a coiled ribbon in Figure 10.4, is formed by interactions between neighboring amino acids that twist the backbone into a righthanded helix in which the N-H in each peptide group is hydrogen-bonded with the C-O in the peptide group located four amino acids further along the helix. In contrast, the β sheet, represented as parallel "flat" ribbons in Figure 10.4, is formed by interactions between amino acids in distant parts of the polypeptide chain; the backbones of the polypeptide chains are held flat and rigid (forming a "sheet"), because alternate N-H groups in one polypeptide backbone are hydrogen-bonded with alternate C-O groups in the polypeptide backbone of the adjacent chain. In each polypeptide backbone, alternate C-O and N-H groups are free to form hydrogen bonds with their counterparts in a different polypeptide backbone on the opposite side, so a β sheet can consist of multiple aligned segments in the same (or different) polypeptide chains. Other types of interactions also are important in protein folding—for example, covalent bonds may form between the sulfur atoms of pairs of cysteines in different parts of the polypeptide. However, the rules of folding are so complex that, except for the simplest proteins, the final shape of a protein cannot usually be predicted from the amino acid sequence alone.

Many protein molecules consist of more than one polypeptide chain. When this is the case, the protein is said to contain **subunits.** The subunits may be identical or different. For example, hemoglobin, the

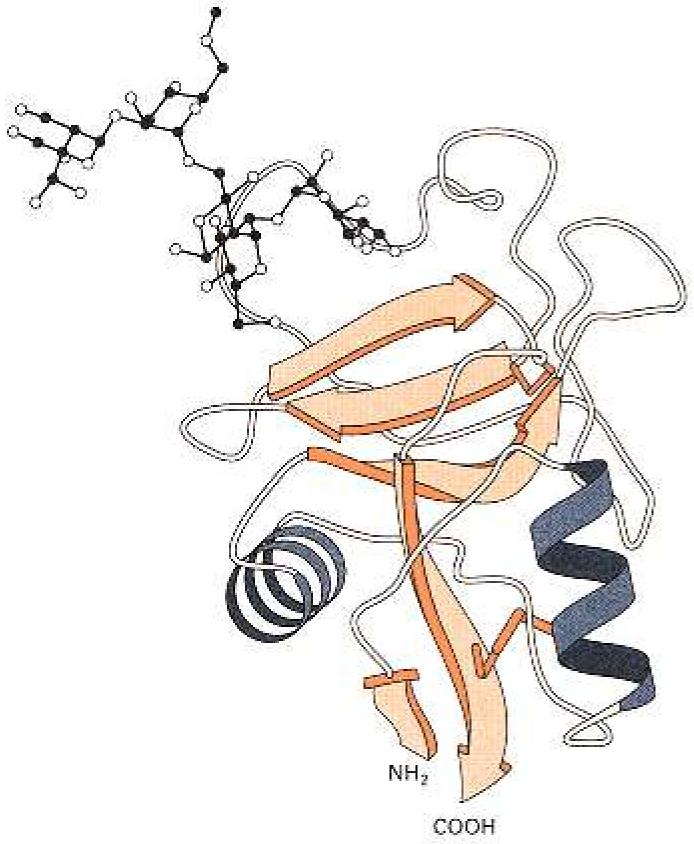


Figure 10.4
A "ribbon" diagram of the path of the backbone of a polypeptide, showing the ways in which the polypeptide is folded. Arrows represent parallel β sheets, each of which is held to its neighboring β sheet by hydrogen bonds. Helical regions are shown as coiled ribbons. The polypeptide chain in this example is a mannose-binding protein. The stick figure at the upper left shows a molecule of mannose bound to the protein.
[Adapted from William I. Weis, Kurt Drickamer, and Wayne A. Hendrickson. 1992. *Nature*, 360: 127.]

oxygen carrier of blood, consists of four subunits: two copies of each of two different polypeptides, which are designated the α chain and the β chain. (The use of α and β as names of the polypeptide chains has nothing to do with the α helices and β sheets that are found within the polypeptides.)

10.2— Relations between Genes and Polypeptides

It took half a century to find out that genes control the structure of proteins. In the early 1900s, Archibald Garrod suggested that hereditary human diseases, such as phenylketonuria, result from inborn errors of metabolism (Chapter 1). Support for this idea came in the 1940s when George Beadle and Edward Tatum demonstrated, using *Neurospora crassa*, that genes govern the ability of the fungus to synthesize amino acids, purines, and vitamins. That genes control metabolism by determining protein structure was demonstrated when it was shown, in the early 1950s, that the allele for sickle-cell anemia brings about a change in the charge of the hemoglobin molecule by causing substitution of an uncharged valine for a negatively charged glutamic acid at residue number 6 in the β-globin chain.

Most genes contain the information for the synthesis of only one polypeptide chain. Furthermore, the *sequence* of nucleotides in a gene determines the *sequence* of amino acids in a polypeptide. This point was first proved by studies of the tryptophan synthase gene *trpA* in *E. coli*, a gene in which many mutations had been obtained and accurately mapped. The effects of numerous mutations on the amino acid sequence of the enzyme were determined by directly analyzing the amino acid sequences of the wildtype and mutant enzymes. Each mutation was found to result in a single amino acid substituting for the wildtype amino acid in the enzyme; more important, *the order of the mutations in the genetic map was the same as the order of the affected amino acids in the polypeptide chain* (Figure 10.5). This attribute of genes and

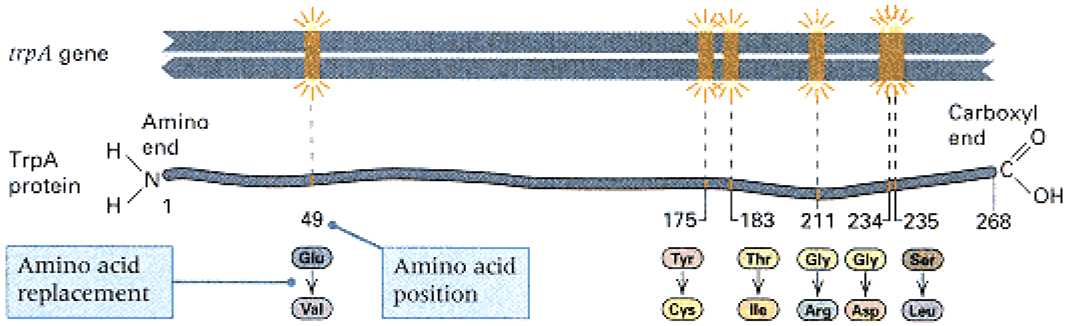


Figure 10.5
Correlation of the positions of mutations in the genetic map of the *E. coli* trpA gene with positions of amino acid replacements in the TrpA protein.

polypeptides is called **colinearity,** which means that the sequence of base pairs in DNA determines the sequence of amino acids in the polypeptide in a colinear, or point-to-point, manner. Colinearity is found almost universally in prokaryotes. However, we will see later that, in eukaryotes, noninformational DNA sequences interrupt the continuity of most genes; in these genes, the order of mutations along a gene (but not their spacing) correlates with the respective amino acid substitutions.

What Are the Minimal Genetic Functions Needed for Life?

The bacterium *Mycoplasma genitalium* belongs to a large group of bacteria, called mycoplasmas, that lack a cell wall. Mycoplasmas are free-living organisms that are parasites on a wide range of plant and animal hosts, including human beings. *M. genitalium,* which exists in parasitic association with ciliated epithelial cells of the genital and respiratory tracts of primates, is thought to have the smallest genome of all self-replicating organisms. The entire sequence of the genome has been determined, which enables us to see what constitutes a minimal functional gene set for a cell. The *M. genitalium* genome is a circular DNA molecule 580 kb in length (only about 3.5 times larger than that of the bacteriophage T4), and it encodes 471 genes.

The entire gene set of *M. genitalium* is depicted in Figure 10.6. The cellular processes in which these gene products participate are summarized in Table 10.1. A substantial fraction of the genome is devoted to macromolecular syntheses

Table 10.1 Summary of functions of 471 genes of *Mycoplasma genitalium*

Function	**Number of Genes**	**Percent***	**Number of Function**	**Genes**	**Percent***
DNA replication	32	10	Nucleoside & nucleotide synthesis	19	6
Transcription	12	4	Salvage (degradative pathwats)	10	
Translation	101	32	Othermetabolism		
Cell envelope	17	5	(lipids, cofactors, amino acids, intermediary metabolism)	18	6
Transport of small molecules	34	11	Regulatory	7	2
Energy metabolism	31	10	Other	27	8
ATP-proton force generation	8		Hypothetical or unknown	152	
Glycolysis					
Cell processes					
(cell division, secretion, stress response)	21	7			

*Percent among all genes with identified functions. Data from C. M. Fraser, J. D. Gacayne, O. White, M. D. Adams, R. A, Clayton, R. D. Fleischmann, and 23 other authors, *Science*, 1995, 270; 397.

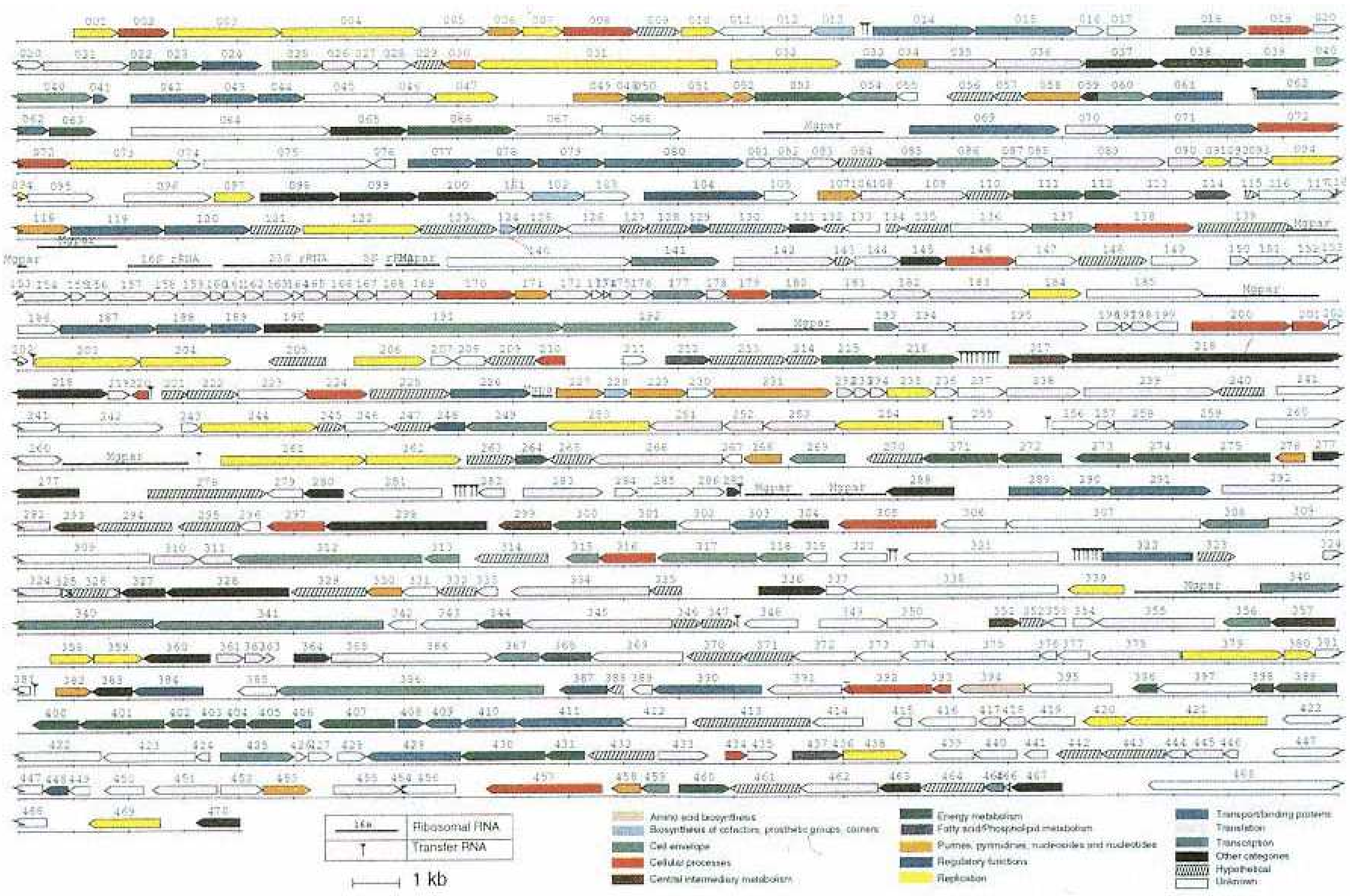


Figure 10.6
Arrangement of coding sequenes in M. *gtenitalium* as determined from the complete DNA sequence of the genome. The genes are color-code according to the function of the gene product. Each arrowhead denotes the direction of transcription.
[Figure design by O. White, courtesy of C. M. Fraser, J. D. Gocayne, O. White, M. D. Adams, R. A. Clayton, R. D. Fleischmann, and 23 other authors, 1995. *Science* 270: 397.]

(DNA, RNA, protein), cell processes, and energy metabolism. There are very few genes for biosynthesis of small molecules. However, genes that encode proteins for salvaging and/or for transporting small molecules make up a substantial fraction of the total, which underscores the fact that the bacterium is parasitic. The remaining genes are largely devoted to formation of the cellular envelope and evasion of the immune system of the host.

10.3— Transcription

The first step in gene expression is the synthesis of an RNA molecule copied from the segment of DNA that constitutes the gene. The basic features of the production of RNA are described in this section.

General Features of RNA Synthesis

The essential chemical characteristics of the enzymatic synthesis of RNA resemble those of DNA synthesis (Chapter 5).

1. The precursors in the synthesis of RNA are the four ribonucleoside 5'-triphosphates—adenosine triphosphate (ATP), guanosine triphosphate (GTP),

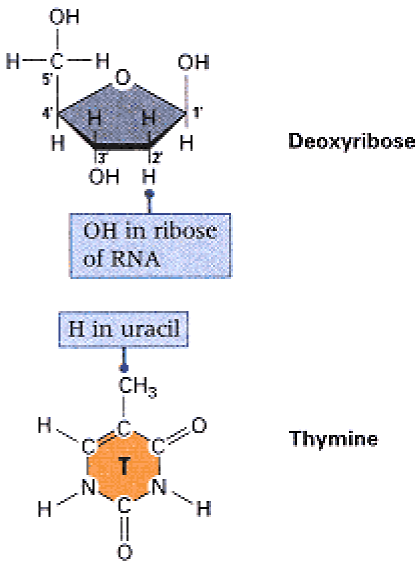


Figure 10.7
Differences in the structures of ribose and deoxyribose and in those of uracil and thymine.

cytidine triphosphate (CTP), and uridine triphosphate (UTP). They differ from the DNA precursors only in that the sugar is ribose rather than deoxyribose and the base uracil (U) replaces thymine (T) (Figure 10.7).

2. In the synthesis of RNA, a sugar-phosphate bond is formed between the 3'-hydroxyl group of one nucleotide and the 5'-triphosphate of the next nucleotide in line (Figure 10.8A and B). This is the same chemical bond as in the synthesis of DNA, but the enzyme is different. The enzyme used in transcription is **RNA polymerase** rather than DNA polymerase.

3. The linear order of bases in an RNA molecule is determined by the sequence of bases in the DNA template. Each base added to the growing end of the RNA chain is chosen for its ability to basepair with the DNA template strand. Thus the bases C, T, G, and A in a DNA strand cause G, A, C, and U, respectively, to be added to the growing end of an RNA molecule.

4. Nucleotides are added only to the 3'-OH end of the growing chain; as a result, the 5' end of a growing RNA molecule bears a triphosphate group. Note that the 5'-to-3' direction of RNA chain growth is the same as that in DNA synthesis.

A significant difference between DNA polymerase and RNA polymerase is that *RNA polymerase is able to initiate chain growth without a primer.* Furthermore,

> Each RNA molecule produced in transcription derives from a single strand of DNA, because in any particular region of the DNA, only one strand serves as a template for RNA synthesis.

The implications of this statement are shown in Figure 10.8C.

The synthesis of RNA can be described as consisting of four discrete stages.

1. ***Promoter recognition*** RNA polymerase binds to DNA within a base sequence from

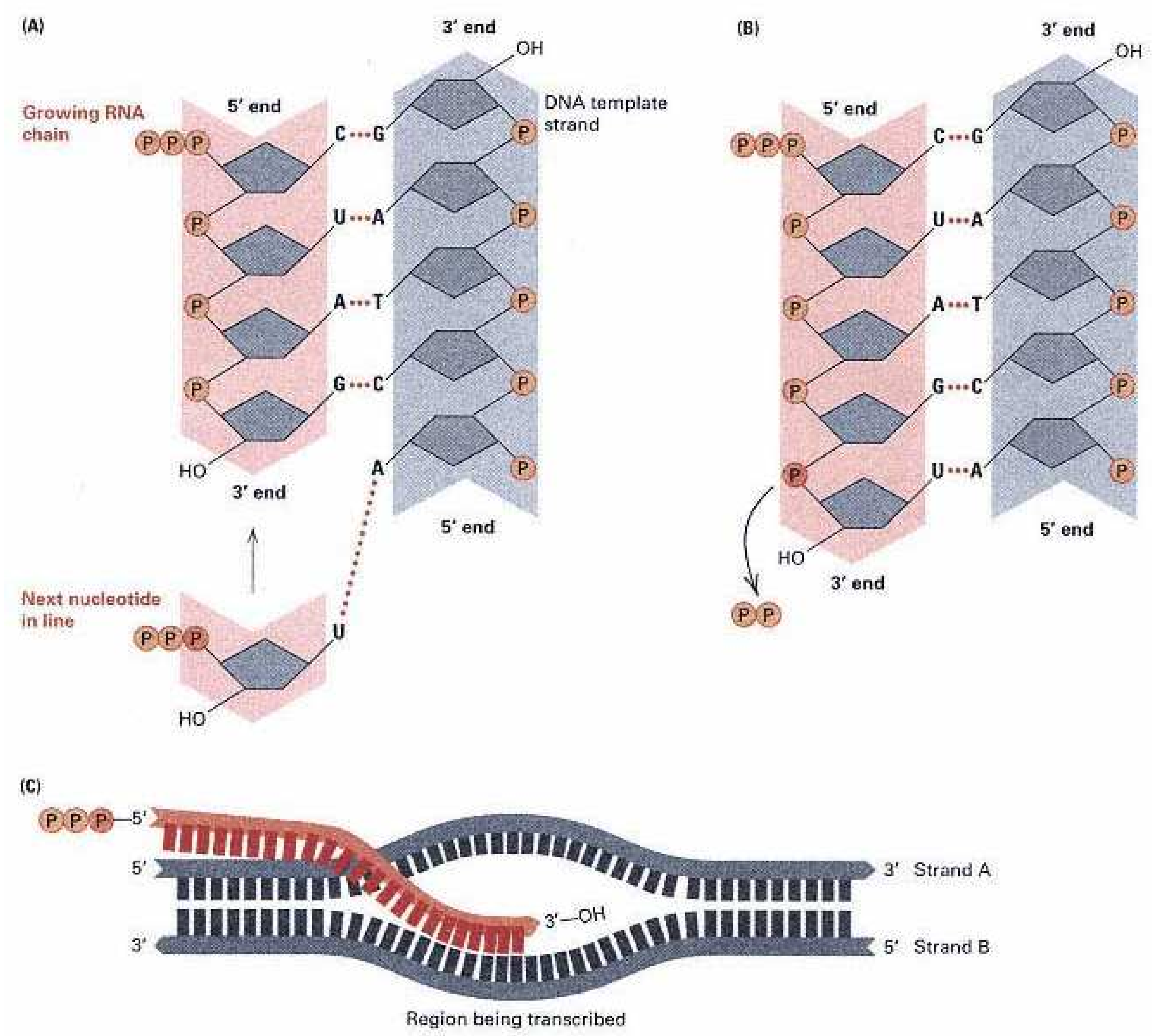


Figure 10.8
RNA synthesis. (A) The polymerization step in RNA synthesis. The incoming nucleotide forms hydrogen bonds (red dots) with a DNA base. The -OH group in the growing RNA chain reacts with the orange P in the next nucleotide in line (B). (C) Geometry of RNA synthesis. RNA is copied from only one strand of a segment of a DNA molecule—in this example, strand B—without the need for a primer. In this region of the DNA, RNA is not copied from strand A. However, in a different region (for example, in a different gene) strand A might be copied rather than strand B. Because RNA elongates in the 5'-to-3' direction, its synthesis moves along the DNA template in the 3'-to-5' direction; that is, the RNA molecule is antiparallel to the DNA strand being copied.

20 to 200 bases in length called a **promoter.** Many promoter sequences have been isolated and their base sequences determined. Although there is substantial sequence variation among promoter regions (in part corresponding to different strengths of the promoters in binding with the RNA polymerase) certain sequence patterns or "motifs" are quite frequent. Two such patterns often found in promoter regions in *E. coli* are illustrated in Figure 10.9. Each pattern is defined by a **consensus sequence** of bases determined from the actual sequences by majority rule: Each

Connection One Gene, One Enzyme

George W. Beadle and Edward L. Tatum 1941

Stanford University, Stanford,
California

Genetic Control of Biochemical Reactions in Neurospora

How do genes control metabolic processes? The suggestion that genes control enzymes was made very early in the history of genetics, most notably by the British physician Archibald Garrod in his 1908 book Inborn Errors of Metabolism. *But the precise relationship between genes and enzymes was still uncertain. Perhaps each enzyme is controlled by more than one gene, or perhaps each gene contributes to the control of several enzymes. The classic experiments of Beadle and Tatum showed that the relationship is usually remarkably simple: One gene codes for one enzyme. The pioneering experiments united genetics and biochemistry, and for the "one gene—one enzyme" concept, Beadle and Tatum were awarded a Nobel Prize in 1958 (Joshua Lederberg shared the prize for his contributions to microbial genetics). Because we now know that some enzymes contain polypeptide chains encoded by two (or occasionally more) different genes, a more accurate statement of the principle is "one gene, one polypeptide." Beadle and Tatum's experiments also demonstrate the importance of choosing the right organism. Neurospora had been introduced as a genetic organism only a few years earlier, and Beadle and Tatum realized that they could take advantage of the ability of this organism to grow on a simple medium composed of known substances.*

From the standpoint of physiological genetics the development and functioning of an organism consist essentially of an integrated system of chemical reactions controlled in some manner by genes. . . . In investigating the roles of genes, the physiological geneticist usually attempts to determine the physiological and biochemical bases of already known hereditary traits. . . . There are, however, a number of limitations inherent in this approach. Perhaps the most serious of these is that

These preliminary results appear to us to indicate that the approach may offer considerable promise as a method of learning more about how genes regulate development and function.

the investigator must in general confine himself to the study of non-lethal heritable characters. Such characters are likely to involve more or less non-essential socalled "terminal" reactions. . . . A second difficulty is that the standard approach to the problem implies the use of characters with visible manifestations. Many such characters involve morphological variations, and these are likely to be based on systems of biochemical reactions so complex as to make analysis exceedingly difficult. . . . Considerations such as those just outlined have led us to investigate the general problem of the genetic control of development and metabolic reactions by reversing the ordinary procedure and, instead of attempting to work out the chemical bases of known genetic characters, to set out to determine if and how genes control known biochemical reactions. The ascomycete *Neurospora* offers many advantages for such an approach and is well suited to genetic studies. Accordingly, our program has been built around this organism. The procedure is based on the assumption that x-ray treatment will induce mutations in genes concerned with the control of known specific chemical reactions. If the organism must be able to carry out a certain chemical reaction to survive on a give medium, a mutant unable to do this will obviously be lethal on this medium. Such a mutant can be maintained and studied, however, if it will grow on a medium to which has been added the essential product of the genetically blocked reaction. . . . Among approximately 2000 strains [derived from single cells after x-ray treatment], three mutants have been found that grow essentially normally on the complete medium and scarcely at all on the minimal medium. One of these strains proved to be unable to synthesize vitamin B_6 (pyridoxine). A second strain turned out to be unable to synthesize vitamin B_1 (thiamine). A third strain has been found to be unable to synthesize para-aminobenzoic acid. . . . These preliminary results appear to us to indicate that the approach may offer considerable promise as a method of learning more about how genes regulate development and function. For example, it should be possible, by finding a number of mutants unable to carry out a particular step in a given synthesis, to determine whether only one gene is ordinarily concerned with the immediate regulation of a given specific chemical reaction.

Source: Proceedings of the National Academy of Sciences of the USA 27: 499–506

base in the consensus sequence is the base most often observed at that position in actual sequences. Any particular sequence may resemble the consensus sequence very well or very poorly.

The consensus sequences in the promoter regions in *E. coli* are TTGACA, centered approximately 35 base pairs upstream from the transcription start site (+1), and TATAAT, centered approximately 10 base pairs upstream from the +1 site. The -10 sequence, which is called the **TATA box,** is similar to sequences found at corresponding positions in many eukaryotic promoters. The positions of the promoter sequences determine where the RNA polymerase begins synthesis, and an A or G is often the first nucleotide in the transcript.

The strength of the binding of RNA polymerase to different promoters varies greatly, which causes differences in the extent of expression from one gene to another. Most of the differences in promoter strength result from variations in the -35 and -10 promoter elements and in the spacing between them. Promoter strength among *E. coli* genes differs by a factor of 10^4, and most of the variation can be attributed to the promoter sequences themselves. In general, the more closely the promoter elements resemble the consensus sequence, the stronger the promoter.

Mutations that change the base sequence in a promoter can alter the strength of the promoter; changes that result in less resemblance to the consensus sequence lower the strength, whereas those with greater resemblance to the consensus increase the strength. Furthermore, there are promoters that differ greatly from the consensus sequence in the -35 region. These promoters typically require accessory proteins to activate transcription by RNA polymerase. In eukaryotes, in addition to the promoter sequences, there are also other DNA sequences called *enhancers* that interact with the promoter to determine the level of transcription.

2. ***Chain initiation*** After the initial binding step, the RNA polymerase "melts" (locally denatures) the DNA double helix, causing the strand that is to be transcribed to separate from its partner strand and become accessible to the polymerase. The RNA polymerase then initiates RNA synthesis at a nearby transcription start site, denoted the +1 site in Figure 10.9. The first nucleoside triphosphate is placed at this site, and synthesis proceeds in a 5'-to-3' direction.

3. ***Chain elongation*** RNA polymerase moves progressively along the transcribed DNA strand, adding nucleotides to the growing RNA chain. Only one DNA strand, the **template strand,** is transcribed.

4. ***Chain termination*** RNA polymerase reaches a chain-termination sequence, and both the newly synthesized RNA molecule and the polymerase are released. Two kinds of termination events are known: those that are self-terminating and depend only on the base sequence in the DNA template, and those that require the presence of a termination protein. In self-termination, which is the most common case, transcription stops when the polymerase encounters a particular sequence of bases in the transcribed DNA strand that is able to fold back upon itself to form a hairpin loop. An example of such a terminator found in *E. coli* is shown in Figure 10.10. The hairpin

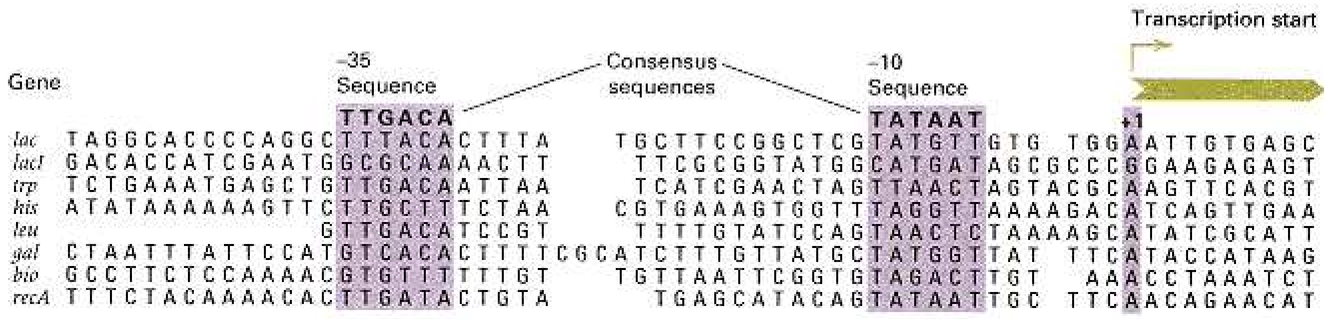


Figure 10.9
Base sequences in promoter regions of several genes in *E. coli*. The consensus sequences located 10 and 35 nucleotides upstream from the transcription start site (+1) are indicated. Promoters vary tremendously in their ability to promote transcription. Much of the variation in promoter strength results from differences between the promoter elements and the consensus sequences at -10 and -35.

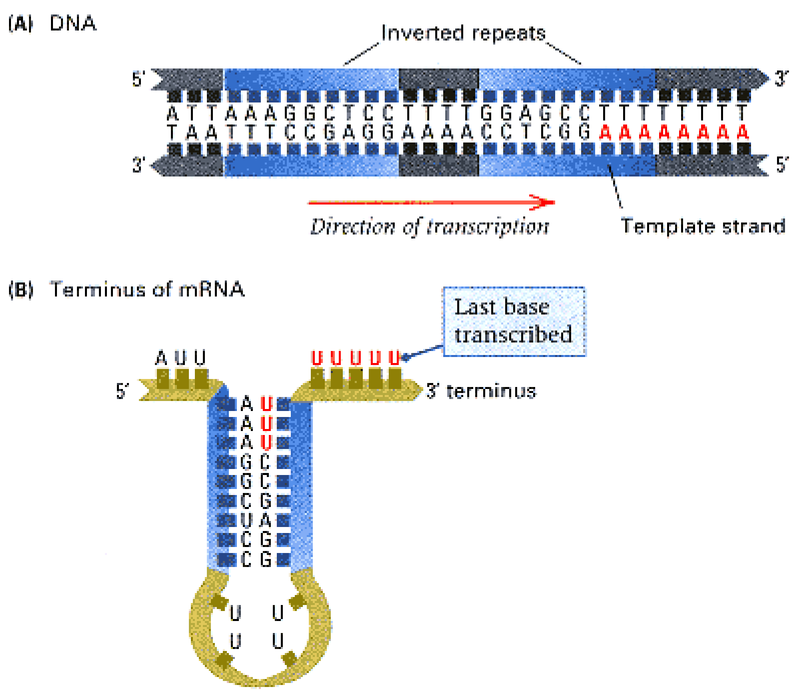


Figure 10.10
(A) Base sequence of the transcription-termination region for the set of tryptophan-synthesizing genes in *E. coli.* The inverted repeat sequences (blue) are characteristic of termination sites. (B) The 3' terminus of the RNA transcript, folded to form a stem-and-loop structure. The sequence of U's found at the end of the transcript in this and many other prokaryotic genes is in red. The RNA polymerase, not shown here, terminates transcription when the loop forms in the transcript.

loop alone is not enough for termination of transcription; the run of U's at the end of the hairpin is also necessary.

Initiation of a second round of transcription need not await completion of the first, because the promoter becomes available once RNA polymerase has polymerized from 50 to 60 nucleotides. For a rapidly transcribed gene; such reinitiation occurs repeatedly, and a gene can be cloaked with numerous RNA molecules in various degrees of completion. The micrograph in Figure 10.11 shows a region of the DNA of the newt *Triturus* that contains tandem repeats of a particular gene. Each gene is associated with growing RNA molecules. The shortest RNA molecules are at the promoter end of the gene; the longest are near the gene terminus.

The existence of promoters was first demonstrated in genetic experiments with *E. coli* by the isolation of particular Lac^- mutations, denoted p^-, that eliminate activity of the *lac* gene but *only when the mutations are adjacent to the gene in the same DNA molecule.* The need for the coupled genetic configuration, also called the *cis* configuration, can be seen by examining a cell with two copies of the gene *lacZ*—for example, a cell containing an F' *lacZ* plasmid, which contains *lacZ* in the bacterial chromosome as well as *lacZ* in the F' plasmid. Transcription of the *lacZ* gene enables the cell to synthesize the enzyme β-galactosidase. Table 10.2 shows that a wildtype *lacZ* gene ($lacZ^+$) is inactive when it and a p^- mutation are present in the same DNA molecule (either in the chromosome or in an F' plasmid); this can be seen by comparing entries 4 and 5. Analysis of the RNA shows that, in a cell with the genotype p^- $lacZ^+$ gene is not transcribed, whereas if the genotype is p^+ $lacZ^-$, a mutant RNA is produced. The p^- mutations are called **promoter mutations.**

Figure 10.11
Electron micrograph of part of the DNA of the newt Triturus viridescens containing tandem repeats of genes being transcribed into ribosomal RNA. The thin strands forming each feather-like array are RNA molecules. A gradient of lengths can be seen for each rRNA gene. Regions in the DNA between the thin strands are spacer DNA sequences, which are not transcribed.
[Courtesy of Oscar Miller.]

Table 10.2 Effect of promoter mutations on transcription of the *lacZ* gene

Genotype	**Transcription of *lacZ*⁺ gene**
1. p^+lacZ^+	Yes
2. p^-lacZ^+	No
3. p^+lacZ^+/p^+lacZ^-	Yes
4. $p^-lacZ^+/p^+lacZ^+/p^+lacZ^-$	No
5. p^+lacZ^+/p^-lacZ^-	Yes

Note: $LacZ^+$ is the wildtype gene; $lacZ^-$ is a mutant that produces a nonfunctional enzyme.

Mutations have also been instrumental in defining the transcription-termination region. Mutations have been isolated that create a new termination sequence upstream from the normal one. When such a mutation is present, an RNA molecule is made that is shorter than the wildtype RNA. Other mutations eliminate the terminator, resulting in a longer transcript.

The best understood RNA polymerase is that of the bacterium *E. coli.* This enzyme consists of five protein subunits and can be easily seen by electron microscopy (Figure 10.12). In *E. coli,* all transcription is catalyzed by this enzyme. Eukaryotic cells have three distinct RNA polymerases, denoted I, II, and III, each of which makes a particular class of RNA molecule. RNA polymerase I catalyzes synthesis of all ribosomal RNA species except 5S RNA; RNA polymerase III catalyzes synthesis of 5S and all of the transfer RNAs. The RNA polymerase II is the enzyme responsible for the synthesis of all RNA transcripts that contain information specifying amino acid sequences. These transcripts are called messenger RNA molecules, which are discussed in the next section. RNA polymerase II also catalyzes synthesis of most small nuclear RNAs involved in RNA splicing, which is discussed in Section 10.4.

Messenger RNA

Amino acids do not bind directly to DNA. Consequently, intermediate steps are needed for arranging the amino acids in a polypeptide chain in the order determined by the DNA base sequence. This process begins with transcription of the base sequence of the template strand of DNA into the base sequence of an RNA molecule. In prokaryotes, this RNA molecule, which is called **messenger RNA,** or **mRNA,** is

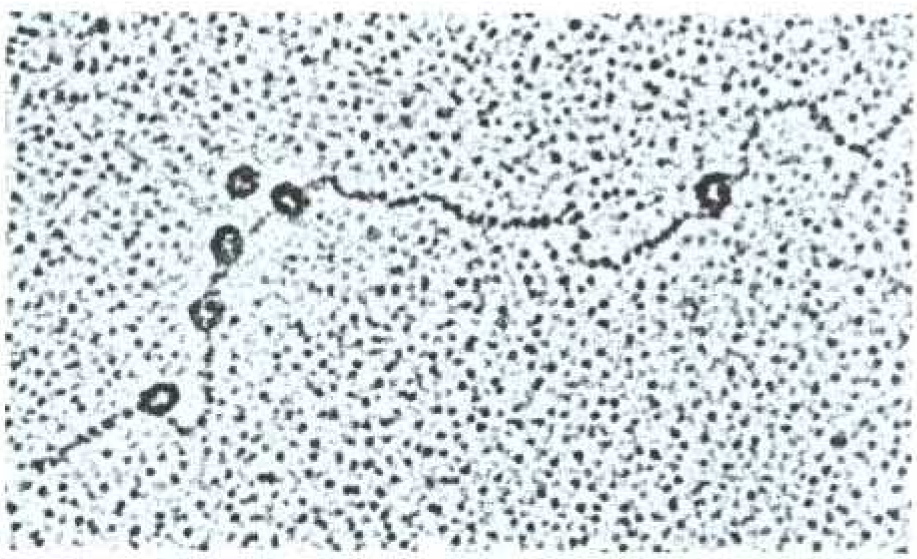


Figure 10.12
E. coli RNA polymerase molecules bound to DNA.
[Courtesy of Robley Williams.]

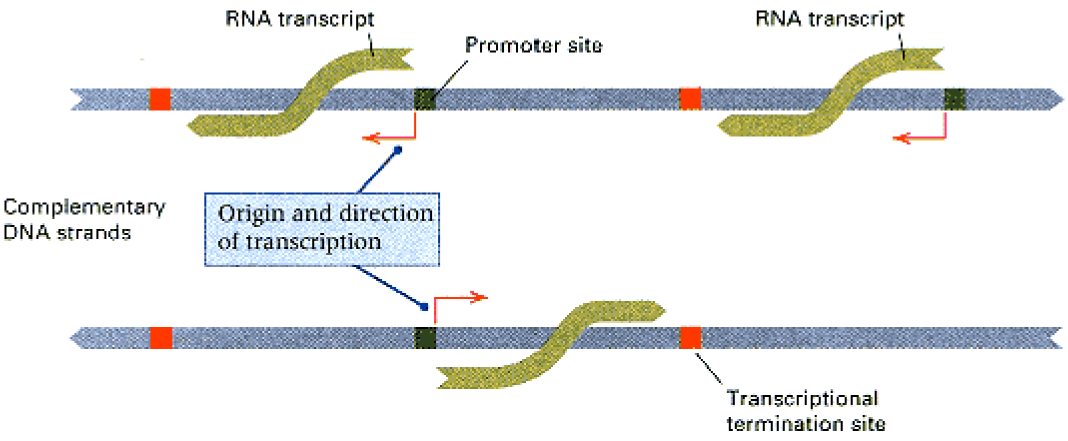


Figure 10.13
A typical arrangement of promoters (green) and termination sites (red) in a segment of a DNA molecule. Promoters are present in both DNA strands. Termination sites are usually located such that transcribed regions do not overlap.

used directly in polypeptide synthesis. In eukaryotes, the RNA molecule is generally processed before it becomes mRNA. The amino acid sequence is then determined by the base sequence in mRNA by the protein-synthesizing machinery of the cell.

Not all base sequences in an mRNA molecule are translated into the amino acid sequences of polypeptides. For example, translation of an mRNA molecule rarely starts exactly at one of its ends and proceeds to the other end; instead, initiation of polypeptide synthesis may begin many nucleotides downstream from the 5' end of the mRNA. The untranslated 5' segment of RNA is called a **leader** and in some cases contains regulatory sequences that affect the rate of protein synthesis. The leader is followed by a **coding sequence,** also called an **open reading frame,** or **ORF,** which specifies the order in which the amino acids are present in the polypeptide chain. A typical coding sequence in an mRNA molecule is between 500 and 3000 bases long (depending on the number of amino acids in the protein). Like the leader sequence, the 3' end of an mRNA molecule following the coding sequence is not translated.

The template strand of each gene is only one of the two DNA strands present in the gene, but which DNA strand is the template can differ from gene to gene along a DNA molecule. That is, except in some small viruses, *not all mRNA molecules are transcribed from the same DNA strand.* Thus in an extended segment of a DNA molecule, mRNA molecules would be seen growing in either of two directions (Figure 10.13), depending on which DNA strand functions as a template.

In prokaryotes, most mRNA molecules are degraded within a few minutes after synthesis. In eukaryotes, a typical lifetime is several hours, although some last only minutes, and others persist for days. In both kinds of organisms, the degradation enables cells to dispose of molecules that are no longer needed. The short lifetime of prokaryotic mRNA is an important factor in regulating gene activity (Chapter 11).

10.4—
RNA Processing

The process of transcription is very similar in prokaryotes and eukaryotes, but there are major differences in the relation between the transcript and the mRNA used for polypeptide synthesis. In prokaryotes, the immediate product of transcription (the **primary transcript**) is mRNA; by contrast, *the primary transcript in eukaryotes must be converted into mRNA.* This conversion, which is called **RNA processing,** usually consists of two types of events: modification of the ends and excision of untranslated sequences embedded *within* coding sequences. These events are illustrated diagrammatically in Figure 10.14.

Each end of a eukaryotic transcript is processed. The 5' end is altered by the addition of a modified guanosine, 7-methyl guanosine, in an uncommon 5'-to-5' (instead of 3'-to-5') linkage; this terminal group is called a **cap.** The 3' terminus of a eukaryotic mRNA molecule is usually modified by the addition of a polyadenosine sequence (the **poly-A tail**) of as many as 200 nucleotides. The 5' cap is necessary for the mRNA to bind with the ribosome to begin protein synthesis, and the poly-A tail helps to determine mRNA stability.

A second important feature peculiar to the primary transcript in eukaryotes, also shown in Figure 10.14, is the presence of segments of RNA, called **introns** or **intervening sequences,** that are excised from the primary transcript. Accompanying the excision of introns is a rejoining of the coding segments (**exons**) to form the mRNA molecule. The excision of the introns and the joining of the exons is called **RNA splicing.** The mechanism of RNA splicing is illustrated schematically in Figure 10.15. Figure 10.15A shows the consensus sequence found at the 5' (**donor**) end and at the 3' (**acceptor**) end of most introns. The symbols are N, any nucleotide; R, any purine (A or G); Y, any pyrimidine (C or U); and S, either A or C. In the first step of

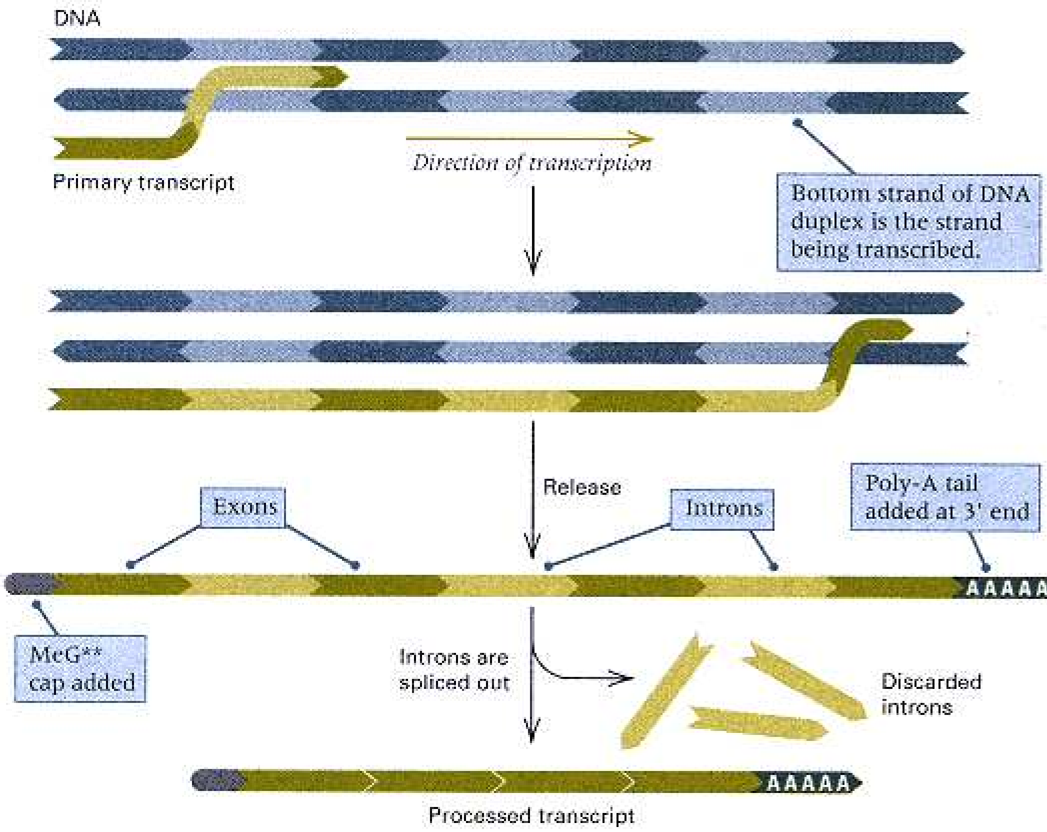


Figure 10.14
A schematic drawing showing the production of eukaryotic mRNA. The primary transcript is capped before it is released from the DNA. MeG denotes 7-methylguanosine (a modified form of guanosine), and the two asterisks indicate two nucleotides whose riboses are methylated. The 3' end is usually modified by the addition of consecutive adenines. Along the way, the introns are excised. These reactions take place within the nucleus.

splicing, the 2'-OH of the adenosine (A) at the branch site, which is located a short distance upstream from a run of prymidines (Y) near the acceptor site, attacks the phosphodiester bond at the donor splice site junction. The attack results in cleavage at the donor splice site and formation of a branched molecule (Figure 10.15B) known as a **lariat** because it has a loop and a tail. The A-G linkage at the "knot" of the lariat is unusual in being 2'-to-5' (instead of the usual 3'-to-5'). In the final step of splicing (Figure 10.15C), the 3'-OH of the guanosine of the donor exon attacks the phosphodiester bond at the acceptor splice site, freeing the lariat intron and joining the donor and acceptor exons together. The lariat intron is rapidly degraded into individual nucleotides by nucleases.

RNA splicing takes place in nuclear particles known as **spliceosomes.** These abundant particles are composed of protein and several types of specialized small RNA molecules ranging from 100 to 200 bases in length. The specificity of splicing comes from the small RNAs, some of which contain sequences that are complementary to the splice junctions, but numerous spliceosome proteins also are required for splicing. One model for the process is illustrated in Figure 10.16, in which U1, U2, and U5 are designations for three different types of small nuclear RNAs. The ends of the intron are brought together by U1, which forms base pairs with nucleotides in the intron at both the 5' and the 3' ends. The ends of the exons are brought together by U5, which forms base pairs with nucleotides in the exons at both the donor splice site and the acceptor splice site. The black arrow in Figure 10.16 indicates the initial attack of the branch site A on the donor splice site

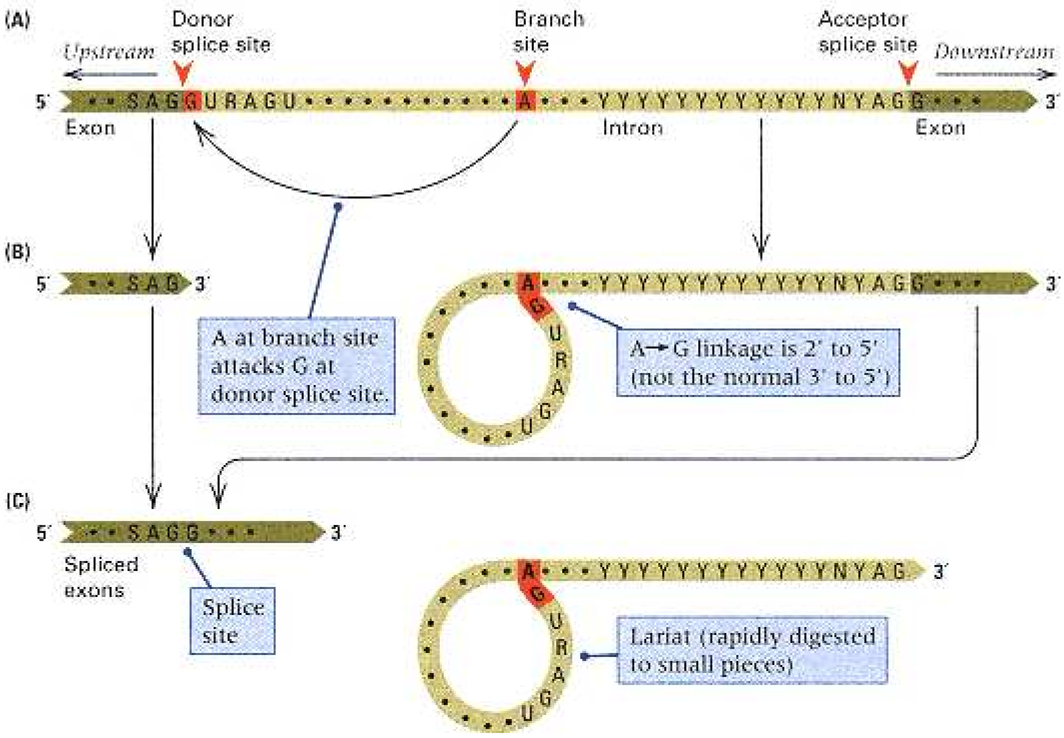


Figure 10.15
A schematic diagram showing removal of one intron from a primary transcript. The A nucleotide at the branch site attacks the terminus of the 5' exon, cleaving the exon-intron junction and forming a loop connected back to the branch site. The 5' exon is later brought to the site of cleavage of the 3' exon, a second cut is made, and the exon termini are joined and sealed. The loop is released as a lariat-shaped structure that is degraded. Because the loop includes most of the intron, the loop of the lariat is usually very much longer than the tail.

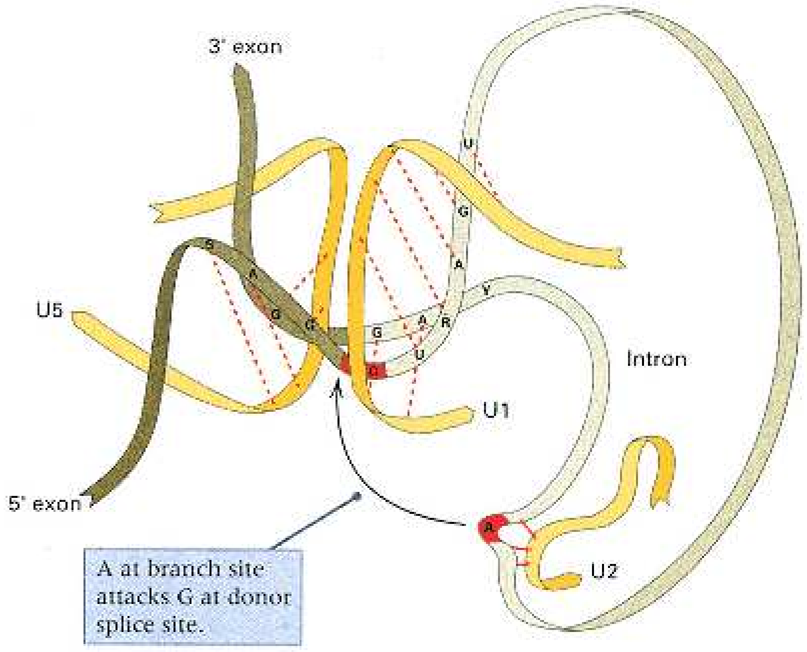


Figure 10.16
Model for RNA splicing. U1, U2, and U5 are small nuclear RNA molecules. The exons are shown in dark green, and the intron is shown in light green. The base pairs that form with nucleotides in the small nuclear RNAs are indicated by dashed (U1 and U5) or solid (U2) lines between the RNA strands. The arrow shows the first step, in which the adenosine of the branch site, held in place by U2, attacks the phosphodiester bond at the splice donor site, resulting in cleavage at the splice donor site and formation of a lariat structure. The process of splicing also requires additional small nuclear RNAs, as well as numerous proteins. Together, these components form the spliceosome.
[Modified from J. A. Steitz. 1992. *Science*, 257:888.]

junction. Note that the branch site is held in place by U2.

Introns are also present in some genes in organelles, but the mechanisms of their excision differ from those of introns in nuclear genes because organelles do not contain spliceosomes. In one class of organelle introns, the intron contains a sequence coding for a protein that participates in removing the intron that codes for it. The situation is even more remarkable in the splicing of a ribosomal RNA precursor in the ciliate *Tetrahymena.* In this case, the splicing reaction is intrinsic to the folding of the precursor; that is, the RNA precursor is *self-splicing* because the folded precursor RNA creates its own RNA-splicing activity. The self-splicing *Tetrahymena* RNA was the first example found of an RNA molecule that could function as an enzyme in catalyzing a chemical reaction; such enzymatic RNA molecules are called **ribozymes.**

The existence and the positions of introns in a particular primary transcript are readily demonstrated by renaturing the transcribed DNA with the fully processed mRNA molecule. The DNA-RNA hybrid can then be examined by electron microscopy. An example of adenovirus mRNA (fully processed) and the corresponding DNA are shown in Figure 10.17.

The DNA copies of the introns appear as single-stranded loops in the hybrid molecule, because no corresponding RNA sequence is available for hybridization.

The number of introns per RNA molecule varies considerably from one gene to the next. For example, 2 introns are present in the primary transcript of human α-globin, and 52 introns occur in collagen RNA. Furthermore, within a particular RNA molecule, the introns are widely distributed and have many different sizes (Figure 10.18). In human beings and other mammals, most introns range in size from 100 to 10,000 base pairs, and in the processing of a typical primary transcript, the amount of discarded RNA ranges from about 50 percent to nearly 90 percent of the primary transcript. Genes in lower eukaryotes, such as yeast, nematodes, and fruit flies, generally have fewer introns than genes in mammals, and the introns tend to be much smaller.

Most introns appear to have no function in themselves. An artificial gene that lacks a particular intron usually functions normally. In those cases in which an intron seems to be required for function, it is usually not because the interruption of the gene is necessary, but because the intron happens to include certain nucleotide sequences that regulate the timing or tissue specificity of transcription. The implication is that many mutations in introns, including small deletions and insertions, should have essentially no effect on gene function, and this is the case. Moreover, the nucleotide sequence of a particular intron is found to undergo changes (including small deletions and insertions) extremely rapidly in the course of evolution, and this lack of sequence conservation is another indication that most of the nucleotide sequences present within introns are not important.

Mutations that affect any of the critical splicing signals do have important consequences, because they interfere with the splicing reaction. Two possible outcomes are illustrated in Figure 10.19. In Figure 10.19A, the intron with the mutated splice site fails to be removed, and it remains in the processed mRNA. The result

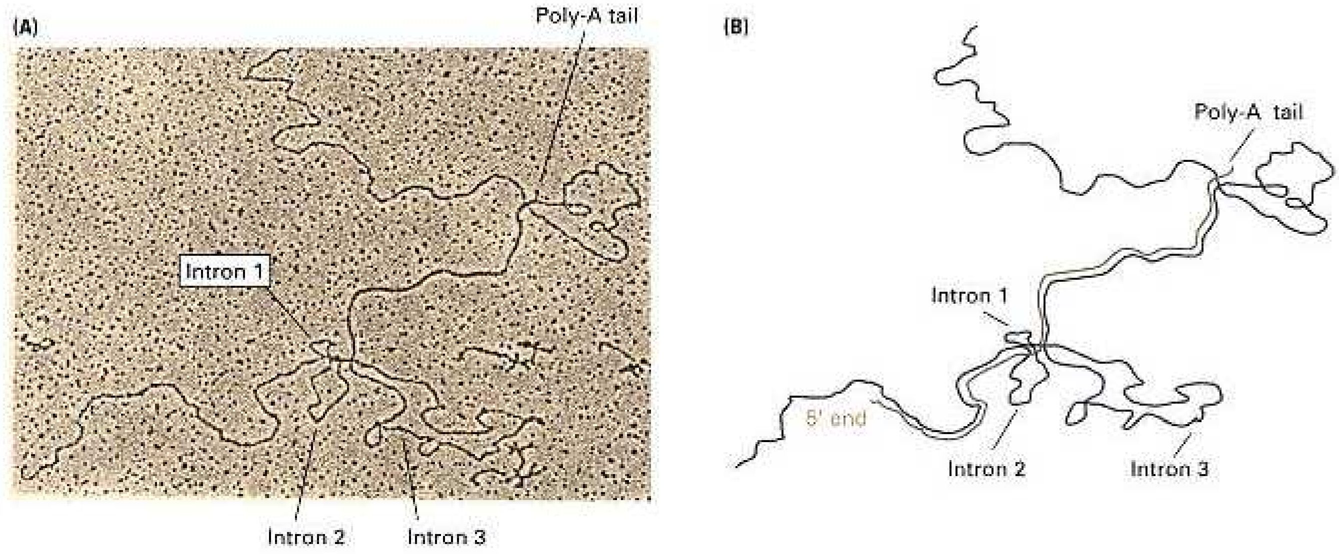


Figure 10.17
(A) An electron micrograph of a DNA-RNA hybrid obtained by annealing a single-stranded segment of adenovirus DNA with one of its mRNA molecules. The loops are single-stranded DNA. (B) An interpretive drawing. RNA and DNA strands are shown in red and blue, respectively. Four regions do not anneal, creating three single-stranded DNA segments that correspond to the introns and the poly-A tail of the mRNA molecule.
[Electron micrograph courtesy of Tom Broker and Louise Chow.]

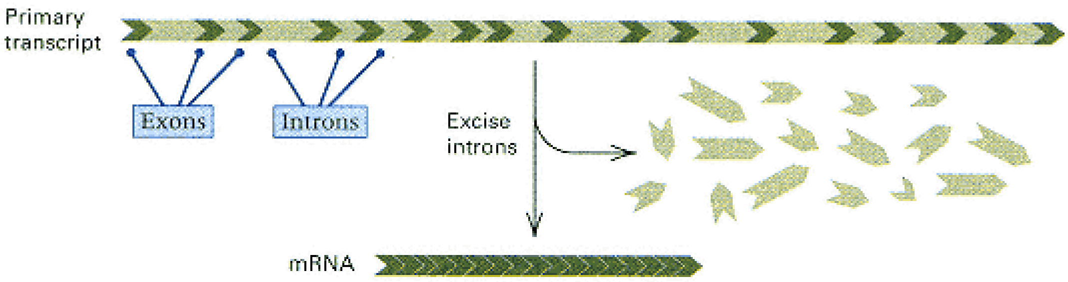


Figure 10.18
A diagram of the primary transcript and the processed mRNA of the conalbumin gene. The 16 introns, which are excised from the primary transcript, are shown in light green. The exons range in size from 29–331 bp (average 138 bp); the introns range in size from 124–1313 bp (average 512 bp). Approximately 75 percent of the primary transcript consists of introns.

is the production of a mutant protein with a normal sequence of amino acids up to the splice site but an abnormal sequence afterward. Most introns are long enough that, by chance, they contain a stop sequence that terminates protein synthesis, and once a stop is encountered, the protein grows no further. A second kind of outcome is shown in Figure 10.19B. In this case, splicing does occur, but at an alternative splice site. (The example shows the alternative site downstream from the mutation, but alternative

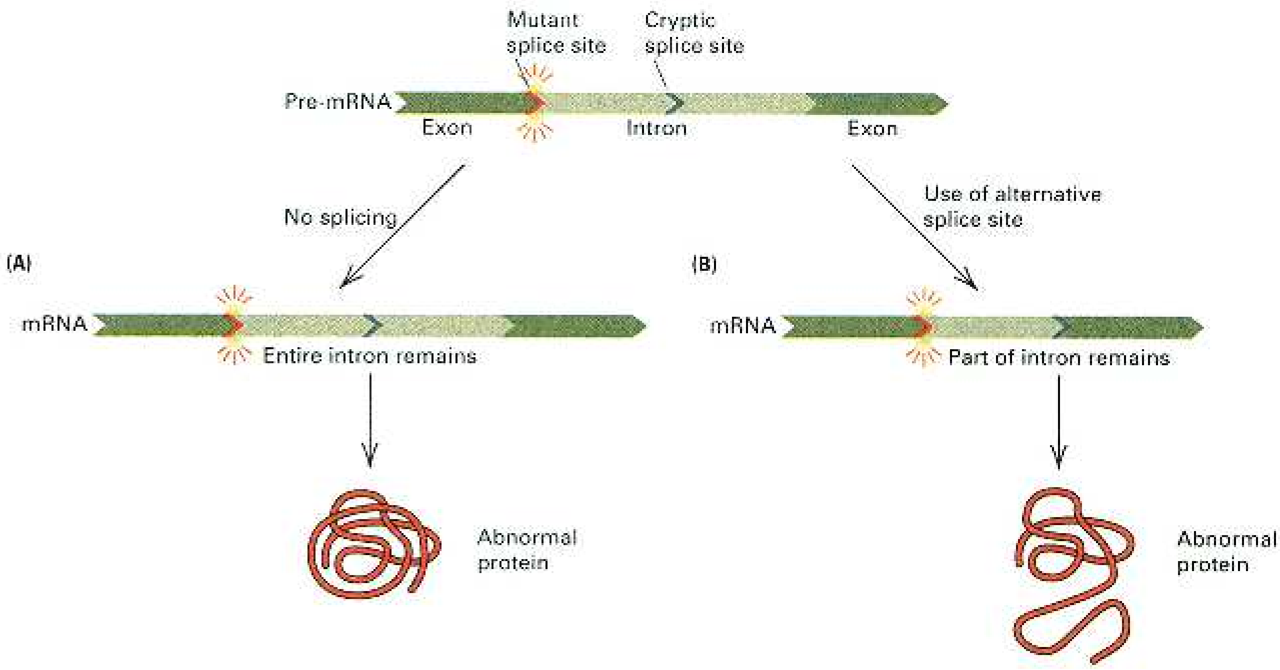


Figure 10.19
Possible consequences of mutation in the donor splice site of an intron. (A) No splicing occurs, and the entire intron remains in the processed transcript. (B) Splicing occurs at a downstream cryptic splice site, and only the upstream part of the original intron still remains in the processed transcript. Neither outcome results in a normal protein product.

sites can also be upstream.) The alternative site is called a **cryptic splice site** because it is not normally used. The cryptic splice site is usually a poorer match with the consensus sequence and is ignored when the normal splice site is available. The result of using the alternative splice site is again an incorrectly processed mRNA and a mutant protein. In some splice-site mutations, both outcomes (Figure 10.19A and B) can occur: Some transcripts leave the intron unspliced, whereas others are spliced at cryptic splice sites.

Although introns are not usually essential in regulating gene expression, they may play a role in gene evolution. In some cases, the exons in a gene code for segments of the completed protein that are relatively independent in their folding characteristics. For example, the central exon of the β-globin gene codes for the segment of the protein that folds around an iron-containing molecule of heme. Relatively autonomous folding units in proteins are known as folding **domains,** and the correlation between exons and domains found in some genes suggests that the genes were originally assembled from smaller pieces. In some cases, the ancestry of the exons can be traced. For example, the human gene for the low-density lipoprotein receptor that participates in cholesterol regulation shares exons with certain blood-clotting factors and epidermal growth factor. The model of protein evolution through the combination of different exons is called the **exon-shuffle** model. The mechanism for combining exons from different genes is not known. Although some genes support the model, in other genes the boundaries of the folding domains do not coincide with exons.

The evolutionary origin of introns is unknown. On the one hand, introns may be an ancient feature of gene structure. The existence of self-splicing RNAs means that introns could have existed long before the evolution of the spliceosome mechanism, and therefore some introns may be as old as the genes themselves. Furthermore, the finding that some genes have introns in the same places in both plants and animals suggests that the introns may have been in place before plants and animals became separate lineages. If introns are ancient, then exon shuffling might have been important early in evolution by creating new genes with novel combinations of exons. It has even been suggested that all forms of early life had introns in their genes and that today's prokaryotes, which lack introns, lost their introns in their evolution. On the other hand, it has also been argued that introns arose relatively late in evolution and became inserted into already existing genes, particularly in vertebrate genomes.

10.5— Translation

The synthesis of every protein molecule in a cell is directed by an mRNA originally copied from DNA. Protein production includes two kinds of processes: (1) information-transfer processes in which the RNA base sequence determines an amino acid sequence, and (2) chemical processes in which the amino acids are linked together. The complete series of events is called **translation.**

The main ingredients necessary for translation are as follows:

• ***Messenger RNA*** Messenger RNA is needed to bring the ribosomal subunits together (described below) and to provide the coding sequence of bases that determines the amino acid sequence in the resulting polypeptide chain.

• ***Ribosomes*** These components are particles on which protein synthesis takes place. They move along an mRNA molecule and align successive transfer RNA molecules; the amino acids are attached one by one to the growing polypeptide chain by means of peptide bonds. Ribosomes consist of two subunit particles. In *E. coli,* their sizes are 30S (the small subunit) and 50S (the large subunit). The counterparts in eukaryotes are 40S and 60S. (The S stands for *Svedberg unit,* which measures the rate of sedimentation of a particle in a centrifuge and so is an indicator of size.) Together, the small and large particles form a functional ribosome. An electron

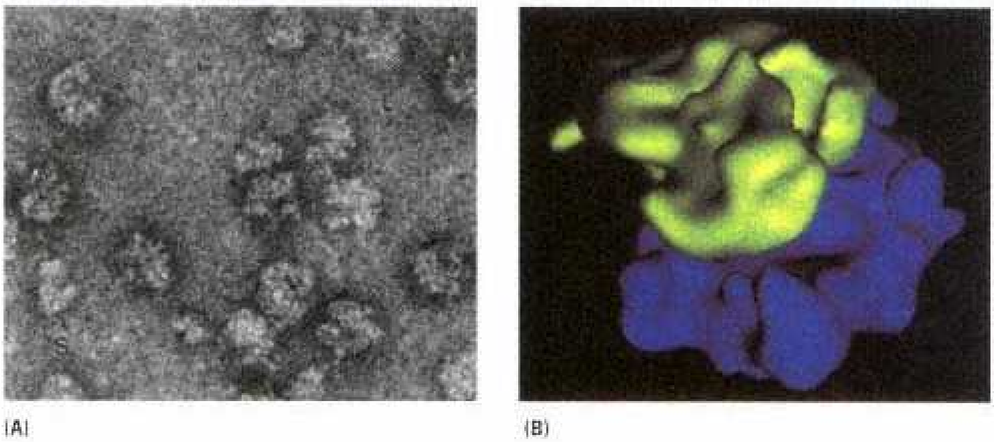


Figure 10.20
Ribosomes. (A) An electron micrograph of 70S ribosomes from E. coli. The 70S ribosome consists of one small subunit of size 30S and one large subunit of size 50S. (B) A three-dimensional model of the E. coli 70S ribosome based on high-resolution electron microscopy. The 30S subunit is in light green, and the 50S subunit is in dark blue.
[A, courtesy of James Lake; B, courtesy of J. Frank, A. Verschoor, Y. Li, J. Zhu, R. K. Lata, M. Radermacher et al. 1995. Biochemistry and Cell Biology. 73: 357.]

micrograph and a model of an *E. coli* ribosome are shown in Figure 10.20.

• ***Transfer RNA, or tRNA*** The sequence of amino acids in a polypeptide is determined by the base sequence in the mRNA by means of a set of adaptor molecules, the tRNA molecules, each of which is attached to a particular amino acid. Each group of three adjacent bases in the mRNA forms a **codon** that binds to a particular group of three adjacent bases in the tRNA (an **anticodon**), bringing the attached amino acid into line for addition to the growing polypeptide chain.

• ***Aminoacyl tRNA synthetases*** This set of enzymes catalyzes the attachment of each amino acid to its corresponding tRNA molecule. A tRNA attached to its amino acid is called an **aminoacylated tRNA** or a **charged tRNA.**

• ***Initiation, elongation, and termination factors*** Polypeptide synthesis can be divided into three stages—(1) initiation, (2) elongation, and (3) termination. Each stage requires specialized proteins.

In prokaryotes, all of the components for translation are present throughout the cell; in eukaryotes, they are located in the cytoplasm, as well as in mitochondria and chloroplasts.

In overview, the process of translation is that an mRNA molecule binds to a ribosome. The aminoacylated tRNAs are brought along sequentially, one by one, to the ribosome that is translating the mRNA molecule. Peptide bonds are made between successively aligned amino acids, each time joining the amino group of the incoming amino acid to the carboxyl group of the amino acid at the growing end. Finally, the chemical bond between the last tRNA and its attached amino acid is broken, and the completed polypeptide is removed.

Initiation

The main features of the initiation step in polypeptide synthesis are the binding of mRNA to the small subunit of the ribosome and the binding of a charged tRNA bearing the first amino acid (Figure 10.21A). The ribosome includes three sites for tRNA molecules. They are called the **E (exit)** site, the **P (peptidyl)** site, and the **A (aminoacyl)** site. In the initiation of translation,

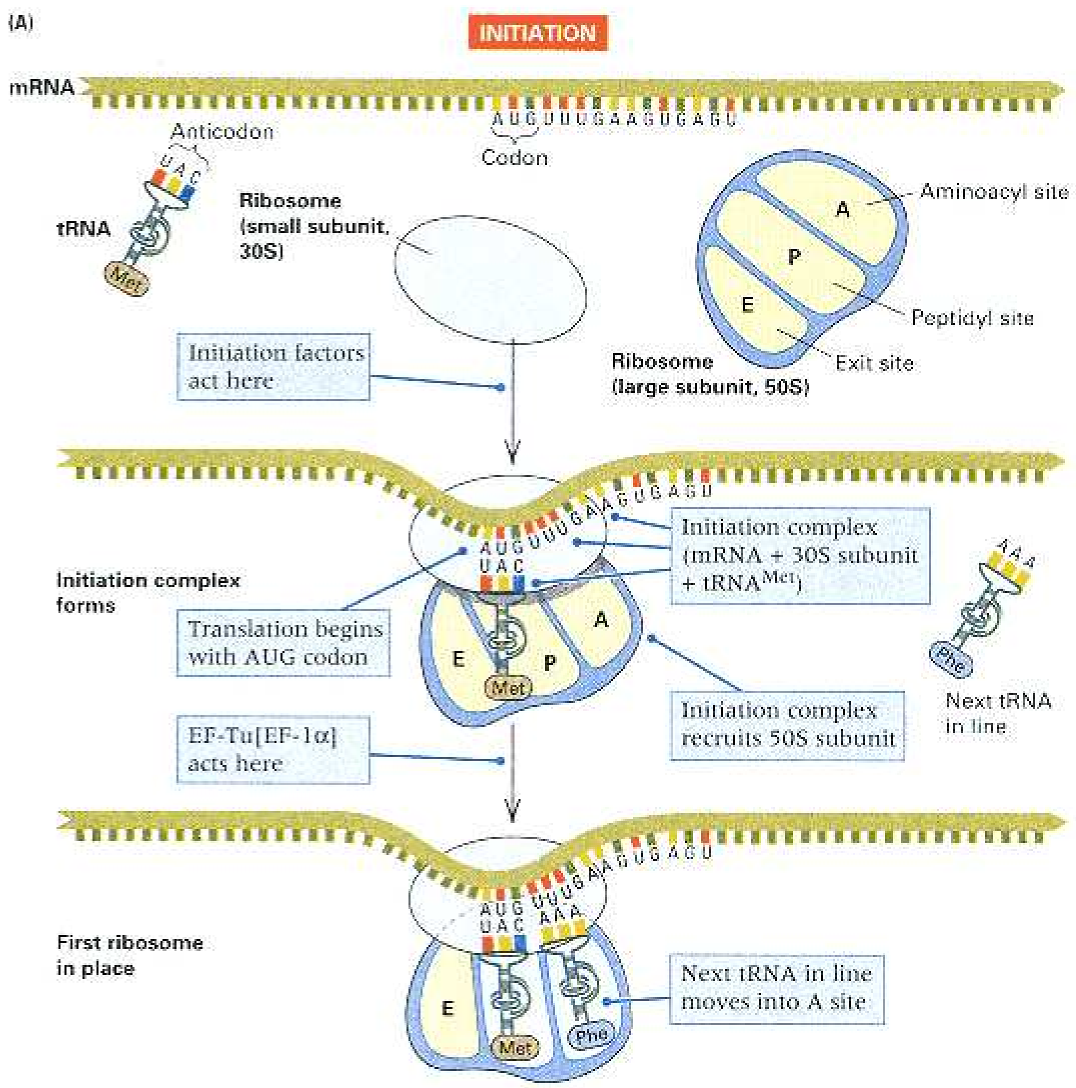
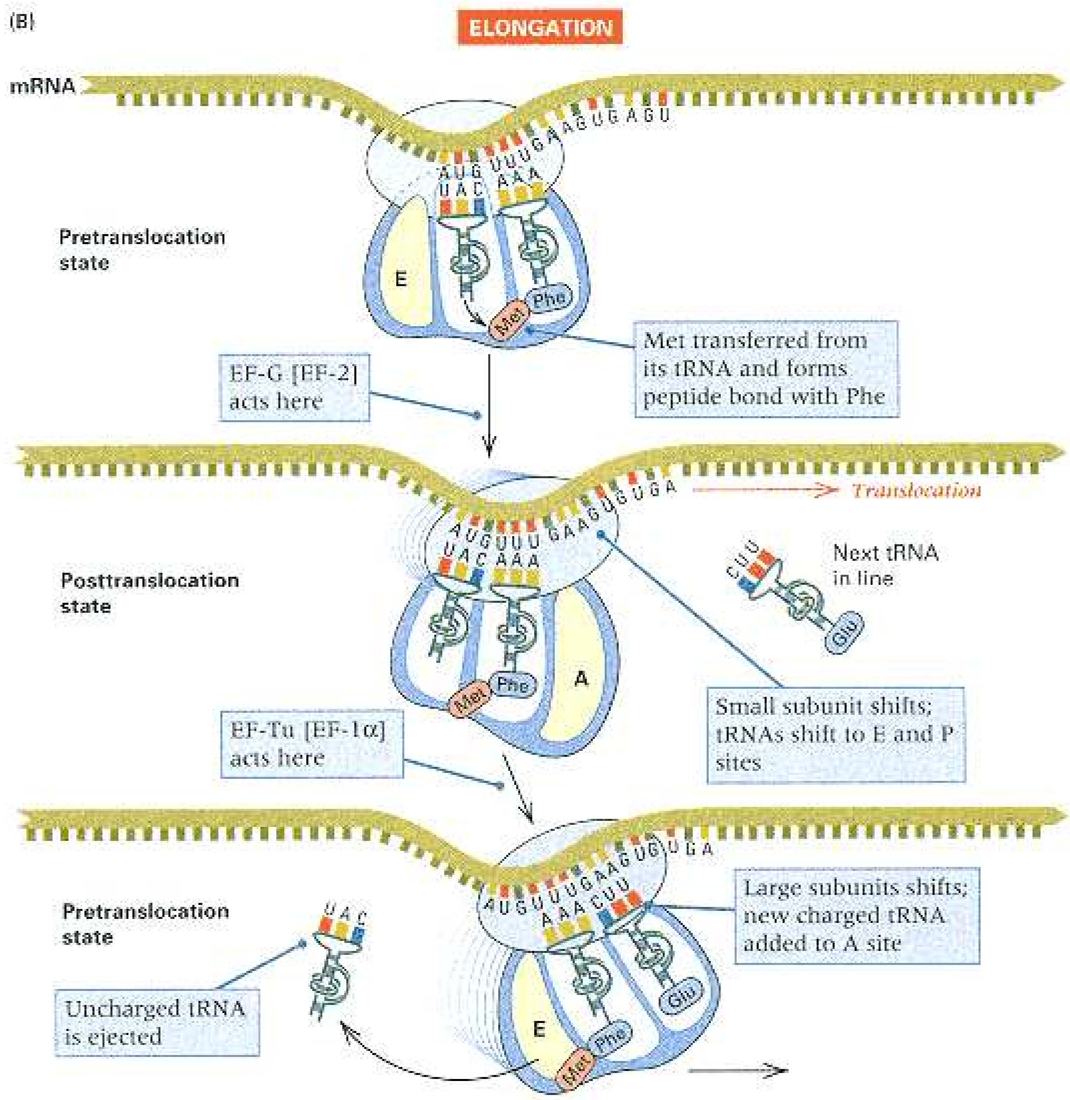


Figure 10.21
Initiation of protein synthesis. (A) The initiation complex—consisting of the mRNA, one 30S ribosomal subunit, and $tRNA^{Met}$—recruits a 50S ribosomal subunit in which the $tRNA^{Met}$ occupies the P (peptidyl) site of the ribosome. A second charged tRNA (in this example, $tRNA^{Phe}$) joins the complex in the A (aminoacyl) site.

(figure continued on next page)

two initiation factors (IF-1 and IF-3) interact with the 30S subunit at the same time that another initiation factor (IF-2) binds with a special initiator tRNA charged with methionine. (In prokaryotes, the initiator tRNA actually carries formylmethionine, yielding $tRNA^{fMet}$.) These components come together and combine with an mRNA. In prokaryotes, the mRNA binding is facilitated by hydrogen bonding between the 16S RNA present in the 30S subunit and the **ribosome-binding site** of the mRNA; in eukaryotes, the 5' cap on the mRNA is instrumental. Together, the 30S + $tRNA^{Met}$ + mRNA complex recruits a 50S subunit, in which the $tRNA^{Met}$ is positioned in the P site and aligned with the AUG initiation codon, forming the 70S initiation complex (Figure 10.21A). The tRNA binding is accomplished by hydrogen bonding between the AUG codon in the mRNA and the three-base anticodon in

(figure continued from previous page)



(B) First steps in elongation. The methionine is transferred from the tRNA^{Met} onto the amino group of phenylalanine (the attacking group), resulting in cleavage of the bond between methionine and its tRNA and peptide bond formation in a concerted reaction catalyzed by peptidyl transferase in the 50S subunit. Then the ribosome shifts one codon along the mRNA to the next in line.

the tRNA. In the assembly of the completed ribosome, the initiation factors dissociate from the complex.

Elongation

The elongation stage of translation consists of three processes: bringing each new aminoacylated tRNA into line, forming the new peptide bond to elongate the polypeptide, and moving the ribosome along the mRNA so that the codons can be translated successively. The first step in elongation is illustrated in Figure 10.21B. A key role is played by the elongation factor EF-Tu, although a second protein, EF-Ts, is also required. (The eukaryotic counterpart of EF-Tu is called EF- 1α.) The EF-Tu, bound with guanosine triphosphate (EF-Tu-GTP), brings the next aminoacylated tRNA into the A site on the 50S subunit, which in this example is tRNA^{Phe}. This processes requires the hydrolysis of GTP to GDP, and once the GDP is formed, the EF-Tu-GDP has low affinity for the ribosome and diffuses away, becoming available for reconversion into EF-Tu-GTP.

Once the A site is filled, a **peptidyl transferase** activity catalyzes a concerted reaction in which the bond connecting the methionine to the tRNA^{Met} is transferred to the amino group of the phenylalanine, forming the first peptide bond. Peptidyl transferase activity is not due to a single

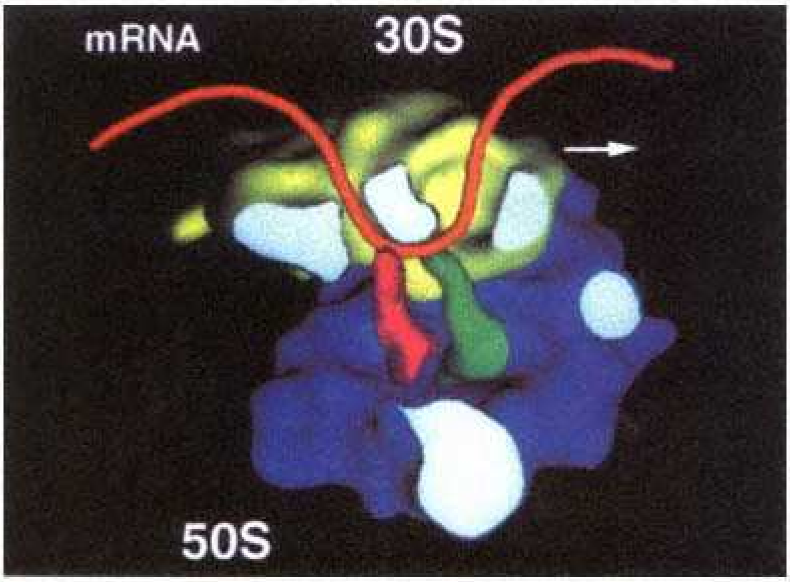


Figure 10.22
A model of a 70S ribosome with some parts cut away to show the orientation of the mRNA relative to the 30S and 50S subunits. The P and A tRNA sites are indicated in red and dark green, respectively. This is the pretranslocation state, in which the E site is unoccupied.
[Courtesy of J. Frank, A. Verschoor, Y. Li, J. Zhu, R. K. Lata, M. Radermacher *et al*. 1995. *Biochemistry and Cell Biology* 73: 357.]

molecule but requires several components of the 50S subunit, including several proteins and an RNA component (called 23S) of the 50S subunit. Some evidence indicates that the actual catalysis is carried out by the 23S RNA, which would suggest that 23S is an example of a ribozyme at work.

Figure 10.22 shows a cutaway view of a 70S ribosome and the bound tRNA molecules in the P site and the A site. The 30S subunit, in light green at the top, binds the mRNA and moves along it in the direction indicated by the arrow. The 50S subunit, in blue at the bottom, contains the tRNA binding sites in the P and A sites. The P site is in red, to the left, and the A site is in dark green, to the right.

Note in Figure 10.21B that the relative positions of the 30S and 50S ribosomal subunits are shifted from one panel to the next. The configuration of the subunits in the top panel is called the **pretranslocation state.** In the middle panel, the 30S subunit shifts one codon to the right. This event is called *translocation.* After translocation, the ribosome is said to be in the *post-translocation state.* In the next step of polypeptide synthesis, shown in the bottom panel in Figure 10.21B, the 50S sub-unit shifts one step over to the right, which reconfigures the ribosome back into the pretranslocation state.

The term **translocation,** as applied to protein synthesis, means the movement of the 30S subunit one codon further along the mRNA. With each successive translocation of the 30S subunit, one more amino acid is added to the growing polypeptide chain. The entire cycle of charged tRNA addition, peptide bond formation, and translocation is elongation.

The repetitive steps in elongation are outlined in Figure 10.23. Starting with a ribosome in the pretranslocation state of Figure 10.23A, the elongation factor EF-G binds with the ribosome. (The eukaryotic counterpart of EF-G is called EF-2.) Like EF-Tu, the EF-G comes on in the form EF-G-GTP and, in fact, binds to the same ribosomal site as EF-Tu-GTP. Hydrolysis of the GTP to GDP yields the energy to shift the tRNAs in the P and A sites to the E and P sites, respectively, as well as to translocate the 30S subunit one codon along the mRNA (red arrow). The ribosome is thereby converted to the **posttranslocation state** (Figure 10.23B), and the EF-G-GDP is released.

At this stage, EF-Tu-GTP comes into play again, and four events happen, as indicated in Figure 10.23C:

• The next aminoacylated tRNA is brought into line (in this case, $tRNA^{Val}$).

• The uncharged tRNA is ejected from the E site.

• In a concerted reaction, the bond connecting the growing polypeptide chain to the tRNA in the P site is transferred to the amino group of the amino acid in the A site, forming the new peptide bond.

• The ribosome transitions to the *pretranslocation* state.

Also, the EF-Tu-GDP is released, making room for the EF-G-GTP, whose function is shown in Figure 10.23D:

• *Translocation* of the 30S ribosome one codon further along the mRNA and return of the ribosome to its *postranslocation* state.

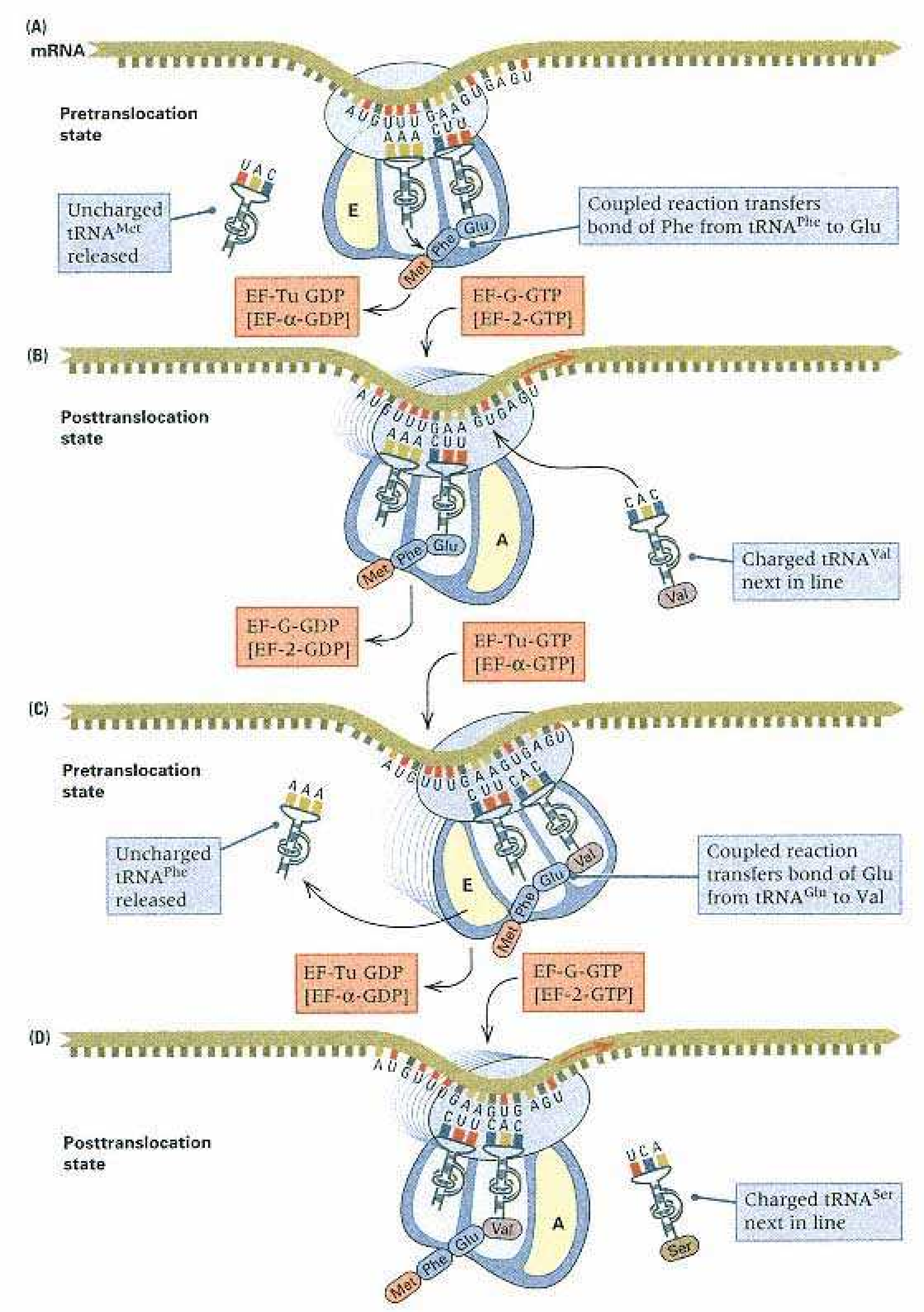


Figure 10.23
Elongation cycle in protein synthesis. (A) Pretranslocation state. (B) Posttranslocation state, in which an uncharged tRNA occupies the E site and the polypeptide is attached to the tRNA in the P site. (C) The function of EF-Tu is to release the uncharged tRNA and bring the next charged tRNA into the A site. A peptide bond is formed between the polypeptide and the amino acid held in the A site, in this case Val. Simultaneously, the 50S subunit is shifted relative to the 30S subunit, forming the pretranslocation state. (D) The function of EF-G is to translocate the 30S ribosome to the next codon, once again generating the posttranslocation state.

After these steps, the EF-G-GDP is released for regeneration into EF-G-GTP for use in another cycle. The translocation state of the ribosome is not depicted separately in Figure 10.23 because it happens so rapidly. In effect, the ribosome shuttles between the pretranslocation and posttranslocation states.

You will note that Figure 10.23D is essentially identical to Figure 10.23B except that the ribosome is one codon farther to the right and the polypeptide is one amino acid greater in length. Hence the ribosome is again available for EF-Tu-GTP to start the next round of elongation. Polypeptide elongation may therefore be considered as a cycle of events repeated again and again. The steps B → C → B → C (or, equivalently, C → D → C → D) are carried out repeatedly until a termination codon is encountered.

In Figure 10.23D, for example, the configuration of the ribosome is such that the $tRNA^{Gly}$ and $tRNA^{Val}$ are occupying the E and P sites, respectively. The aminoacylated tRNA that corresponds to the codon AGU is brought into line, which is $tRNA^{Ser}$, and the bond connecting the polypeptide to $tRNA^{Val}$ is transferred to the amino group of Ser, creating a new peptide bond and elongating the polypeptide by one amino acid. At the same time, the ribosome is converted to the pretranslocation state in preparation for translocation. The elongation cycle happens relatively fast. Under optimal conditions, *E. coli* synthesizes a polypeptide at the rate of about 20 amino acids per second; in eukaryotes, the rate of elongation is about 15 amino acids per second.

Termination

The elongation steps of protein synthesis are carried out repeatedly until a stop codon for termination is reached. The stop codons are UAA, UAG, and UGA. No tRNA exists that can bind to a stop codon, so the tRNA holding the polypeptide remains in the P site (Figure 10.24). Specific release factors act to cleave the polypeptide from the tRNA to which it is attached as well as to disassociate the 70S ribosome from the mRNA, after which the individual 30S and 50S subunits are recycled to initiate translation of another mRNA. The release factor RF-1 recognizes the stop codons UAA and UAG, whereas release factor RF-2 recognizes UAA and UGA. A third release factor, RF-3, is also required for translational termination.

Monocistronic and Polycistronic mRNA

The process of selecting the correct AUG initiation codon is of some importance in understanding many features of gene expression. In prokaryotes, mRNA molecules commonly contain information for the amino acid sequences of several different polypeptide chains; such a molecule is called a **polycistronic mRNA.** (*Cistron* is a term often used to mean a base sequence that encodes a single polypeptide chain.) In a polycistronic mRNA, each polypeptide coding region is preceded by its own ribosome-binding site and AUG initiation codon. After the synthesis of one polypeptide is finished, the next along the way is translated (Figure 10.25). The genes contained in a polycistronic mRNA molecule often encode the different proteins of a metabolic pathway. For example, in *E. coli*, the ten enzymes needed to synthesize histidine are encoded by one polycistronic mRNA molecule. The use of polycistronic mRNA is an economical way for a cell to regulate the synthesis of related proteins in a coordinated manner. For example, in prokaryotes, the usual way to regulate the synthesis of a particular protein is to control the synthesis of the mRNA molecule that codes for it (Chapter 11). With a polycistronic mRNA molecule, the synthesis of several related proteins can be regulated by a single signal, so that appropriate quantities of each protein are made at the same time; this is termed **coordinate regulation.**

In eukaryotes, the 5' terminus of an mRNA molecule binds to the ribosome, after which the mRNA molecule slides along the ribosome until the AUG codon nearest the 5' terminus is in contact with the ribosome. Then protein synthesis begins. There is no mechanism for initiating polypeptide synthesis at any AUG other than the first one encountered. Eukaryotic mRNA is always monocistronic (Figure 10.25).

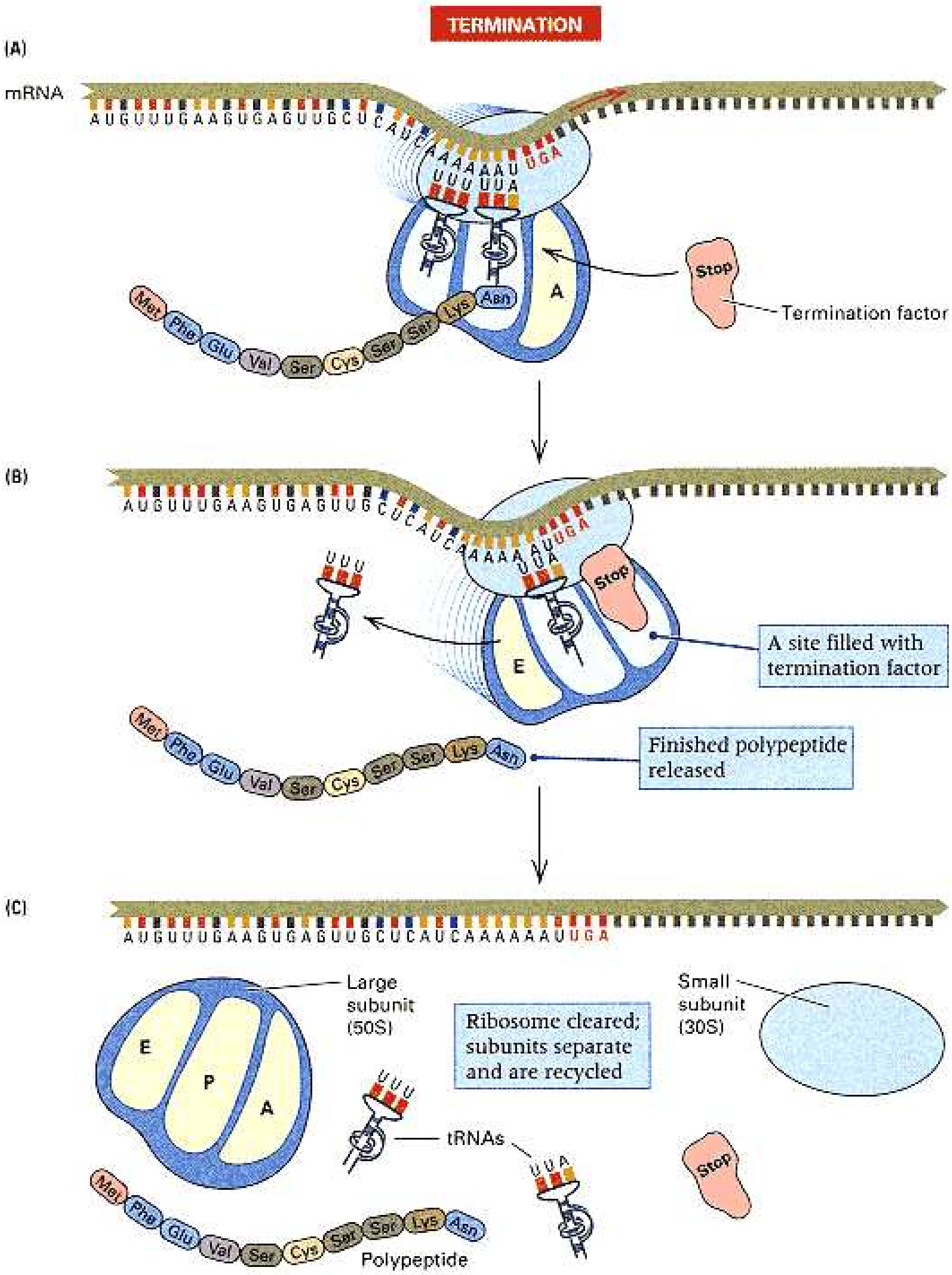


Figure 10.24
Termination of protein synthesis. When a stop codon is reached, no tRNA can bind to that site, which causes the release of the newly formed polypetide and the remaining bound tRNA.

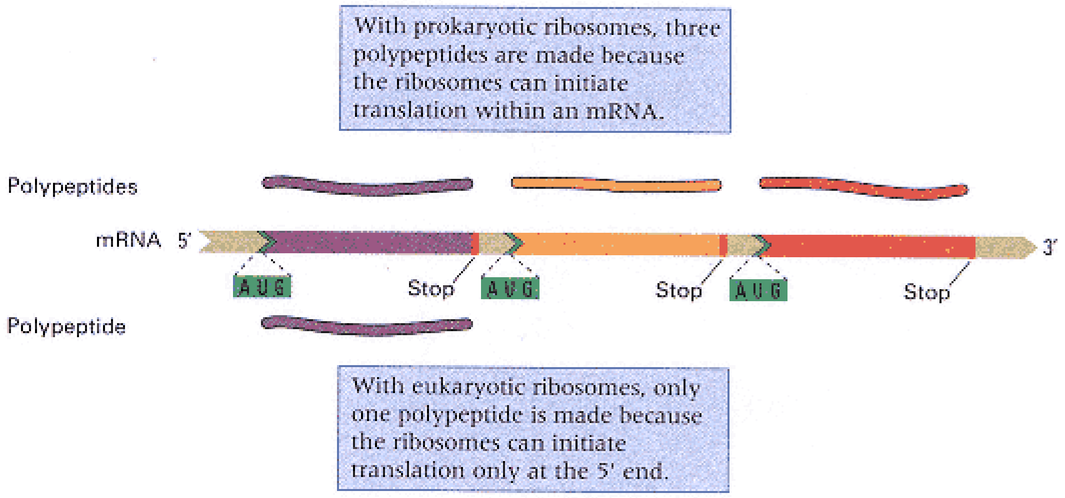


Figure 10.25
Different products are translated from a three-cistron mRNA molecule by the ribosomes of prokaryotes and eukaryotes. The prokaryotic ribosome translates all of the genes, but the eukaryotic ribosome translates only the gene nearest the 5' terminus of the mRNA. Translated sequences are shown in purple, yellow, and orange; stop codons in red; the ribosome binding sites in green; and the spacer sequences in light green.

The definitive feature of translation is that it proceeds in a particular direction along the mRNA and the polypeptide:

> The mRNA is translated from an initiation codon to a stop codon in the 5'-to-3' direction. The polypeptide is synthesized from the amino end toward the carboxyl end by the addition of amino acids. one by one, to the carboxyl end.

For example, a polypeptide with the sequence

$$NH_2-\mathrm{Met}-\mathrm{Pro}-\cdots-\mathrm{Gly}-\mathrm{Ser}-COOH$$

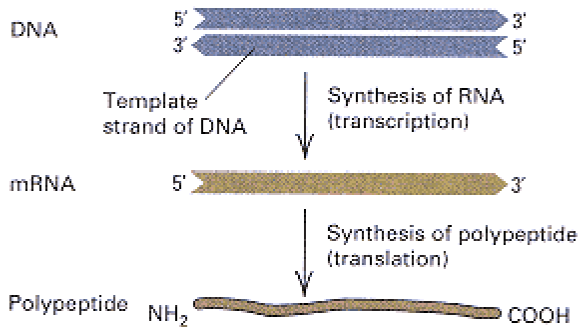


Figure 10.26
Direction of synthesis of RNA with respect to the coding strand of DNA, and of synthesis of protein with respect to mRNA.

would start with methionine, and serine would be the last amino acid added to the chain. The directions of synthesis are illustrated schematically in Figure 10.26.

In writing nucleotide sequences, by convention, we place the 5' end at the left and, in writing amino acid sequences, we place the amino end at the left. Polynucleotides are generally written so that both synthesis and translation proceed from left to right, and polypeptides are written so that synthesis proceeds from left to right. This convention is used in all of the following sections concerning the genetic code.

10.6— The Genetic Code

The four bases in DNA—A, T, G, and C—are sufficient to specify the 20 amino acids in proteins because each codon is three bases in length. Each sequence of three adjacent bases in mRNA is a codon that specifies a particular amino acid (or chain termination). The **genetic code** is the list of all codons and the amino acid that each one encodes. Before the genetic code was determined experimentally, it was reasoned that if all codons were assumed to

Connection Messenger Light

Sydney Brenner,[1] Francois Jacob,[2] and Matthew Meselson[3] 1961

[1]Cavendish Laboratory, Cambridge,
England.
[2]Institute Pasteur, Paris, France.
[3]California Institute of Technology,
Pasadena, California.
An Unstable Intermediate Carrying Information from Genes to Ribosomes for Protein Synthesis

Brenner and Jacob were guest investigators at the California Institute of Technology in 1961. At that time there was great interest in the mechanisms by which genes code for proteins. One possibility, which seemed reasonable at the time, was that each gene produced a different type of ribosome, differing in its RNA, which in turn produced a different type of protein. Francois Jacob and Jacques Monod had recently proposed an alternative, which was that the informational RNA ("messenger RNA") is actually an unstable molecule that breaks down rapidly. In this model, the ribosomes are nonspecific protein-synthesizing centers that synthesize different proteins according to specific instructions they receive from the genes through the messenger RNA. The key to the experiment is density-gradient centrifugation, which can separate macromolecules made "heavy" or "light" according to their content of ^{15}Nor ^{14}N, respectively. (This techinque is described in Chapter 5.) The experiment is a purely biochemical proof of an issue absolutely critical for genetics—that genes code for proteins through the intermediary of a relately short-lived messenger RNA.

A large amount of evidence suggests that genetic information for protein structure is encoded in deoxyribonucleic acid (DNA) while the actual assembling of amino acids into proteins occurs in cytoplasmic ribonucleoprotein particles called ribosomes. The

The results also suggest that the messenger RNA may be large enough to code for long polypeptide chains

fact that proteins are not synthesized directly on genes demands the existence of an intermediate information carrier. . . . Jacob and Monod have put forward the hypothesis that ribosomes are non-specialized structures which receive genetic information from the gene in the form of an unstable intermediate of "messenger." We present here the results of experiments on phage-infected bacteria which give direct support to this hypothesis. . . . When growing bacteria are infected with T2 bacteriophage, synthesis of DNA stops immediately, to resume 7 minutes later, while protein synthesis continues at a constant rate; in all likelihood, the protein is genetically determined by the phage. . . . Phage-infected bacteria therefore provide a situation in which the synthesis of a protein is suddenly switched from bacterial to phage control. . . . It is possible to determine experimentally [whether an unstable messenger RNA is produced] in the following way: Bacteria are grown in heavy isotopes so that all cell constituents are uniformly labelled "heavy." They are infected with phage and transferred immediately to a medium containing light isotopes so that all constituents synthesized after infection are "light." The distribution of new RNA and new protein, labelled with radioctive isotopes, is then followed by density gradient centrifugation of purified ribosomes. . . . We may summarize our findings as follows: (1) After phage infection no new ribosomes can be detected. (2) A new RNA with a relatively rapid turnover is synthesized after phage infection. This RNA, which has a base composition corresponding to that of the phage DNA, is added to per-existing ribosomes, from which it can be detached in a cesium chloride gradient by lowering the magnesium concentration. (3) Most, if not all, protein synthesis in the infected cell occurs in pre-existing ribosomes. . . . The results also suggest that the messenger RNA may be large enough to code for long polypeptide chains. . . . It is a prediction of the messenger RNA hypothesis that the messenger RNA should be a simple copy of the gene, and its nucleotide composition should therefore correspond to that of the DNA. This appears to be the case in phageinfected cells. . . . If this turns out to be universally true, interesting implications for the coding mechanisms will be raised.

Source:Nature 190: 576–581

have the same number of bases, then each codon would have to contain at least three bases. Codons consisting of pairs of bases would be insufficient, because four bases can form only $4^2 = 16$ pairs; triplets of bases would suffice, because four bases can form $4^3 = 64$ triplets. In fact, the genetic code is a triple code, and all 64 possible codons carr information of some sort. Most amino acids are encoded by more than one codon. Furthermore, in the translation of mRNA molecules, the codons do not overlap but are used sequentially (Figure 10.27).

Genetic Evidence for a Triplet Code

Although theoretical considerations suggested that each codon must contain at least three letters, codons having mor than three letters could not be ruled out.

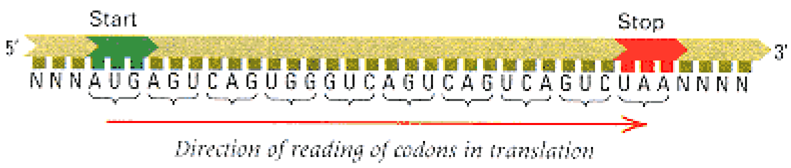


Figure 10.27
Bases in an RNA molecule are read sequentially in the 5'-to-3' direction, in groups of three.

The first widely accepted proof for a triplet code came from genetic experiments using *rII* mutants of bacteriophage T4 that had been induced by replication in the presence of the chemical *proflavin.* These experiments were carried out in 1961 by Francis Crick and collaborators. Proflavin-induced mutations typically resulted in total loss of function. Because proflavin is a large planar molecule, it was suspected that it caused mutations that inserted or deleted a base pair by interleaving between base pairs in the double helix. Analysis of the properties of these mutations led directly to the deduction that the code is read three nucleotides at a time from a fixed point; in other words, there is a **reading frame** to each mRNA. Mutations that delete or add a base pair shift the reading frame and are called **frameshift mutations.** Figure 10.28 illustrates the profound effect of a frameshift mutation on the amino acid sequence of the polypeptide produced from the mRNA of the mutant gene.

The genetic analysis of the structure of the code began with an *rII* mutation called FCO, which was arbitrarily designated (+), as if it had an inserted base pair. (It could also arbitrarily have been designated (-),

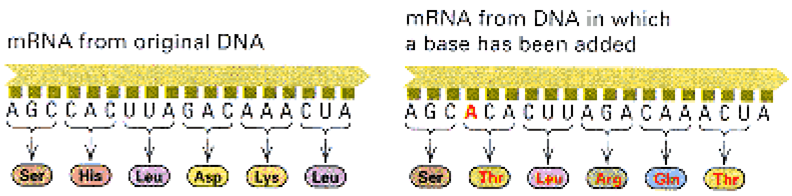


Figure 10.28
The change in the amino acid sequence of a protein caused by the addition of an extra base, which shifts the reading frame. A deleted base also shifts the reading frame.

as if it had a deleted base pair. Calling it (+) was a lucky guess, however, because when FCO was sequenced, it did turn out to have a single-base insertion.) If FCO has a (+) insertion, then it should be possible to revert the FCO allele to "wildtype" by deletion of a nearby base. Selection for r^+ revertants was carried out by isolating plaques formed on a lawn of an *E. coli* strain K12 that was lysogenic for phage λ. The basis of the selection is that *rII* mutants are unable to propagate in K12(λ). Analysis of the revertants revealed that each still carried the original FCO mutation that reversed the effects of the FCO mutation. The suppressor mutations could be separated by recombination from the original mutation by crossing each revertant to wildtype; each suppressor mutation proved to be an *rII* mutation that, by itself, would cause the *r* (rapid lysis) phenotype. If FCO had an inserted base, then the suppressors should all result in deletion of a base pair; hence each suppressor of FCO was designated (-). Three such revertants and their consequences for the translational reading frame, illustrated using ordinary three-letter words, are illustrated in Figure 10.29. The (-) mutations are designated $(-)^1$, $(-)^2$, and $(-)^3$, and those parts of the mRNA translated in the correct reading frame are indicated in green.

Each of the individual (-) suppressor mutations could, in turn, be used to select other "wildtype" revertants, with the expectation that these revertants would carry new suppressor mutations of the (+) variety, because the (-) (+) combination should yield a phage able to form plaques on K12 (λ).

Various double mutant combinations were made. Usually any (+) (-) combination, or any (-) (+) combination, resulted in a wildtype phenotype, whereas (+) (+) and (-) (-) double mutant combinations always resulted in the mutant phenotype. The truly telling result came when triple mutants were made. Usually, the (+) (+) (+) and (-) (-) (-) triple mutants yielded the wildtype phenotype!

The phenotypes of the various (+) and (-) combinations were interpreted in terms of a reading frame. The initial FCO mutation, a + 1 insertion, shifts the reading frame, resulting in incorrect amino acid sequence from that point on and thus a nonfunctional protein (Figure 10.29). Deletion of a base pair nearby will restore the reading frame, although the amino acid sequence encoded between the two mutations will be different and incorrect. In (+) (+) or (-) (-) double mutants, the reading frame is shifted by two bases; the protein made is still nonfunctional. However, in the (+) (+) (+) and (-) (-) (-) triple mutants, the reading frame is restored, though all amino acids encoded within the region bracketed by the outside mutations are incorrect; the protein made is one amino acid longer for (+) (+) (+) and one amino acid shorter for (-) (-) (-) (Figure 10.29).

The genetic analysis of the (+) and (-) mutations strongly supported the following conclusions:

• Translation of an mRNA starts from a fixed point.

• There is a single reading frame maintained throught the process of translation.

• Each codon consists of three nucleotides.

Crick and his colleagues also drew other inferences from these experiments. First, in the genetic code, most codons must function in the specification of an amino acid. Second, each amino acid must be specified by more than one codon. They reasoned that if each amino acid had only one codon, then only 20 of the 64 possible codons could be used for coding amino acids. In this case, most frameshift mutations should have affected one of the remaining 44 "noncoding" codons in the

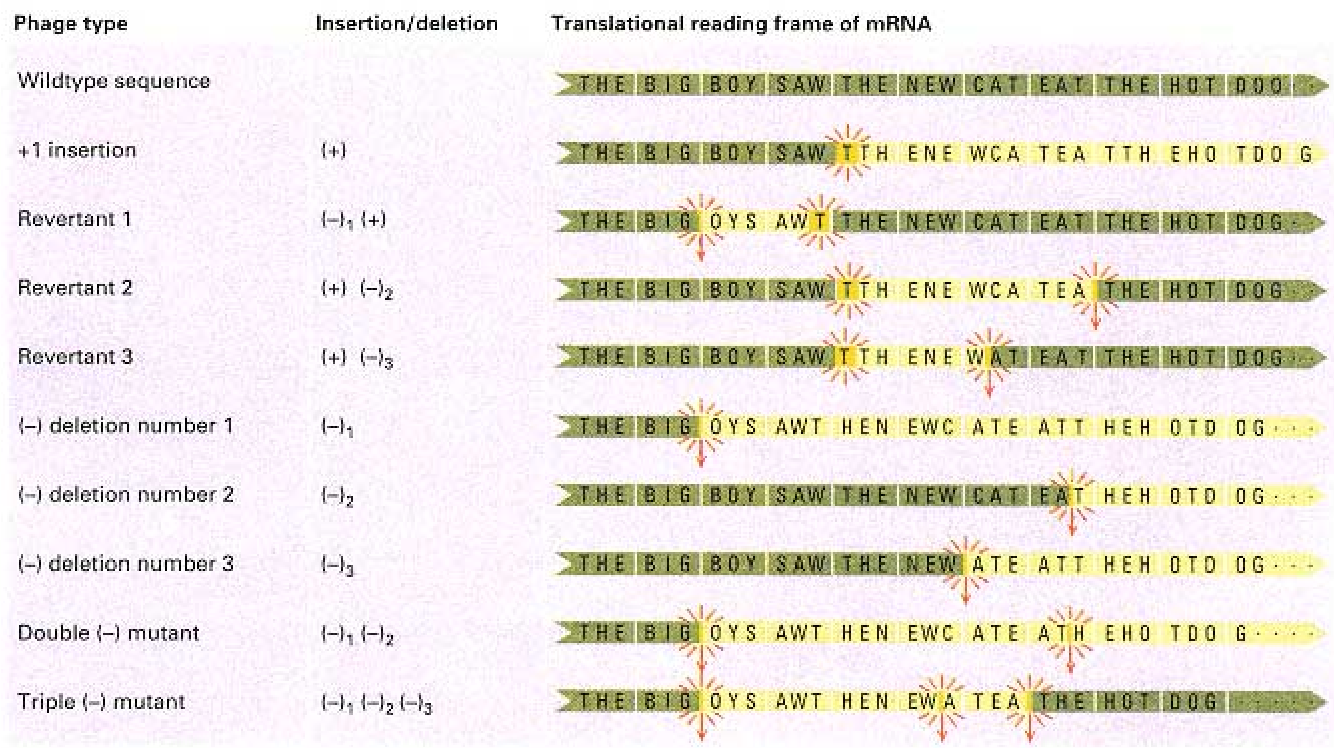


Figure 10.29
Interpretation of the *rII* frameshift mutations showing that combinations of appropriately positioned single-base insertions (+) and single-base deletions (-) can restore the correct reading frame (green). The key finding was that a combination of three single-base deletions, as shown in the bottom line, also restores the correct reading frame (green). Two single-base deletions do not restore the reading frame. These classic experiments gave strong genetic evidence that the genetic code is a triplet code.

Connection Uncles and Aunts

Francis H. C.Crick, Leslie Barnett, Sydney Brenner, and R. J. Watts-Tobin, 1961

Cavendish Laboratory,
Cambridge, England
General Nature of the Genetic Code for Proteins

No other paper affords a better demonstration of the power of mutational analysis in the hands of clever researchers. The issue was this: How many bases are needed to code for one amino acid in a protein? Crick et al. answered the question by using single-base insertions and deletions in the rIIB cistron (Seymour Benzer's name for a region of DNA that codes for a single polypeptide chain). In the laboratory they referred to the (+) and (-) mutations as "uncles" and "aunts" because it was not known, in any particular case, whether a mutation was truly an insertion or truly a deletion. The paper is written as if FCO were an insertion mutation, solely for the sake of simplicity. (This guess later proved to be right.) The presentation is unusual in another respect also. The paper starts by assuming its conclusion in order to describe the results, and then demonstrates how the conclusion was arrived at. Try to imagine writing it in the usual way. Any such effort would be bound to be less clear. (Incidentally, the suggestion that the use of synthetic polynucleotides would solve the coding problem within a year was not far wrong. By 1968, Marshall W. Nirenberg, Robert W. Holley, and Har Gobind Khorana would be awarded a Nobel Prize for their deciphering of the genetic code table.)

In this article we report genetic experiments which suggest that the genetic code [is one in which] a group of three bases (or, less likely, a multiple of three bases) codes for one amino acid. . . . Our genetic experiments have been carried out on the *B* cistron of the *rII* region of the bacteriophage T4. . . . We report here our work on the mutant *FCO*. This mutant was originally produced by the action of proflavin. . . . which we have previously argued acts as a mutagen because it adds or deletes a base

Fortunately, we have convincing evidence that the coding ratio is in fact 3 or a multiple of 3.

or bases. . . . If an acridine mutant is produced by, say, adding a base, it should revert to "wildtype" by deleting a base. Our work on *FCO* shows that it usually reverts not by reversing the original mutation but by producing a second mutation at a nearby point on the genetic map. . . . A genetic map of 18 suppressors of *FCO* shows that they scatter over a region about, say, one-tenth of the size of the *B* cistron. . . . In all we have isolated about eighty independent *rllB* mutants, all suppressors of *FCO,* or suppressors of suppressors, or suppressors of suppressors of suppressors. . . . Although we have no direct evidence that the *B* cistron produces a polypeptide chain (probably through an RNA intermediate), in what follows we shall assume this to be so. To fix ideas, we imagine that the string of nucleotides is read, triplet by triplet, from a starting point on the left of the *B* cistron. We now suppose that, for example, the mutant *FCO* was produced by the insertion of an additional base in the wildtype sequence. This additional of a base at the *FCO* site will mean that the reading of all the triplets to the right of *FCO* will be shifted along one base, and will therefore be incorrect. Thus the amino-acid sequence of the protein which the *B* cistron is presumed to produce will be completely altered from that point onwards. This explains why the function of the gene is lacking. . . . We now postulate that a suppressor of *FCO* (for example, *FC1*) is formed by deleting a base. Thus when the *FC1* mutation is present by itself, all triplets to the right of *FC1* will be read incorrectly and thus the function will be absent. However, when both mutations are present in the same piece of DNA, then although the reading of triplets between *FCO* and *FC1* will be altered, the original reading will be restored to the rest of the gene. . . . So far we have spoken as if the evidence supported a triplet code, but this was simply for illustration. . . . Fortunately, we have convincing evidence that the coding ratio is in fact 3 or a multiple of 3. This we have obtained by constructing triple mutants of the form (+ with + with +) or (- with - with-). One must be careful not to make shifts across the "unacceptable" region of *rllB,* but this we can avoid by a proper choice of mutants. . . . It is possible by various devices, either chemical or enzymatic, to synthesize polyribonucleotides with defined or partially defined sequences. If these will produce specific polypeptides, the coding problem is wide open for experimental attack . . . and the] genetic code may well be solved within a year.

Source: Nature 192: 1227–1232

reading frame, and hence a nearby frameshift of the opposite polarity mutation should not have suppressed the original mutation. Consequently, the code was deduced to be **degenerate,** which means that more than one codon ca specify a particular amino acid.

Elucidation of the Base Sequences of the Codons

Polypeptide synthesis can be carried out in *E. coli* cell extracts obtained by breaking cells open. Various component can be isolated, and a functioning protein-synthesizing system can be reconstituted by mixing ribosomes. tRNA molecules, mRNA molecules, and various protein factors. If radioactive amino acids are added to the extract, then radioactive polypetides are made. Synthesis continues for only a few minutes because mRNA is rapidly degraded by nucleases in the mixture. The elucidation of the genetic code began with the observation that when the degradation of mRNA was allowed to go to completion and the synthetic polynucleotide polyuridylic acid (poly-U) was added to the mixture as an mRNA molecule, a polypeptide consisting only of phenylalanine (Phe-Phe-Phe- . . .) was synthesized. From this simple result, and knowledge that the code is a triplet code, it was concluded that UUU must be a codon for the amino acid phenylalanine. Variations on this basic experiment identified other codons. For example, when a long sequence of guanines was added at the terminus of the poly-U, the polyphenylalanine was terminated by a sequence of glycines, indicating that GGG is a glycine codon (Figure 10.30). A trace of leucine or

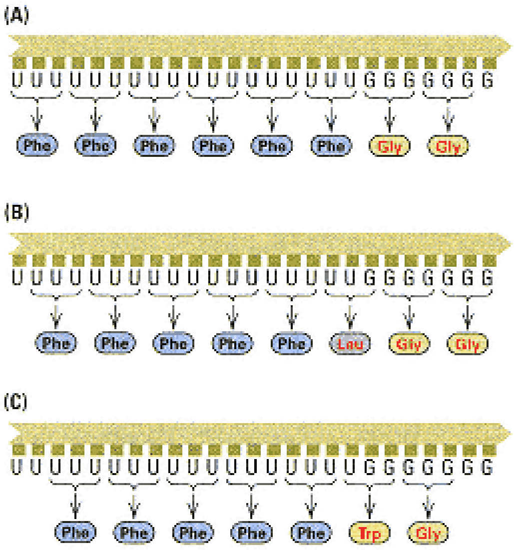


Figure 10.30
Polypeptide synthesis using UUUU . . . UUGGGGGGG as an mRNA in three different reading frames, showing the reasons for the incorporation of glycine, leucine, and tryptophan.

tryptophan was also present in the glycineterminated polyphenylalanine. Incorporation of these amino acids was directed by the codons UUG and UGG at the transition point between U and G. When a single guanine was added to the terminus of a poly-U chain, the polyphenylalanine was terminated by leucine. Thus UUG is a leucine codon, and UGG must be a codon for tryptophan. Similar experiments were carried out with poly-A, which yielded polylysine, and with poly-C, which produced polyproline.

Other experiments led to a complete elucidation of the code. Three codons

UAA UAG UGA

were found to be stop signals for translation, and one codon, **AUG,** which encodes methionine, was shown to be the initiation codon. AUG also codes for internal methionines, but it uses a different tRNA to do so.

A Summary of the Code

The *in vitro* translation experiments, which used components isolated from the bacterium *E. coli*, have been repeated with components obtained from many species of bacteria, yeast, plants, and animals. The standard genetic code deduced from these experiments is considered to be nearly universal because the same codon assignments can be made for nuclear genes in almost all organisms that have been examined. However, some minor differences in codon assignments are found in certain protozoa and in the genetic codes of organelles.

The standard code is shown in Table 10.3. Note that four codons—the three stop codons and the start codon—are signals. Altogether, 61 codons specify amino acids, and in many cases several codons direct the insertion of the same amino acid into a polypeptide chain. This feature confirms the inference from the *rII* frameshift mutations that the genetic code is redundant (degenerate). In a redundant genetic cod, some amino acids are encoded by two or more different codons. In the actual genetic code, all amino acids except tryptophan and methionine are specified by more than one codon. The redundancy is not random. For example, with the exception of serine, leucine, and arginine, all

Table 10.3 The standard genetic code

First position (5' end)	Second position: U	C	A	G	Third position (3' end)
	U	C	A	G	
U	UUU Phe F	UCU Ser S	UAU Tyr Y	UGU Cys C	U
	UUC Phe F	UCC Ser S	UAC Tyr Y	UGC Cys C	C
	UUA Leu L	UCA Ser S	UAA Stop	UGA Stop	A
	UUG Leu L	UCG Ser S	UAG Stop	UGG Trp W	G
C	CUU Leu L	CCU Pro P	CAU His H	CGU Arg R	U
	CUC Leu L	CCC Pro P	CAC His H	CGC Arg R	C
	CUA Leu L	CCA Pro P	CAA Gln Q	CGA Arg R	A
	CUG Leu L	CCG Pro P	CAG Gln Q	CGG Arg R	G
A	AUU Ile I	ACU Thr T	AAU Asn N	AGU Ser S	U
	AUC Ile I	ACC Thr T	AAC Asn N	AGC Ser S	C
	AUA Ile I	ACA Thr T	AAA Lys K	AGA Arg R	A
	$\boxed{\text{AUG Met}}$ M	ACG Thr T	AAG Lys K	AGG Arg R	G
G	GUU Val V	GCU Ala A	GAU Asp D	GGU Gly G	U
	GUC Val V	GCC Ala A	GAC Asp D	GGC Gly G	C
	GUA Val V	GCA Ala A	GAA Glu E	GGA Gly G	A
	GUG Val V	GCG Ala A	GAG Glu E	GGG Gly G	G

Note: Each amino acid is given its conventional abbreviation in both the single-letter and three-letter format. The codon AUG, which codes for methionine (boxed) is usually used for initiation. The codons are conventinally written with the 5' base on the left and the 3' base on the right.

Note: Each amino acid is given its conventional abbreviation in both the single-letter and three-letter format. The codon AUG, which codes for methionine (boxed) is usually used for initiation. The codons are conventinally written with the 5' base on the left and the 3' base on the right.

codons that correspond to the same amino acid are in the same box of Table 10.3; that is, *synonymous codons usually differ only in the third base.* For example, GGU, GGC, GGA, and GGG all code for glycine. Moreover, in all cases in which two codons code for the same amino acid, the third base is either A or G (both purines) or T or C (both pyrimidines).

The codon assignments shown in Table 10.3 are completely consistent with all chemical observations and with the amino acid sequences of wildtype and mutant proteins. In virtually every case in which a mutant protein differs by a single amino acid from the wildtype form, the amino acid substitution can be accounted for by a single base change between the codons corresponding to the two different amino acids. For example, substitution of glutamic acid by valine, which occurs in sickle-cell hemoglobin, results from a change from GAG to GUG in codon six of the β-globin mRNA.

Mutations that change the nucleotide sequence of a gene may differ in their consequences, and a special terminology is used to describe them. A **missens** mutation results in the replacement of one amino acid by another. For example, the change from GAG to GTG in the DNA of the gene for β-globin results in the replacement of glutamic acid by valine in the sickle-cell hemoglobin molecule; the mutation is therefore a missense mutation. In contrast, a **silent mutation** is one that does not change the amino acid sequence. Silent mutations often result from changes in the third codon position; for example, a mutation that changes an AAA codon into an AAG codon is silent because both codons specify lysine. A most interesting class of mutations consists of changes that convert a codon that specifies an amino acid into a chain-terminating codon. A mutation of this type is called a **nonsense mutation,** and it results in premature termination of the polypeptide chain. An example of a nonsense mutation is found in the β-globin gene, in which a mutation from AAG to TAG in the seventeenth codon results in a truncated polypeptide only 16 amino acids in length. This mutation is one of several types associated with the disease β-thalassemia.

Transfer RNA and Aminoacyl-tRNA Synthetase Enzymes

The decoding operation by which the base sequence within an mRNA molecule becomes translated into the amino acid sequence of a protein is accomplished by aminoacylated, or charged, tRNA molecules, each of which is linked to the correct amino acid by an aminoacyl-tRNA synthetase.

The tRNA molecules are small, single-stranded nucleic acids ranging in size from about 70 to 90 nucleotides. Like all RNA molecules, they have a 3'-OH terminus, but the opposite end terminates with a 5'- monophosphate rather than a 5'-triphosphate, because tRNA molecules are cut from a larger primary transcript. Internal complementary base sequences form short double-stranded regions, causing the molecule to fold into a structure in which open loops are connected to one another by double- stranded stems (Figure 10.31). In two dimensions, a tRNA molecule is drawn as a

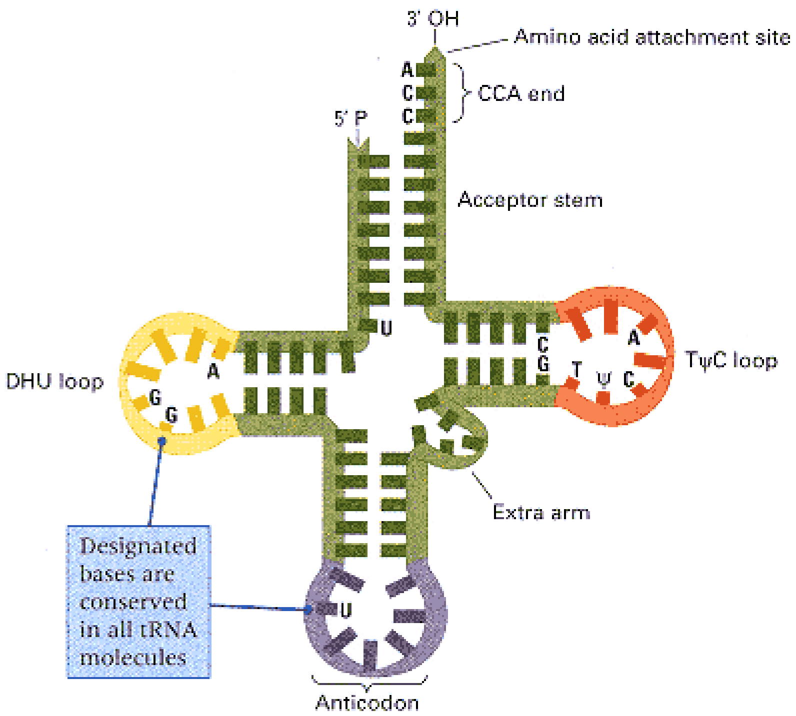


Figure 10.31
A tRNA cloverleaf configuration. The heavy black letters indicate a few bases that are conserved in the sequence of all tRNA molecules. The labeled loop regions are those found in all tRNA molecules.
DHU refers to a base, dihydrouracil, found in one loop; the Greek letter Ψ is a symbol for the unusual base pseudouridine.

planar cloverleaf. Its three-dimensional structure is more complex, as is shown in Figure 10.32, in which part A shows a skeletal model of a yeast tRNA molecule that carries phenylalanine and part B is an interpretive drawing. Note how the Tψ*C* loop and the DHU loop are in close proximity. When viewed from either side, the folded structure roughly resembles the diagrammatic representations of the tRNAs used in Figures 10.21, 10.23, and 10.24.

Particular regions of each tRNA molecule are used in the decoding operation. One region is the anticodon sequence, which consists of three bases that can form base pairs with a codon sequence in the mRNA. No normal tRNA molecule has an anticodon complementary to any of the stop codons UAG, UAA, and UGA, which is why these codons are stop signals. A second critical site is at the 3' terminus of the tRNA molecule, where the amino acid attaches. A specific aminoacyl-tRNA synthetase matches the amino acid with the anticodon. At least one, and usually only one, aminoacyl-synthetase exists for each amino acid. To make the correct attachment, the synthetase must be able to distinguish one tRNA molecule from another. The necessary distinction is provided by recognition regions that encompass many parts of the tRNA molecule.

Figure 10.33 shows the three-dimensional structure of the seryl-tRNA synthetase complexed with its tRNA. On binding with the tRNA, a part of the protein makes contact with the variable part of the Tψ*C* loop of the tRNA and guides the acceptor stem into the active pocket of the enzyme. These interactions depend primarily on recognition of the shape of the $tRNA^{Ser}$ through contacts with the backbone and only secondarily on interactions that are specific to the anticodon.

Redundancy and Wobble

Several features of the genetic code and of the decoding system suggest that something is missing in the explanation of codon-anticodon binding. First, the code is highly redundant. Second, the identity of the third base of a codon is often unimportant. In some cases, any nucleotide will do; examples include proline (Pro), threonine (Thr), and glycine (Gly). In other cases, either purine (A or G) or either pyrimidine (U or C) in the third position codes for the same amino acid; examples include histidine (His), glutamine (Gln), and tyrosine (Tyr).

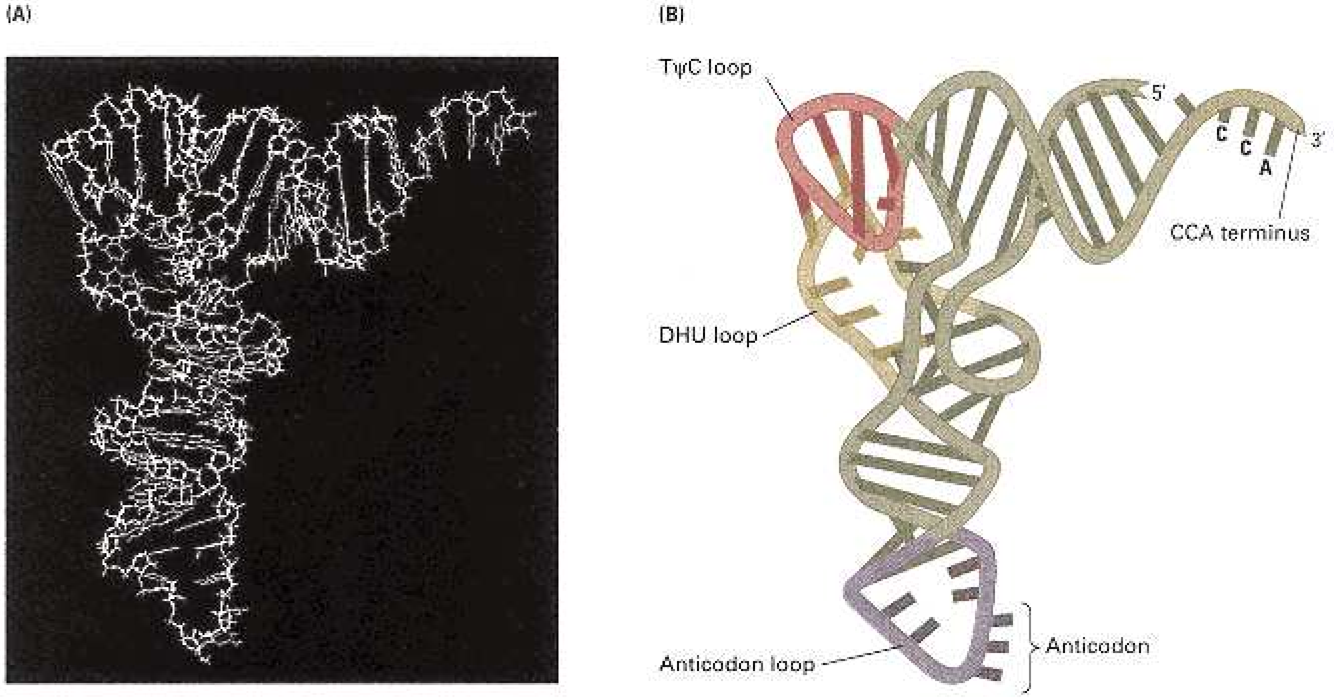


Figure 10.32
Yeast phenylalanine tRNA (called tRNA^{Phe}). (A) A skeletal model. (B) A schematic diagram of the three-dimensional structure of yeast tRNA^{Phe}.
[Courtesy of Sung-Hou Kim.]

Third, the number of distinct tRNA molecules that have been isolated from a single organism is less than the number of codons; because all codons are used, the anticodons of some tRNA molecules must be able to pair with more than one codon. Experiments with several purified tRNA molecules showed this to be the case.

To account for these observations, the **wobble** concept was advanced in 1966 by Francis Crick. He proposed that the first two bases in a codon form base pairs with

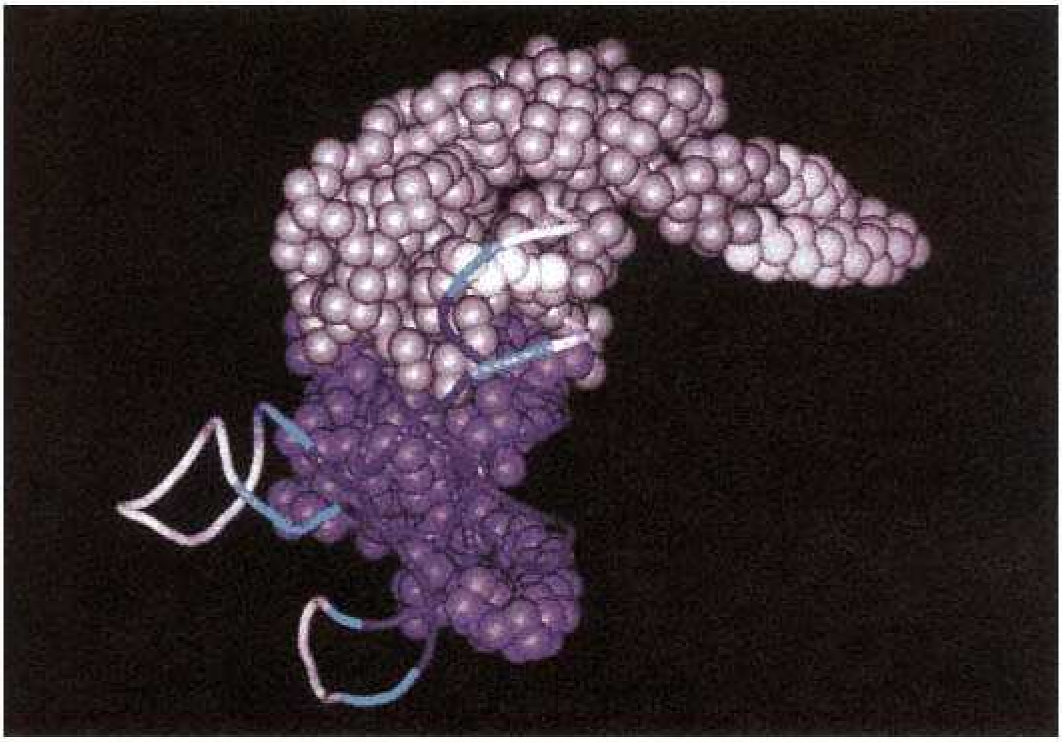


Figure 10.33
Three-dimensional structure of seryl-tRNA synthetase (solid spheres) complexed with its tRNA. Note that there are many points of contact between the enzyme and the tRNA. The molecules are from Thermus thermophilus.
[Courtesy of Stephen Cusack. From V. Biou, A. Yaremchuk, M.

Tukalo, and S. Cusack, 1996. Science 263:1404.]

the tRNA anticodon according to the usual rules (A-U and G-C) but that the base at the 5' end of the anticodon is less spatially constrained than the first two and can form hydrogen bonds with more than one base at the 3' end of the codon. He suggested the pairing rules given in Table 10.4. Evidence has confirmed the wobble concept and indicates that the pairings given in Table 10.4 are largely true for *E. coli*.

On the other hand, analysis of tRNAs in the yeast *Saccharomyces cerevisiae* has indicated that wobble is more restricted in yeast than in *E. coli*. Table 10.5 summarizes the wobble rules for *E. coli* and yeast. The yeast rules may hold for other eukaryotes as well. In yeast, single tRNAs can recognize the pairs of related codons ending in U or C. However, separate tRNAs are needed for codons that end in A or G. Thus at least three tRNAs are required for amino acids such as proline and glycine, which are specified by a set of four codons. A total of 46 tRNAs are needed to decode mRNA molecules in yeast. As noted, there are no tRNAs corresponding to the stop codons.

Nonsense Suppression

As described in Section 10.6, mutations can occur that result in premature chain termination during translation. For whimsical historical reasons, the stop codons are referred to as the amber (UAG), ochre (UAA), and UGA codons. A remarkable observation was that some phage mutants bearing nonsense mutations in the gene encoding the major head protein were able to propagate in some bacterial strains but not in others. How could this happen? On further analysis, it turned out that the strains able to support growth of the phage nonsense mutants carry suppressor mutations that act by changing the way the mRNA is read, not by changing the nucleotide sequence of the phage gene. The suppressor mutations proved to be mutant tRNA genes. These suppressor tRNA genes act by reading a stop codon as though it were a signal for a specific amino acid. The amino acid is inserted at that position and translation continues. So long as the inserted amino acid is compatible with the function of the protein, the effects of the original mutations are suppressed and plaques can be produced. In *E. Coli*,

Table 10.4 Allowed pairings due to wobble

First base in anticodon (5' position)	Allowed bases in third codon position (3' position)
A	U
C	G
U	A or G
G	C or U
I	A or C or U

"amber suppressors" recognize and suppress only amber mutations because the anticodon of the altered tRNA pairs only with the UAG codon; on the other hand, "ochre suppressors" suppress both ochre and amber mutations because the mutant tRNA can recognize and suppress both UAG and UAA. In yeast there is more specificity because of the more stringent wobble rules: Ochre suppressors suppress only ochre mutations, and amber suppressors suppress only amber mutations.

We can illustrate nonsense suppression by examining a chain-termination codon formed by mutation of the tyrosine codon UAC to the stop codon UAG (Figure 10.34A and B). Such a mutation can be suppressed by a mutant leucine tRNA molecule. In *E. coli,* $tRNA^{Leu}$ has the anticodon 3'-AAC-5', which pairs with the codon 5'-UUG-3'. A suppressor mutation in the $tRNA^{Leu}$ gene produces an altered tRNA

Table 10.5 Wobble rules for tRNAs of *E. coli* and *Saccharomyces cerevisiae*

Third position of codon	First position of anticodon	
	E. coli	**Yeast (*S. cerevisiae*)**
U	A, G, or I	G or I
C	G or I	G or I
A	U or I	U*

G	C or U	C

Note: I indicates inosine, which is structurally similar to adenosine except that the $-NH_2$ is replaced with -OH. U* indicates a modified uridine.

Source: Data from C. Guthrie and J. Abelson. 1982. *The Molecular Biology of the Yeast Saccharomyces: Metabolism and Gene Expression*, edited by J. N. Strathern, E. W. Jones, and J. R. Broach, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, p. 487.

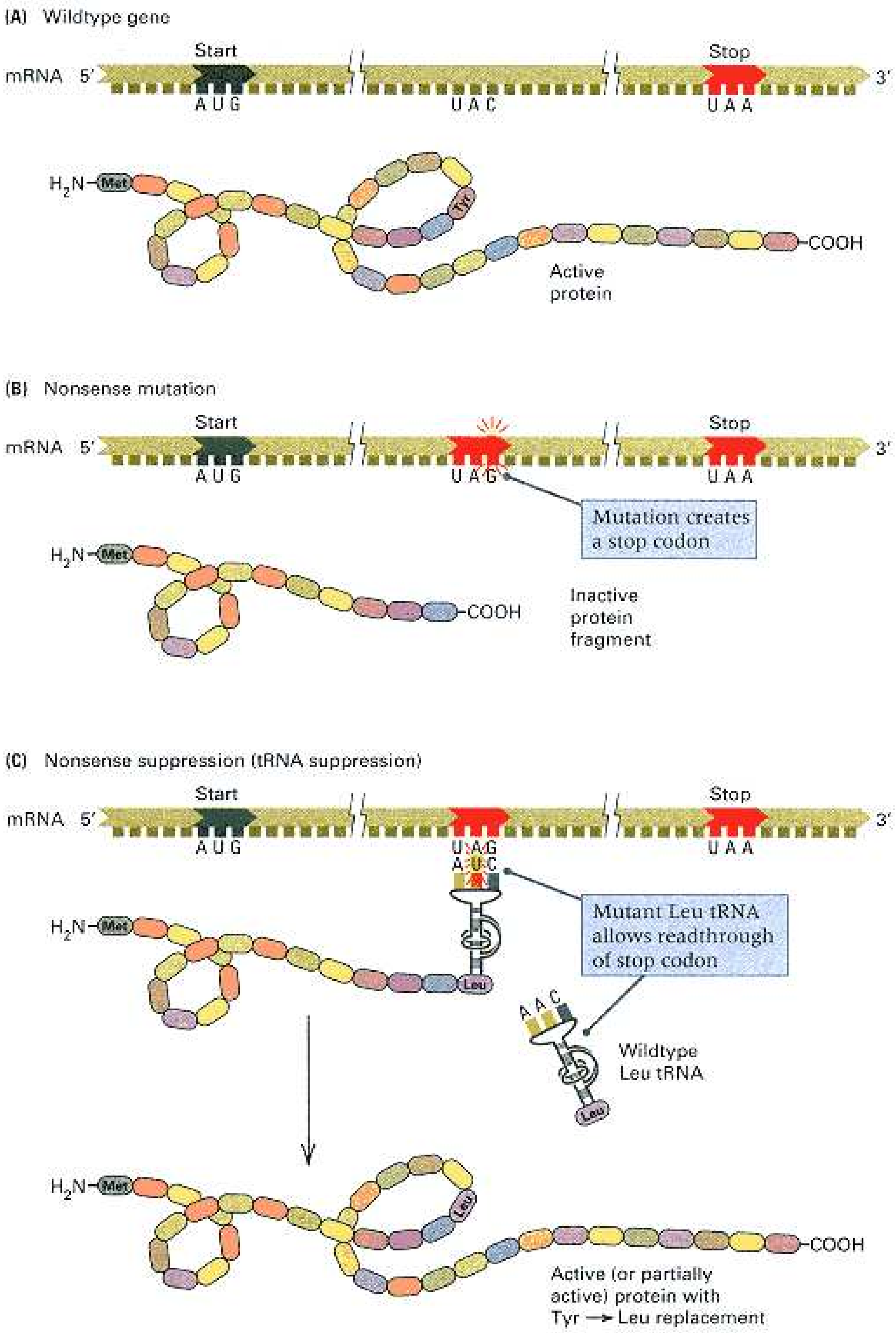


Figure 10.34
The mechanism of suppression by a suppressor tRNA molecule. (A) The wildtype gene. (B) A UAC $\rightarrow$ UAG chain-termination mutation leads to an inactive, prematurely terminated protein. (C) A mutation in the tRNA^{Leu} gene produces an altered tRNA molecule, which has a codon complementary to a UAG stop codon but can still be charged with leucine. This tRNA molecule allows the protein to be completed but with a leucine at the site of the original tyrosine. Suppression will be achieved if the substitution restores activity to the protein.

with the anticodon AUC; this tRNA molecule is still charged with leucine but responds to the stop codon UAG rather than to the normal leucine codon UUG. Thus, in a cell that contains this suppressor tRNA, the mutant protein is completed and suppression occurs as long as the mutant protein can tolerate a replacement of leucine for tyrosine (Figure 10.34C). Many suppressor tRNA molecules of this type have been observed. Each suppressor is effective against only some nonsense mutations, because the resulting amino acid replacement may not yield a functional protein.

In *E. coli*, there are three classes of tRNA suppressors: those that suppress only UAG (amber suppressors), those that suppress both UAA and UAG (ochre suppressors), and those that suppress only UGA. They share the following properties:

1. The original mutant gene still contains the mutant base sequence (UAG in Figure 10.34).

2. The suppressor tRNA suppresses all chain-termination mutations with the same stop codon, provided that the amino acid inserted is an acceptable amino acid at the site.

3. A cell can survive the presence of a tRNA suppressor *only* if the cell contains two or more copies of the same tRNA gene. Taking the example in Figure 10.34, if only one $tRNA^{Leu}$ gene were present in the genome and if it were mutated, then the normal leucine codon UUG would no longer be read as a sense codon, and all polypeptide chains would terminate wherever a UUG codon occurred. However, *multiple copies of most tRNA genes exist*, so if one copy is mutated to yield a suppressor tRNA, a normal copy nearly always remains.

4. Any chain-termination codon can be translated by a suppressor tRNA mutation that recognizes that codon. For example, translation of UAG by insertion of an amino acid would prevent termination of all wildtype mRNA reading frames terminating in UAG.

> However, the anticodon of the suppressor tRNA usually binds rather weakly to the stop codon, so the stop codon often results in termination anyway.

Suppressed tRNA mutations are very useful in genetic analysis because they allow nonsense mutations to be identified through their ability to be suppressed. This is important because nonsense alleles usually result in a truncated and completely inactive protein and so are considered true loss-of-function alleles. Suppressor tRNAs have been widely used for genetic analysis in prokaryotes, yeast, and even nematodes, because their other harmful effects are tolerated by the organism. In higher organisms, suppressor tRNAs have such severe harmful effects that they are of limited usefulness.

The Sequence Organization of a Typical Prokaryotic mRNA Molecule

Most prokaryotic mRNA molecules are **polycistronic,** which means that they contain sequences specifying the synthesis of several proteins (Figure 10.25). A polycistronic mRNA molecule must therefore posses a series of start and stop codons for use in translation. If an mRNA molecule encodes three proteins, then the minimal coding requirement would be the sequence

AUG (start)/protein 1/stop—

    AUG/protein 2/stop—AUG/protein 3/stop

The stop codons might be UAA, UAG, or UGA. Actually, such an mRA molecule is probably never so simple in that (1) the leader sequence preceding the first start signal may be several hundred bases long and (2) spacer sequences containing ribosome binding sites are usually present between one stop codon and the next start codon.

10.7—
Overlapping Genes

The idea that two or more reading frames might exist within a single coding segment of DNA was not considered for many years. The reason is that a mutation in a gene that overlaps another gene often produces defects in both gene products, but double mutations are very rare. Furthermore, the existence of overlapping reading frames was thought to place severe constraints on the amino acid sequences of two proteins translated from the same part of an mRNA molecule. However, because the code is highly redundant, the constraints are not so rigid.

If genes contained multiple reading frames, then a single DNA segment would be utilized with maximal efficiency. However, a disadvantage is that evolution might be more difficult because random mutations would rarely improve the function of both proteins. Nevertheless, some cases of **overlapping genes** have been found. Most examples are in transposable elements or in small viruses in which there is a premium on packing the largest amount of genetic information into a small DNA molecule. Some of the best examples of overlapping genes occur in the *E. coli* phage ϕX174.

hage ϕX174 contains a single strand of DNA consisting of 5386 nucleotides. If a single reading frame were used, at most 1795 amino acids could be encoded in the sequence, and with an average protein size of about 400 amino acids, only 4 or 5 proteins could be made. However, ϕX174 makes 11 proteins containing a total of more than 2300 amino acids. This paradox was resolved when it was shown that translation occurs in several reading frames from three mRNA molecules (Figure 10.35). For example, the sequence for protein B is contained totally in the sequence for protein A' but is translated in a different reading frame; similarly, the sequence for protein E is included within the sequence for protein D. Protein K is initiated near the end of gene *A'*, includes the base sequence of gene *B,* and terminates in gene *C;* synthesis is not in phase with either gene *A'* or gene *C.* Of note is protein A', which is formed by initiating translation within the mRNA for protein A using the same reading frame, so that it terminates at the same stop codon as protein A. Thus, the amino acid sequence of A' is identical with a segment of protein A. In total, six different proteins obtain some or all of their primary structure from shared base sequences in ϕX174.

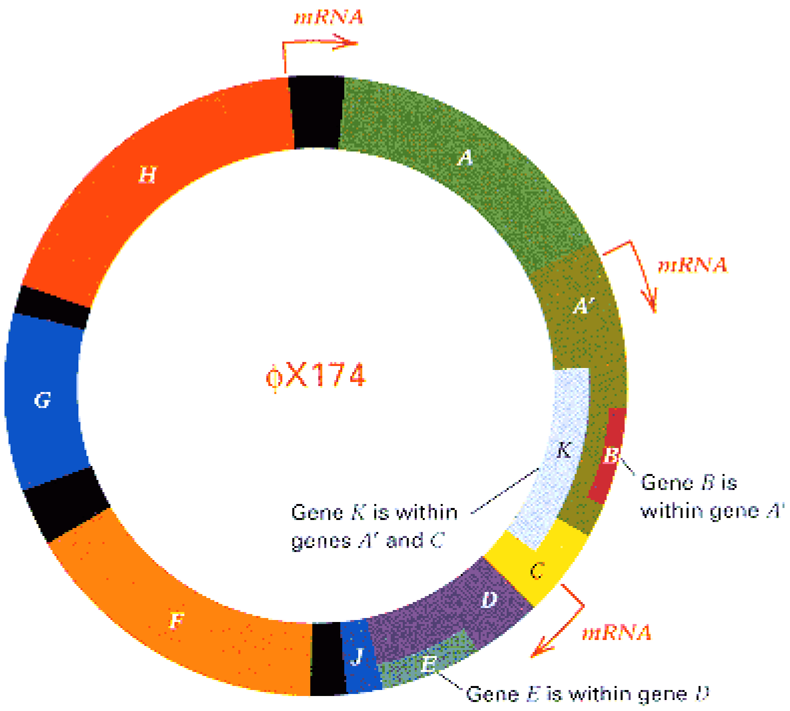


Figure 10.35
Physical map of *E. coli* phage ϕX174. The red arrows show the start points for synthesis of the major mRNA transcripts, and the uppercase letters indicate the regions from which different protein products are translated. The solid black regions are untranslated spacers.

10.8—
Complex Translation Units

In most prokaryotes and eukaryotes, the unit of translation is almost never simply one ribosome traversing an mRNA molecule. Rather, it is a more complex structure, of which there are several forms. Two examples are given in this section.

After about 25 amino acids have been joined together in a polypeptide chain, an AUG initiation codon is completely free of the ribosome, and a second initiation complex can form. The overall configuration is that of two ribosomes moving along the mRNA at the same speed. When the second ribosome has moved along a distance similar to that traversed by the first, a third ribosome can attach to the initiation site. The process of movement and reinitiation continues until the mRNA is covered with ribosomes at a density of about one ribosome per 80 nucleotides. This large translation unit is called a **polysome,** and this is the usual form of the translation unit. An electron micrograph of a polysome is shown in Figure 10.36.

An mRNA molecule being synthesized has a free 5' terminus. Because translation takes place in the 5'-to-3' direction, the mRNA is synthesized in a direction appropriate for immediate translation. That is, the ribosome-binding site (in prokaryotes) and the 5' terminus (in eukaryotes) is transcribed first, followed in order by the initiating AUG codon, the region encoding the amino acid sequence, and finally the stop codon. Thus in prokaryotes, in which no nuclear envelope separates the DNA and the ribosome, the initiation complex can form before the mRNA is released from the DNA. This allows the simultaneous occurrence, or **coupling,** of transcription and translation. Figure 10.37 shows an electron micrograph of a DNA molecule with a number of attached mRNA molecules, each associated with ribosomes (Figure 10.37A), and an interpretation (Figure 10.37B). Transcription of DNA is beginning in the upper left part of the micrograph. The lengths of the polysomes increase with distance from the transcription initiation site, because the mRNA is farther from that site and hence of greater length because the process of transcription has been going on for a longer time. *Coupled transcription and translation does not take place in eukaryotes,* because the mRNA is synthesized and processed in the nucleus and later transported through the nuclear envelope to the cytoplasm where the ribosomes are located.

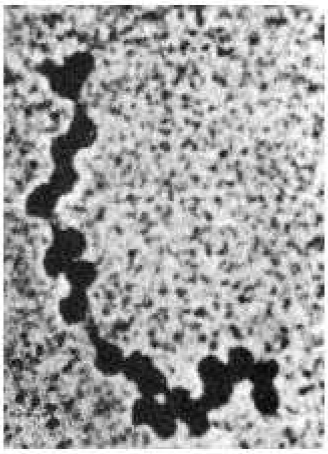


Figure 10.36
Electron micrograph of *E. coli* polysomes.
[Courtesy of Barbara Hamkalo.]

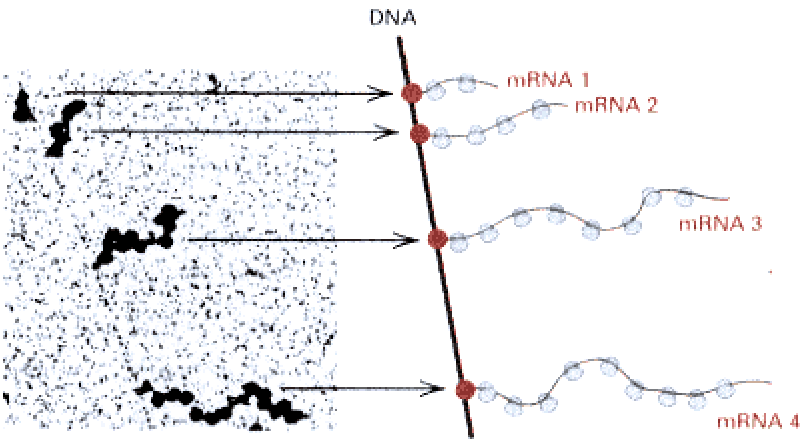


Figure 10.37
Visualization of transcription and translation. The photograph shows transcription of a section of the DNA of *E. coli* and translation of the nascent mRA. The dark spots are ribosomes, which coat the mRNA. An interpretation of the electron micrograph is at the right. Each mRNA has ribosomes attached along its length. The large red dots are the RNA polymerase molecules; they are too small to be seen in the photo. The length of each mRNA is equal to the distance that each RNA polymerase has progressed from the transcription-initiation site.
[Electron micrograph courtesy of O. L. Miller, B. A. Hamkalo, and C. A. Thomas. 1977. *Science* 169: 392.]

10.9—
The Overall Process of Gene Expression

In this chapter, the main features of the process of gene expression have been described. The mechanisms of gene expressions are complex. Nonetheless, the basic process is a simple one:

> A base sequence in a DNA molecule is converted into a complementary base sequence in an intermediate molecule (mRNA), and then the base sequence in the mRNA is converted into an amino acid sequence of a polypeptide chain using tRNA molecules, each charged with the correct amino acid.

Both of these steps, which have a multitude of substeps, utilize the simplest of principles: (1) The rules of base pairing provide the base sequence of the mRNA, and (2) a two-ended molecule (tRNA), with an amino acid attached at one end and able to base-pair with RNA bases at the other, translates each set of three bases into one amino acid. Various recognition regions are needed to ensure that the correct base sequence is read and that the correct amino acid is put in the appropriate position in the protein. As is always the case when the information in nucleic acid molecules is used, base sequences provide the information for the first process. That is, specific sequences in the DNA are recognized as the beginning (promoter) and end (transcription-termination site) of a gene, and these sequences are recognized by an enzyme (RNA polymerase) that makes the copy of the gene that is used by the protein-synthesizing machinery. To ensure that the correct amino acid sequence is assembled, one codon (AUG) is used to tell the system where to start reading, and a stop codon defines the end of the polypeptide chain. Particular recognition sites in tRNA enable the aminoacyl tRNA synthetase enzymes to connect amino acids to the tRNA molecules with the correct anticodons.

An essential feature of the entire process of gene expression is that both DNA and RNA are scanned by molecules that move in a single direction. That is, RNA polymerase moves along the DNA as it polymerizes nucleotides, and the ribosome and the mRNA move with respect to one another as different amino acids are brought in for covalent linking.

Chapter Summary

The flow of information from a gene to it product is from DNA to RNA to protein. The properties of the different protein products of genes are determined by the sequence of amino acids of the polypeptide chain and by the way in which the chain is folded. Each gene is usually responsible for the synthesis of a single polypeptide.

Gene expression begins with the enzymatic synthesis of an RNA molecule that is a copy of one strand of the DNA segment corresponding to the gene. This process is called transcription and is carried out by the enzyme RNA polymerase. This enzyme joins ribonucleoside triphosphates by the same chemical reaction used in DNA synthesis. RNA polymerase differs from DNA polymerase in that a primer is not needed to initiate synthesis. Transcription is initiated when RNA polymerase binds to a promoter sequence. Each promoter consists of several subregions, of which two are the polymerase binding site and the polymerization start site. Polymerization continues until a termination site is reached. The product of transcription is an RNA molecule. In prokaryotes, this molecule is used directly as messenger RNA (mRNA) in polypeptide synthesis. In eukaryotes, the RNA is processed: Noncoding sequences called introns are removed, the exons are spliced together, and the termini are modified by formation of a 5' cap and usually by addition of a poly-A tail at the 3' end.

After mRNA is formed, polypeptide chains are synthesized by translation of the mRNA molecule. Translation is the successive reading of the base sequence of an mRNA molecule in groups of 3 bases called codons. There are 64 codons; 61 correspond to the 20 amino acids, of which 1 (AUG) is a start codon. The remaining 3 codons (UAA, UAG, and UGA) are stop codons. The code is highly redundant: Many amino acids have several codons. The codons in mRNA are recognized by tRNA molecules, which contain a 3-base sequence complementary to a codon and called an anticodon. When used in polypeptide synthesis, each tRNA molecule possesses a terminally bound amino acid (aminoacylated, or charged, tRNA). The correct amino acid is attached to each tRNA species by specific enzymes called aminoacyl-tRNA synthetases.

Polypeptides are synthesized on particles called ribosomes. Synthesis begins with the formation of a 30S ribosomal subunit + charged tRNA + mRNA complex, which recruits a 50S subunit to complete the mature 70S ribosome. (In eukaryotes, the 40S and 60S subunits come together to from the 80S ribosome.) Next, charged tRNA molecules are successively brought to the A site on the 50S ribosome by elongation factor EF-Tu (EF- 1 α in eukaryotes). These are hydrogen-bonded to the mRNA in the 30S subunit by a codon-anticodon interaction, and the 50S subunit is shifted to the pretranslocation state. As each charged tRNA is brought aboard, its amino acid is attached by a peptide bond to the growing polypeptide chain. Translocation of the 30S ribosome one codon farther along the tRNA is the function of elongation factor EF-G (EF-2 in eukaryotes), converting the ribosome to the posttranslation state and shifting the uncharged tRNA to the E site and the polypetidyl tRNA (the one carrying the incomplete polypeptide) to the P site.

The elongation process continues until a stop codon in the mRNA is reached. No tRNAs for the stop codons exist. Instead, specific release factors cleave the complete polypeptide from the last polypetidly tRNA and free the ribosome components for reuse in translation.

Several ribosomes can translate an mRNA molecule simultaneously, forming a polysome. In prokaryotes, translation often begins before synthesis of mRNA is completed; in eukaryotes, this does not occur because mRNA is made in the nucleus, whereas the ribosomes are located in the cytoplasm. Prokaryotic mRNA molecules are often polycistronic, encoding several different polypeptides. Translation proceeds sequentially along the mRNA molecule from the start codon nearest the ribosome-binding site, terminating at stop codons and reinitiating at the next start codon. This is not possible in eukaryotes, because only the AUG site nearest the 5' terminus of the mRNA can be used to initiate polypeptide synthesis; thus eukaryotic mRNA is monocistronic.

Key Terms

amino terminus

aminoacyl-tRNA synthetase

aminoacylated-tRNA

anticodon

AUG

cap

carboxyl terminus

chain elongation

chain initiation

chain termination

chaperone

charged tRNA

coding sequece

codon

colinearity

consensus sequence

coordinate regulation

coupled transcription-translation

cryptic splice site

degenerate code

exon

exon shuffle

folding domain

gene product

gentic code

inosine (I)

intervening sequence

intron

lariat structure

leader

messenger RNA (mRNA)

missense mutation

monocistronic mRNA

mRNA

nonsense mutation

open reading frame (ORF)

overlapping genes

peptide bond

peptidyl transferase

poly-A tail

polycistronic mRNA

polypeptide chain

polysome

pretranslocation state

posttranslocation state

primary transcript

R group

reading frame

ribosome

ribosome-binding site

ribozyme

RNA polymerase

RNA processing

RNA splicing

silent mutation

splice acceptor

splice donor

spliceosome

start codon

stop codon

TATA box

template strand

transcription

transfer RNA (tRNA)

translation

translocation

triplet code

tRNA

uncharged tRNA]

frameshift mutation	promoter	wobble
gene expression	protein subunit	

Review the Basics

• Is the DNA strand that serves as the template for RNA polymerase transcribed in the 5&inch; → or the 3&inch; → direction? Which end of the mRNA molecule is translated first? Which end of the polypeptide encoded in the mRNA is synthesized first?

• What is the difference between the reaction catalyzed by DNA polymerase and that catalyzed by RNA polymerase?

• What are the principal characteristics of the standard genetic code?

• What is a primary transcript and how does a primary transcript differ from mRNA in prokaryotes? In eukaryotes?

• Give an example of overlapping genes. Why do you suppose overlapping genes are unusual except in certain phages and viruses in which there is a premium on small genome size?

• What are the roles of the different types of RNA molecules that are necessary for protein synthesis?

• How do prokaryotes and eukaryotes differ in the mechanism for selecting an AUG codon as a start for polypeptide synthesis?

• What is the consequence when an incorrect nucleotide is inserted into the new DNA strand during replication if it is not corrected by the proofreading function of DNA polymerase or other repair mechanisms prior to the next replication? What is the consequence when an incorrect nucleotide is inserted into an RNA molecule during transcription?

• What is a polysome and what is its role in polypeptide synthesis?

• Which of the following is the mechanism by which polypeptide chain termination takes place: (1) mRNA synthesis stops at a chain-termination codon; (2) the tRNA corresponding to a chain-termination codon cannot be charged with an amino acid; (3) chain-termination codons have no tRNA molecules that bind with them, but they interact with specific release-factor proteins instead.

Guide to Problem Solving

Problem 1: The following is the nucleotide sequence of a strand of DNA.

```
TACGTCTCCAGCGGAGATCTTTTCCGGTCGCAACTGAGGTTGATC
```

The strand is transcribed from left to right and codes for a small peptide.

(a) Which end is the 3' end and which the 5' end?

(b) What is the sequence of the complementary DNA strand?

(c) What is the sequence of the transcript?

(d) What is the amino acid sequence of the peptide?

Chapter 10 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

http://www.jbpub.com/genetics

Select the link to ***Genetics: Principles and Analysis*** and then choose the link to ***GeNETics on the web.*** You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. The **ribosomal RNA** genes of *E. coli* are clustered together in seven transcriptional units, each called an *operon.* The RNA coding genes are greater than 99 percent identical from one operon to the next. Each 23S ribosomal RNA (2904 nucleotides in length) is encoded in an *rrl* gene; each 5S ribosomal RNA (120 nucleotides in length) is encoded in an *rrf* gene. One 23S and one 5S molecule are included in each 50S (large) ribosomal subunit. The 30S (small) ribosomal subunit contains one molecule of 16S RNA (1542 nucleotides in length), which is encoded in any of the *rrs* genes. Search at the keyword site for Name *rrn* and Type *operon.* Follow the links to learn the map position of each ribosomal RNA operon and the direction of transcription. You will see that some other genes are also included in the ribosomal RNA operons. What are these genes? Does their inclusion make any sense from the standpoint of translation? If assigned to do so, draw a map of the *E. coli* chromosome showing the name and location of each rRNA operon and the direction of transcription.

2. More about the various forms of **RNA polymerase,** including three-dimensional structural representations, can be found at this site. What common shape is found in the RNA polymerase holo- enzyme from *E. coli* and PolII from yeast? If assigned to do so, write one paragraph describing the difference in subunit composition between the *E. coli* holoenzyme and the core enzyme.

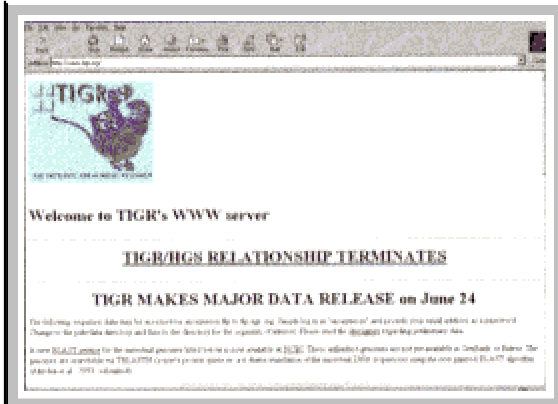
3. The gene *CCA1* in yeast is critical for formation of a mature **transfer RNA** molecule ready for charging with the correct amino acid. Search this keyword site for the gene *CCA1* to find out what it does. If assigned to do so, diagram an immature tRNA showing what reaction is catalyzed by the *cca1* gene product. Locate the gene on the genetic map and retrieve its DNA and amino acid sequence.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 10, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 10.



(e) In which direction along the transcript does translation occur?

(f) Which is the amino (-NH_2) and which the carboxyl (-COOH) end of the peptide?

Answer: Because the DNA strand is transcribed from left to right, and the first nucleotides incorporated into an RNA transcript form its 5' end, the 3' end of the DNA strand must be at the left. The original strand (answer to problem 1 (a), the complementary strand (answer to problem 1 (b)), the transcript (answer to problem 1(c)), and the peptide (answer to problem 1(d)) are shown in the accompanying table.

(e) Translation goes from 5'-to-3' along the mRNA, so in this example, translation also goes from left to right.

(f) The amino end of the peptide is synthesized first (Met) and the carboxyl end last (Asn).

Problem 2: Using the sequence given in Problem 1, determine what effects the following mutations would have on the peptide produced. Each of the mutations affects the TTTT in the middle of the strand.

(a) TTTT $\rightarrow$ TCTT (substitution of C for T)

(b) TTTT $\rightarrow$ TATT (substitution of A for T)

(c) `TTTT` → `TTTTT` (one-base insertion)

(d) `TTTT` → `TTT` (one-base deletion)

Answer: In problems of this type, it is best to derive the sequence of the mutant mRNA and then the amino acid sequence of the peptide. The sequence alignments are shown in the table below.

(a) The mutation changes a GAA codon into a GAG codon, which still codes for glutamic acid, so no change in the peptide results.

(b) The mutation changes a GAA codon into a GAU codon, which changes glutamic acid into aspartic acid in the peptide.

(c) The insertion introduces a new nucleotide into the mRNA and shifts the reading frame downstream from the site of the mutation. In this case, the normal sequence starting with Lys is replaced with Lys-Gly-Gln-Arg, and the UGA stop codon following the Arg codon results in premature termination of the peptide.

(d) The deletion again shifts the reading frame downstream from the site of the mutation. In this case, translation continues because a stop codon is not reached immediately, but the entire amino acid sequence downstream from the mutation is altered.

Problem 3: What anticodon sequence would pair with the codon 5'-AUG-3', assuming only Watson-Crick base pairing?

Answer: With Watson-Crick base pairing only, the base pairing is the usual A with U and G with C. However, the orientations of the codon and anticodon are antiparallel, so the anticodon sequence is 3'-UAC-5'.

Problem 4: What amino acids can be present at the site of a UAA codon that is suppressed by a suppressor tRNA created by a mutant base in the anticodon, assuming only Watson-Crick base pairing?

Answer: The nonmutant tRNA must be able to pair with the codon at two sites, and the mutant base in the anticodon allows the third site to pair as well. Therefore, the amino acids that can be inserted into the suppressed site are those whose codons differ from UAA in a single base. These codons are AAA (Lys), CAA (Gln), GAA (Glu), UUA (Leu), UCA (Ser), UAC (Tyr), and UAU (Tyr).

Answer to Problem 1(a)—1(d)

(a)	`3'-TACGTCTCCAGCGGAGACCTTTTCCGGTCGCAACTGAGGTTGATC-5'`
(b)	`5'-ATGCAGAGGTCGCCTCTGGAAAAGGCCAGCGTTGACTCCAACUAG-3'`
(c)	`5'-AUGCAGAGGUCGCCUCUGGAAAAGGCCAGCGUUGACUCCAACUAG-3'`
(D)	`MetGlnArgSerproLeuGluLysAlaSerValAspSerAsn`

Sequence Alignments for Answer to Problem 2(a)-2(d)

(a)	DNA	`3'-TACGTCTCCAGCGGACCTCTTCCGGTCGCAACTGAGGTTGATC-5'`
	mRNA	`5'-AUGCAGAGGUCGCCUCUGGAGAAGGCCAGCGUUGACUCCAACUAG-3'`
	Peptide	`MetGlnArgSerproLeuGluLysAlaSerValAsSerAsn`

(b)	DNA	3'-TACGTCTCCAGCGGAGACCTATTCCGGTCGCAACTGAGGTTGATC-5'
	mRNA	5'AUGCAGAGGUCGCCUCUGGAUAAGGCCAGCGUUGACUCCAACUAG-3'
	Peptide	MetGlnArgSerproLeuAspLysAlaSerValAspSerAsn
(c)	DNA	3'-TACGTCTCCAGCGGAGACCTTTTCCGGTCGCAACTGAGGTTGATC-5'
	mRNA	5'-AUGCAGAGGUCGCCUCUGGAAAAGGCCAGCGUUGACUCCAACUAG-3'
	Peptide	metGlnArgSerproLeuGluLysGlyGlnArg
(d)	DNA	3-TACGTCTCCAGCGGAGACCTTTCCGGTCGCAACTGAGGTTGATC-5'
	mRNA	5-AUGCAGAGGUCGCCUCUGGAAAGGCCAGCGUUGACUCCACUAG-3'
	Peptide	MetGlnArgSerproLeuGluArgProAlaLeyThrProThr

Analysis and Applications

10.1 What are the translation initiation and stop codons in the genetic code? In a random sequence of four ribonucleotides, what is the probability that any three adjacent nucleotides will be a start codon? A stop codon? In an mRNA molecule of random sequence, what is the average distance between stop codons?

10.2 A part of the coding strand of a DNA molecule that codes for the 5' end of an mRNA has the sequence 3'-TTTTACGGGAATTAGAGTCGCAGGATG-5'. What is the amino acid sequence of the polypeptide encoded by this region, assuming that the normal start codon is needed for initiation of polypeptide synthesis?

10.3 Poly-U codes for polyphenylalanine. If a G is added to the 5' end of the molecule, the polyphenylalanine has a different amino acid at the amino terminus, and if a G is added to the 3' end, there is a different amino acid at the carboxyl terminus. What are the amino acids?

10.4 The synthetic polymer, poly-A, is used as an mRNA molecule in an *in vitro* protein-synthesizing system that does not need a special start codon. Polylysine is synthesized. A single guanine nucleotide is added to one end of the poly-A. The resulting polylysine has a glutamic acid at the amino terminus. Was the G added to the 3' or the 5' end of the poly-A?

10.5 What polypeptide products are made when the alternating polymer GUGU . . . is used in an *in vitro* protein-synthesizing system that does not need a start codon?

10.6 What polypeptide products are made when the alternating polymer GUCGUC . . . is used in an *in vitro* protein-synthesizing system that does not need a start codon?

10.7 Some codons in the genetic code were determined experimentally by the translation of random polymers. If a ribonucleotide polymer is synthesized that contains 3 4 A and 1 4 C in random order, which amino acids would the resulting polypeptide contain, and in what frequencies?

10.8 How many different sequences of nine ribonucleotides would code for the amino acids Met—His—Thr? For Met—Arg—Thr? Using the symbol Y for any pyrimidine, R for any purine, and N for any nucleotide, what are the sequences?

10.9 At one time, it was considered that the genetic code might be one in which the codons overlapped. For example, with a two-base overlap, the codons in the mRNA sequence CAUCAU would be translated as CAU AUC UCA CAU rather than as CAU CAU. How is this hypothesis affected by the observation that mutant proteins usually differ from the wildtype protein by a single amino acid?

10.10 What codons could pair with the anticodon 5'-IAU-3'? (I stands for inosine.) What amino acid would be incorporated?

10.11 Two possible anticodons could pair with the codon UGG, but only one is actually used. Identify the possible anticodons, and explain why one of them is not used.

10.12 Two *E. coli* genes, *A* and *B*, are known from mapping experiments to be very close to each other. A deletion mutation is isolated that eliminates the activity of both *A* and *B* Neither the A nor the B protein can be found in the mutant, but a novel protein is isolated in which the amino-terminal 30 amino acids are identical to those of the *B* gene product and the carboxyl-terminal 30 amino acids are identical to those of the *A* gene product.

(a) With regard to the 5'-to-3' orientation of the nontranscribed DNA strand, is the order of the genes *A B* or *B A*?

(b) Can you make any inference about the number of bases deleted in the coding regions?

10.13 The nontranscribed sequence at the beginning of a gene reads

```
5'-ATGCATCCGGGCTCATTAGTCT . . .-3'
```

Two mutations are studied. Mutation X has an insertion of a G immediately after the underlined G, and mutation Y has a deletion of the red A. What is the amino acid sequence of each of the following?

(a) the wildtype polypeptide

(b) the polypeptide in mutant X

(c) the polypeptide in mutam Y

(d) the polypeptide in a recombinant organism containing both mutations

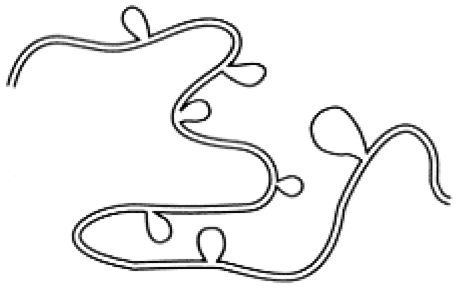
10.14 The amino terminus of a wildtype enzyme in yeast has the amino acid sequence

Met—Leu—His—Tyr—Met—Gly—Asp—Tyr—Pro

A mutant, X, is found that contains an inactive enzyme with the sequence Met—Gly—Asp—Tyr—Pro at the amino terminus and the wildtype sequence at the carboxyl terminus. A second mutant, Y, also lacks enzyme activity, but there is no trace of a full-length protein. Instead, mutant Y makes a short peptide containing just three amino acids. What single-base changes can account for the features of mutation X and mutation Y? What is the sequence of the tripeptide produced by mutant Y?

10.15 Protein synthesis occurs with high fidelity. In prokaryotes, incorrect amino acids are inserted at the rate of approximately 10^{-3} (that is, one incorrect amino acid per 1000 translated). What is the probability that a polypeptide of 300 amino acids has exactly the amino acid sequence specified in the mRNA?

10.16 A DNA fragment containing a particular gene is isolated from a eukaryotic organism. This DNA fragment is mixed with the corresponding mRNA isolated from the organism, denatured, renatured, and observed by electron microscopy. Heteroduplexes of the type shown in the accompanying figure are observed. How many introns does this gene contain?



10.17 If the DNA molecule shown here is transcribed from left to right, what are the sequence of the mRNA and the amino acid sequence? What are the sequence of the mRNA and the amino acid sequence if the segment in red is inverted?

```
5'-AGACTTCAGGCTCAACGTGGT-3'
3'-TCTGAAGTCCGAGTTGCACCA-5'
```

10.18 You are given the nontemplate-strand nucleotide sequence of a part of an exon of an active gene.

```
5'-TAACGTATGCTTGACCTCCAAGCAATCGATGCCAGCTCAAGG-3'
```

Assuming the standard genetic code, what is the amino acid sequence in the polypeptide chain? What tells you that you have identified the correct reading frame?

Challenge Problems

10.19 In performing an evolutionary analysis, biologists often consider the 6-fold degenerate serine (Ser) as two separate amino acids—a 4-fold and a 2-fold degenerate class—even though the amino acid is the same. Taking into account what you know about translation and the genetic code, why does it make sense to do this for serine but not for other 6-fold degenerate amino acids?

10.20 For two different frameshift mutations in the second codon of a gene, the amino terminal sequences of the mutant proteins are

Mutant 1: Met—Lys—UAG

Mutant 2: Met—Ile—Val—UAA

Mutant 1 has a single-nucleotide addition, and mutant 2 has a single-nucleotide deletion. Furthermore, the first five amino acids of the wildtype protein are known to be Met- (Asn, Val, Ser, Lys), where the parentheses mean that the order of the amino acids is unknown. Using the information provided by the frameshift mutations, determine the first five codons in the wildtype gene as well as the nature of each frameshift mutation.

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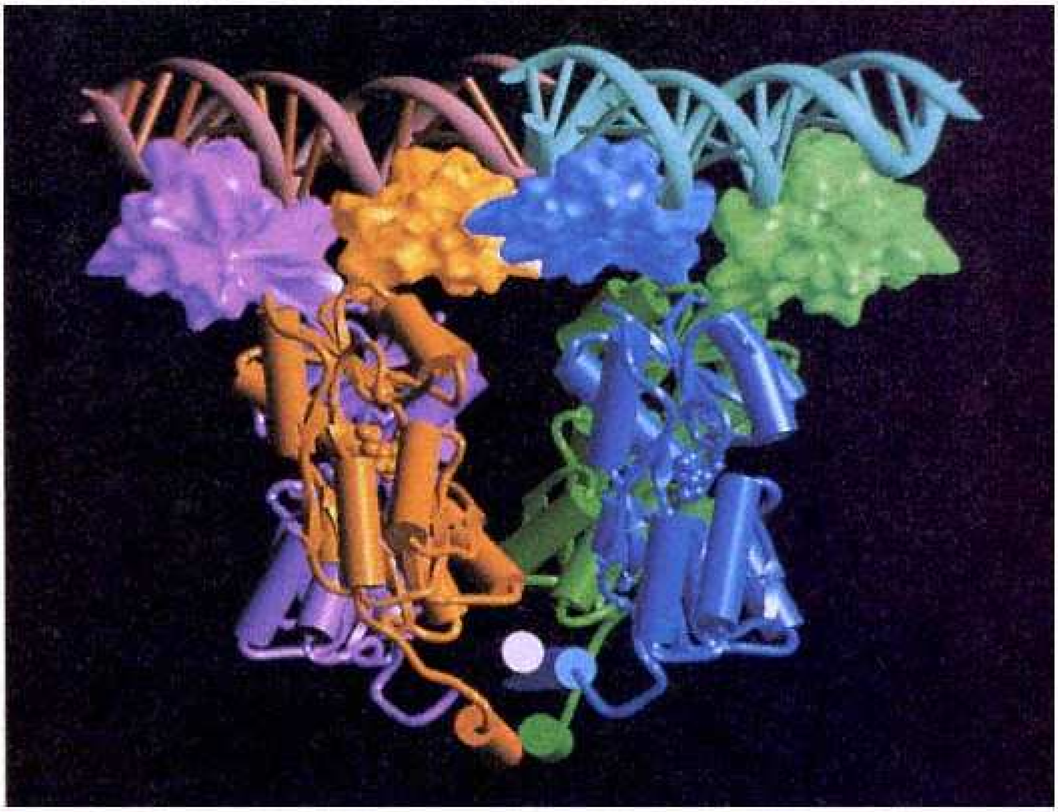

In E. coli, production of the enzymes necessary for growth on the sugar lactose is controlled by a transcriptional repressor protein, composed of four identical polypeptide subunits which interacts with DNA sequences comprising the lac operator. This model shows how the operator sequences (red and blue double helices running across the top) are contracted by the repressor subunit shown below.
[Courtesy of Thomas A. Steitz.]

Chapter 11— Regulation of Gene Activity

CHAPTER OUTLINE

11-1 Transcriptional Regulation in Prokaryotes

11-2 Lactose Metabolism and the Operon

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Inducible and Constitutive Synthesis and Repression

The Repressor

The Operator Region

The Promoter Region

The Operon Model of Transcriptional Regulation

Positive Regulation of the Lactose Operon

11-3 Regulation of the Tryptophan Operon Attenuation

11-4 Regulation in Bacteriophage λ

11-5 Regulation in Eukaryotes

Differences in Genetic Organization of Prokaryotes and Eukaryotes

11-6 Alteration of DNA

Gene Dosage and Gene Amplification

Programmed DNA Rearrangements

Antibodies and Antibody Variability

Gene Splicing in the Origin of T-Cell Receptors

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11-7 Transcriptional Regulation in Eukaryotes

Galactose Metabolism in Yeast

Yeast Mating Type

Transcriptional Activator Proteins

Hormonal Regulation

Transcriptional Enhancers

The Logic of Combinatorial Control

Enhancer-Trap Mutagenesis

Alternative Promoters

PRINCIPLES

• Genes can be regulated at any level, including transcription, RNA processing, translation, and post-translation.

• Control of transcription is an important mechanism of gene regulation.

• Transcriptional control can be negative ("on unless turned off") or positive ("off unless turned on"); many genes include regulatory regions for both types of regulation.

• Most genes have multiple, overlapping regulatory mechanisms that operate at more than one level, from transcription through post-translation.

• In prokaryotes, the genes coding for the enzymes in a metabolic pathway are often clustered in the genome and controlled jointly by a regulatory protein that binds with an "operator" region at the 5' end of the cluster. This type of gene organization is known as an operon.

• In eukaryotes, genes are not organized into operons. Genes at dispersed locations in the genome are coordinately controlled by one or more "enhancer" DNA sequences located near each gene that interact with transcriptional activator proteins that enable transcription of each nearby gene to occur.

CONNECTIONS

CONNECTION: Operator? Operator?

François Jacob, David Perrin, Carmen Sanchez, and Jacques Monod 1960
The operon: A group of genes whose expression is coordinated by an operator

CONNECTION: Sex-Change Operations

James B. Hicks, Jeffrey N. Strathern, and Ira Herskowitz 1977
The cassette model of mating-type interconversion

Not all genes are expressed continuously. The level of gene expression may differ from one cell type to the next or according to stage in the cell cycle. For example, the genes for hemoglobin are expressed at high levels only in precursors of the red blood cells. The activity of genes varies according to the functions of the cell. A vertebrate animal, such as a mouse, contains approximately 200 different types of cells with specialized functions. With minor exceptions, all cell types contain the same genetic complement. The cell types differ only in which genes are active. In general, the synthesis of particular gene products is controlled by mechanisms collectively called **gene regulation.**

In many cases, gene activity is regulated at the level of transcription, either through signals originating within the cell itself or in response to external conditions. For example, many gene products are needed only on occasion, and transcription can be regulated in an on-off manner that enables such products to be present only when external conditions demand. However, the flow of genetic information is regulated in other ways also. Control points for gene expression include the following:

1. *DNA rearrangements,* in which gene expression changes depending on the position of DNA sequences in the genome.

2. *Transcriptional regulation* of the synthesis of RNA transcripts by controlling initiation or termination.

3. *RNA processing,* or regulation through RNA splicing or alternative patterns of splicing.

4. *Translational control* of polypeptide synthesis.

5. *Stability of mRNA,* because mRNAs that persist in the cell have longer-lasting effects than those that are degraded rapidly.

6. *Post-translational control,* which includes a great variety of mechanisms that affect enzyme activity, activation, stability, and so on.

The regulatory systems of prokaryotes and eukaryotes are somewhat different from each other. Prokaryotes are generally free-living unicellular organisms that grow and divide indefinitely as long as environmental conditions are suitable and the supply of nutrients is adequate. Their regulatory systems are geared to provide the maximum growth rate in a particular environment, except when such growth would be detrimental. Prokaryotes can also use the coupling between transcription and translation (Chapter 10) for regulation, but the absence of introns eliminates RNA splicing as a possible control point.

The requirements of multicellular eukaryotes are different from those of prokaryotes. In a developing organism, not only must a cell grow and divide, but the progeny cells must also undergo considerable changes in morphology and biochemistry and then each maintain its altered state. Furthermore, during embryonic development, most eukaryotic cells are challenged less by the environment than are bacteria in that the composition and concentration of the growth medium does not change drastically with time. Finally, in an adult organism, growth and cell division in most cell types have stopped, and each cell needs only to maintain itself and its specialized characteristics.

In this chapter, we consider the basic mechanisms of the regulation of transcription and RNA processing. The examples we use are those in which the regulation is well understood.

11.1—
Transcriptional Regulation in Prokaryotes

In bacteria and phages, on-off gene activity is often controlled through transcription. Synthesis of a particular mRNA takes place only when the gene product is needed, and when the gene product is not needed, mRNA synthesis occurs at greatly reduced levels. In discussing transcription, we use the term *off* for convenience, but remember that this usually means "very low." In bacteria, few examples are known of a system being switched completely off. When transcription is in the "off" state, a basal level of gene expression almost always remains, often averaging one transcriptional event or fewer per cell generation; hence there is very little synthesis of the gene product. Extremely low levels of expression are also found in certain classes of genes in eukaryotes, including many genes that participate in embryonic development. Regulatory mechanisms other than the on-off type also are known in both prokaryotes and eukaryotes; in these examples, the level of expression of a gene may be modulated in gradations from high to low according to conditions in the cell.

In bacterial systems, when several enzymes act in sequence in a single metabolic pathway, usually either all or none of these enzymes are produced. This **coordinate regulation** results from control of the synthesis of one or more polycistronic mRNA molecules encoding all of the gene products that function in the same metabolic pathway. This type of regulation is not found in eukaryotes because eukaryotic mRNA is monocistronic, as we saw in Chapter 10.

Several mechanisms of regulation of transcription are common. The particular one used often depends on whether the enzymes being regulated act in degradative or biosynthetic metabolic pathways. For example, in a multistep degradative (catabolic) system, the availability of the molecule to be degraded helps determine whether the enzymes in the pathway will be synthesized. In the presence of the molecule, the enzymes of the degradative (catabolic) pathway are synthesized; in its absence, they are not. Such a system, in which the presence of a small molecule results in enzyme synthesis, is said to be **inducible.** The small molecule is called the **inducer.** The opposite situation is often found in the control of the synthesis of enzymes that participate in biosynthetic (anabolic) pathways; in these cases, the final product of the pathway is frequently the regulatory molecule. In the presence of the final product, the enzymes of the biosynthetic pathway are not synthesized; in its absence, they are synthesized. Such a system, in which the presence of a small molecule results in failure to synthesize enzymes, is said to be **repressible.** The small molecule that participates in the regulation is called the **co-repressor.**

The molecular mechanisms for each of the regulatory patterns vary quite widely but usually fall into one of two major categories—**negative regulation** and **positive regulation.** In a negatively regulated system (Figure 11.1A), a **repressor** protein

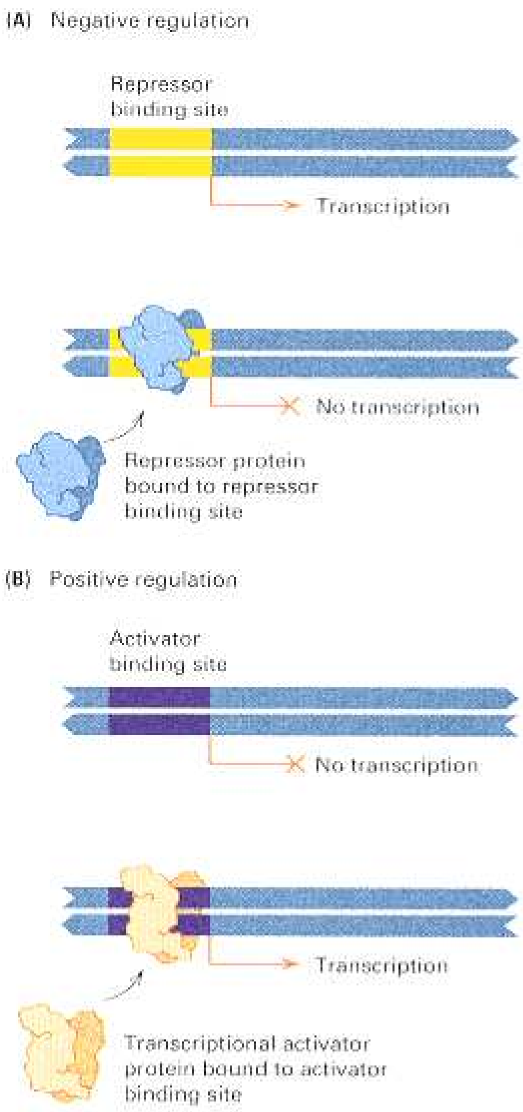

Figure 11.1
The distinction between negative and positive regulation. (A) In negative regulation, the "default" state of the gene is one in which transcription takes place. The binding of a repressor protein to the DNA molecule prevents transcription. (B) In positive regulation, the default state is one in which transcription does not take place. The binding of a transcriptional activator protein stimulates transcription. A single genetic element may be regulated both positively and negatively; in such a case, transcription requires the binding of the transcriptional activator and the absence of repressor binding.

present in the cell prevents transcription. In an inducible system that is negatively regulated, the repressor protein acts by itself to prevent transcription. The inducer antagonizes the repressor, allowing the initiation of transcription. In a repressible system, an **aporepressor** protein combines with the co-repressor molecule to form the functional repressor, which prevents transcription. In the absence of the co-repressor, the aporepressor is unable to prevent transcription. On the other hand, in a positively regulated system (Figure 11.1B), mRNA synthesis only takes place if a regulatory protein binds to a region of the gene that activates transcription. Such a protein is usually referred to as a **transcriptional activator.** Negative and positive regulation are not mutually exclusive, and some systems are both positively and negatively regulated, utilizing two regulators to respond to different conditions in the cell. Negative regulation is more common in prokaryotes, positive regulation in eukaryotes.

A degradative system may be regulated either positively or negatively. In a biosynthetic pathway, the final product usually negatively regulates its own synthesis; in the simplest type of negative regulation, absence of the product increases its synthesis (through production of the necessary enzymes), and presence of the product decreases its synthesis (through repression of enzyme synthesis). Even in a system in which a single protein molecule (not necessarily an enzyme), is translated from a monocistronic mRNA molecule, the protein may be **autoregulated,** which means that the protein regulates its own transcription. In negative autoregulation, the protein inhibits transcription, and high concentrations of the protein result in less transcription of the mRNA that codes for the protein. In positive autoregulation, the protein stimulates transcription: As more protein is made, transcription increases to the maximum rate. Positive autoregulation is a common way for weak induction to be amplified. Only a weak signal is necessary to get production of the protein started, but then the positive autoregulation stimulates the production to the maximum level.

The next two sections are concerned with several systems of regulation in prokaryotes. These serve as an introduction to the remainder of the chapter, which deals with regulation in eukaryotes.

11.2— Lactose Metabolism and the Operon

Metabolic regulation was first studied in detail in the system in *E. coli* responsible for degradation of the sugar lactose, and most of the terminology used to describe regulation has come from genetic analysis of this system.

Lac$^-$ Mutants

In *E. coli,* two proteins are necessary for the metabolism of lactose. They are the enzyme **β-galactosidase,** which cleaves lactose (a β-galactoside) to yield galactose and glucose; and a transporter molecule, **lactose permease,** which is required for the entry of lactose into the cell. The existence of two different proteins in the lactose-utilization system was first shown by a combination of genetic experiments and biochemical analysis.

First, hundreds of mutants unable to use lactose as a carbon source, designated Lac$^-$ mutants, were isolated. Some of the mutations were in the *E. coli* chromosome and others were in an F' *lac,* a plasmid carrying the genes for lactose utilization. By performing F' × F$^-$ matings, investigators constructed partial diploids with the genotypes F' *lac*$^-$/*lac*$^+$ and F' *lac*$^+$/*lac*$^-$. (The genotype of the plasmid is given to the left of the slash and that of the chromosome to the right.) It was observed that all of these diploids always had a Lac$^+$ phenotype (that is, they made β-galactosidase and permease); thus none produced an inhibitor that prevented functioning of the *lac* genes. Other partial diploids were then constructed in which both the F' *lac* plasmid and the chromosome carried a *lac*$^-$ allele. These were tested for the Lac$^+$ phenotype, with the result that all of the mutants initially isolated could be placed into two complementation groups, called *lacZ* and *lacY,* a result that implies that the *lac* system consists of at least two genes. Complementation is indicated by the observation that the partial diploids

$F'\ lacY^-\ lacZ^+/lacY^+\ lacZ^-$

and

$F'\ lacY^+\ lacZ^-/lacY^-\ lacZ^+$

had a Lac+ phenotype, producing both β-galactosidase and permease. However, the genotypes

$F'\ lacY^-\ lacZ^+/lacY^-\ lacZ^+$

and

$F'\ lacY^+\ lacZ^-/lacY^+\ lacZ^-$

had the Lac- phenotype because they were unable to synthesize the permease and the β-galactosidase, respectively. Hence the *lacZ* gene codes for the β-galactosidase and the *lacY* gene for the permease. (A third gene that participates in lactose metabolism was later discovered; it was not included among the early mutants because it is not essential for growth on lactose.) A final important result—that the *lacY* and *lacZ* genes are adjacent—was deduced from a high frequency of cotransduction observed in genetic mapping experiments.

Inducible and Constitutive Synthesis and Repression

The on-off nature of the lactose-utilization system is evident in the following observations:

1. If a culture of Lac+ *E. coli* is growing in a medium that does not include lactose or any other β-galactoside, then the intracellular concentrations of β-galactosidase and permease are exceedingly low: roughly one or two molecules per bacterial cell. However, if lactose is present in the growth medium, then the number of each of these molecules is about 10^3-fold higher.

2. If lactose is added to a Lac+ culture growing in a lactose-free medium (also lacking glucose, a point that will be discussed shortly), then both β-galactosidase and permease are synthesized nearly simultaneously, as shown in Figure 11.2. Analysis of the total mRNA present in the cells before and after the addition of lactose shows that almost no *lac* mRNA (the polycistronic mRNA that codes for β-galactosidase and permease) is present before lactose is added and that the addition of lactose triggers synthesis of *lac* mRNA.

These two observations led to the view that transcription of the lactose genes is **inducible transcription** and that lactose is an *inducer* of transcription. Some analogs of lactose are also inducers, such as a sulfur-containing analog denoted IPTG (isopropyl-thiogalactoside), which is convenient for experiments because it induces but is not cleaved by β-galactosidase, so this inducer is stable in the cell whether or not the β-galactosidase enzyme is present.

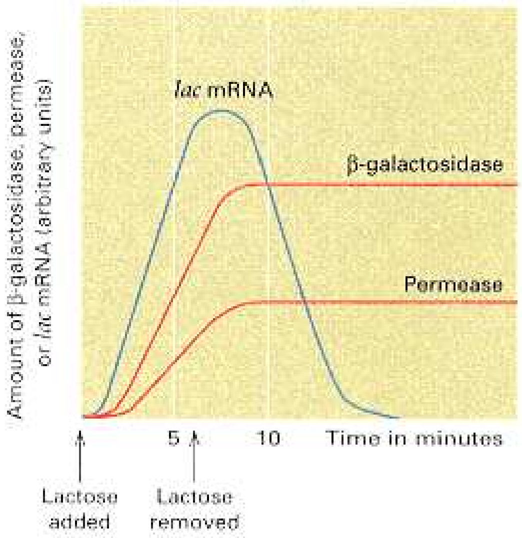

Figure 11.2
The "on-off" nature of the *lac* system. The lac mRNA appears soon after lactose of another inducer is added; β-galactosidase and permease appear at nearly the same time but are delayed with respect to mRNA synthesis because of the time required for translation. When lactose is removed, no more lac mRNA is made, and the amount of lac mRNA decreases because of the degradationb mRNA already present. Both β-galactosidase and permease are stable proteins: their amounts remain constant even when synthesis ceases. However, their concentration per cell gradually decreases as a result of repeated cell divisions.

Mutants were also isolated in which lac mRNA was synthesized (and hence β-galactosidase and permease produced) in both the presence and the absence of an inducer. The mutants that eliminated regulation provided the key to understanding induction; because of their constant synthesis, the mutants were termed **constitutive.** Mutants were also obtained that failed to produce lac mRNA (and hence β-galactosidase and permease) even when the inducer was present. These uninducible mutants fell into two classes, lac^s and $lacP^-$. The characteristics of the mutants are shown in Table 11.1 and discussed in the following sections.

L The Repressor

In Table 11.1, genotypes 3 and 4 show that *lacl*$^-$ mutations are recessive. In the absence of inducer, a *lacl*$^+$ cell does not make *lac*mRNA, whereas the mRNA is made in a *lacl*$^-$ mutant. These results suggest that

> The *lacl* gene is a regulatory gene whose product is the repressor protein that keeps the system turned off. Because the repressor is necessary to shut off mRNA synthesis, regulation by the repressor is negative regulation.

A *lacl*$^-$ mutant lacks the repressor and hence is constitutive. Wildtype copies of the repressor are present in a *lacl*$^+$ *lacl*$^-$ partial diploid, so transcription is repressed. It is important to note that the single *lacl*$^+$ gene prevents synthesis of *lac* mRNA from both the F' plasmid and the chromosome. Therefore, the repressor protein must be diffusible within the cell to shut off mRNA synthesis from both DNA molecules present in a partial diploid.

On the other hand, genotypes 7 and 8 indicate that the *lacl*^{s} mutations are dominant and act to shut off mRNA synthesis from both the F' plasmid and the chromosome, whether or not the inducer is present (the superscript in *lacl*^{s} signifies *super-repressor.* the *lacl*^{s} mutations result in repressor molecules that fail to recognize and bind the inducer and thus permanently shut off *lac* mRNA synthesis. Genetic mapping experiments placed the *lacl* gene adjacent to the *lacZ* gene and established the gene order *lacl lacZ lacY.* How the *lacl* repressor prevents synthesis of *lac* mRNA will be explained shortly.

The Operator Region

Entries 1 and 2 in Table 11.1 show that *lacO*^{c} mutants are dominant. However, the dominance is evident only in certain Combinations of *lac* mutations, as can be seen by examining the partial diploids shown in entries 5 and 6. Both combinations are Lac^+ because a functional *lacZ* gene is present. However, in the combination shown in entry 5, synthesis of β-galactosidase is inducible even though a *lacO*^{c} mutation is present. The difference between the two combinations in entries 5 and 6 is that in entry 5, the *lacO*^{c} mutation is present in the same DNA molecule as the *lacZ*$^-$ mutation whereas in entry 6,*lacO*^{c} is contained in the same DNA molecule as *lacZ*$^+$. The key feature of these results is that

> A *lacO*^{c} mutation causes constitutive synthesis of β-galactosidase only when the *lacO*^{c} and *lacZ*$^+$ alleles are contained in the same DNA molecule.

The *lacO*^{c} is said to be *cis*-dominant, because only genes in the *cis* configuration (in the same DNA molecule as that containing the mutation) are expressed in dominant fashion. Confirmation of this conclusion comes from an important biochemical observation: The mutant enzyme coded by the *lacZ*$^-$ sequence is synthesized constitutively in a *lacO*^{c} *lacO*$^+$ *lacZ*$^+$ partial diploid (entry 5), whereas the wildtype enzyme (coded by the *lacZ*$^+$ sequence) is synthesized only if an inducer is added. All *lacO*^{c} mutations are located between the *lacl* and *lacZ* genes; hence the gene order of the four genetic elements of the *lac* system is

An important feature of all *lacO*^{c} mutations is that they cannot be complemented (a characteristic feature of all *cis*-dominant mutations); that is, a *lacO*$^+$ allele cannot alter the consitutive activity of a *lacO*^{c}

mutation. This observation implies that the *lacO* region does not encode a diffusible product and must instead define a site in the DNA that determined whether synthesis of the product of the adjacent *lacZ* gene is inducible or constitutive. The *lacO* region is called the **operator.** In the next section we will see that the operator is in fact a *binding site* in the DNA for the repressor protein.

The Promoter Region

Entries 11 and 12 in Table 11.1 show that $lacP^-$ mutations, like $lacO^c$ mutations, are *cis*-dominant. The *cis*-dominance can be seen in the partial diploid in entry 11. The genotype in entry 11 is uninducible, in contrast to the partial diploid of entry 12, which is inducible. The difference between the two genotypes is that in entry 11, the $lacP^-$ mutation is in the same DNA molecule with $lacZ^+$, whereas in entry 12, the$lacP^-$ mutation is combined with $lacZ^-$. This observation means that a wildtype $lacZ^+$ remains inexpressible in the presence of $lacP^-$; no *lac* mRNA is transcribed from that DNA molecule. The $lacP^-$ mutations map between *lacl* and *lacO,* and the order of the five genetic elements of the *lac* system is

As expected because of the *cis*-dominance of $lacP^+$ allele on another DNA molecule cannot supply the missing function to a DNA molecule carrying a $LacP^-$ mutation. Thus*lacP,* like *lac),* must define a site that determines whether synthesis of *lac* mRNA will take place. Because synthesis does not occur if the site is defective or missing, *lacP* defines an essential site for mRNA synthesis. The *lacP* defines an essential sitefor mRNA synthesis. The *lacP* region is called the **promoter.** It is a site at which RNA polymerase binding takes place to allow initiation of transcription.

The Operon Model of Transcriptional Regulation

The genetic regulatory mechanism of the *lac* system was first explained by the **operon model** of François Jacob and Jacques Monod, which is illustrated in Figure 11.3 (The figure uses the abbreviations *i, o, p, z, y,* and a for *lacl, lacO, lacP, lacZ,*

Table 11.1 Characteristics of partial diploids containing several combinations of *lacI*, *lacO* and *lacP* alleles

Genotype	Synthesis of *lac* mRNA	Lac phenotype
1. F' $lacO^c\ lacZ^+/lacO^+\ lacZ^+$	Constitutive	+
2. F' $lacO^+\ lacZ^+\ /lacO^C\ lacZ^+$	Constitutive	+
3. F' $lacI^-\ lacZ^+\ /lacI^+\ lacZ^+$	Inducible	+
4. F' $lacI^+\ lacZ^+\ /lacI^-\ lacZ^+$	Inducible	+
5. F' $lacO^c\ lacZ^-\ /lacO^+\ lacZ^+$	Inducible	+
6. F' $lacO^c\ lacZ^+\ /lacO^+\ lacZ^-$	Constitutive	+
7. F' $lacI^s\ lacZ^+\ /lacI^+\ lacZ^-$	Uninducible	+
8. F' $lacI^+\ lacZ^+\ /lacI^s\ lacZ^Z$	Uninducible	-
9. F' $lacP^-\ lacZ^+\ /lacP^+\ lacZ^+$	Inducible	-
10. F' $lacP^-\ lacZ^+\ /lacP^-\ lacZ^+$	Inducible	+
11. F' *lacP*sup>+ $lacZ^-\ /lacP^-\ lacZ^+$	Uninducible	-
12. F' $lacp^+\ lacZ^+\ /lacP^-\ lacZ^+$	Inducible	+

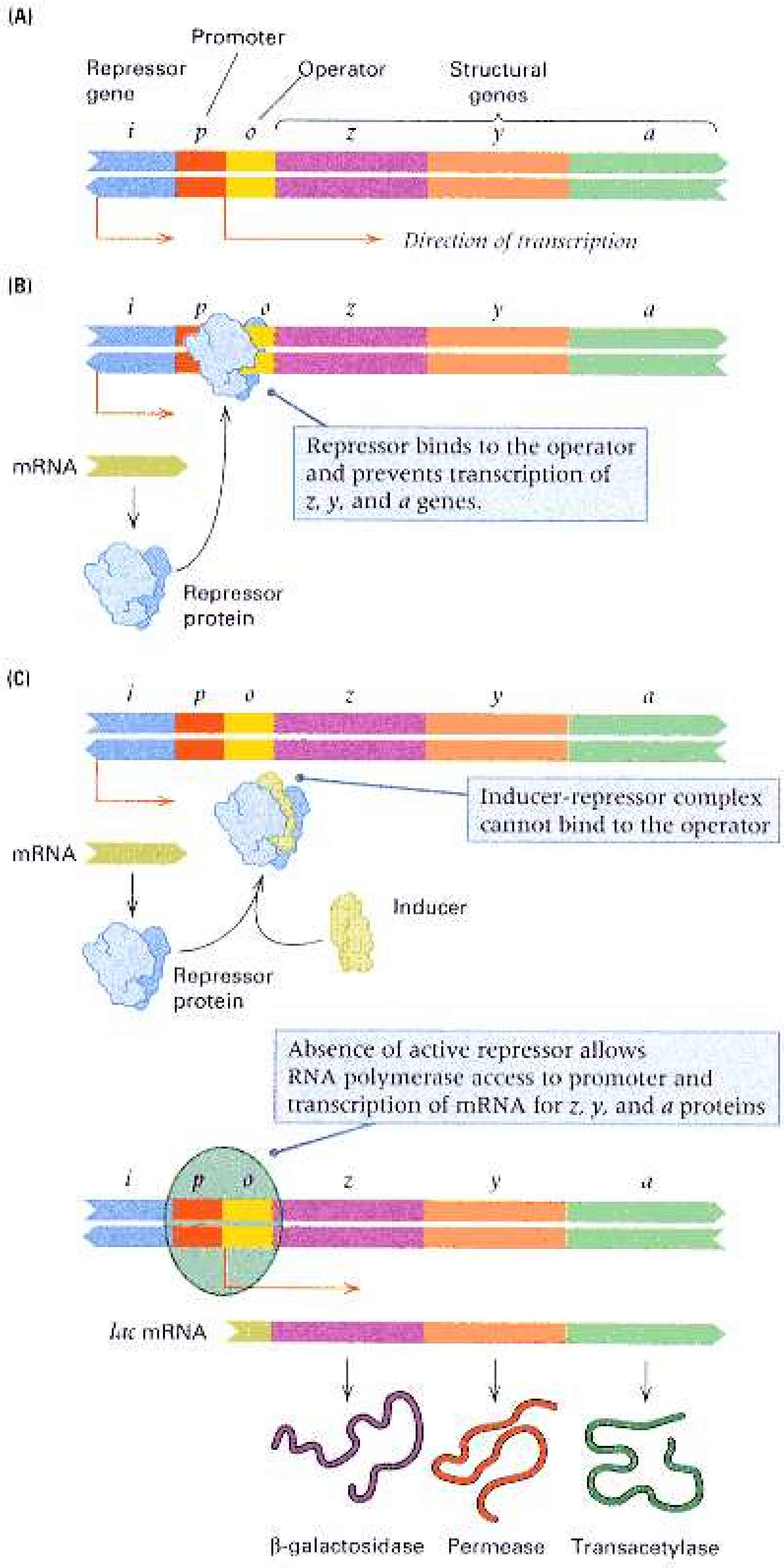

Figure 11.3
(A) A map of the *lac* operon, not drawn to scale. The *p* and *o* sites are actually much smaller than the other regions and together comprise only 83 base pairs (B) A diagram of the *lac* operon in the repressed state. (C) A diagram of the *lac* operon in the induced state. The inducer alters the shape of the repressor so that the repressor can no longer bind to the operator. The common abbreviations *i, p, o z, y,* and *a* are used instead of *lacI, lacO,* and so on. The*vlacA* gene is not essential for lactose utilization.

lacY,and *lacA*)The operon model has the following features:

1. The lactose-utilization system consists of two kinds of components—*structural genes* (*lacZ* and *lacY*), which encode proteins needed for the transport and metabolism of lactose, and *regulatory elements* (the repressor gene *lacI,* the promoter *lacP,* and the operator *lacO*).

2. The products of the *lacZ* and *lacY* genes are coded by a single polycistronic mRNA molecule. (A third protein, encoded by *lacA*, is also translated from the mRNA. This protein is the enzyme transacetylase; it is used in the metabolism of certain β-galactosides other than lactose and will not be of further concern here.) The linked structural genes, together with *lacP* and *lacO,* constitute the ***lac* operon.**

3. The promoter mutations (*lacP*$^-$) eliminate the ability to synthesize *lac* mRNA.

4. The product of the *lacI* gene is a repressor, which binds to a unique sequence of DNA bases constituting the operator.

5. When the repressor is bound to the operator, initiation of transcription of *lac* mRNA by RNA polymerase is prevented.

6. Inducers stimulate mRNA synthesis by binding to and inactivating the repressor. In the presence of an inducer, the operator is not bound with the repressor, and the promoter is available for the initiation of mRNA synthesis.

Note that regulation of the operon requires that the *lacO* operator either overlap or be adjacent to the promoter of the structural genes, because binding with the repressor prevents transcription. Proximity of *lacI* to *lacO* is not strictly necessary, because the *lacI* repressor is a soluble protein and is therefore diffusible throughout the cell. The presence of inducer has a profound effect on the DNA binding properties of the repressor; the inducer-repressor complex has an affinity for the operator that is approximately 10^3 smaller than that of the repressor alone.

The ratio of the numbers of copies of β-galactosidase, permease, and transacetylase

Connection Operator? Operator?

François Jacob, David Perrin, Carmen Sanchez, and Jacques Monod, Institute Pasteur, Paris, France. 1960

The Operon: A Group of Genes Whose Expression Is Coordinated by an Operator (original in French)

How is gene expression controlled? Before Jacob and Monod and their collaborators addressed this question experimentally, it was all a matter of speculation. Prior to this report, the researchers had previously discovered the i (lacl) gene that controls expression of the β-galactosidase (z) and permease (y) genes needed for lactose utilization. They also had strong reason to believe that lacl produces a regulatory protein. How does the regulatory protein work? Here they give evidence that it works by directly binding to a DNA "operator" adjacent to the genes it regulates. Furthermore, the z and y genes are adjacent and are controlled coordinately by the same "operator" upstream from z. The discovery was immediately recognized as fundamental. Jacob and Monod, along with André Lwoff, were awarded the Nobel Prize in 1965. We now know that coordinate regulation via operons is restricted to bacteria. However, the underlying principle—that regulatory genes often control their target genes by direct binding to DNA—is valid for all organisms.

The analysis of different bacterial systems leads to the conclusion that, in the synthesis of certain proteins, there is a dual genetic determination involving two types of genes with distinct functions: one (the gene for structure) is responsible for the structure of the protein molecule, and the other (the regulatory gene) governs the expression of the former through the intermediary action of a repressor. The regulatory genes that have so far been identified show the remarkable property of exercising a *coordinated effect*, each governing the expression of several genes for structure, closely linked together, and corresponding to enzyme proteins belonging to the *same biochemical pathway*. To explain this effect, it seems necessary to invoke a new type of genetic entity, called an "operator," which

It seems necessary to invoke a new type of genetic entity, called an "operator," which would be (a) adjacent to the group of genes and would control their activity; and (b) would be sensitive to the repressor produced by a particular regulatory gene.

would be (a) adjacent to the group of genes and would control their activity; and (b) would be sensitive to the repressor produced by a particular regulatory gene. In the presence of the repressor, the expression of the group of genes would be inhibited through the mediation of the repressor. This hypothesis leads to some distinctive predictions concerning mutations that could affect the structure of the operator. (1) Certain mutations affecting an operator would be manifested by the loss of the capacity to synthesize the proteins determined by the group of linked genes "coordinated" by that operator. . . . (2) Other mutations, for example involving a loss of sensitivity (affinity) of the operator for the corresponding repressor, would be manifested by the constitutive synthesis of the protein determined by the coordinated genes. . . . We have studied certain mutations affecting the metabolism of lactose in *Escherichia coli* that act simultaneously on the synthesis of β-galactosidase [the product of the *z* gene] and galactoside permease [the product of the *y* gene]. . . . The *i* gene is the regulatory gene synthesizing a repressor specific for the system. The genes *i, z* and *y* are closely linked. . . . Constitutive mutants (o^c) have now been isolated. [In partially diploid genotypes] only the allele of *z* or *y* that is *cis* with respect to o^c Is constitutively expressed. . . . Other mutants have been isolated that have lost the ability to synthesize both the permease and the β-galactosidase. . . . These mutants are recessive. . . . Genetic analysis shows that these mutations (o^o) are extremely closely linked to the o^c mutations and that the order of the *lac* region is *i-o-z-y*. . . . The remarkable properties of the o^c and o^o mutations are inexplicable according to the "classical" concept of the genes for structure [*z* and *y*] and distinguish them equally from mutations affecting the regulatory gene *i*. On the other hand, they conform to the predictions arising from the hypothesis of the operator.

Source: Comptes Rendus des Séances de I'Academie des Sciences 250: 1727–1729. Translated in E. A. Adelberg, 1966. *Papers on Bacterial Genetics. Boston: Little Brown.*

is 1.0: 0.5: 0.2 when the operon is induced. These differences are partly due to the order of the genes in the mRNA: Downstream cistrons are less likely to be translated owing to failure of reinitiation when an upstream cistron has finished translation.

The operon model is supported by a wealth of experimental data and explains many of the features of the *lac* system, as well as numerous other negatively regulated genetic systems in prokaryotes. One aspect of the regulation of the *lac* operon—the effect of glucose—has not yet been discussed. Examination of this feature indicates that the *lac* operon is also subject to positive regulation, as we will see in the next section.

Positive Regulation of the Lactose Operon

The function of β-galactosidase in lactose metabolism is to form glucose by cleaving lactose. (The other cleavage product, galactose, also is ultimately converted into glucose by the enzymes of the galactose operon.) If both glucose and lactose are present in the growth medium, activity of the *lac* operon is not needed. In fact, in the presence of glucose, no β-galactosidase is formed until virtually all of the glucose in the medium has been consumed. The lack of synthesis of β-galactosidase is a result of the lack of synthesis of *lac* mRNA. No *lac* mRNA is made in the presence of glucose, because in addition to an inducer to inactivate the *lacI* repressor, another element is needed for initiating *lac* mRNA synthesis; the activity of this element is regulated by the concentration of glucose.

The inhibitory effect of glucose on expression of the *lac* operon is indirect. The small molecule *cyclic adenosine monophosphate* **(cAMP),** shown in Figure 11.4, is widely distributed in animal tissues, and in multicellular eukaryotic organisms, in which it is important in mediating the action of many hormones. It is also present in *E. coli* and many other bacteria, where it has a different function. Cyclic AMP is synthesized by the enzyme *adenyl cyclase*, and the concentration of cAMP is regulated indirectly by glucose metabolism. When bacteria are growing in a medium containing glucose, the cAMP concentration in the cells is quite low. In a medium containing glycerol or any carbon source that cannot

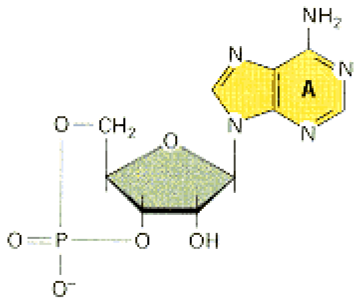

Figure 11.4
Structure of cyclic AMP.

Table 11.2 Concentration of cyclic AMP in cells growing in media with the indicated carbon sources

Carbon source	cAMP concentration
Glucose	Low
Glycerol	High
Lactose	High
Lactose + glucose	Low
Lactose + glycerol	High

enter the biochemical pathway used to metabolize glucose (the glycolytic pathway), or when the bacteria are otherwise starved of an energy source, the cAMP concentration is high (Table 11.2). Glucose levels help regulate the cAMP concentration in the cell, and *cAMP regulates the activity of the lac operon* (as well as that of several other operons that control degradative metabolic pathways).

E. coli (and many other bacterial species) contain a protein called the *cyclic AMP receptor protein* **(CRP),** which is encoded by a gene called *crp*. Mutations of either the *crp* or the adenyl cyclase gene prevent synthesis of *lac* mRNA, which indicates that both CRP function and cAMP are required for *lac* mRNA synthesis. CRP and cAMP bind to one another, forming a complex denoted **cAMP-CRP,** which is an active regulatory element in the *lac* system. The requirement for cAMP-CRP is independent of the *lacI* repression system, because *crp* and adenyl cyclase mutants are unable to make *lac* mRNA even if a *lacI*$^{-}$ or a *lacO*^{c} mutation is present. The reason is that the cAMP-CRP complex must be bound to a base sequence in the DNA in the promoter region in order for transcription to occur (Figure 11.5). Unlike the repressor, which is a *negative* regulator, the cAMP-CRP complex is a *positive* regulator. The positive and negative regulatory systems of the *lac* operon are independent of each other.

Experiments carried out *in vitro* with purified *lac* DNA, *lac* repressor, cAMP-CRP, and RNA polymerase have established two further points:

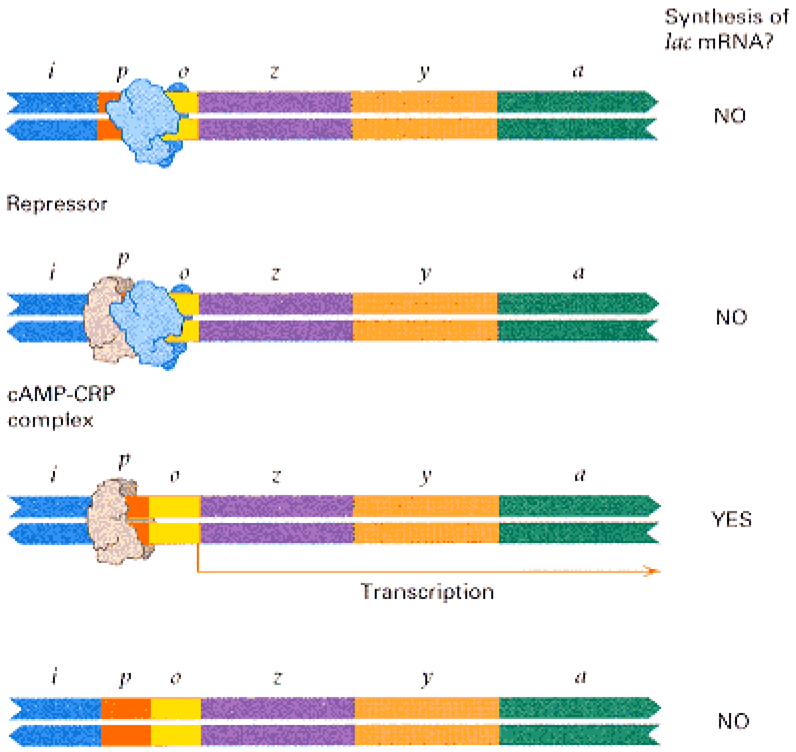

Figure 11.5
Four regulatory states of the *lac* operon. The *lac* mRNA is synthesized only if cAMP-CAP is present and the repressor is absent.

1. In the absence of the cAMP-CRP complex, RNA polymerase binds only weakly to the promoter, but its binding is stimulated when cAMP-CRP is also bound to the DNA. The weak binding rarely leads to initiation of transcription, because the correct interaction between RNA polymerase and the promoter does not occur.

2. If the repressor is bound to the operator, then RNA polymerase cannot stably bind to the promoter.

These results explain how lactose and glucose function together to regulate transcription of the *lac* operon. The relationship of these elements to one another, to the start of transcription, and to the base sequence in the region is depicted in Figure 11.6.

A great deal is also known about the three-dimensional structure of the regulatory states of the *lac* operon. Figure 11.7 shows that there is actually a 93-base-pair loop of DNA that forms in the operator region when it is in contact with the repressor. This loop corresponds to the *lac* operon region -82 to + 11 (numbered as in Figure 11.6). The DNA region in red corresponds, on the right-hand side, to the operator region centered at + 11 and, on the left-hand side, to a second repressor-binding site immediately upstream and adjacent to the CRP binding site. The *lac* repressor tetramer (violet) is shown bound to these sites. The DNA loop is formed by the region between the repressor-binding sites and includes, in medium blue, the CRP binding site, to which the CAP protein (dark blue) is shown bound. The DNA regions in green are the -10 and -35 sites in the *lacP* promoter indicated in Figure 11.6. In this configuration, the *lac* operon is not transcribed. Removal of the repressor opens up the loop and allows transcription to occur.

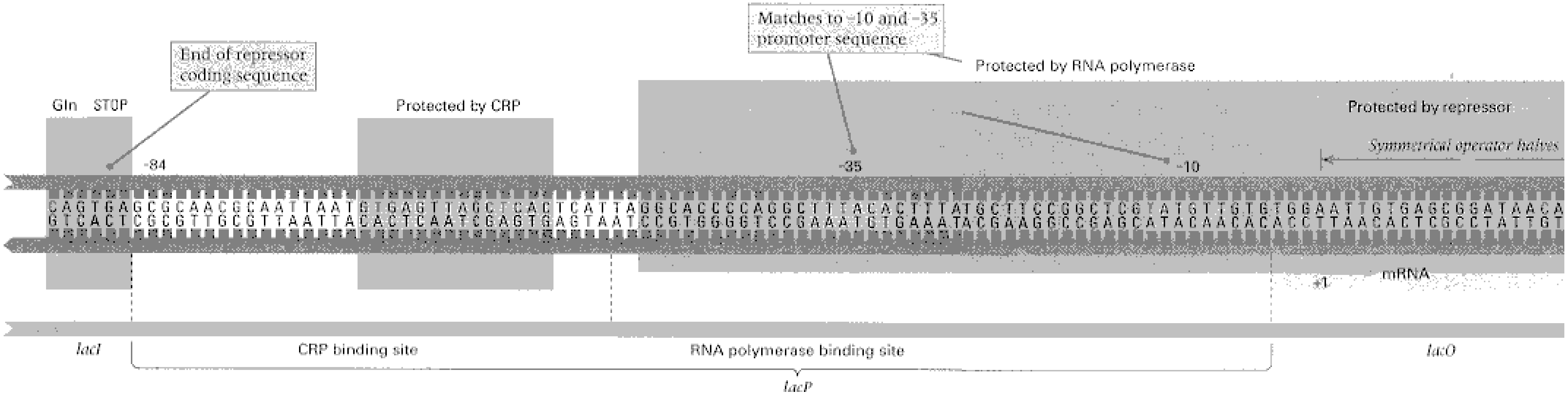

Figure 11.6
The base sequence of the control region of the lac operon. Sequences protected from DNase digestion by binding of the stipulated proteins are indicated in the upper part. The end of the lacI gene is shown at the extreme left; the ribosome binding site is the site at which the ribosome binds to the lac mRNA. The consensus sites for CRP binding and for RNA polymerase promoter binding are indicated along the bottom.

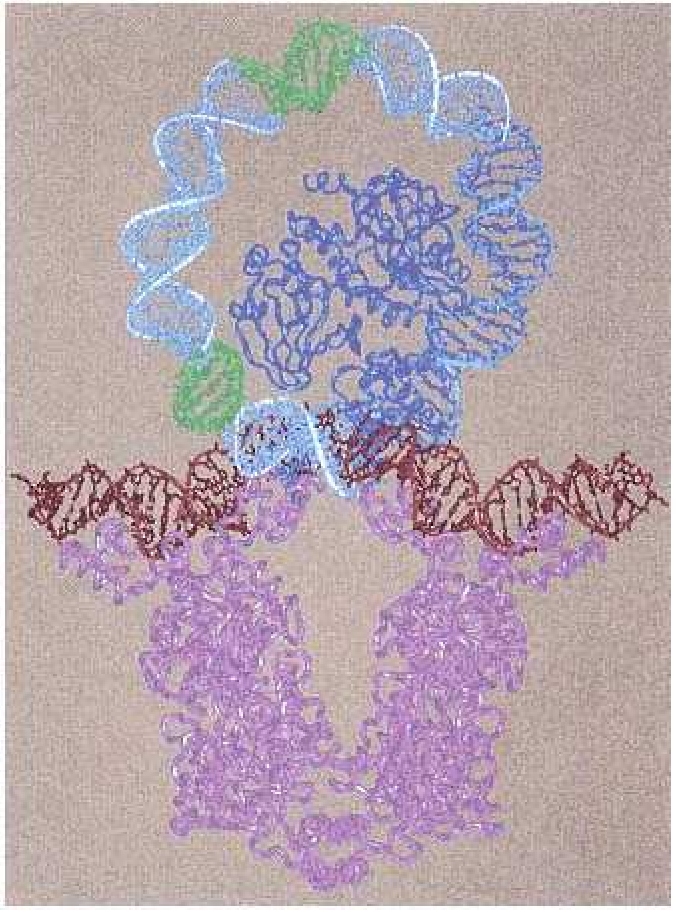

Figure 11.7
Structure of the *lac* operon repression loop. The *lac* repressor, shown in violet, binds to two DNA regions (red) consisting of the symmetrical operator region indicated in Figure 11.6 and a second region immediately upstream from the CRP binding site. Within the loop is the CRP binding site (medium blue), shown bound with CAP protein (dark blue). The -10 and -35 promoter regions are in green
[Courtesy of
Mitchell Lewis; from M. Lewis, G. Chang N. C. Horton,
M. A. Kercher, H. C. Pace, M. A. Schumacher, R. G. Brennan,
and P. Lu. 1996. *Science* 271: 1247.]

11.3— Regulation of the Tryptophan Operon

The tryptophan (*trp*) operon of *E. coli* contains structural genes for enzymes that synthesize the amino acid tryptophan. This operon is regulated in such a way that when adequate tryptophan is present in the growth medium, transcription of the operon is repressed; however, when the supply of tryptophan is insufficient, transcription takes place. Regulation in the *trp* operon is similar to that of the *lac* operon because mRNA synthesis is regulated negatively by a repressor. However, it differs from regulation of *lac* in that tryptophan acts as a co-repressor, which stimulates binding of the repressor to the *trp* operator to shut off synthesis. The *trp* operon is a repressible rather than an inducible operon, although both the *lac* and the *trp* operons are negatively regulated. Furthermore, because the *trp* operon codes for a set of biosynthetic enzymes rather than degradative enzymes, neither glucose nor cAMP-CRP functions in regulation of the *trp* operon.

A simple on-off system, as in the *lac* operon, is not optimal for a biosynthetic pathway. For example, a situation may arise in which some tryptophan is present in the growth medium, but the amount is not enough to sustain optimal growth. Under these conditions, it is advantageous to synthesize tryptophan, but at less than the maximum possible rate. Cells adjust to this situation by means of a regulatory mechanism in which *the amount of transcription in the derepressed state is determined by the concentration of tryptophan in the cell.* This regulatory mechanism is found in many operons responsible for amino acid biosynthesis.

Tryptophan is synthesized in five steps, each requiring a particular enzyme. The genes coding for these enzymes are adjacent and in the same linear order in the *E. coli* chromosome as the order in which the enzymes function in the biosynthetic pathway. The genes are called *trpE*, *trpD*, *trpC*, *trpB*, and *trpA*, and the enzymes are translated from a single polycistronic mRNA molecule. The *trpE* coding region is the first one translated. Upstream (on the 5' side) of *trpE* are the promoter, the operator, and two regions called the *leader* and the *attenuator*, which are designated *trpL* and *trpa* (not *trpA*), respectively (Figure 11.8). The repressor gene, *trpR*, is located quite far from this operon.

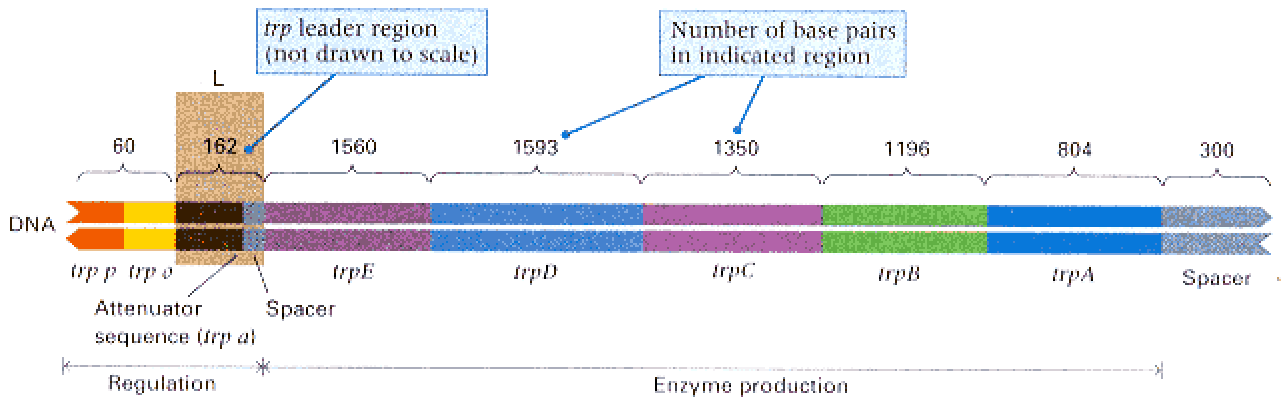

Figure 11.8
The *E. coli trp* operon. For clarity, the regulatory region is enlarged with respect to the coding region. The actual size of each region is indicated by the numbers of base pairs. Region *L* is the leader.

The regulatory protein of the *trp* operon is the product of the *trpR* gene. Mutations in either this gene or the operator cause constitutive initiation of transcription of *trp* mRNA, as in the *lac* operon. The *trpR* gene product is called the *trp* **aporepressor.** It does not bind to the operator unless it is first bound to tryptophan; that is, the aporepressor and the tryptophan molecule join together to form the active *trp* repres-

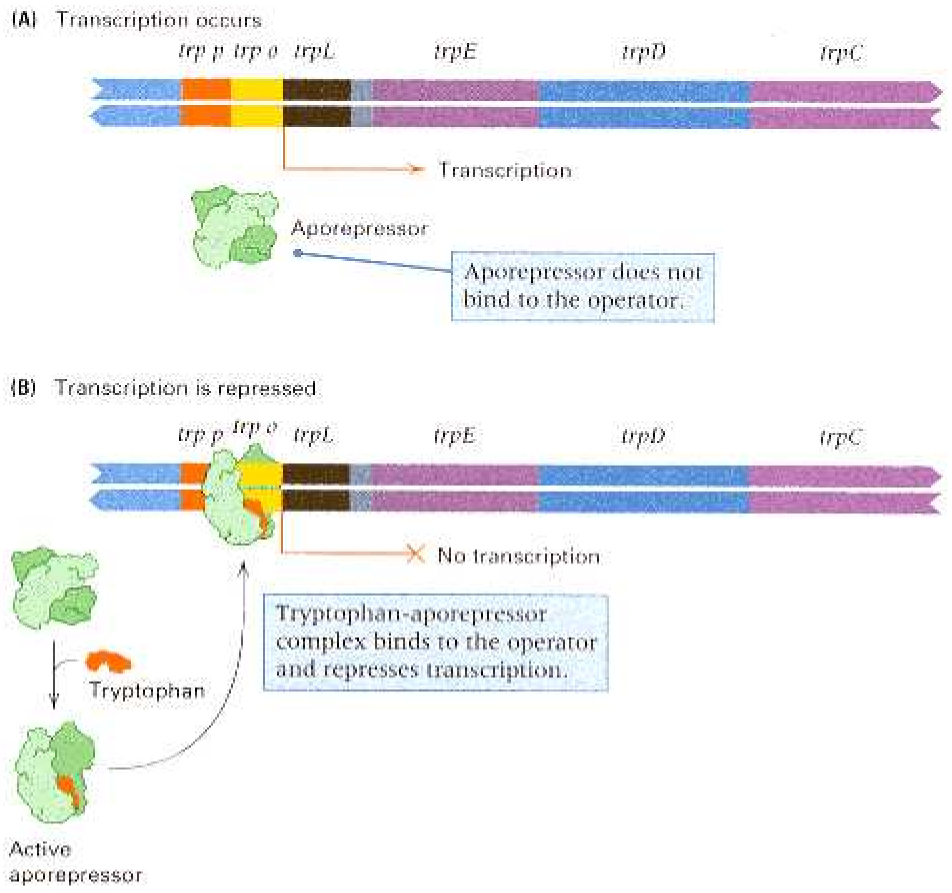

Figure 11.9
Regulation of the *E. coli trp* operon. (A) By itself, the *trp* aporepressor protein does not bind to the operator, and transcription occurs. (B) In the presence of sufficient tryptophan, the combination of aporepressor and tryptophan forms the active repressor that binds to the operator, and transcription is repressed.

sor, which binds to the operator. The reaction scheme is outlined in Figure 11.9. When there is not enough tryptophan, the aporepressor adopts a three-dimensional conformation unable to bind with the *trp* operator, and the operon is transcribed (Figure 11.9A). On the other hand, when tryptophan is present at high enough concentration, some molecules bind with the aporpressor and cause it to change conformation

into the active repressor. The active repressor binds with the *trp* operator and prevents transcription (Figure 11.9B). Thus only when tryptophan is present in sufficient amounts is the active repressor molecule formed. This is the basic on-off regulatory mechanism.

Attenuation

In the *on* state, a still more sensitive regulation of transcription is exerted by the internal concentration of tryptophan. This type of regulation is called **attenuation,** and it uses translation to control transcription. In the presence of even small concentrations of intracellular tryptophan, translation of part of the leader region of the mRNA immediately after its synthesis results in termination of transcription before the first structural gene of the operon is transcribed.

Attenuation results from interactions between DNA sequences present in the leader region of the *trp* transcript. In wild-type cells, transcription of the *trp* operon is often initiated. However, in the presence of even small amounts of tryptophan, most of the mRNA molecules terminate in a specific 28-base region within the leader sequence. The result of termination is an RNA molecule containing only 140 nucleotides that stops short of the genes

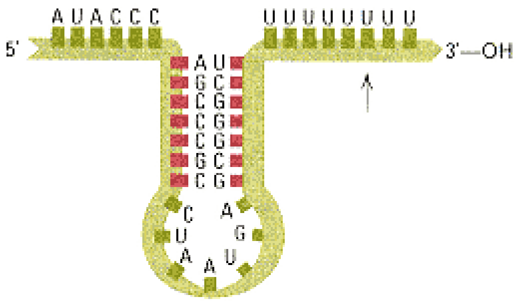

Figure 11.10
The terminal region of the *trp* attenuator sequence. The arrow indicates the final uridine in attenuated RNA Nonattenuated RNA continues past that base. The bases in red letters form the hypothetical stem sequence that is shown.

coding for the *trp* enzymes. The 28-base region in which termination occurs is called the **attenuator.** The base sequence of this region (Figure 11.10) contains the usual features of a termination site, including a potential stem-and-loop configuration in the mRNA followed by a sequence of eight uridines.

The leader sequence, shown in Figure 11.11, contains several notable features.

1. An AUG codon and a downstream UGA stop codon in the same reading frame defining a region that codes for a polypeptide consisting of only 14 amino acids, which is called the **leader polypeptide.**

2. Two adjacent tryptophan codons that are located in the leader polypeptide at positions 10 and 11. We will see the significance of these repeated codons shortly.

Figure 11.11
The sequence of bases in the *trp* leader mRNA, showing the leader polypeptide, the two tryptophan codons (red letters), and the beginning of the TrpE protein. The numbers 23 and 91 are the numbers of bases in the

sequence that, for clarity, are not shown.

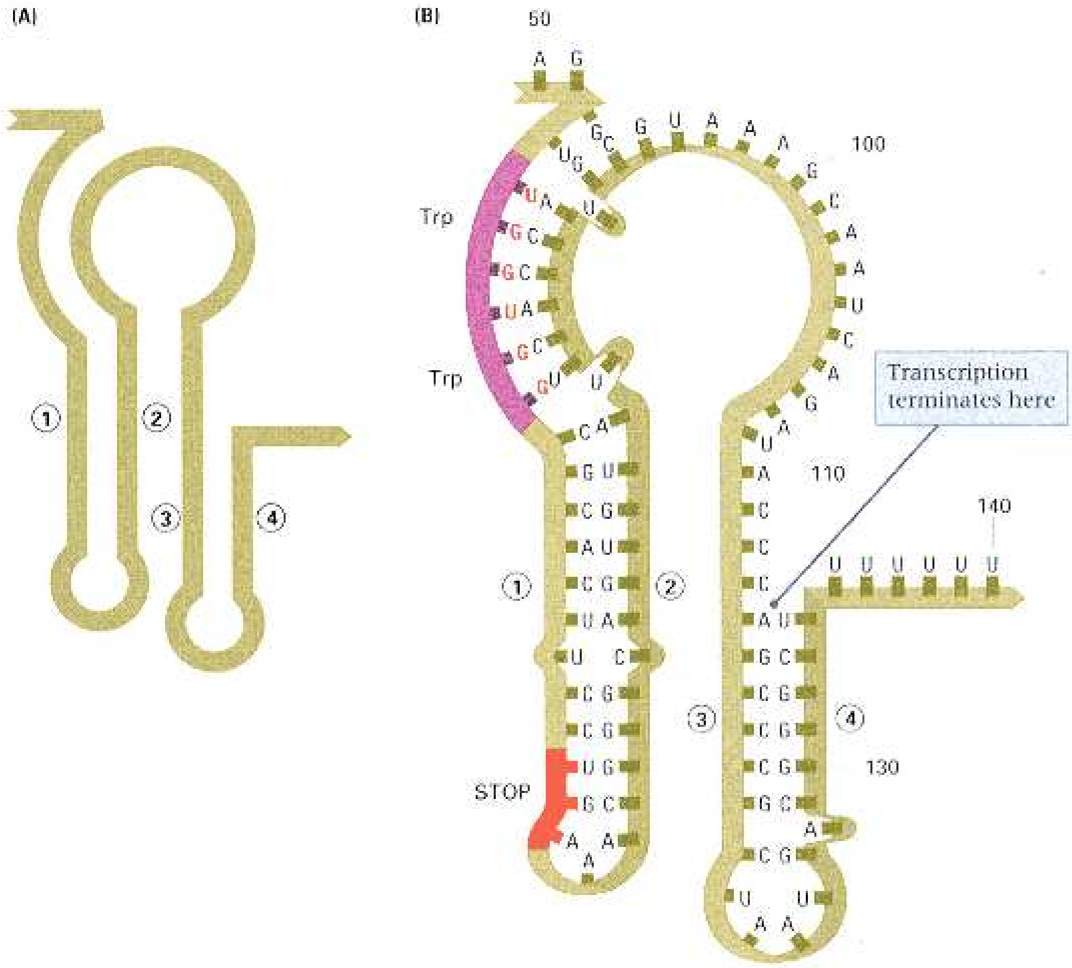

Figure 11.12
(A) Diagram of the transcript of the *trp* leader region, showing the proposed foldback structure in which a sequence of bases in region 1 can base-pair with a sequence in region 2 and a sequence of bases in region 3 can base-pair with a sequence in region 4. (B) Details of the structure. Note the two Trp codons in the 1-2 loop.

3. Four segments of the leader RNA—denoted in Figure 11.12 as regions 1, 2, 3, and 4—that are capable of base-pairing with each other. In one configuration, region 1 pairs with region 2, and region 3 with region 4. The details of this configuration are shown in Figure 11.12. When pairing takes place in this configuration, transcription is terminated at the run of uridines preceding nucleotide 140. This type of pairing occurs in purified *trp* leader mRNA.

4. An alternative type of pairing can also take place, in which region 2 pairs with region 3. The potential for this type of base pairing is apparent in Figure 11.12B in the nearly complementary sequence of bases present in regions 2 and 3.

Through the alternative modes of base pairing (essentially either 3-4 or 2-3), the sequence organization of the *trp* leader mRNA makes possible regulation of trans-

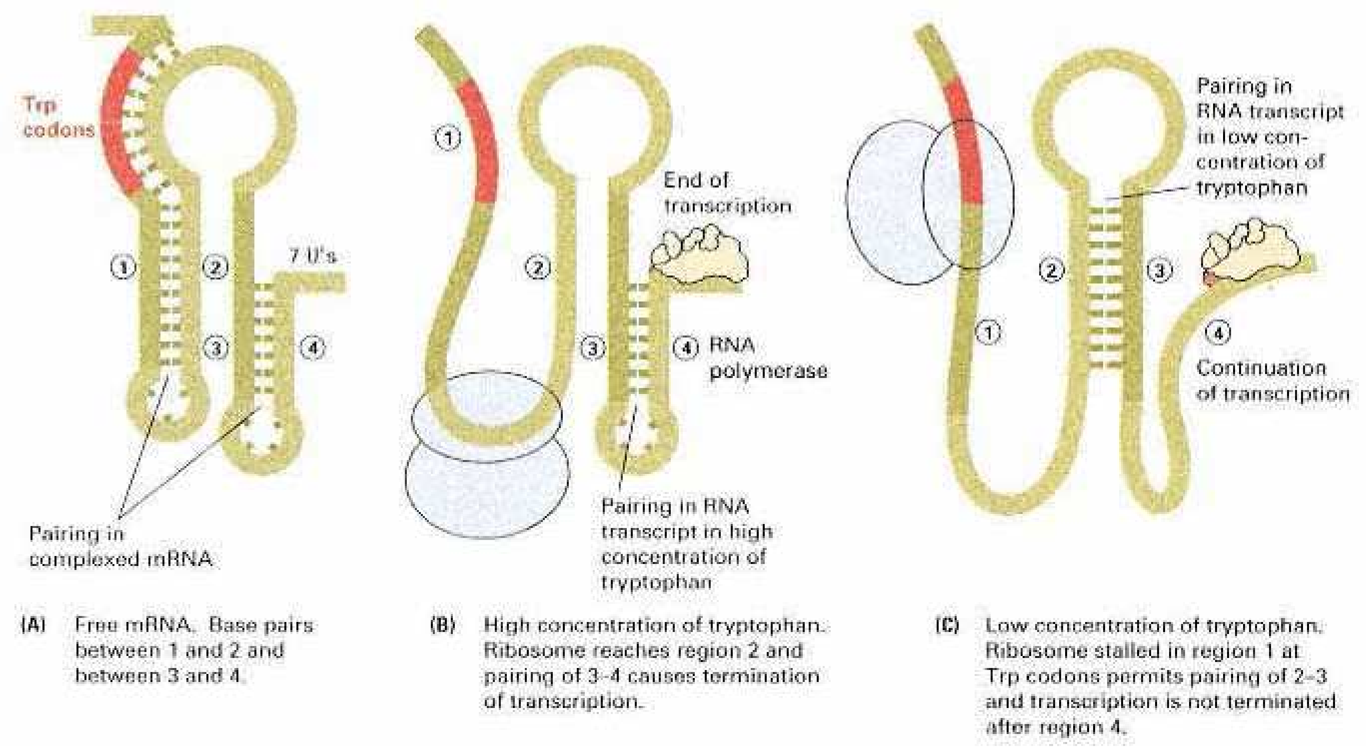

Figure 11.13
The explanation for attenuation in the *E. coli* trp operon. The tryptophan codons in part A are those highlighted in red in Figure 11.11.

scription through translation of the leader polypeptide. The mechanism is shown in Figure 11.13. As the leader region is transcribed, translation of the leader polypeptide is initiated. Because there are two tryptophan codons in the coding sequence, the translation of the sequence is sensitive to the concentration of charged $tRNA^{Trp}$. If the supply of tryptophan is adequate for translation, the ribosome passes through the Trp codons and into region 2 (Figure 11.13B). Because the presence of a ribosome eliminates the possibility of base pairing in a region of about 10 bases on each side of the codons being translated, the presence of a ribosome in region 2 prevents its becoming paired with region 3. In this case, region 3 pairs with region 4 and forms the terminator shown in Figure 11.13B (and in detail in Figure 11.12B), and transcription is terminated at the run of uridines that follows region 4.

On the other hand, when the level of charged $tRNA^{Trp}$ is insufficient to support translation, the translation of the leader peptide is stalled at the tryptophan codons (Figure 11.13C). The stalling prevents the ribosome from proceeding into region 2, which is then free to pair with region 3. Pairing of regions 2 and 3 prevents formation of the terminator structure, so the complete *trp* mRNA molecule is made, including the coding sequences for the structural genes.

In summary, attentuation is a finetuning mechanism of regulation superimposed on the basic negative control of the *trp* operon:

> When charged tryptophan tRNA is present in amounts that support translation of the leader polypeptide, transcription is terminated, and the *trp* enzymes are not synthesized. When the level of charged tryptophan

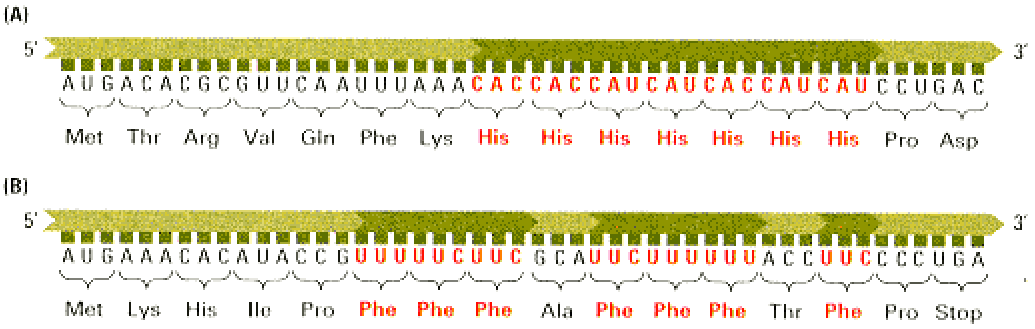

Figure 11.14
Amino acid sequence of the leader peptide and base sequence of the corresponding segment of mRNA from the histidine operon (A) and the phenylalanine operon (B). The repetition of these amino acids is emphasized in red letters.

tRNA is too low, transcription is not terminated, and the *trp* enzymes are made. At intermediate concentrations, the fraction of transcription initiation events that result in completion of *trp* mRNA depends on how frequently translation is stalled, which in turn depends on the intracellular concentration of charged tryptophan tRNA.

Many operons responsible for amino acid biosynthesis (for example, the leucine, isoleucine, phenylalanine, and histidine operons) are regulated by attenuators that function by forming alternative paired regions in the transcript. In the histidine operon, the coding region for the leader polypeptide contains seven adjacent histidine codons (Figure 11.14A). In the phenylalanine operon, the coding region for the leader polypeptide contains seven phenylalanine codons divided into three groups (Figure 11.14B). This pattern, in which codons for the amino acid produced by enzymes of the operon are present at high density in the leader peptide mRNA, is characteristic of operons in which attenuation is operative. Through these codons, the cell monitors the level of aminoacylated tRNA charged with the amino acid that is the end product of each amino acid biosynthetic pathway. Note that

> Attenuation cannot take place in eukaryotes because transcription and translation are uncoupled; transcription takes place in the nucleus and translation in the cytoplasm.

Regulation of the *lac* and *trp* operons exemplifies some of the important mechanisms that control transcription of genes in prokaryotes. In the following section, we will see that similar mechanisms are used in the control of genes in bacteriophages.

11.4— Regulation in Bacteriophage λ

When Jacob and Monod proposed the operon model and negative regulation by repression, they suggested that the model could account not only for regulation in inducible and repressible operons for metabolic enzymes but also for the lysogenic cycle of temperate bacteriophages. They proposed that λ bacteriophage was kept quiescent and prevented from replicating within bacterial lysogens by a repressor. This explanation ultimately proved to be correct, although the biochemical route to achievement of the repressed, lysogenic state is more complicated than was initially thought.

When λ bacteriophage infects *E. coli*, each infected cell can undergo one of two possible outcomes: (1) a lytic infection, resulting in lysis and production of phage particles, or (2) a lysogenic infection, resulting in integration of the λ molecule into the *E. coli* chromosome and formation of a lysogen. Because of this dichotomy, λ normally produces turbid (not completely clear) plaques on a lawn of *E. coli*. The initial infection and lysis do produce a cleared region in the bacterial lawn, but a few lysogens grow within the cleared region, partially repopulating the cleared zone and producing a turbid plaque.

Mutations in regulatory genes in λ were first identified in phage mutants that give clear rather than turbid plaques. The mutants proved to fall into four classes: λ*vir*, cI^-, cII^-, and $cIII^-$. The characteristics of these mutants are shown in Table 11.3. The genetic positions of the *cI*, *cII*, and *cIII* regions are shown in the simplified genetic map of λ bacteriophage in Figure 11.15, in which the genes are grouped by functional categories. Recall that, upon infection, the λ DNA molecule circularizes, bringing the *R* and *A* genes adjacent to one another.

Among the mutants in Table 11.3, the "clear" mutants proved to be analogous to *lacI* and *lacO* mutants in *E. coli*. The λ*vir* mutant is dominant to the wildtype λ^+ in mixed infection, as indicated by the combination of infecting phage designated 1 in Table 11.3. This combination of phage carries out a productive infection and prevents lysogeny by the wildtype λ^+. The λ*vir* mutant is therefore analogous to the $lacO^c$ mutation. However, λ*vir* proves to be a double mutant, bearing mutations in two different operators, O_L and O_R, as depicted in Figure 11.16. The cl^- mutations are recessive, as can be seen at entry 2 in Table 11.3. These cl^- mutations are analogous to $lacI^-$ mutations in that the cl^+ gene encodes the λ repressor, which is diffusible.

Table 11.3 Characteristics of mixed infections containing several combinations of λ*vir*, cI^- cII^-, and $cIII^-$ mutants

Infecting phages	Clear or turbid plaques
1. λ*vir* + λ^+	Clear
2. cI^- + cI^+	Turbid
3. cII^- + cII^+	Turbid
4. $cIII^-$ + $cIII^+$	Turbid

The cII^+ and $cIII^+$ genes encode not for repressor but rather for proteins needed in establishing lysogeny. The $cIII^-$ and $cIII^-$ mutations are also recessive in mixed infections (entries 3 and 4 in Table 11.3).

The molecular basis on which the decision between the lysogenic and the lytic cycle is determined is summarized in Figure 11.16. Upon infection of *E. coli* by λ, the λ molecule circularizes, and RNA polymerase binds at P_L and P_R and initiates transcription of the *N* and *cro* genes. N protein acts to prevent termination of the transcripts from P_L and P_R, allowing production of cII protein; cII protein activates transcription at P_E and P_I, thus allowing production of the cI and int proteins. The cI protein shuts down further transcription from P_L and P_R and stimulates transcription at P_M, increasing its own synthesis. Lysogeny is achieved if the concentration of cI protein reaches levels high enough to prevent transcription from P_L and P_R and to allow int protein to catalyze site-specific recombination between the circular λ molecule and the *E. coli* chromosome at their respective attachment (*att*) sites.

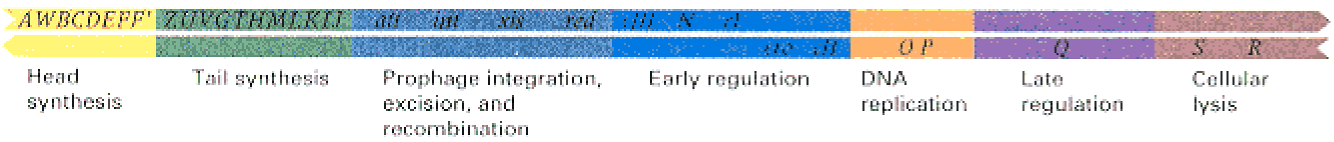

Figure 11.15
Genetic map of λ bacteriophage. The map is drawn to emphasize the functional organization of genes within the phage genome and to draw attention to the regulatory features. For a more detailed map, see Figure 8.24.

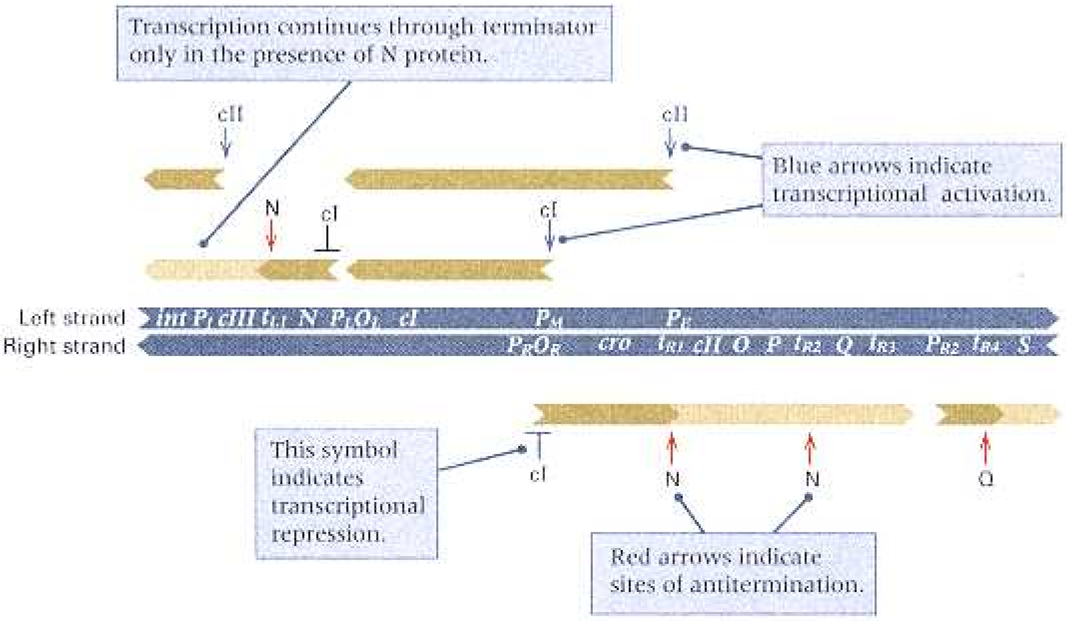

Figure 11.16
Genetic and transcriptional map of the control region of bacteriophage λ as expressed in the early stages of a lysogenic infection. The green arrows show the origin, direction, and extent of transcription. Light green arrows indicate portions of transcripts that are synthesized as a result of antitermination activity of the N or Q proteins. The sites of antitermination activity are indicated with red arrows pointing to the interior of a transcript. Blue arrows pointing to the origin of a transcript indicate transcriptional activation by cI or cII proteins; the sites of transcriptional repression by cI protein are indicated.

The alternative pathway to lysogeny, which leads to lytic development, takes place when the cro protein dominates. The cro protein also can bind to O_R and, in doing so, blocks transcription from P_M. If this occurs, then the concentration of repressor cannot rise to the levels required to block transcription from P_L and P_R. Transcription will continue from P_R and P_{R2}, and N and Q proteins will prevent termination in the rightward (and also leftward) transcripts. Because the λ DNA molecule is in a circular configuration, rightward transcription moves through genes *S* and *R* and thence through the head and tail genes (*A* through *J*, see Figures 11.15 and 11.16). The production of proteins needed for cellular lysis and formation of phage particles ensues, followed by phage assembly and cellular lysis to release phage.

In λ regulation, the cro and cI proteins compete for binding to O_L and O_R (each operator has three subsites that participate in the competition, but for our purposes this level of detail is unnecessary). If cro wins, the lytic cycle results in that cell; if cI wins, a lysogen is formed. Hence

> The cI and cro proteins function as a genetic switch; cI turns on lysogeny and cro turns on the lytic cycle.

The details that determine whether cI or cro controls the fate of a particular infection are quite complex. This complexity is apparent in Figure 11.17, where the major regulatory components and their interactions are shown in a form analogous to a wiring diagram used by electrical engineers. The key promoters are indicated in yellow, the proteins are encircled, and the interactions are shown in red. The multiple feedback loops and interactions are evident. The lesson from Figure 11.17 is that over the course of hundreds of millions of years, the regulation of even apparently

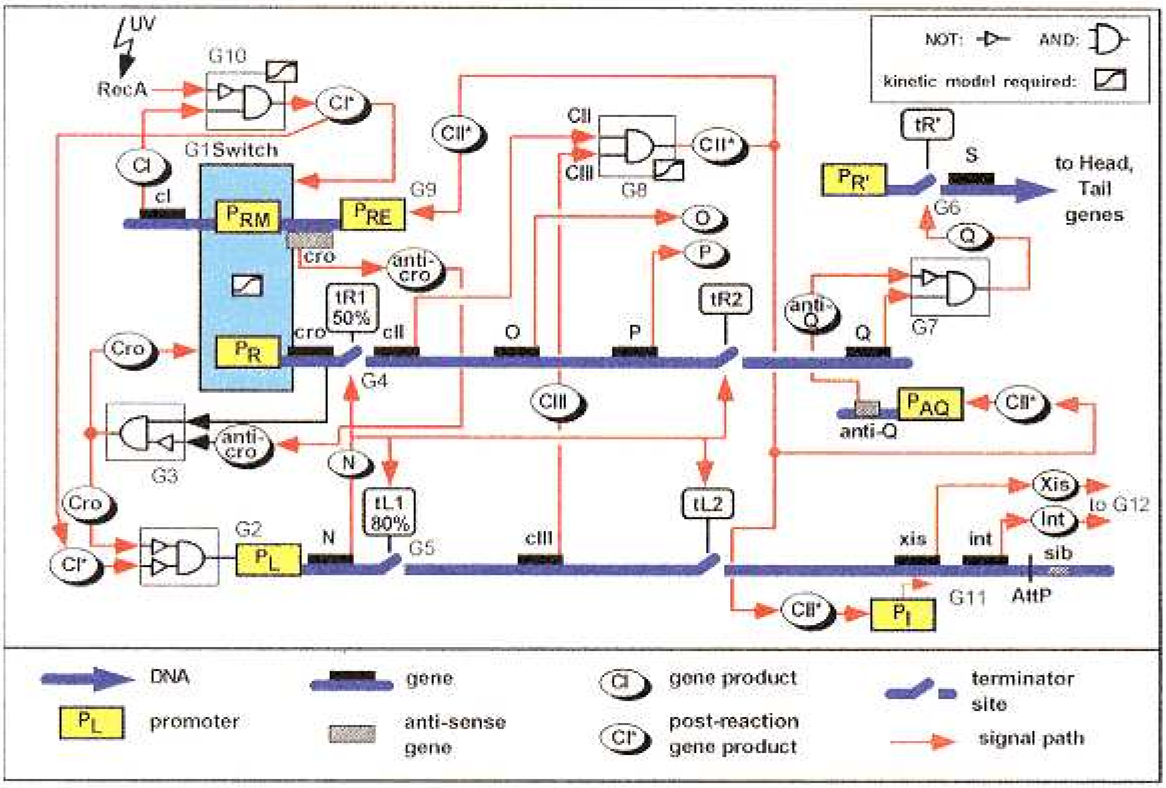

Figure 11.17
Genetic "circuit" determining the phage λ lysis-versus-lysogeny decision. The λ early promoters are shown in yellow, major protein components in circles, and the various regulatory interactions as red arrows. (Proteins CI, CII, and CIII are called cI, cII, and cIII in Figure 11.16.) The orientation of the operaons is for convenience in representation and does not reflect their orientation in the genome.
[Courtesy of Lucy Shapiro. From H. H. McAdams and L. Shapiro. 1995. *Science* 269:650.]

"simple" systems such as lysogeny in phage λ may evolve great complexity, ultimately comprising layer upon layer of checks and balances.

11.5— Regulation in Eukaryotes

Eukaryotic cells and organisms have different needs for regulation than prokaryotic cells. At the cellular level, eukaryotic cells are compartmentalized and can sequester and mobilize small molecules intracellularly, which can serve to damp environmental change. At the organismal level, multicellular eukaryotes have elaborate developmental programs and numerous specialized cell types. Within the organism, the environment of the cells may not change drastically in time. During development of the organism, cells differentiate for the following reasons:

• As a result of sequential changes in gene activity that are programmed in the genome

• In response to molecular signals released by other cells

• In response to physical contact with other cells

• In response to changes in the external environment

After cells have differentiated, they remain genetically quite stable, producing particular substances either at a constant rate or in response to external stimuli such as hormones, nutrient concentrations, or temperature changes.

The great complexity of multicellular eukaryotes requires a wide variety of genetic regulatory mechanisms. On the whole, these mechanisms are not understood as thoroughly as are those in prokaryotes. However, many important examples of different types of mechanisms have been studied in animals as diverse as mammals (especially the mouse), birds (usually the chicken), amphibians (toads of the genus *Xenopus*), insects (*Drosophila*), nematode worms (*Caenorhabditis elegans*), echinoderms (the sea urchin), and ciliates (*Tetrahymena*), as well as in yeast and other fungi. These examples reveal the general features of eukaryotic gene regulation that are discussed in the following sections.

Differences in Genetic Organization of Prokaryotes and Eukaryotes

Numerous differences exist between prokaryotes and eukaryotes with regard to transcription and translation, and in the spatial organization of DNA, as described in Chapters 6 and 10. Here are some of those most relevant to regulation:

1. In a eukaryote, usually only a single type of polypeptide chain can be translated from a completed mRNA molecule. Thus polycistronic mRNA of the type seen in prokaryotes is not found in eukaryotes.

2. The DNA of eukaryotes is bound to histones, forming chromatin, and to numerous nonhistone proteins. Only a small fraction of the DNA is bare. In bacteria, some proteins are present in the folded chromosome, but most of the DNA is free.

3. A significant fraction of the DNA of eukaryotes consists of moderately or highly repetitive nucleotide sequences. Some of the repetitive sequences are repeated in tandem copies, but others are not. Bacteria contain little repetitive DNA other than duplicated rRNA (ribosomal RNA) and tRNA genes and a few transposable elements.

4. A large fraction of eukaryotic DNA is untranslated; most of the nucleotide sequences do not code for proteins. Unicellular eukaryotes, such as yeast, are exceptions to this generalization, as are "lower" multicellular eukaryotes, such as *Drosophila* and *C. elegans,* and even certain vertebrates, such as the pufferfish, *Fugu rubripes,* with its relatively small (for a vertebrate) genome of 400 Mb.

5. Some eukaryotic genes are expressed and regulated by the use of mechanisms for rearranging certain DNA segments in a controlled way and for increasing the number of specific genes when needed.

6. Genes in eukaryotes are split into exons and introns, and the introns must be removed in the processing of the RNA transcript before translation begins.

7. In eukaryotes, mRNA is synthesized in the nucleus and must be transported through the nuclear envelope to the cytoplasm, where it is utilized. Bacterial cells do not have a nucleus separated from the cytoplasm.

We shall see in the following sections how some of these features are incorporated into particular modes of regulation.

11.6—
Alteration of DNA

Some genes in eukaryotes are regulated by alteration of the DNA. For example, certain sequences may be amplified or rearranged in the genome, or the bases may be chemically modified. Some of the alterations are reversible, but others permanently change the genome of the cells. However, the permanent changes take place only in somatic cells, so they are not genetically transmitted to the offspring through the germ line.

Gene Dosage and Gene Amplification

Some gene products are required in much larger quantities than others. One means of maintaining particular ratios of certain gene products (other than by differences in transcription and translation efficiency, as discussed earlier) is by **gene dosage.** For example, if two genes, *A* and *B*, are transcribed at the same rate and the translation efficiencies are the same, then 20 times as much of product A can be made as of product B if there are 20 copies of gene *A* per copy of gene *B.* The histone genes exemplify a gene-dosage effect: To synthesize the huge amount of histone required to form chromatin, most cells contain hundreds of times as many copies of histone genes as of genes required for DNA replication. In this case, the high expression is automatic because the repeated genes are part of the normal chromosome complement.

In some cases, gene dosage is increased temporarily by a process called **gene amplification,** in which the number of genes increases in response to some signal. An example of gene amplification is found in the development of the oocytes of the toad *Xenopus laevis.* The formation of an egg from its precursor, the oocyte, is a complex process that requires a huge amount of protein synthesis. To achieve the necessary rate, a very large number of ribosomes are needed. Ribosomes contain molecules of rRNA, and the number of rRNA genes in the genome is insufficient to produce the required number of ribosomes for the oocyte in a reasonable period of time. In the development of the occyte, the number of rRNA genes increases by about 4000-fold. The precursor to the oocyte, like all somatic cells of the toad, contains about 600 rRNA-gene (rDNA) units; after amplification, about 2×10^6 copies of each unit are present. This large amount enables the oocyte to synthesize 10^{12} ribosomes, which are required for the protein synthesis that occurs later during early development of the embryo, at a time when no ribosomes are being formed.

Before amplification, the 600 rDNA units are arranged in tandem. During amplification, which occurs over a 3-week period in which the oocyte develops from a precursor cell, the rDNA no longer consists of a single contiguous DNA segment containing 600 rDNA units but instead forms a large number of small circles and replicating rolling circles. The rolling-circle replication accounts for the increase in the number of copies of the genes. The precise mechanism of excision of the circles from the chromosome and formation of the rolling circles is not known.

When the occyte is mature, no more rRNA needs to be synthesized until well after fertilization and into early development, at which time 600 copies are sufficient. The excess rDNA serves no purpose and is slowly degraded by intracellular enzymes. Following fertilization, the chromosomal DNA replicates and mitosis ensues, occurring repeatedly as the embryo develops. During this period, the extra chromosomal rDNA does not replicate; degradation continues, and by the time several hundred cells have formed, none of this extra rDNA remains. Amplification of rRNA genes during oogenesis occurs in many organisms, including insects, amphibians, and fish.

Some protein-coding genes also undergo amplification. For example, in *Drosophila* females, the genes that produce chorion proteins (a component of the sac that encloses the egg) are amplified in follicle cells just before maturation of the egg. The amplification enables the cells to produce a large amount of protein in a short time. In some cases, amplification occurs in abnormal regulation. For example, a gene called N-*myc* is frequently amplified in human tumor cells in the disease neuroblastoma, and the degree of amplification is correlated with progress of the disease and tendency of the tumor to spread. N-*myc* is the normal cellular counterpart of a viral oncogene. (Oncogenes are discussed in Section 7.8).

Programmed DNA Rearrangements

Rearrangement of DNA sequences in the genome is an unusual but important mechanism by which some genes are regulated. An example is the phenomenon known as **mating-type interconversion** in yeast. As we saw in Chapter 4, yeast has two mating types, denoted **a** and α. Mating between haploid **a** and haploid α cells produces the **a**α diploid, which can undergo meiosis to produce four-spored asci that contain haploid **a** and α spores in the ratio 2: 2. If a single yeast spore of either the **a** or the α genotype is cultured in isolation from other spores, then mating between progeny cells would not be expected because the progeny cells would have the mating type of the original parent. However, *S. cerevisiae* has a mating system called **homothallism,** in which some cells undergo a conversion into the opposite mating type that allows matings between cells in what would otherwise be a pure culture of one mating type or the other.

The outlines of mating-type interconversion are shown in Figure 11.18. An original haploid spore (in this example, α) undergoes germination to produce two progeny cells. Both the mother cell (the original parent) and the daughter cell have mating-type α, as expected from a normal mitotic division. However, in the next cell division, a switching (interconversion) of mating type takes place in both the mother cell and its *new* progeny cell, in which the original α mating type is replaced with the **a** mating type. After this second cell division is complete, the α and **a** cells are able

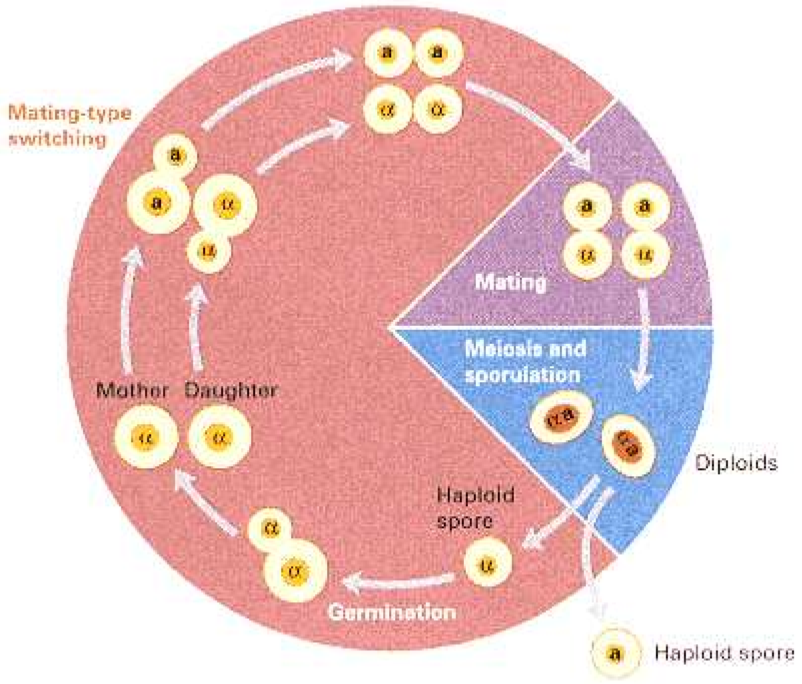

Figure 11.18
Mating-type switching in the yeast Saccharomyces cerevisiae. Germination of a spore (in this example, one of mating type α) forms a mother cell and a bud that grows into a daughter cell. In the next division, the mother cell and its new daughter cell switch to the opposite mating type (in this case, a). The result is two α and two a cells. Cells of opposite mating type can fuse to form aα diploid zygotes. In a similar fashion, germination of an a spore is accompanied by switching to the α mating type.

to undergo mating because they now are of opposite mating types. Fusion of the nuclei produces the **a**α diploid, which undergoes mitotic divisions and later sporulation to again produce **a** and α haploid spores.

The genetic basis of mating-type inter-conversion is DNA rearrangement as outlined in Figure 11.19. The gene that controls mating type is the *MAT* gene in chromosome III, which can have either of two allelic forms, **a** or α. If the allele in a haploid cell is *MAT***a,** then the cell has mating-type **a;** if the allele is *MAT*α, then the cell has mating-type α. However, both genotypes normally contain both **a** and α genetic information in the form of unexpressed **cassettes** present in the same chromosome. The *HML*α cassette contains the α DNA sequence about 200 kb away from the *MAT* gene, and the *HMR***a** cassette contains the **a** DNA sequence about 150 kb away from *MAT* on the other side. (Figure 11.19 shows the relative positions of the genes in the chromosome.) When mating-type interconversion occurs, a specific endonuclease, encoded by the *HO* gene elsewhere in the genome, is produced and cuts both strands of the DNA in the *MAT* region. The double-stranded break initiates a process in which genetic information in the unexpressed cassette that contains the opposite mating type becomes inserted into *MAT.* In this process, the DNA sequence in the donor cassette is duplicated, so the mating type becomes converted, but the same genetic information is retained in unexpressed form in the cassette. The terminal regions of *HML*, *MAT,* and *HMR* are identical (illustrated in light blue and dark blue in Figure 11.19), and these regions are critical in making possible recognition of the regions for interconversion. The unique part of the α region is 747 base pairs in length; that of the **a** region is 642 base pairs long. The molecular details of the conversion process are similar to those of the double-strand gap mechanism of recombination, which is discussed in Chapter 13.

Figure 11.19 illustrates two sequential mating-type interconversions. In the first, an α cell (containing the *MAT*α allele) undergoes conversion into **a,** using the DNA sequence contained in the *HMR***a** cassette. The converted cell has the genotype

*MAT***a.** In a later generation, a descendant **a** cell may become converted into mating-type α, using the unexpressed DNA sequence contained in *HML*α. This cell has the genotype *MAT*α. Mating-type switches can occur repeatedly in the lineage of any particular cell.

Antibodies and Antibody Variability

Another important example of programmed DNA rearrangement takes place in vertebrates in cells that form the immune system. In this case, the precursor cells contain numerous DNA sequences that can serve as alternatives for various regions in the final gene. In the maturation of each cell, a combination of the alternatives is created by DNA cutting and rejoining, producing a great variety of possible genes that enable the immune system to recognize and attack most bacteria and viruses.

It has been estimated that a normal mammal is capable of producing more than 10^8 different antibodies, each of which can combine specifically with a particular antigen. Antibodies are proteins, and each unique antibody has a different amino acid sequence. If antibody genes were conventional in the sense that each gene codes for a single polypeptide, then mammals would need more than 10^8 genes for the production of antibodies. This is considerably more genes than are present in the entire genome. In fact, mammals use only a few hundred genes for antibody production, and the huge number of different antibodies derives from remarkable events that take place in the DNA of certain somatic cells. These events are discussed in this section.

Although an individual organism is capable of producing a vast number of different antibodies, only a fraction of them are synthesized at any one time. Antibodies are produced by a type of white blood cell called a B cell. Each B cell can produce a single type of antibody, but the antibody is not secreted until the cell has been stimulated by the appropriate antigen. Once stimulated, the B cell undergoes successive mitoses and eventually produces a clone of identical cells that secrete the antibody. Moreover, antibody secretion may continue even if the antigen is no longer pres-

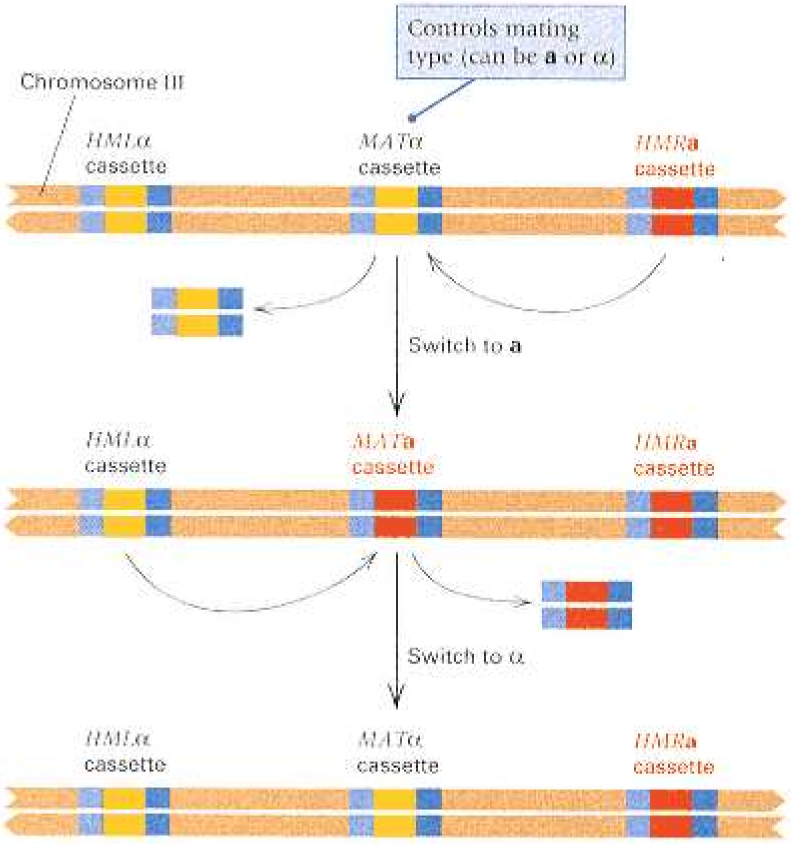

Figure 11.19
Genetic basis of mating-type interconversion. The mating type is determined by the DNA sequence present at the *MAT* locus. The *HML* and *HMR* loci are cassettes that contain unexpressed mating-type genes, either α or a. In the interconversion from α to a, the α genetic information present at *MAT* is replaced with the a genetic information from *HMR*a. In the switch from a to α, the a genetic information at *MAT* is replaced with the α genetic information from *HML*α.

ent. In this manner, organisms produce antibodies only to the antigens to which they have been exposed.

The five distinct classes of antibodies known are designated IgG, IgM, IgA, IgD, and IgE (Ig stands for **immunoglobulin**). These classes serve specialized functions in the immune response and exhibit certain structural differences. However, each contains two types of polypeptide chains differing in size: a large one called the **heavy (H) chain** and a small one called the **light (L) chain.**

Immunoglobulin G (IgG) is the most abundant class of antibodies and has the simplest molecular structure. Its molecular organization is illustrated in Figure 11.20.

Connection Sex-Change Operations

James B. Hicks, Jeffrey N. Strathern, and Ira Herskowitz, University of Oregon, Eugene, Oregon 1977

The Cassette Model of Mating-type Interconversion

*Mating in yeast requires cells of opposite mating types, **a** and α, to come together and fuse. Both **a** and α cells release signaling substances into the medium that prepare the opposite cell type for mating. In the **a**α diploid cell, genes specific for the diploid phase of the life cycle are expressed, and those specific for the haploid phase of the life cycle are turned off. Remarkably, yeast cells can change their mating type. In homothallic cells, the switch can take place in every generation. In heterothallic cells, it takes place at a frequency of 10^{-6}. This paper proposed a very bold hypothesis, later confirmed experimentally, that all yeast cells contain, in addition to the information for the expressed mating type at the* MAT *locus, both α and **a** genetic information in unexpressed cassettes at* HML *and* HMR *genetically linked to, but distinct from,* MAT. *The* HO *gene that distinguishes homothallic from heterothallic yeast codes for a site-specific endonuclease that cleaves within the mating-type locus and initiates the information-transfer process from either* HMLα *or* HMR***a**. *This is the physical basis of mating-type interconversion.*

Studies of mating-type interconversion in the yeast *Saccharomyces cerevisiae* have led us to propose a new mechanism of gene control involving mobile genes. . . . The mating-type locus of *S. cerevisiae* exists in two states, **a** or α, which control the ability of yeast cells to mate and sporulate. . . . It is clear that the **a** and α alleles are distinct entities, as they are codominant—an **a**/α diploid differs from a [rarely formed] **a**/**a** or α/α diploid. . . . The **a** allele

The mating-type locus is viewed as analogous to a playback head of a tape recorder which can give expression to whatever cassette of information is plugged into it.

thus is not simply the absence of α, and the α allele is not simply the absence of **a.** . . . In homothallic strains, changes to opposite mating types occur frequently, as often as every generation. These strains carry a dominant nuclear gene (*HO*), unlinked to the mating-type locus. . . . *HO* cells that have sustained a change in mating type are fully capable of continuing to change mating type. However, when the *HO* gene is removed by genetic crosses, the new mating type is stable . . . Cells with a defect at the α mating-type locus can be converted to functional **a** cells. However, these **a** cells are then observed to switch to become functional α cells. In other words, a functional α mating-type locus can be restored through the mating-type locus can be restored through the mating-type interconversion process. We explain this recovery by proposing that yeast cells contain an additional copy (or copies) of the mating-type locus information. Specifically, we propose that yeast cells contain a silent (unexpressed) copy of **a** information and a silent copy of α information and that the *HO* gene activates this information by inserting it (or a copy) into the mating-type locus. Genetic studies have revealed the existence of two loci [*HML*α and *HMR**a***] in addition to *HO* which are necessary for mating-type inter-conversion and which we propose are the silent α and **a** information. . . . To summarize, we propose that cell type in *S. cerevisiae* is regulated by a locus [*MAT*] into which various blocs of information can be inserted. The mating-type locus is viewed as analogous to a playback head of a tape recorder which can give expression to whatever cassette of information is plugged into it.

Source: DNA Insertion Elements, Plasmids, and Episomes, eds. Ahmad I. Bukhari, James A. Shapiro, and Sankar L. Adhya. Cold Spring Harbor, New York: Cold Spring Harbor Laboratory, pp. 457–462.

An IgG molecule consists of two heavy and two light chains held together by disulfide bridges (two joined sulfur atoms) and has the overall shape of the letter Y. The sites on the antibody that carry its specificity and combine with the antigen are located in the upper half of the arms above the fork of the Y. Each IgG molecule with a different antigen specificity has a different amino acid sequence for the heavy and light chains in this part of the molecule. These specificity regions are called the **variable regions** (blue pointers in Figure 11.20) of the heavy and light chains.

The remaining regions of the polypeptide are the **constant regions,** which are called constant because they have virtually the same amino acid sequence in all IgG molecules.

Initial understanding of the genetic mechanisms responsible for variability in the amino acid sequences of antibody polypeptide chains came from cloning a gene for the light chain of IgG. The critical observation was made by comparing the nucleotide sequence of the gene in embryonic cells or germ cells with that in mature antibody-producing cells. In the genome of a B cell that was actively producing the antibody, the DNA segments corresponding to the constant and variable regions of the light chain were found to be very close together, as expected of DNA that codes for different parts of the same polypeptide. However, in embryonic cells, these same DNA sequences were located far apart. Similar results were obtained for the variable and constant regions of the heavy chains: Segments encoding these regions were close together in B cells but widely separated in embryonic cells.

Extensive DNA sequencing of the genomic region that codes for antibody proteins revealed not only the reason for the different gene locations in B cells and germ cells but also the mechanism for the origin of antibody variability. Cells in the germ line contain a small number of genes corresponding to the constant region of the light chain, which are close together along the DNA. Separated from them, but on the same chromosome, is another cluster consisting of a much larger number of genes that correspond to the variable region of the light chains. In the differentiation of a B cell, one gene for the constant region is spliced (cut and joined) to one gene for the variable region, and this splicing produces a complete light-chain antibody gene. A similar splicing mechanism yields the constant and variable regions of the heavy chains.

The formation of a finished antibody gene is slightly more complicated than this description implies, because light-chain genes consist of three parts and heavychain genes consist of four parts. Gene splicing in the origin of a light chain is illustrated in Figure 11.21. For each of two parts of the variable region, the germ line contains multiple coding sequences called the **V (variable)** and **J (joining)** regions. In the differentiation of a B cell, a deletion makes possible the joining of one the V regions with one of the J regions. The DNA joining process is called **combinatorial joining** because it can create many combinations of the V and J regions. When transcribed, this joined V-J sequence forms the 5' end of the light-chain RNA transcript. Transcription continues on through the DNA region coding for the constant (C) part of the gene. RNA splicing subsequently attaches the C region, creating the light-chain mRNA.

Combinatorial joining also takes place in the genes for the antibody heavy chains.

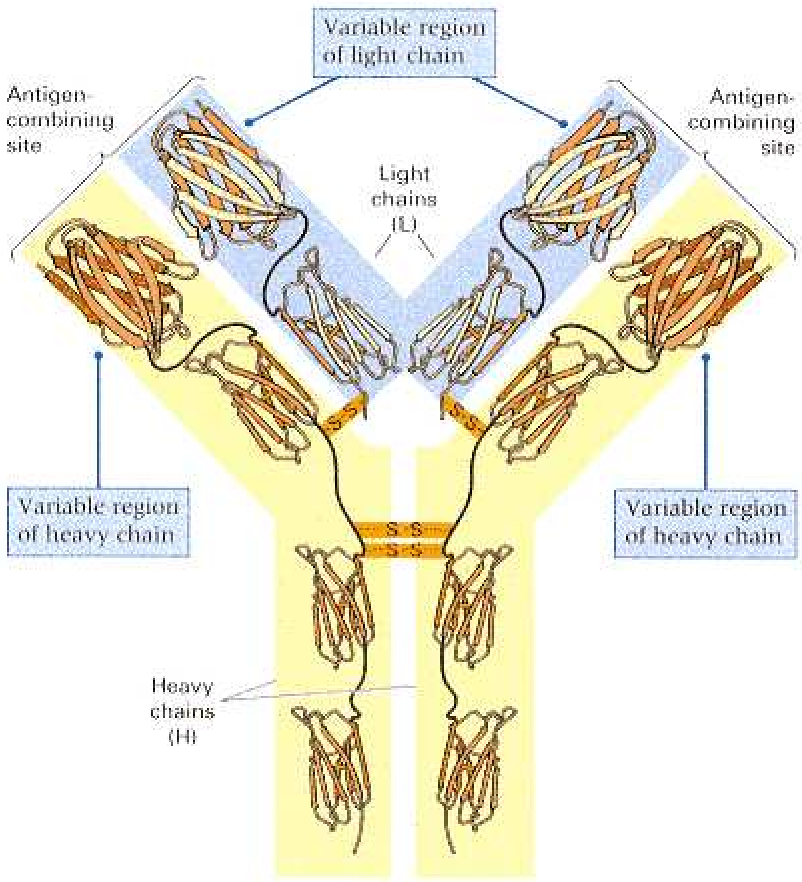

Figure 11.20
Structure of the immunoglobulin G (IgG) molecule showing the light chains (L, shaded blue) and heavy chains (H, shaded yellow). Variable and constant regions are indicated.

In this case, the DNA splicing joins the heavy-chain counterparts of V and J with a third set of sequences, called D (for diversity), located between the V and J clusters.

The amount of antibody variability that can be created by combinatorial joining is calculated as follows: In mice, the light chains are formed from combinations of about 250 V regions and 4 J regions, giving 250 × 4 = 1000 different chains. For the heavy chains, there are approximately 250 V, 10 D, and 4 J regions, producing 250 × 10 × 4 = 10,000 combinations. Because any light chain can combine with any heavy chain, there are at least 1000 × 10,000 = 10^7 possible types of antibodies. The number of DNA sequences used for antibody production is quite small (about 500), but the number of possible antibodies is very large.

The value of 10^7 different antibody types is an underestimate, because there

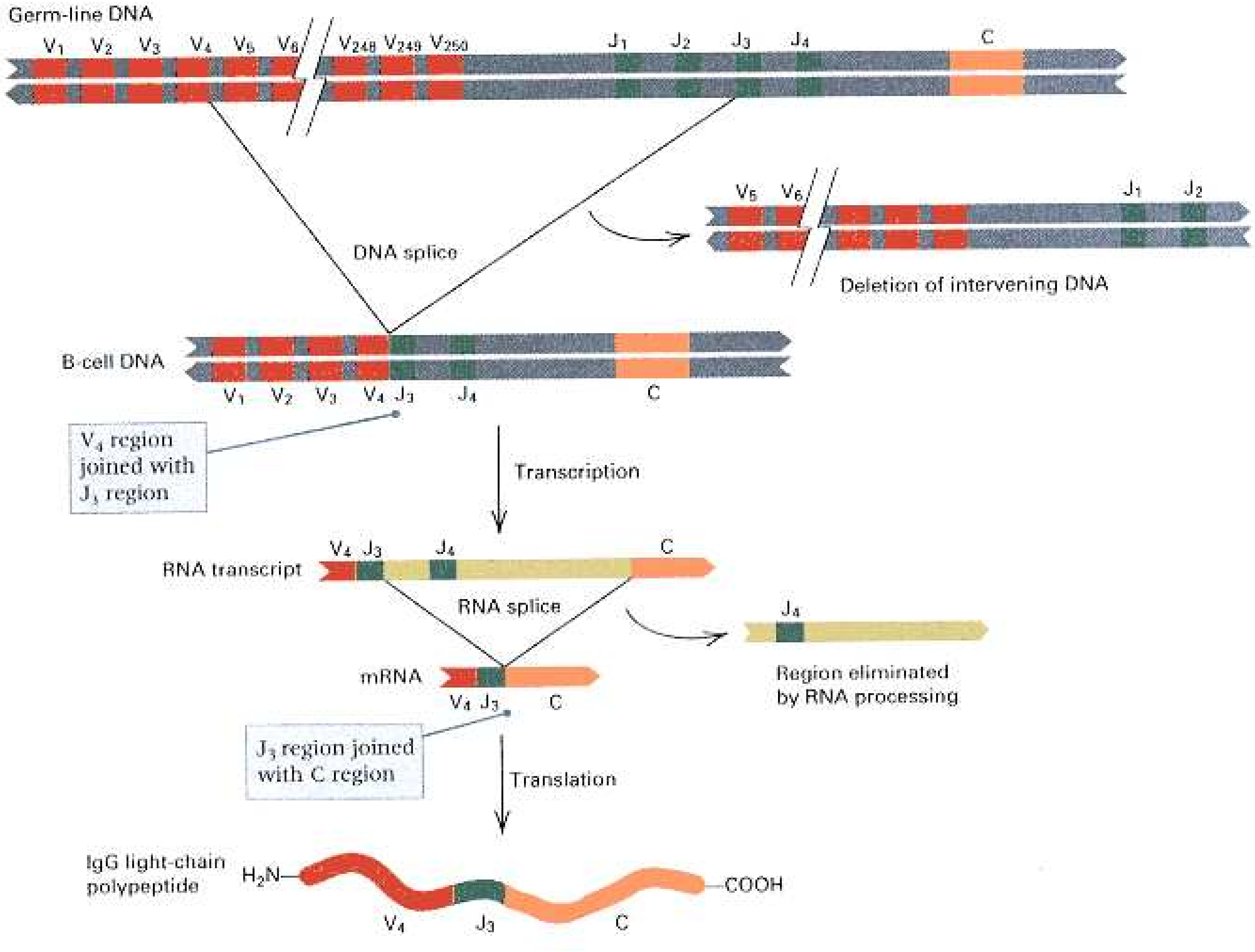

Figure 11.21
Formation of a gene for the light chain of an antibody molecule. One variable (V) region is joined with one randomly chosen J region by deletion of the intervening DNA.
The remaining J regions are eliminated from the RNA transcript during RNA processing.

are two additional sources of antibody variability:

1. The junction for V-J (or V-D-J) splicing in combinatorial joining can be between different nucleotides of a particular V-J combination in light chains (or a particular V-D or D-J combination in heavy chains). The different splice junctions can generate different codons in the spliced gene. For example, a particular combination of V and J sequences can be spliced in five different ways. At the splice junction, the V sequence contains the nucleotides CATTTC, and the J sequence contains CTGGGTG. The splicing event determines the codons for amino acids 97, 98, and 99 in the completed antibody light chain, and it can occur in any of the following ways, the last two of which result in altered amino acid sequences:

Spliced sequence	**97**	**98**	**99**
CA**CTGGGTG**	His	Trp	Val
CAT**TGGGTG**	His	Trp	Val
CATT**GGGTG**	His	Trp	Val
CATTT**GGTG**	His	Leu	Val
CATTTG**GTG**	His	Phe	Val

In this manner, variability in the junction of V-J joining can result in polypeptides that differ in amino acid sequence.

2. The V regions are susceptible to a high rate of *somatic mutation*, which occurs in B-cell development. These mutations allow different B-cell clones to produce different polypeptide sequences, even if they have undergone exactly the same V-J joining. The mechanism for this high mutation rate is unknown.

Gene Splicing in the Origin of T-Cell Receptors

Immunity is also mediated by a different type of white blood cell called a T cell. A T cell carries an antigen receptor on its surface that combines with an antigen, stimulating the T cell to respond. Like the antibody molecules produced in B cells, the T-cell receptors are highly variable in amino acid sequence, enabling the T cells to respond to many antigens. Although the polypeptide chains in T-cell receptors are different from those in antibody molecules, they have a similar organization in that they are formed from the aggregation of two pairs of polypeptide chains. A particular T cell may carry either of two types of receptors. The majority carry the $\alpha\beta$ receptor, composed of polypeptide chains designated α and β, and the rest carry the $\gamma\delta$ receptor, composed of chains designated γ and δ. Each receptor polypetide includes a variable region and a constant region. As their variability and similarity in organization suggest, the T-cell receptor genes are formed by somatic rearrangement of components analogous to those of the V, D, J, and C regions in the B cells. For example, in the mouse, the β chain of the T-cell receptor is spliced together from one each of approximately 20 V regions, 2 D regions, 12 J regions, and 2 C regions. Note that there are far fewer V regions for T-cell receptor genes than there are for antibody genes, yet T-cell receptors seem able to recognize just as many foreign antigens as B cells. The extra variation results from a higher rate of somatic mutation in the T- cell receptor genes.

DNA Methylation

In most eukaryotes, a small proportion of the cytosine bases are modified by the addition of a methyl (CH_3) group to the number-5 carbon atom (Figure 11.22). The cytosines are incorporated in their normal, unmodified form in the course of DNA replication, but they are modified later by an enzyme called a **DNA methylase.** Cytosines are modified preferentially in 5'-CG-3' dinucleotides. When a CG dinucleotide that is methylated in both strands undergoes DNA replication, the result is two daughter molecules, each of which contains one parental strand with a methylated CG and one daughter strand with an unmethylated CG. The DNA methylase recognizes the half-methylation in these molecules and methylates the cytosines in the daughter strands. Methylation of CG dinucleotides in the sequence CCGG can easily be detected by the use of the restriction enzymes *MspI* and *HpaII.* Both enzymes cleave the sequence CCGG. However, *MspI* cleaves regardless of whether the interior C is methylated, whereas *HpaII* cleaves only unmethylated DNA. Therefore, *MspI* restriction sites that are not cleaved by *HpaII* are sites at which the interior C is methylated (Figure 11.23).

Many eukaryotic genes have CG-rich regions upstream of the coding region, providing potential sites for methylation that may affect transcription. A number of observation suggest that high levels of methylation are associated with genes for which the rate of transcription is low. One example is the inactive X chromosome in mammalian cells, which is extensively methylated. Another example is the *Ac* transposable element in maize. Certain *Ac* elements lose activity of the transposase gene without any change in DNA sequence. These elements prove to have heavy methylation in a region particularly rich in the CG dinucleotides. Return to normal activity of the methylated *Ac* elements coincides with loss of methylation through the action of demethylating enzymes in the nucleus.

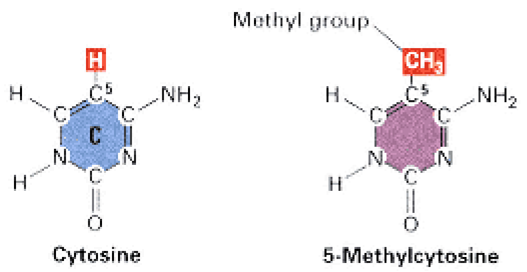

Figure 11.22
Structures of cytosine and 5-methylcytosine.

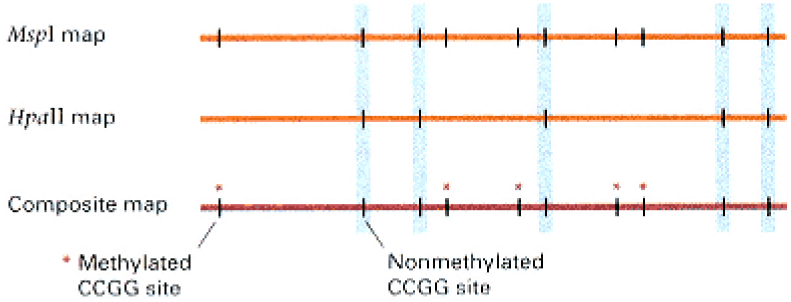

Figure 11.23
Detection of methylated cytosines in CCGG sequences by means of restriction enzymes. The enzyme *MspI* cleaves all CCGG sites regardless of methylation, whereas *HpaII* cleaves only nonmethylated sites. The positions of the methylated sites are determined by comparing the restriction maps.

Although there is a correlation between methylation and gene inactivity, it is possible that heavy methylation is a result of gene inactivity rather than a cause of it. However, treatment of cells with the cytosine analog *azacytidine* reverses methylation and can restore gene activity. For example, some clones of rat pituitary tumor cells express the gene for prolactin, whereas other related clones do not. The gene is methylated in the nonproducing cells but is not methylated in the producers. Reversal of methylation in the nonproducing cells with azacytidine results in prolactin expression. On the other hand, not all organisms exhibit methylation. For example, *Drosophila* DNA is not methylated. In organisms in which the DNA is methylated, methylation increases susceptibility to certain kinds of mutations, which are discussed in Chapter 13.

11.7—
Transcriptional Regulation in Eukaryotes

Many eukaryotic genes code for essential metabolic enzymes or cellular components and are expressed constitutively at relatively low levels in all cells; they are called **housekeeping genes.** The expression of other genes differs from one cell type to the next or among different stages of the cell cycle; these genes are often regulated at the level of transcription. In prokaryotes, the levels of expression in induced and uninduced cells may differ by a thousandfold or more. Such extreme levels of induction are uncommon in eukaryotes, except for some genes in lower eukaryotes such as yeast. Most eukaryotic genes are induced by factors ranging from 2 to 10.

In this section, we consider some components of transcriptional regulation in eukaryotes.

Galactose Metabolism in Yeast

We will introduce transcriptional regulation in eukaryotes by examining the control of galactose metabolism in yeast and comparing it with the *lac* operon in *E. coli*. The first steps in the biochemical pathway for galactose degradation are illustrated in Figure 11.24. Three enzymes, encoded by

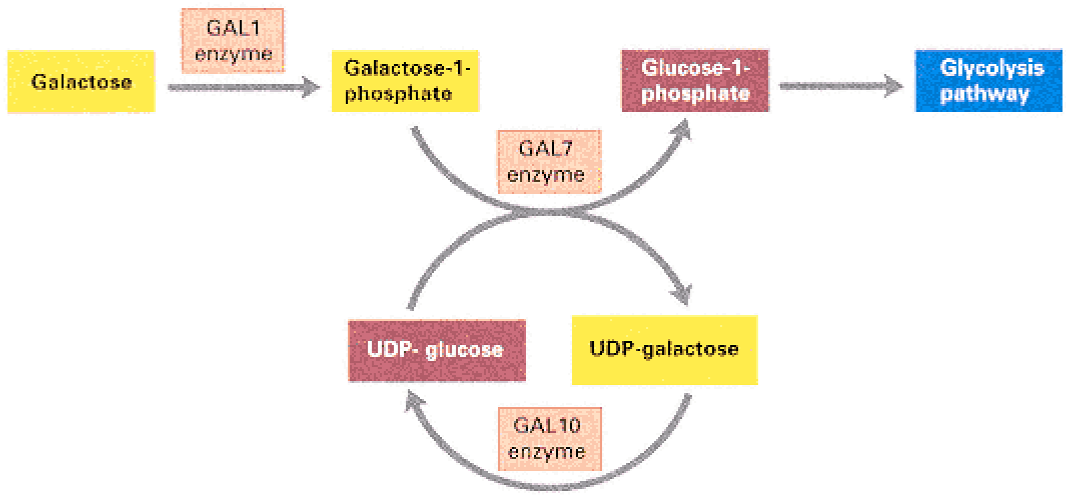

Figure 11.24
Metabolic pathway by which galactose is converted to glucose- 1 -phosphate in the yeast *Saccharomyces cerevisiae.*

Figure 11.25
The linked *GAL* genes of *Saccharomyces cerevisiae.* Arrows indicate the transcripts produced. The *GAL1* and *GAL10* transcripts come from divergent promoters, *GAL7* from its own promoter.

the genes *GAL1, GAL7,* and *GAL10,* are required for conversion of galactose to glucose-1-phosphate. These three genes are tightly linked, as shown in Figure 11.25. Despite the tight linkage of the three genes, the genes are not part of an operon; the mRNAs are monocistronic. The *GAL1* and *GAL10* mRNAs are synthesized from divergent promoters lying between the genes, and *GAL7* mRNA is synthesized from its own promoter. The mRNAs are synthesized only when galactose is present as inducer; the genes are thus inducible.

Constitutive and uninducible mutants have been observed. In two types of mutants, *gal80* and *GAL81*^{c}, the mutants synthesize *GAL1, GAL7,* and *GAL10* mRNAs constitutively. Another type of mutant, *gal4*, does not synthesize the mRNAs whether or not galactose is present; it is uninducible. The characteristics of the mutants are shown in Table 11.4. The terms *cis* and *trans* are of no help in interpreting these results, for the regulatory genes are unlinked to the genes they regulate: *GAL1, GAL7* and *GAL10* are on chromosome II, *GAL80* is on chromosome XIII, and *GAL4* and *GAL81* are on chromosome XVI.

The *gal80* mutation is recessive (entry 1 of Table 11.4). Thus, superficially, it behaves like a *lacI*$^-$ mutation. The wildtype *GAL80* allele does indeed encode a protein, called a "repressor," that is a negative regulator of transcription. However, the *GAL80* protein acts not by binding to an operator but by binding to, and inactivating, a transcriptional activator protein. The activator is the product of the *GAL4* gene.

The wildtype *GAL4* allele encodes a protein that is required for transcription of the three *GAL* genes. The *gal4* mutation is therefore recessive. In the absence of the GAL4 protein, the *GAL* genes are all uninducible. The GAL4 protein is a positive regulatory protein that activates transcription of the three *GAL* genes. However, it does so by activating transcription of three different mRNAs starting at three different sites upstream of each of the activated genes. The GAL4 protein bound with its target site in the DNA is shown in Figure 11.26, in which the GAL4 protein (a dimer) is shown in blue and the DNA molecule in red. The small yellow spheres represent ions of zinc, which are essential components in the DNA binding.

The GAL80 "repressor" protein acts by binding to GAL4 protein and sequestering it so that GAL4 is not free activate transcription. The inducer (galactose) eliminates the ability of GAL80 protein to bind to GAL4 freeing the GAL4 protein to activate transcription.

The constitutive mutation, *GAL81*^{c}, is dominant; it results in constitutive synthesis of all three mRNAs. However, because it does not map near *GAL1, GAL7,* and *GAL10* and is a single mutation, it cannot be an operator mutation comparable to *lacO*^{c}. The *GAL81*^{c} mutation does not define a separate regulatory gene but instead maps to a position within the *GAL4* gene. The mutation gives rise to a GAL4 protein that no longer binds to GAL80 protein. Hence the GAL4 protein produced by the *GAL81*^{c} allele cannot be sequestered by GAL80 and is able to activate transcription in the absence of galactose, whether or not wildtype GAL4 protein is also present.

The main point of these comparisons is that the superficial similarity between the constitutive and uninducible mutations in the *lac* operon of *E. coli* and the *GAL* genes of yeast are not indicative of similar molecular regulatory mechanisms. However, some physiological similarities remain in

Table 11.4 Characteristics of diploids containing various combinations of *gal180, gal4,* and *GAL81*^{c} mutations.

Genotype	Synthesis of *GAL1. GAL7,* and *GAL10* mRNAs	Gal phenotype
1. *ga180 GAL1/GAL80 GAL 1*	Inducible	+
2. *gal4 GAL1/GAL4 GAL 1*	Inducible	+

3. *GAL81*^{c} *GAL1*/*GAL 81 GAL1*	Constitutive	+

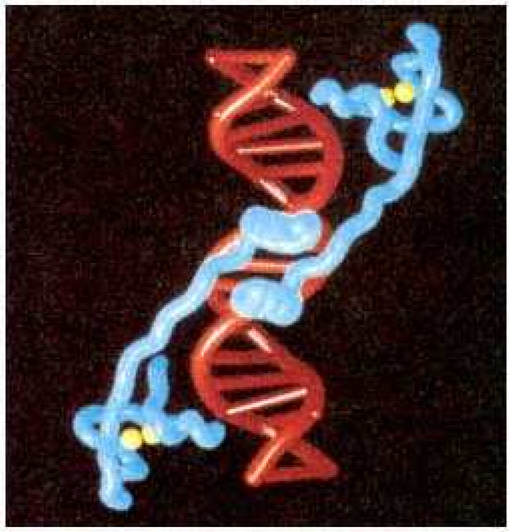

Figure 11.26
Three-dimensional structure of the GAL4 protein (blue) bound to DNA (red). The protein is composed of two polypeptide subunits held together by the coiled regions in the middle. The DNA-binding domains are at the extreme ends, and each physically contacts three base pairs in the major groove of the DNA. The zinc ions in the DNA-binding domains are shown in yellow.
[Courtesy of Dr. Stephen C. Harrison. See also R. Marmorstein, M. Carey, M. Ptashne, and S. C. Harrison. 1992. DNA recognition by GAL4: Structure of a protein-DNA complex. *Nature* 356: 408–414.]

that, in both prokaryotes and eukaryotes, the genes for a particular metabolic (or developmental) pathway are expressed in a coordinated manner in response to a signal. The principle at work is that alternative molecular mechanisms can be employed to achieve similar ends.

Yeast Mating Type

As mentioned earlier, the mating type of a yeast cell is controlled by the allele of the *MAT* gene that is present (refer to Figure 11.19 for the genetic basis of mating-type interconversion in yeast). Both *MAT***a** (mating-type **a**) and *MAT*α (mating-type α) express a set of haploid-specific genes. They differ in that *MAT***a** expresses a set of **a**-specific genes and *MAT*α expresses a set of α-specific genes. The haploid-specific genes that cells of both mating types express include *HO,* which encodes the HO endonuclease used in mating-type interconversion, and *RME1,* which encodes a repressor of meiosis-specific genes. The functions of the mating-type-specific genes include (1) secretion of a mating peptide that arrests cells of the opposite mating type before DNA synthesis and prepares them for cell fusion, and (2) production of a receptor for the mating peptide secreted by the opposite mating type. Therefore, when **a** and α cells are in proximity, they prepare each other for mating and undergo fusion.

Regulation of mating type is at the level of transcription according to the regulatory interactions diagrammed in Figure 11.27. These regulatory interactions were originally proposed on the basis of the phenotypes of various types of mutants, and most of the details have been confirmed by direct molecular studies. The symbols **a**sg, αsg, and *hsg* represent the **a**-specific genes, the α-specific genes, and the haploid- specific genes, respectively; each set of genes is represented as a single segment (lack of a "sunburst" indicates that transcription does not take place). In a cell of mating-type **a** (Figure 11.27A), the *MAT***a** region is transcribed and produces a polypeptide called **a**1. By itself, **a**1 has no regulatory activity, and in the absence of any regulatory signal, **a**sg and *hsg* are transcribed, but not αsg. In a cell of mating- type α (Figure 11.27B), the *MAT*α region is transcribed, and two regulatory proteins denoted α1 and α2 are produced: α1 is a *positive regulator* of the α-specific genes, and α2 is a *negative regulator* of the **a**-specific genes. The result is that αsg and *hsg* are transcribed, but transcription of **a**sg is turned off. Both α1 and α2 bind with particular DNA sequences upstream from the genes that they control.

In the diploid (Figure 11.27C), both *MAT***a** and *MAT*α are transcribed, but the only polypeptides produced are **a**1 and

$\alpha 2$. The reason is that the **a**1 and $\alpha 2$ polypeptides combine to form a negative regulatory protein that represses transcription of the $\alpha 1$ gene in *MAT*α and of the haploid-specific genes. The $\alpha 2$ polypeptide acting alone is a negative regulatory protein that turns off **a**sg. Because $\alpha 1$ is not produced, transcription of αsg is not turned on. In sum, then, the α*sg* are not turned on because $\alpha 1$ is absent, the **a**sg are turned off because $\alpha 2$ is present, and the *hsg* are turned off by the $\alpha 2$/**a**1 complex. This ensures that meiosis

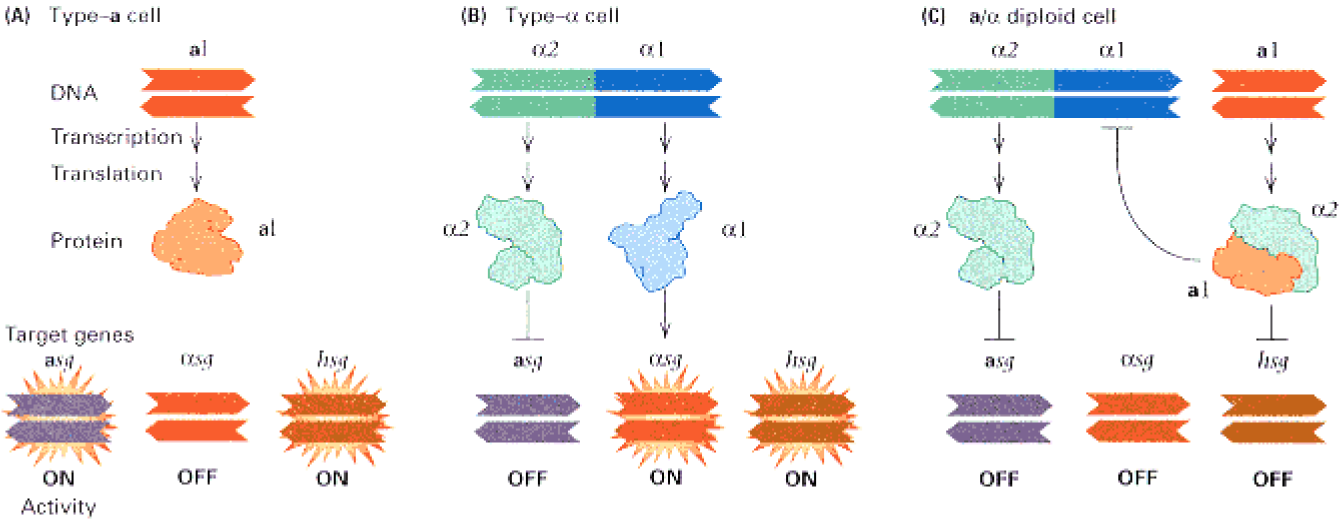

Figure 11.27
Regulation of mating type in yeast. The symbols **a**sg, *αsg*, and *hsg* denote sets of **a**-specific genes, α-specific genes, and haploid-specific genes, respectively. Sets of genes represented with a "sunburst" are *on*, and those unmarked are *off.* (A) In an **a** cell, the **a**1 peptide is inactive, and the sets of genes manifest their basal states of activity (**a**sg and *hsg on* and *αsg off*), so the cell is an **a** haploid. (B) In an αcell, the α2 peptide turns the asg *off* and the α peptide turns the α*sg on*, so the cell is an α haploid. (C) In an **a** α diploid, the α2 and **a**1 peptides form a complex that turns the *hsg off*, the α2 peptide turns the **a**sg *off*, and the α*σγ* manifest their basal activity of *off*, so physiologically the cell is non-**a**, non-α, and nonhaploid (that is, it is a normal diploid).

can occur (because expression of *RME1* is turned off) and that mating type switching ceases (because the HO endonuclease is absent). Thus the homothallic **a**α diploid is stable and able to undergo meiosis. The result is that the **a**α diploid does not transcribe either the mating-type-specific set of genes or the haploid-specific genes.

The repression of transcription of the haploid-specific genes mediated by the **a**1/α2 protein is an example of negative control of the type already familiar from the *lac* and *trp* systems in *E. coli.* The interesting twist in the yeast example is that the α2 protein has a regulatory role of its own in repressing transcription of the **a**-specific genes. Why does the α2 protein, on its own, not repress the haploid-specific genes as well? The answer lies in the specificity of its DNA binding. By itself, the α2 protein has low affinity for the target sequences in the haploid-specific genes. However, the **a1**/α2 heterodimer has both high affinity and high specificity for the target DNA sequences in the haploid-specific genes. The three-dimensional structure of the **a1**/α2 protein in complex with target DNA is shown in Figure 11.28. Upon binding, the **a1**/α2 complex produces a pronounced 60° bend in the DNA molecule, which may play a role in transcriptional repression.

Transcriptional regulation of the mating-type genes includes negative control (**a**-specific genes and haploid-specific genes) and positive control (α-specific genes). Although the regulation of transcription in eukaryotes is both positive and negative, positive regulation is more usual. The regulatory proteins required are the subject of the next section.

Transcriptional Activator Proteins

The α1 protein that functions in the activation of the α-specific genes is an example of a **transcriptional activator protein,** which must bind with an upstream DNA sequence in order to prepare a gene for transcription. We have already seen an example of another transcriptional activator in the case of the GAL4 protein (Figure 11.26). Some transcriptional activator proteins work by direct interaction with one or more proteins present in large complexes

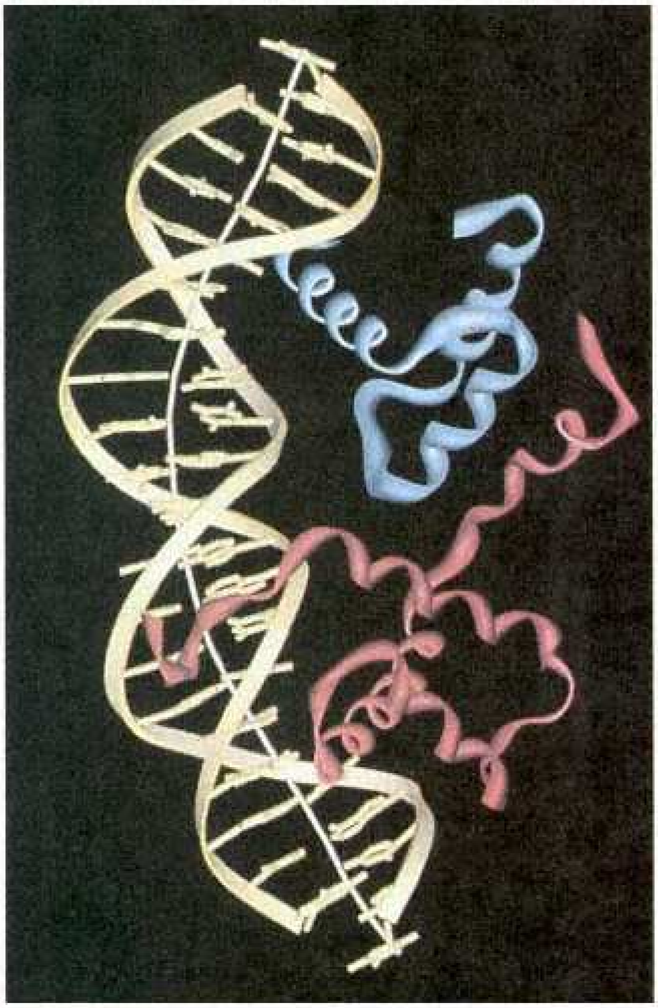

Figure 11.28
Structure of the **a**1/α2 protein bound with DNA. The **a**1 subunit is shown in blue, the α2 subunit in red. Contact with the DNA target results in a sharp bend in the DNA.
[Courtesy of Cynthia Wolberger. From T. Li, M. R. Stark, A. D. Johnson, and C. Wolberger. 1995. *Science* 270: 262.]

of proteins needed for transcription, among them RNA polymerase II (PolII), and attract the transcription complexes to the promoter of the gene to be activated. Other transcriptional activator proteins may initiate transcription by an already assembled transcription complex. In either case, the activator proteins are essential for the transcription of genes that are positively regulated.

Many transcriptional activator proteins can be grouped into categories on the basis

of characteristics that their amino acid sequences share. For example, one category has a **helix-turn-helix** motif, which consists of a sequence of amino acids forming a pair of α-helices separated by a bend; the helices are so situated that they can fit neatly into the groves of a double-stranded DNA molecule. The helix-turn-helix motif is the basis of the DNA-binding ability, although the sequence specificity of the binding results from other parts of the protein. The α2 protein that regulates yeast mating type has a helix-turn-helix motif, as do many other transcriptional activator proteins in both prokaryotes and eukaryotes.

A second large category of transcriptional activator proteins includes a DNA- binding motif that is called a **zinc finger** because the folded structure incorporates a zinc ion. An example is the GAL4 transcriptional activator protein in yeast. The protein functions as a dimer composed of two identical GAL4 polypeptides oriented with their zinc-binding domains at the extreme ends (Figure 11.26 shows the zinc ions in yellow). The DNA sequence recognized by the protein is a symmetrical sequence, 17 base pairs in length, which includes a CCG triplet at each end that makes direct contact with the zinc-containing domains.

A more detailed illustration of the DNA- binding domain of the GAL4 protein is shown in Figure 11.29. Each of two zinc ions (Zn^{2+}) is chelated by bonds with four cysteine residues in characteristic positions at the base of a loop that extends for an additional 841 amino acids beyond those shown. The amino acids marked by a red asterisk are the sites of mutations that result in mutant proteins unable to activate transcription. Replacements at amino acid positions 15 (Arg $\rightarrow$ Gln), 26 (Pro $\rightarrow$ Ser), and 57 (Val $\rightarrow$ Met) are particularly interesting because they provide genetic evidence that zinc is necessary for DNA binding. In particular, the mutant phenotypes can be rescued by extra zinc in the growth medium because the molecular defect reduces the ability of the zinc finger part of the molecule to chelate zinc; extra zinc in the medium overcomes the defect and restores the ability of the mutant activator protein to attach to its particular binding sites in the DNA.

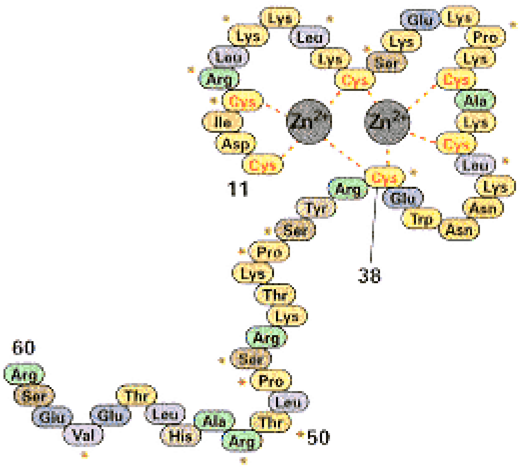

Figure 11.29
DNA-binding domain present in the GAL4 transcriptional activator protein in yeast. The four cysteine residues bind a zinc ion and form a peptide loop called a zinc finger. The zinc finger is a common motif found in DNA-binding proteins. The amino acids marked by a red asterisk have been identified by mutations as sites at which amino acid replacements can abolish the DNA-binding activity of the protein. The result is that the target genes cannot be activated.

Hormonal Regulation

Among the known regulators of transcription in higher eukaryotes, the **hormones**—small molecules or polypeptides that are carried from hormone-producing cells to target cells—have perhaps been studied in most detail. One class of hormones consists of small molecules synthesized from cholesterol; these steroid hormones include the principal sex hormones. Many of the steroid hormones act by turning on the transcription of specific sets of genes. If a hormone regulates transcription, then it must somehow signal the DNA. The signaling mechanism for the steroid hormone cortisol is outlined in Figure 11.30. The hormone penetrates a target cell through diffusion, because steroids are hydrophobic (nonpolar) molecules that pass freely through the cell membrane into the cytoplasm. There it encounters a receptor molecule that is complexed with another protein called Hsp82, which functions to mask the receptor. Once cortisol binds to the receptor, it liberates the receptor from Hsp82, and the hormone-receptor complex migrates to the nucleus where it binds to its DNA target sequences to activate transcription. Nontarget cells do not contain the receptors and so are unaffected by the hormone.

A well-studied example of induction of transcription by a hormone is stimulation of the synthesis of ovalbumin in the chicken oviduct by the steriod sex hormone **estrogen.** When hens are injected with estrogen, oviduct tissue responds by synthesizing ovalbumin mRNA. This synthesis continues as long as estrogen is administered. When the hormone is withdrawn,

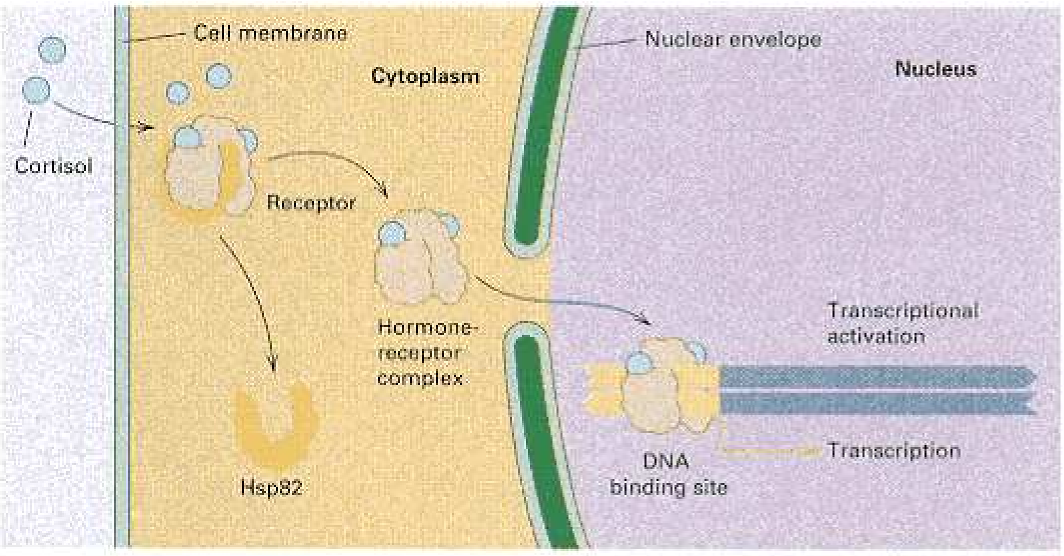

Figure 11.30
A schematic diagram showing how a steroid hormone reaches a DNA molecule and triggers transcription by binding with a receptor to form a transcriptional activator. Entry into the cytoplasm is by passive diffusion.

the rate of synthesis decreases. Before injection of the hormone, and 60 hours after the injections have stopped, no ovalbumin mRNA is detectable. When estrogen is given to hens, only the oviduct synthesizes mRNA because other tissues lack the hormone receptor.

Transcriptional Enhancers

Hormone receptors and other transcriptional activator proteins bind with particular DNA sequences known as **enhancers.** Enhancer sequences are typically rather short (usually fewer than 20 base pairs) and are found at a variety of location around the gene affected. Most enhancers are upstream of the transcriptional start site (sometimes many kilobases away), others are in introns within the coding region, and a few are even located at the 3' end of genes. One of the most thoroughly studied enhancers is in the mouse mammary tumor virus and determines transcriptional activation by the glucocorticoid steroid hormone. The consensus sequence of the enhancer is AGAQCAGQ, in which Q stands for either A or T. The virus contains five copies of the enhancer positioned throughout its genome (Figure 11.31), providing five binding sites for the hormone-receptor complex that activates transcription.

Enhancers are essential components of gene organization in eukaryotes because they enable genes to be transcribed only when proper transcriptional activators are present. Some enhancers respond to molecules outside the cell—for example, steroid hormones that form receptor-hormone complexes. Other enhancers respond to

molecules that are produced inside the cell (for example, during development), and these enhancers enable the genes under their control to participate in cellular differentiation or to be expressed in a tissue-specific manner. Many genes are under the control of several different enhancers, so they can respond to a variety of different molecular signals, both external and internal.

Figure 11.32 illustrates several of the genetic elements found in a typical eukaryotic gene. The transcriptional complex binds to the promoter (*P*) to initiate RNA synthesis. The coding regions of the gene (exons) are interrupted by one or more intervening sequences (introns) that are eliminated in RNA processing. Transcription is regulated by means of enhancer elements (numbered 1 through 6) that respond to different molecules that serve as induction signals. The enhancers are located both upstream and downstream of the promoter, and some (in this example, enhancer 1) are present in multiple copies.

Many enhancers stimulate transcription by means of **DNA looping,** which refers to physical interactions between relatively distant regions along the DNA. The mechanism is illustrated in Figure 11.33. The factors necessary for transcription include a transcriptional activator protein that interacts with at least one protein subunit present in one or more large, multisubunit protein complexes. The protein factors in these complexes are known as **general transcription factors** because they are associated with the transcription of many different genes. The general transcription factors in eukaryotes have been highly con-

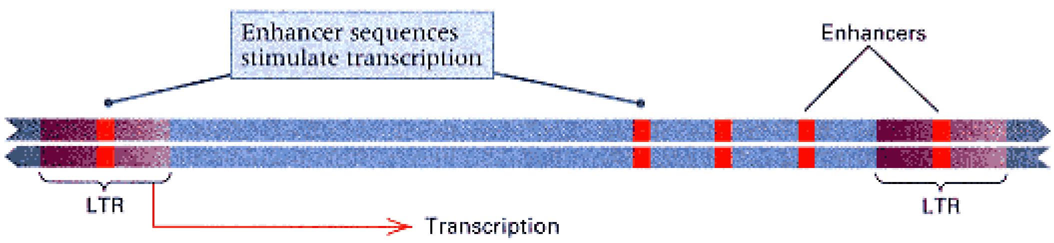

Figure 11.31
Positions of enhancers (orange) in the mouse mammary tumor virus that allow transcription of the viral sequence to be induced by glucocorticoid steroid hormone. LTR stands for long terminal repeat, a DNA sequence found at both ends of the virus.

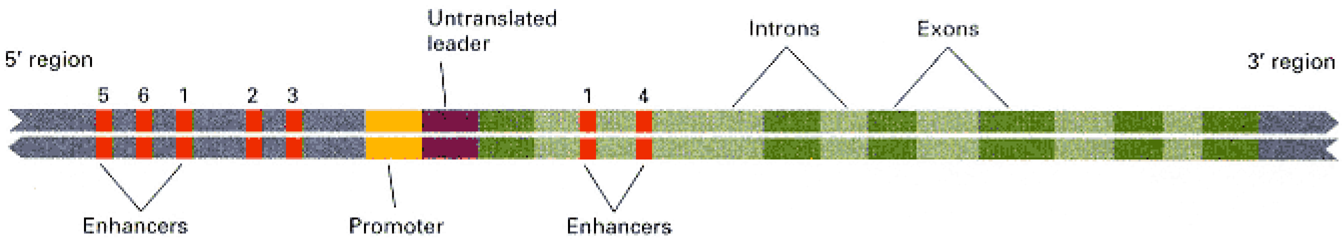

Figure 11.32
Schematic diagram of the organization of a typical gene in a higher eukaryote. Scattered throughout the sequence, but tending to be concentrated near the 5' region, are a number of different enhancer sequences. The enhancers are binding sites for transcriptional activator proteins that make possible temporal and tissue-specific regulation of the gene.

served in evolution. One of the complexes is TFIID, which includes a **TATA-box-binding protein (TBP)** that binds with the promoter in the region of the TATA box. In addition to TBP, the TFIID complex may also include a number of other proteins, called **TBP-associated factors (TAFs),** that act as intermediaries through which the effects of the transcriptional activator are transmitted. (Not all of the TBP is found in association with TAFs.) Transcription also requires an **RNA polymerase holoenzyme,** which consists of PolII (itself composed of 12 protein subunits) combined with at least 9 other protein subunits. In yeast these subunits include the transcription factors TFIIB, TFIIF, and TFIIH, as well as other proteins. Other general transcription factors have also been identified (for example, TFIIA and TFIIE), but it is not known whether these are components of larger complexes or whether they join the transcriptional apparatus as it is being assembled at the promoter.

Illustrated in Figure 11.33 is a mechanism of transcriptional activation called activation by **recruitment.** The key players, shown in Figure 11.33A, are the transcriptional activator protein and the TFIID and RNA polymerase holoenzyme complexes. The actual structures of TFIID and the holoenzyme complexes are not known, but for concreteness they are shown as multisubunit aggregates. To activate transcription (Figure 11.33B), the transcriptional activator protein binds to an enhancer in the DNA and to one of the TAF subunits in the TFIID complex. This interaction attracts ("recruits") the TFIID complex to the region of the promoter (Figure 11.33C). Attraction of the TFIID to the promoter also recruits the holoenzyme (Figure 11.33D) as well as any remaining general transcription factors, and in this manner the transcriptional complex is assembled for transcription to begin.

Experimental evidence for transcriptional activation by recruitment has come from studies of a number of artificial proteins created by fusing a DNA binding domain with one of the protein subunits in TFIID. Such fusion proteins act as transcriptional activators wherever they bind to DNA (provided that a promoter is nearby), because the TFIID is "tethered" to the DNA binding domain and so the "recruitment" of TFIID is automatic. Similarly, fusion proteins that are tethered to a subunit of the holoenzyme can recruit the holoenzyme to the promoter. In this case. TFIID and the remaining general transcription factors are also attracted to the promoter, and the transcriptional complex is assembled. These experiments suggest that a transcriptional activator protein can activate transcription by interacting with subunits of either the TFIIA complex or the holoenzyme.

As Figure 11.33 suggests, the fully assembled transcription complex in eukaryotes is a very large structure. A real example, taken from early development in *Drosophila,* is shown in Figure 11.34. In this case, the enhancers, located a considerable distance upstream from the gene to be activated, are bound by the transcriptional

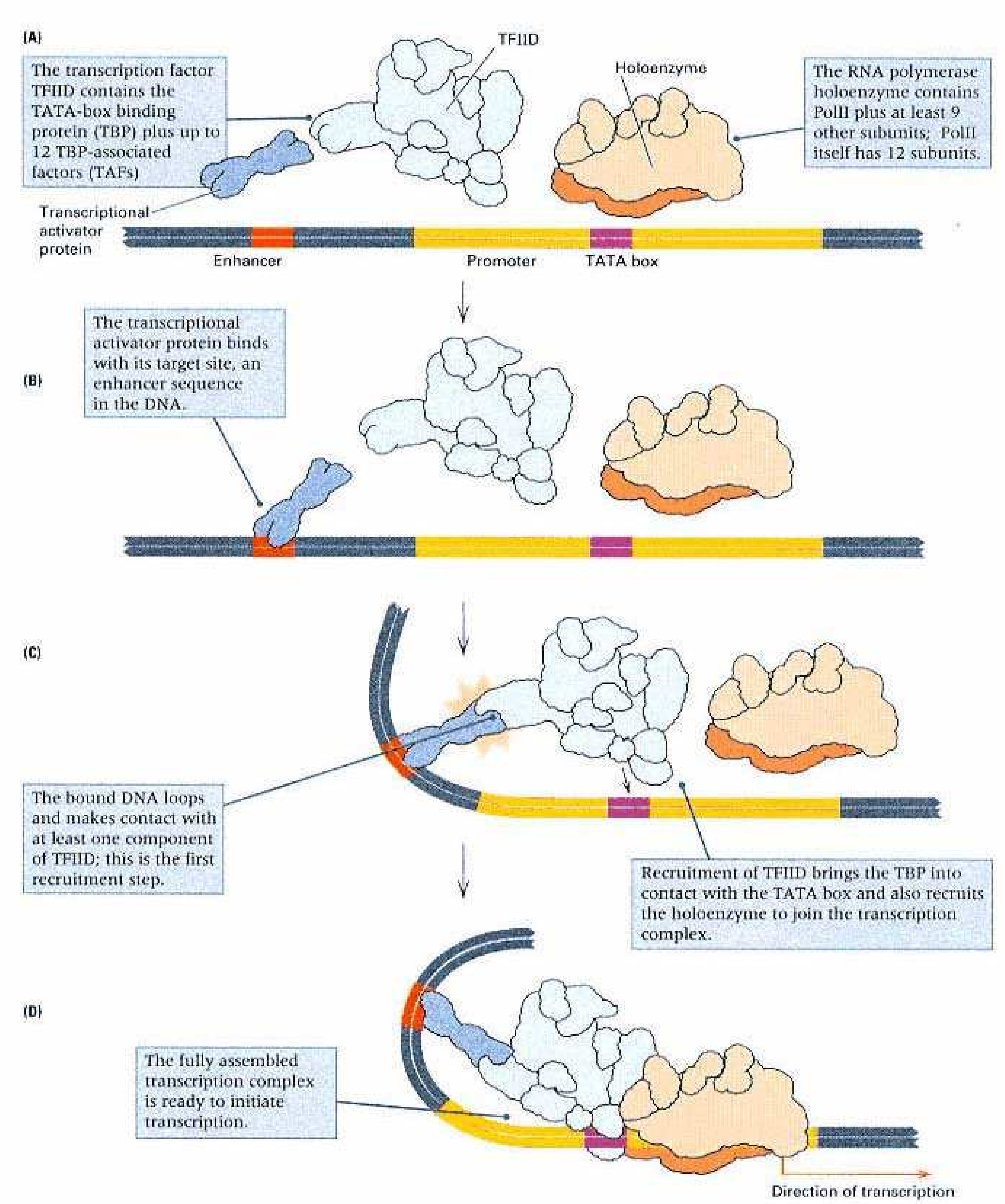

Figure 11.33
Transcriptional activation by recruitment. (A) Relationship between enhancer and promoter and the protein factors that bind to them. (B) Binding of the transcriptional activator protein to the enhancer. (C) Bound transcriptional activator protein makes physical contact with a subunit in the TFIID complex, which contains the TATA-box-binding protein, and attracts ("recruits") the complex to the promoter region. (D) The PolII holoenzyme and any remaining general transcription factors are recruited by TFIID, and the transcription complex is fully assembled and ready for transcription. In the cell, not all of the PolII is found in the holoenzyme, and not all of the TBP is found in TFIID. In this illustration, transcription factors other than those associated with TFIID and the holoenzyme are not shown

activator proteins BCD and HB, which are products of the genes *bicoid (bcd)* and *hunchback (hb)*, respectively; these transcriptional activators function in establishing the anterior-posterior axis in the embryo. (Early *Drosophila* development is discussed in Chapter 12.) Note the position of the TATA box in the promoter of the gene. The TATA box binding is the function of the TBP. The functions of a number of other components of the transcription complex have also been identified. For example, the TFIIH contains both helicase and kinase activity to melt the DNA and to phosphorylate RNA polymerase II. Phosphorylation allows the polymerase to leave the promoter and elongate mRNA. The looping of the DNA effected by the transcriptional activators is an essential feature of the activation process. Transcriptional activation in eukaryotes is a complex process, especially when compared to the prokaryotic RNA polymerase, which consists of a core $\alpha_2\beta\beta'$ tetramer and a σ factor.

The versatility of some enhancers results from their ability to interact with two different promoters in a competitive fashion; that is, at any one time, the enhancer can stimulate one promoter or the other, but not both. An example of this mechanism is illustrated in Figure 11.35, in which P1 and P2 are alternative promoters

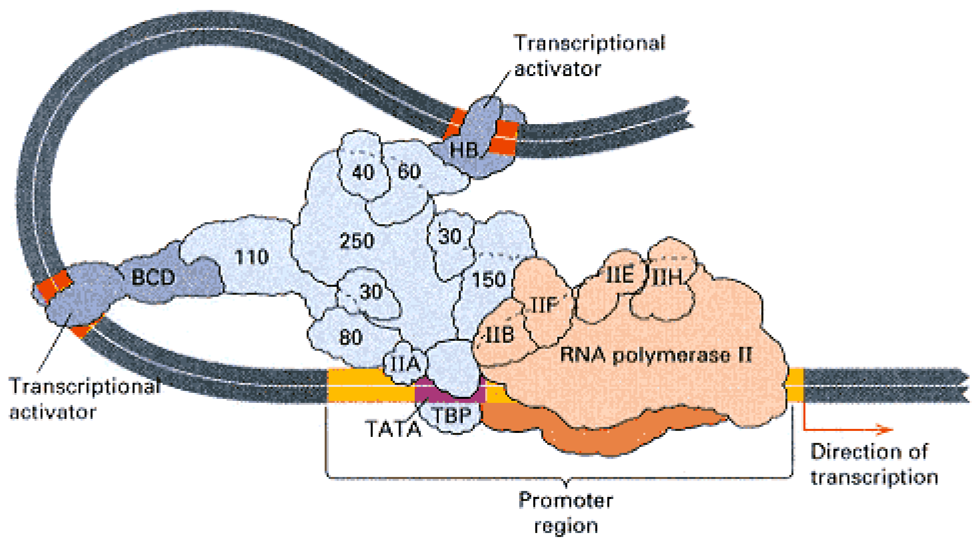

Figure 11.34
An example of transcriptional activation during *Drosophila* development. The transcriptional activators in this example are bicoid protein (BCD) and hunchback protein (HB). The numbered subunits are TAFs (TBP-associated factors) that, together with TBP (TATA-box-binding protein) correspond to TFIID. BCD acts through a 110-kilodalton TAF, and HB, through a 60-kilodalton TAF. The transcriptional activators act via enhancers to cause recruitment of the transcriptional apparatus. The fully assembled transcription complex includes TBP and TAFs, RNA polymerase II, and general transcription factors TFIIA, TFIIB, TFIIE, TFIIF, and TFIIH.

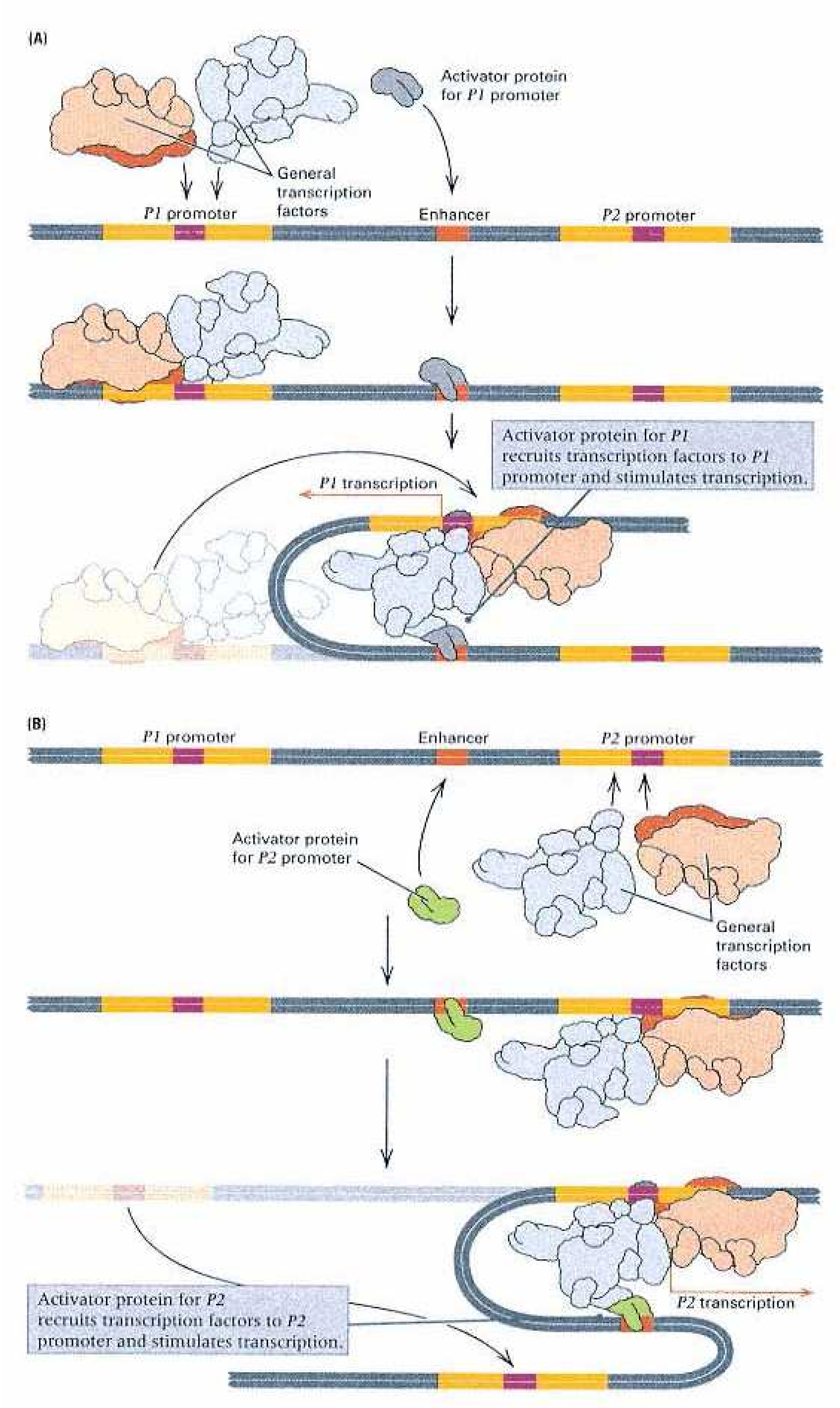

Figure 11.35

Genetic switching regulated by competition for an enhancer. Promoters *P1* and *P2* compete for a single enhancer located between them. When complexed with an appropriate transcriptional activator protein, the enhancer binds preferentially with promoter *P1* (A) or promoter *P2* (B). Binding to the promoter recruits the transcription complex. If either promoter is mutated or deleted, then the enhancer binds with the alternative promoter. The location of the enhancer relative to the promoters is not critical.

that compete for an enhancer located between them (Figure 11.35). When complexed with a transcriptional activator specific for promoter P1, the enhancer binds preferentially with promoter P1 and stimulates transcription (Figure 11.35A). When complexed with a different transcriptional activator specific for promoter P2, the enhancer binds preferentially with promoter P2 and stimulates transcription from it (Figure 11.35B). In this way, competition for the enhancer serves as a sort of switch mechanism for the expression of the P1 or P2 promoter. This regulatory mechanism is present in chickens and results in a change from the production of embryonic β-globin to that of adult β-globin in development. In this case, the embryonic globin gene and the adult gene compete for a single enhancer, which in the course of development changes its preferred binding from the embryonic promoter to the adult promoter. In human beings, enhancer competition appears to control the developmental switch from the fetal γ-globin to that of the adult β-globin polypeptide chains. In persons in whom the β-globin promoter is deleted or altered in sequence and unable to bind with the enhancer, there is no competition for the enhancer molecules, and the γ-globin genes continue to be expressed in adult life when normally they would not be transcribed. Adults with these types of mutations have fetal hemoglobin instead of the adult forms. The condition is called high-F disease because of the persistence of fetal hemoglobin, but the clinical manifestations are very mild.

The Logic of Combinatorial Control

Because the genome of a complex eukaryote contains many possible enhancers that respond to different signals or cellular conditions, in principle each gene could be regulated by its own distinct combination of enhancers. This is called **combinatorial control,** and it is a powerful means of increasing the complexity of possible regulatory states by employing several simple regulatory states in combination.

To consider a simple example, suppose that a gene has a single binding site for only one type of regulatory molecule. Then there are only two regulatory states (call them + or -), which reflect whether or not the binding site is occupied. If the gene has single binding sites for each of two different regulatory molecules, then there are four possible regulatory states: ++, +-, -+, and --. Single binding sites for three different regulatory molecules yield eight combinatorial states, and in general, n different types of regulatory molecules yield 2^n states. If transcription occurs according to the particular pattern of which binding sites are occupied, then a small number of types of regulatory molecules can result in a large number of different patterns of regulation. For example, because eukaryotic cells contain approximately 200 different cell types, each gene would theoretically need as few as 8+/- types of binding sites to specify whether it should be on or off in each cell type, because $2^8 = 256$.

The actual regulatory situation is certainly more complex than the naive calculation based on +/- switches would imply, because genes are not merely on or off; their level of activity is modulated. For example, a gene may have multiple binding sites for an activator protein, and this allows the level of gene expression to be adjusted according to the number of binding sites that are occupied. Furthermore, each cell type is programmed to respond to a variety of conditions both external and internal, so the total number of regulatory states must be considerably greater than the number of cell types. On the other hand, the +/- example demonstrates that combinatorial control is extremely powerful in multiplying the number of regulatory possibilities, and therefore a large number of regulatory states does not necessarily imply a hopelessly complex regulatory apparatus.

Enhancer-Trap Mutagenesis

The ability of enhancers to regulate transcription is the basis of the **enhancer trap,** a genetically engineered transposable element designed to detect tissue-specific expression resulting from insertion of the element near enhancers. A simplified diagram of an enhancer trap based on the transposable *P* element in *Drosophila* is shown in Figure 11.36. The element (Figure 11.36A) contains a weak promoter, unable to initiate transcription without the aid of an enhancer, linked to the β-galactosidase gene from *E. coli;* also shown are the inverted-repeat sequences necessary for transposition. When the enhancer trap transposes and inserts at a random position in the genome, no transcription occurs if the insertion is not near an enhancer (Figure 11.36B). On the other hand, if the insertion is near an enhancer, then the β-galactosidase gene will be transcribed in whatever tissues stimulate the enhancer. Any tissue-specific expression can be detected by the use of staining reagents specific for β-galactosidase; one commonly used stain turns bright blue in the presence of the enzyme. When used in this manner, the β-galactosidase gene is called a **reporter gene** because its expression reveals ("reports") that the gene has been activated.

For example, the enhancer trap has been used to identify genes expressed only in the eye. When the enhancer trap inserts into the genome and comes under the control of eye-specific enhancers, the β-galactosidase is expressed in the eye and nowhere else. The eyes, and only the eyes, of flies with such insertions stain blue. Insertions of the enhancer trap provide an important method for identifying genes that are expressed in particular cells or tissues, because the insertions do not necessarily disrupt the function of the normal gene. Furthermore, the presence of DNA sequences from the *P* element at the site of insertion provides a molecular tag with which to clone the gene.

Alternative Promoters

Some eukaryotic genes have two or more promoters that are active in different cell types. The different promoters result in different primary transcripts that contain the same protein-coding regions. An example from *Drosophila* is shown in Figure 11.37. The gene codes for alcohol dehydrogenase, and its organization in the genome is shown in Figure 11.37; there are three

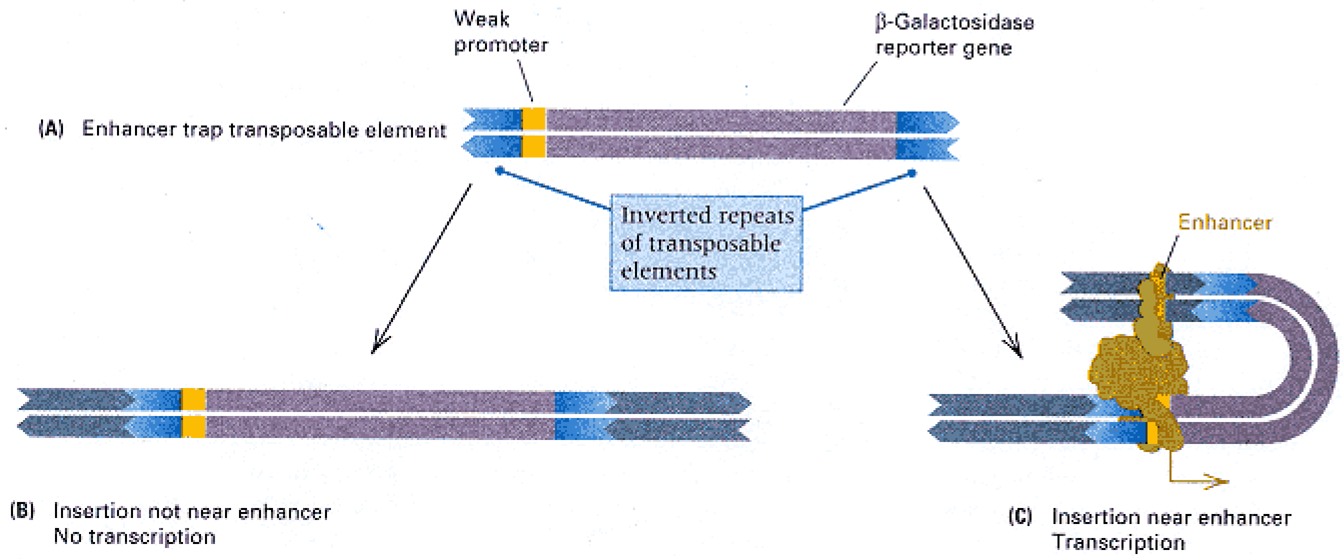

Figure 11.36
Identification of enhancers by genetic means. (A) The enhancer trap, consisting of a transposable element that contains a reporter gene with a weak promoter. (B) If insertion is at a site that lacks a nearby enhancer, then the reporter gene cannot be activated. (C) If insertion is near an enhancer then the reporter gene is transcribed in a temporally regulated or tissue-specific manner determined by the type of enhancer.

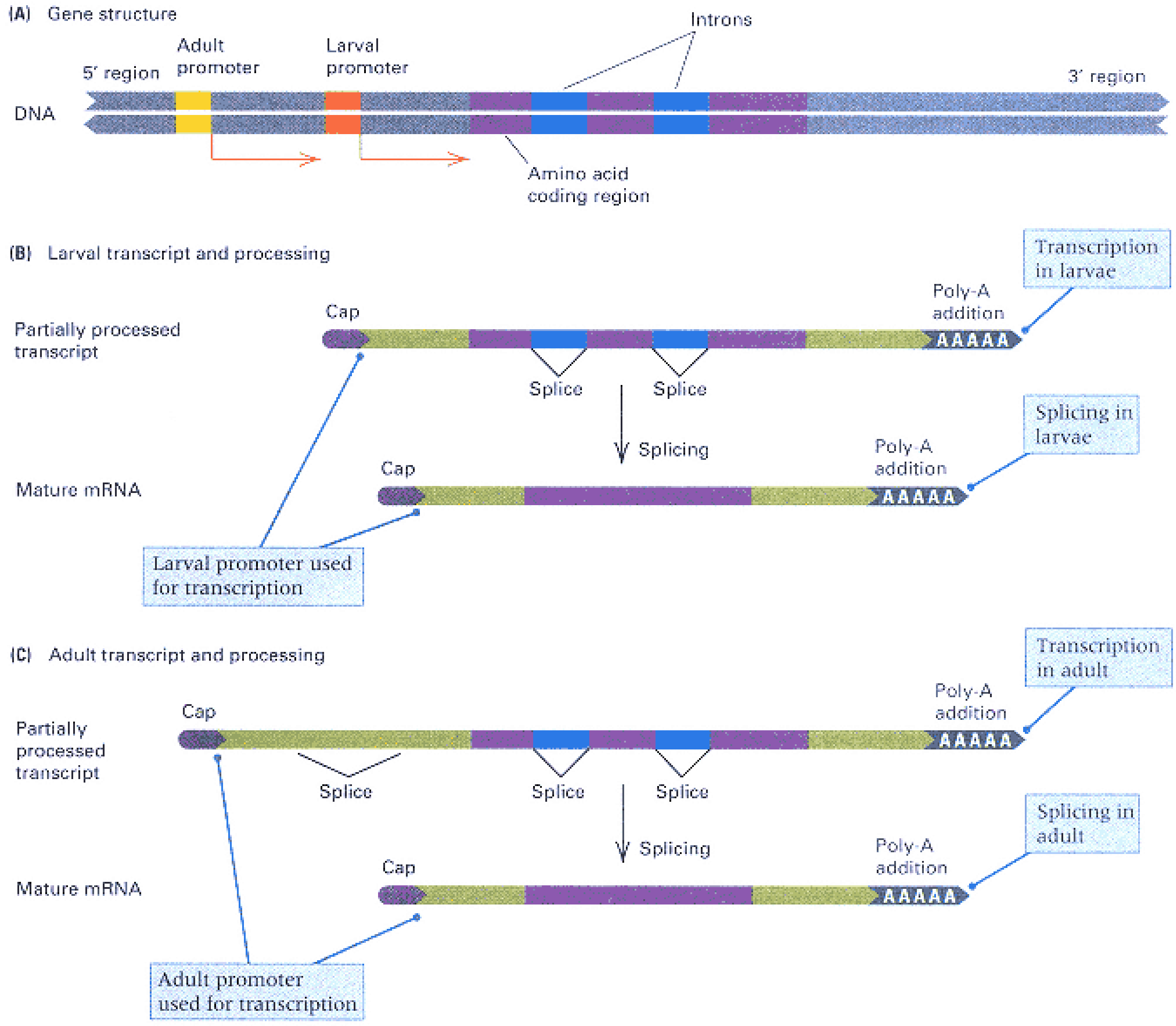

Figure 11.37
Use of alternative promoters in the gene for alcohol dehydrogenase in *Drosophila.* (A) The overall gene organization includes two introns within the amino acid coding region. (B) Transcription in larvae uses the promoter nearest the 5' end of the coding region. (C) Transcription in adults uses a promoter farther upstream, and much of the larval leader sequence is removed by splicing.

protein-coding regions interrupted by two introns. Transcription in larvae (Figure 11.37A) uses a different promoter from that used in transcription in adults (Figure 11.37B). The adult transcript has a longer 5' leader sequence, but most of this sequence is eliminated in splicing. Alternative promoters make possible the independent regulation of transcription in larvae and adults.

11.8— Alternative Splicing

Even when the same promoter is used to transcribe a gene, different cell types can produce different quantities of the protein (or even different proteins) because of differences in the mRNA produced in processing. The reason is that the same transcript can be spliced differently from one cell type to the next. The different splicing patterns may include exactly the same protein-coding exons, in which case the protein is identical but the rate of synthesis differs because the mRNA molecules are not translated with the same efficiency. In other cases, the protein-coding part of the transcript has a different splicing pattern in each cell type, and the resulting mRNA molecules code for proteins that are not identical even though they share certain exons.

In the synthesis of α-amylase in the mouse, different mRNA molecules are produced from the same gene because of different patterns of intron removal in RNA processing. The mouse salivary gland produces more of the enzyme than the liver, although the same coding sequence is transcribed. In each cell type, the same primary transcript is synthesized, but two different splicing patterns are used. The initial part of the primary transcript is shown in Figure 11.38. The coding sequence begins 50 base pairs inside exon 2 and is formed by joining exon 3 and subsequent exons. In the salivary gland, the primary transcript is processed such that exon S is joined with exon 2 (that is, exon L is removed as part of introns 1 and 2). In the liver, exon L is joined with exon 2, and exon S is removed along with intron 1 and the leader L. The exons S and L become alternative 5' leader sequences of the amylase mRNA, and the alternative mRNAs are translated at different rates.

In chicken skeletal muscle, two different forms of the muscle protein myosin are made from the same gene. A different primary transcript is made from each of two promoters, and these transcripts are processed differently to form mRNA molecules that encode distinct forms of the protein. In *Drosophila*, the myosin RNA is processed in four different ways; the precise mode depends on the stage of development of the fly. One class of myosin is found in pupae and another in the later embryo and larval stages. How the mode of processing is varied is not known.

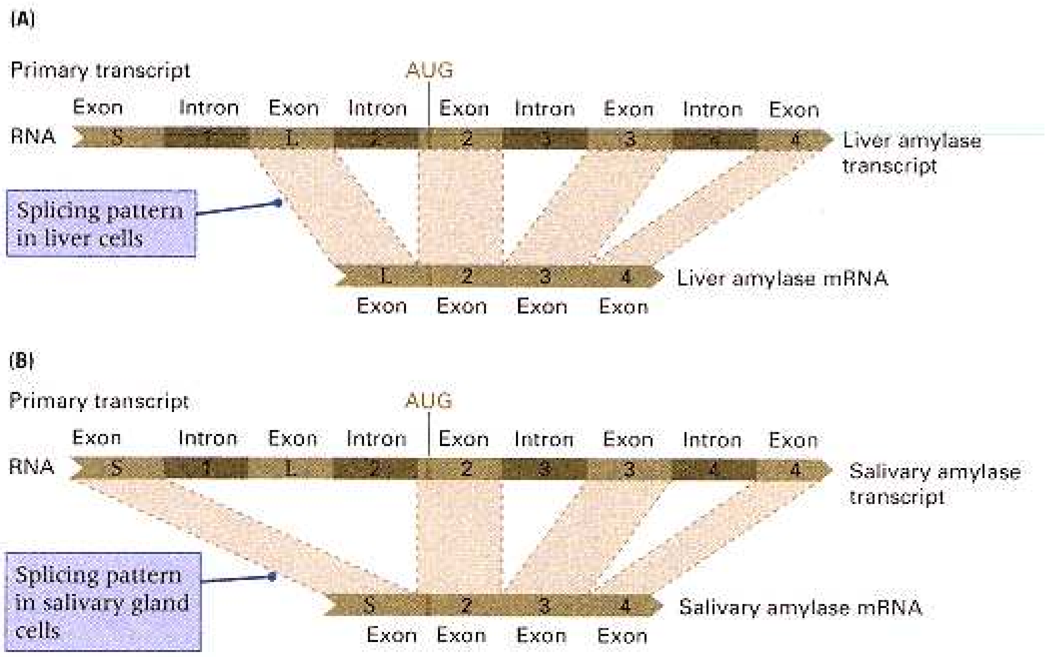

Figure 11.38
Production of distinct amylase mRNA molecules by different splicing events in cells of the salivary gland and liver of the mouse. The leader and the introns are distinguished by color from the exons Exon S and L form part of the untranslated 5' end of the mRNA in salivary glands and liver, respectively. The coding sequence begins at the AUG codon in exon 2. (A Splicing in the liver. Exon L is joined with exons 2, 3, and 4. (B) Splicing in the salivary gland. Exon S is joined with exons 2, 3, and 4.

11.9— Translational Control

In bacteria, most mRNA molecules are translated about the same number of times, with little variation from gene to gene. In eukaryotes, translation is sometimes regulated. The principal types of translational control are

1. Inability of an mRNA molecule to be translated unless a molecular signal is present

2. Regulation of the lifetime of a particular mRNA molecule

3. Regulation of the overall rate of protein synthesis

4. Aborted translation of the principal open reading frame because of the presence of smaller open reading frames upstream in the mRNA

In this section, examples of each of the first three modes of regulation will be presented.

An important example of translational regulation is that of **masked mRNA.** Unfertilized eggs are biologically static, but shortly after fertilization, many new proteins must be synthesized—among them, for example, the proteins of the mitotic apparatus and the cell membranes. Unfertilized sea urchin eggs can store large quantities of mRNA for many months in the form of mRNA-protein particles made during formation of the egg. This mRNA is translationally inactive, but within minutes after fertilization, translation of these molecules begins. Here, the timing of translation is regulated; the mechanisms for stabilizing the mRNA, for protecting it against RNases, and for activation are unknown.

Translational regulation of another type occurs in mature unfertilized eggs. These cells need to maintain themselves but do not have to grow or undergo a change of state. Thus the rate of protein synthesis in eggs is generally low. This is not a consequence of an inadequate supply of mRNA but apparently results from failure to form the ribosome-mRNA complex.

The synthesis of some proteins is regulated by direct action of the protein on the mRNA. For instance, the concentration of one type of antibody molecule is kept constant by self-inhibition of translation; that is, the antibody molecule itself binds specifically to the mRNA that encodes it and thereby inhibits initiation of translation.

A dramatic example of translational control is the extension of the lifetime of silk fibroin mRNA in the silkworm. During cocoon formation, the silk gland of the silkworm predominantly synthesizes a single type of protein, silk fibroin. Because the worm takes several days to construct its cocoon, it is the total amount (not the rate) of fibroin synthesis that must be great; the silkworm achieves this in two ways. First, the silk gland cells become highly polyploid, accumulating thousands of copies of each chromosome. Second, each fibroin gene synthesizes an mRNA molecule that has a very long lifetime. Transcription of the fibroin gene is initiated at a strong promoter, and about 10^4 fibroin mRNA molecules are made in a period of several days. (This synthesis is under transcriptional regulation.) A typical eukaryotic mRNA molecule has a lifetime of about 3 hours before it is degraded. However, fibroin mRNA survives for several days, during which each mRNA molecule is translated repeatedly to yield 10^5 fibroin molecules. Thus each gene is responsible for the synthesis of 10^9 protein molecules in 4 days. Altogether, the silk gland makes about 10^{15} molecules of fibroin in this period. If the lifetime of the mRNA were not extended, then either 25 times as many genes would be needed or synthesis of the required fibroin would take about 100 days.

Another example of an mRNA molecule with an extended lifetime is the mRNA that encodes casein, the major protein of milk, in mammary glands. When the hormone prolactin is received by the gland, the lifetime of casein mRNA increases. Synthesis of the mRNA also continues, so the overall rate of production of casein is markedly increased by the hormone. When the body no longer supplies prolactin, the concentration of casein mRNA decreases because the RNA is degraded more rapidly, and lactation terminates.

11.10— Is There a General Principle of Regulation?

With microorganisms, whose environment frequently changes drastically and rapidly, a general principle of regulation is that bacteria make what they need when it is required and in appropriate amounts. Although such extraordinary efficiency is common in prokaryotes, it is rare in eukaryotes. For example, efficiency appears to be violated by the large amount of DNA in the genome that has no proteincoding function and by the large amount of intron RNA that is discarded.

What is abundantly clear is that there is no universal regulatory mechanism. Many control points are possible, and different genes are regulated in different ways. Furthermore, evolution has not always selected for simplicity in regulatory mechanisms; it sometimes just settles for something that works. If a cumbersome regulatory mechanism were to arise, then it would in time evolve, be refined, and become more effective, but it would not necessarily become simpler. On the whole, regulatory mechanisms include a variety of seemingly *ad hoc* processes, each of which has stood the test of time primarily because it works.

Chapter Summary

Most cells do not synthesize molecules that are not needed. There are important opportunities for controlling gene expression in transcription, RNA processing, and translation. Gene expression can also be regulated through the stability of mRNA, and the activity of proteins can be regulated in a variety of ways after translation. The processes of gene regulation generally differ between prokaryotes and eukaryotes.

In bacteria, the synthesis of most proteins is regulated by controlling the rate of transcription of the genes that code for the proteins. The concentration of a few proteins is autoregulated, usually by direct binding of the protein at or near its promoter.

The synthesis of degradative enzymes needed only on occasion, such as the enzymes required to metabolize lactose, is typically regulated by an off-on mechanism. When lactose is present, the genes that code for the enzymes required to metabolize lactose are transcribed; when lactose is absent, transcription does not occur. Lactose metabolism is negatively regulated. Two enzymes that are needed to degrade lactose—permease, required for the entry of lactose into bacteria, and β-galactosidase, the enzyme that does the degrading—are encoded in a single polycistronic mRNA molecule, *lac* mRNA. Immediately adjacent to the promoter for *lac* mRNA is a regulatory sequence of bases called an operator. A repressor protein is made by a tightly linked gene, and this protein binds tightly to the operator, thereby preventing RNA polymerase from initiating transcription at the promoter. Lactose is an inducer of transcription because it can bind to the repressor and prevent the repressor from binding to the operator. Therefore, in the presence of lactose, there is no active repressor, and the *lac* promoter is always accessible to RNA polymerase. The operator, the promoter, and the structural genes are adjacent to one another and together constitute the *lac* operon. Repressor mutations have been isolated that inactivate the repressor protein, and operator mutations are known that prevent recognition of the operator by an active repressor; such mutations cause continuous production of *lac* mRNA and are said to be constitutive.

When lactose is cleaved by β-galactosidase, the products are glucose and galactose. Glucose is metabolized by enzymes that are made continuously; galactose is broken down by enzymes of the inducible galactose operon. When glucose is present in the growth medium, the enzymes for degrading lactose and other sugars are unnecessary. The general mechanism for preventing transcription of many sugar-degrading operons is as follows: High concentrations of glucose suppress the synthesis of the small molecule cyclic AMP (cAMP). Initiation of transcription of many sugar-degrading operons requires the binding of a protein, called CRP, to a specific region of the promoter. This binding takes place only after CRP has first bound cAMP and formed a cAMP-CRP complex. Only when glucose is absent is the concentration of cAMP sufficient to produce cAMP-CRP and hence to permit transcription of the sugar-degrading operons. Thus, in contrast with a repressor, which must be removed before transcription can begin (negative regulation), cAMP is a positive regulator of transcription.

Biosynthetic enzyme systems exemplify the repressible type of negative transcriptional control. In the synthesis of tryptophan, transcription of the genes encoding the *trp* enzymes is controlled by the concentration of tryptophan in the growth medium. When excess tryptophan is present, it binds with the *trp* aporepressor to form the active repressor that prevents transcription. The *trp* operon is also regulated by attenuation, in which transcription is initiated continually but the transcript forms a hairpin structure that results in premature termination. The frequency of termination of transcription is determined by the availability of charged tryptophan tRNA; with decreasing concentrations of tryptophan, termination occurs less often and the enzymes for tryptophan synthesis are made, thereby increasing the concentration of tryptophan. Attenuators also regulate operons for the synthesis of other amino acids.

Bacteriophage λ adopts either the quiescent lysogenic state or the lytic cycle as a result of competition between two repressors, cI and cro. If cI repressor dominates, the λ genome becomes integrated into the bacterial chromosome. A lysogen is formed, in which cI continues to be synthesized and prevents transcription of all other λ genes. If cro repressor dominates, cI repressor is no longer synthesized, and the lytic cycle ensues. Although there are elements of positive transcriptional regulation in the regulatory circuitry, the predominant mode of regulation is negative.

Eukaryotes employ a variety of genetic regulatory mechanisms, occasionally including changes in DNA. The number of copies of a gene may be increased by DNA amplification; programmed rearrangements of DNA may occur; and in some cases, gene inactivity coincides with the methylation of cytosine bases.

Many genes in eukaryotes are regulated at the level of transcription. Although both negative and positive regulation occur, positive regulation is typical. In the control of yeast mating type by the *MAT* locus, the **a**-specific genes and the haploid-specific genes are negatively regulated and the α-specific genes are positively regulated. In general, positive regulation is effected through transcriptional activator proteins that contain particular structural motifs that bind to DNA—for example, the helix-turn-helix motif or the zinc finger motif. Hormones also can regulate transcription. Steroid hormones bind with receptor proteins to form transcriptional activators.

Transcriptional activators bind to DNA sequences known as enhancers, which are usually short sequences that may be present at a variety of positions around the genes that they regulate. In the recruitment model of transcriptional activation, a transcriptional activator protein interacts directly with one or more protein components of the RNA polymerase holoenzyme and attracts the holoenzyme to the promoter along with other protein complexes, such as TFIID. The fully assembled transcriptional apparatus in eukaryotes consists of a complex assemblage of RNA polymerase II, TATA-box-binding protein (TBP), other general transcription factors (TFIIA, TFIIB, and so forth), and TBP-associated factors (TAFs).

In genetic analysis, enhancers can be identified by the use of transposable elements containing genes that come under the control of enhancers near the sites of insertion. There are many types of enhancers responsive to different transcriptional activators. Combining different type of enhancers provides a large number of possible types of regulation.

Some genes contain alternative promoters used in different tissues; other genes use a single promoter, but the transcripts are spliced in different ways. Alternative splicing can result in mRNA molecules that are translated with different efficiencies, or even in different proteins if there is alternative splicing of the protein-coding exons. Regulation can also be at the level of translation—for example, through masked mRNAs or through factors that affect mRNA stability.

Key Terms

aproepressor	enhancer	masked mRNA
attenuation	enhancer trap	mating-type interconversion
attenuator	estrogen	methylation
autoregulation	gene amplification	negative regulation
β-galactosidase	gene dosage	operator
cAMP-CRP complex	general transcription factor	operon model
cassette	gene regulation	permease

cis-dominant

combinatorial control

combinatorial joining

constant region

consttutive mutant

coordinate regulation

co-repressor

cyclic adenosime monophosphate

(cAMP)

cyclic AMP receptor protein (CRP)

DNA looping

DNA methylase

effector

heavy chain

helix-turn-helix

homothallism

hormone

housekeeping genes

immunoglobulin

inducer

inducible

inducible transcription

lac operon

lactose permease

leader polypeptide

light chain

polyprotein

positive regulation

recruitment

reporter gene

repressible transcription

repressor

RNA polymerase holoenzyme

TATA-box-binding protein (TBP)

TBF-associated factor (TAF)

transcriptional activator protein

variable region

zinc finger

Chapter 11 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

http://www.jbpub.com/genetics

Select the link to ***Genetics: Principles and Analysis*** and then choose the link to ***GeNETics on the web.*** You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor

1. Look up **lactose degradation** at this site for a diagram of the chemical reaction catalyzed by β-galactosidase. Examine the pathway in more detail to see the molecular structures of the substrate and the products of the enzymes. Click on *lacZ* to get more information about the gene itself. If assigned to do so, follow the *Unification link* and pursue further links until you find the amino acid sequence of the β-galactosidase polypeptide (each active enzyme contains four of these polypeptide chains). List, in order, the links you followed to find this information, as well as the number of amino acids in the polypeptide and its molecular weight. Note also which protein database contains this sequence and the entry number.

2. More about the genes that control **mating type** in *Saccharomyces cerevisiae* can be retrieved at this site by examining the map of chromosome 3 and clicking on *HML*, *MAT,* and *HMR.* See if you can follow the links to the Entrez Protein database to find the amino acid sequence of each of the regulatory proteins **a**1, α1, and α2. If assigned to do so, write a short paragraph summarizing

(text box continued on next page)

Review the Basics

• Distinguish between positive regulation and negative regulation of transcription, and give an example of each.

• What is an operon and what is the significance of an operon for gene expression?

• An operon containing genes that encode the enzymes in a metabolic pathway for the synthesis of an amino acid includes a short open reading frame in the leader sequence. This short open reading frame contains multiple codons for the amino acid synthesized by the pathway. What does this observation suggest about regulation of the operon?

• In yeast, which genes control the transcription of haploid-specific genes?

• What is a transcriptional enhancer? What distinguishes it from a promoter?

• What is alternative splicing and what is its significance in gene regulation?

• Distinguish between a repressor and an aporepressor. Give an example of each.

• What is autoregulation? Distinguish between positive and negative autoregulation. Which would be used to amplify a weak induction signal? Which to prevent over-production?

• What term describes a gene that is expressed continuously?

Guide to Problem Solving

Problem 1: A gene *R* codes for a protein that is a negative regulator of transcription of a gene *S.* Is gene *S* transcribed in an R^- mutant? How does the situation differ if the product of *R* is a positive regulator of *S* transcription?

Answer: A negative regulator of transcription is needed to turn transcription off; hence, in the R^- mutant, transcription of *S* is constitutive. The opposite happens in positive regulation. A positive regulator of transcription is needed to turn transcription on, so if the product of *R* is a positive regulator, transcription does not occur in R^- cells.

Problem 2: For each of the following partial diploid genotypes of *E. coli,* state whether β-galactosidase is made and whether its synthesis is inducible or constitutive. The convention for writing partial diploid genotypes is to put the plasmid genes at the left of the slash and the chromosomal genes at the right.

(a) $lacZ^+\ lacY^-/lacZ^-\ lacY^+$

(b) $lacO^c\ lacZ^-\ lacY^+/lacZ^+\ lacY^-$

(c) $lacP^-\ lacZ^+/lacO^c\ lacZ^-$

(text box continued from previous page)

what each entry says about the function of the regulatory molecule.

3. Gene splicing in the origin of **human antibodies** is described at this site, which also offers much other information about the genetics of this system. If assigned to do so, write a one-page summary of the genetics of the immunoglobulin heavy-chain family, and discuss specifically the role of sequence homology and unequal crossing-over in the evolution of the gene families.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 11, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing geneticssites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 11.

(d) $lacI^{+}\ lacP^{-}\ lacZ^{+}/lacI^{-}\ lacZ^{+}$

(e) $lacI^{+}\ lacP^{-}\ lacY^{+}/lacI^{-}\ lacY^{-}$

Answer: The location of the genes does not matter because they function in the same manner whether present in a plasmid or in a chromosome.

(a) In this partial diploid, there are no *cis*-dominant mutations. The plasmid operon is $lacZ^{+}$ and can make the enzyme, whereas the chromosomal operon is $lacZ^{-}$ and cannot. The cell will produce the enzyme from the plasmid gene as long as the operon can be turned on. Turning it on requires the presence of functional regulatory elements, which is the case in this example. (Recall that if a gene is not listed, for example *lacI*, it is assumed to be wildtype.) It is also necessary that the inducer be able to enter the cell, which it can in this example because the chromosomal genotype can supply the *lacY* permease. Thus for this partial diploid, β-galactosidase can be made, and its synthesis is inducible.

(b) This partial diploid has the *cis*-dominant mutation $lacO^{c}$ in the plasmid, so the genes in the plasmid operon are always expressed. However, the *lacZ* gene in the plasmid makes a defective enzyme. The chromosome has a functional *lacZ* gene from which active enzyme can be made, but its synthesis is under the control of a normal operator (because the operator genotype is not indicated, it is wildtype). Hence, enzyme synthesis must be inducible. Thus the partial diploid makes a defective enzyme constitutively and a normal enzyme inducibly, so the overall phenotype is that the cell can be induced to make β-galactosidase.

(c) A promoter mutation, which is *cis*-dominant, is in the plasmid, which means that no *lac* mRNA can be made from this operon. The chromosome contains a *cis*-dominant *lacO*^{c} mutation that drives constitutive synthesis of an mRNA, but the enzyme is defective. Thus there is no way for the cell to make active β-galactosidase.

(d) The plasmid genotype contains a promoter mutation, so no mRNA can be produced from the *lacZ* structural gene. However, the *lacI* gene in the plasmid has its own promoter, so *lac* repressor molecules are present in the cell. The chromosomal genotype alone would make enzyme constitutively because of the *lacI* mutation, but the presence of the functional *lacI* product made by the plasmid means that any synthesis that occurs must be induced. The chromosomal operon can provide both β-galactosidase and permease, so β-galactosidase is inducible in this genotype.

(e) This genotype differs from that in part (d) by the presence of a *lacY*$^-$ mutation in the chromosome. Again, the plasmid contributes only *lacI* repressor to the cell, so any synthesis of the enzyme must be inducible. However, the chromosomal genotype is *lacY*$^-$. Because the *lacY*$^+$ allele in the plasmid cannot be expressed, no inducer can enter the cell, so the cell is unable to make any enzyme.

Problem 3: With regard to mating type in yeast, what phenotypes would each of the following haploid cells exhibit?

(a) a duplication of the mating-type gene giving the genotype *MAT***a**/*MAT*α

(b) a deletion of the *HML*α cassette in a *MAT***a** cell

Answer:

(a) The haploid cell expresses both **a** and α, so the **a**-specific genes, the α-specific genes, and the haploid-specific genes are all inactive. The phenotype is similar to the **a**α diploid. It will not mate, and if it attempts to sporulate, it will self-destruct.

(b) The phenotype is that of a normal **a** haploid, but mating-type switching to α cannot take place.

Analysis and Applications

11.1 The metabolic pathway for glycolysis is responsible for the degradation of glucose and is one of the fundamental energy-producing systems in living cells. Would you expect the enzymes in this pathway to be regulated? Why or why not?

11.2 Among mammals, the reticulocyte cells in the bone marrow extrude their nuclei in the process of differentiation into red blood cells. Yet the reticulocytes and red blood cells continue to synthesize hemoglobin. Suggest a mechanism by which hemoglobin synthesis can continue for a long period of time in the absence of the hemoglobin genes.

11.3 Several eukaryotes are known in which a single effector molecule regulates the synthesis of different proteins coded by distinct mRNA molecules—say, X and Y. At what level in the process of gene expression does regulation occur in each of the following situations?

(a) Neither nuclear nor cytoplasmic RNA can be found for either X or Y.

(b) Nuclear but not cytoplasmic RNA can be found for both X and Y.

(c) Both nuclear and cytoplasmic RNA can be found for both X and Y, but none of it is associated with polysomes.

11.4 Is it necessary for the gene that codes for the repressor of a bacterial operon to be near the structural genes? Why or why not?

11.5 Consider a eukaryotic transcriptional activator protein that binds to an enhancer sequence and promotes transcription. What change in regulation would you expect from a duplication in which several copies of the enhancer were present instead of just one?

11.6 The following questions pertain to the *lac* operon in *E. coli.*

(a) Which proteins are regulated by the repressor?

(b) How does binding of the *lac* repressor to the *lac* operator prevent transcription?

(c) Is production of the *lac* repressor constitutive or induced?

11.7 The permease of *E. coli* that transports the α-galactoside melibiose can also transport lactose, but it is temperature-sensitive: Lactose can be transported into the cell at 30°C but not at 37°C. In a strain that produces the melibiose permease constitutively, what are the phenotypes of *lacZ*$^-$ and *lacY*$^-$ mutants at 30°C and 37°C?

11.8 How do inducers enable transcription to occur in a bacterial operon under negative transcriptional control?

11.9 When glucose is present in an *E. coli* cell, is the concentration of cyclic AMP high or low? Can a mutant with either an inactive adenyl cyclase gene or an inactive *crp* gene synthesize β-galactosidase? Does the binding of cAMP-CRP to DNA affect the binding of a repressor?

11.10 The operon allows a type of coordinate regulation of gene activity in which a group of enzymes with related functions are synthesized from a single polycistronic mRNA. Does this imply that all proteins in the polycistronic mRNA are made in the same quantity? Explain.

11.11 Both repressors and aporepressors bind molecules that are substrates or products of the metabolic pathway encoded by the genes in an operon. Generally speaking, which binds the substrate of a metabolic pathway and which the product?

11.12 Is an attenuator a region of DNA that, like an operator, binds with a protein? Is RNA synthesis ever initiated at an attenuator?

11.13 How do steroid hormones induce transcription of eukaryotic genes?

11.14 In regard to mating-type switching in yeast, what phenotype would you expect from a type α cell that has a deletion of the *HMR***a** cassette?

11.15 What mating-type phenotype would you expect of a haploid cell of genotype *MAT***a',** where the prime denotes a mutation that renders the **a**1 protein inactive? What mating type would you expect in the diploid *MAT***a'**/*MAT*α?

11.16 What mating-type phenotype would you expect from a diploid cell of genotype *MAT***a**/*MAT*α with a mutation in α in which the α2 gene product functions normally in turning off the **a**-specific genes but is unable to combine with the **a**1 product?

11.17 If a wildtype *E. coli* strain is grown in a medium without lactose or glucose, how many proteins are bound to the *lac* operon? How many are bound if the cells are grown in glucose?

11.18 A mutant strain of *E. coli* is found that produces both β-galactosidase and permease constitutively (that is, whether lactose is present or not).

(a) What are two possible genotypes for this mutant?

(b) A second mutant is isolated that produces no active β-galactosidase at any time but produces permease if lactose is present in the medium. What is the genotype of this mutant?

(c) A partial diploid is created from the mutants in parts (a) and (b): When lactose is absent, neither enzyme is made, and when lactose is present, both enzymes are made. What is the genotype of the mutant in part (a)?

11.19 A *lacI*$^{+}$ *lacO*$^{+}$ *lacZ*$^{+}$ *lacY*$^{+}$ Hfr strain is mated with an F^{-} *lacI*$^{-}$ *lacO*$^{+}$ *lacZ*$^{-}$ *lacY*$^{-}$ strain. In the absence of any inducer in the medium, β-galactosidase is made for a short time after the Hfr and F^{-} cells have been mixed. Explain why it is made and why only for a short time.

11.20 What amino acids can be inserted at the site of the UGA codon that is suppressed by a suppressor tRNA?

11.21 An *E. coli* mutant is isolated that is simultaneously unable to utilize a large number of sugars as sources of carbon. However, genetic analysis shows that the operons responsible for metabolism of each sugar are free of mutations. What genotypes of this mutant are possible?

Challenge Problems

11.22 The histidine operon is negatively regulated and contains ten structural genes for the enzymes needed to synthesize histidine. The repressor protein is also coded within the operon—that is, in the polycistronic mRNA molecule that codes for the other proteins. Synthesis of this mRNA is controlled by a single operator regulating the activity of a single promoter. The co-repressor of this operon is tRNA^{His}, to which histidine is attached. This tRNA is not coded by the operon itself. A collection of mutants with the following defects is isolated. Determine whether the histidine enzymes would be synthesized by each of the mutants and whether each mutant would be dominant, *cis*-dominant only, or recessive to its wildtype allele in a partial diploid.

(a) The promoter cannot bind with RNA polymerase.

(b) The operator cannot bind the repressor protein.

(c) The repressor protein cannot bind with DNA.

(d) The repressor protein cannot bind histidyl-tRNA^{His}.

(e) The uncharged tRNA^{His} (that is, without histidine attached) can bind to the repressor protein.

11.23 The attenuator of the histidine operon contains seven consecutive histidines. The relevant part of the attentuator coding sequence is

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5'-AAACACCACCATCATCACCATCATCCTGAC-3'
```

A mutation occurs in which an additional A nucleotide is inserted immediately after the red A. What amino acid sequences are coded by the wildtype and mutant attenuators? What phenotype would you expect of the mutant?

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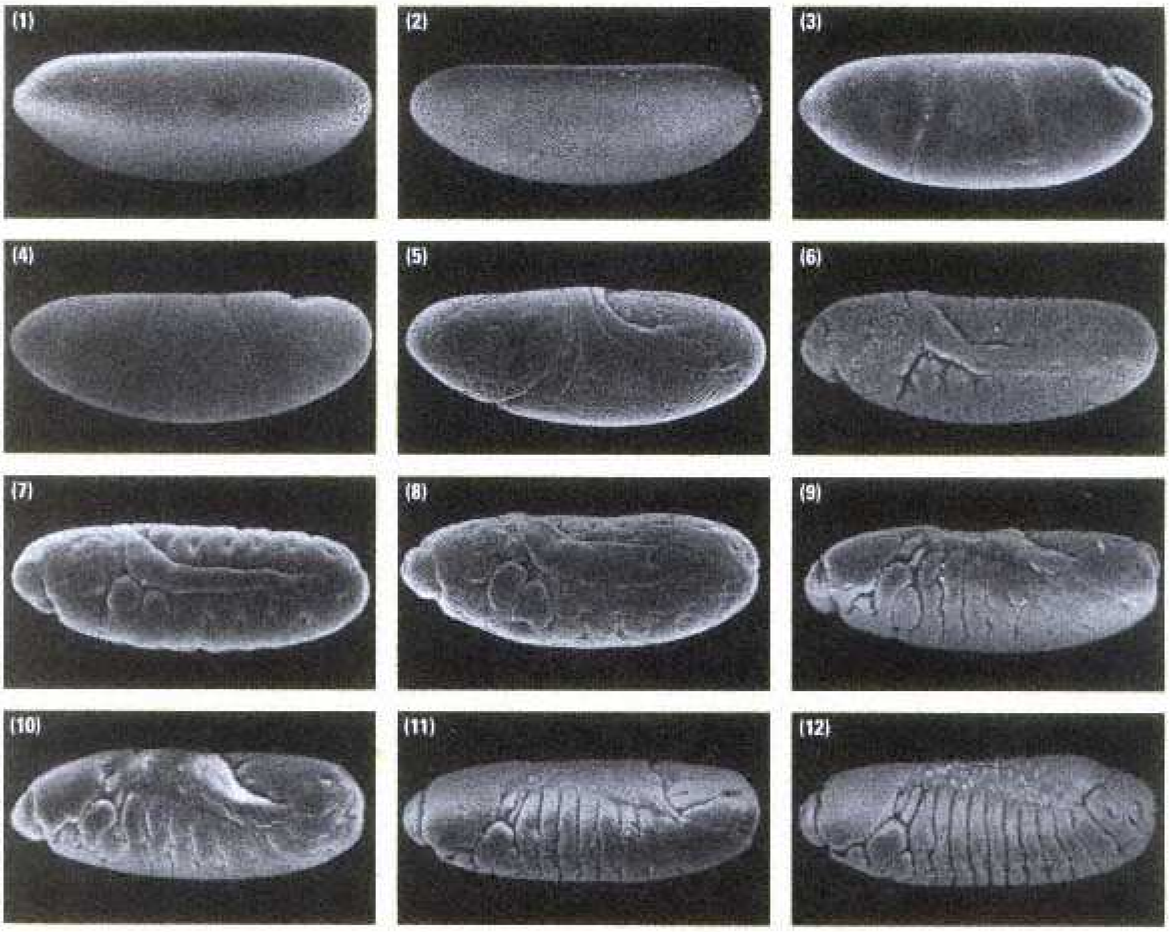


Early development in *Drosophila,* represented by scanning electron micrographs of 12 stages. Each embryo is arranged with the anterior end at the left and the ventral side down. The stages are arranged chronologically from 1 throught 12. The first five stages are the early cleavage divisions, ending with the cellular blastoderm (stage 5) at 3.25 hours after fertilization. Gastrulation (stages 6–8) is followed by formation of the characteristic pattern of segments along the body, ending with stage 12 at 9 hours after fertilization. Compare stages 1–5 with Figure 12.21 and stage 12 with Figure 12.23.
[Courtesy of Thomas C. Kaufman and Rudi Turner.]

Chapter 12— The Genetic Control of Development

CHAPTER OUTLINE

PRINCIPLES

• In animals cells, maternal gene products in the oocyte control the earliest stages of development, including the establishment of the main body axes.

• Developmental genes are often controlled by gradients of gene products, either within cells or across parts of the embryo.

• Regulation of developmental genes is hierarchical—genes expressed early in development regulate the activities of genes expressed later.

• Regulation of developmental genes is combinatorial—each gene is controlled by a combination of other genes.

• Many of the fundamental processes of pattern formation appear to be similar in animals and plants.

CONNECTIONS

CONNECTION: Distinguished Lineages

John E. Sulston, E. Schierenberg, J. G. White, and J. N. Thomson 1983
The embryonic cell lineage of the nematode Caenorhabditis elegans

CONNECTION: Embryo Genesis

Christiane Nüsslein-Volhard and Eric Wieschaus 1980
Mutations affecting segment number and polarity in Drosophila

Understanding gene regulation is central to understanding how an organism as complex as a human being develops from a fertilized egg. In development, genes are expressed according to a prescribed program to ensure that the fertilized egg divides repeatedly and that the resulting cells become specialized in an orderly way to give rise to the fully differentiated organism. The genotype determines not only the events that take place in development, but also the temporal order in which the events unfold.

Genetic approaches to the study of development often make use of mutations that alter developmental patterns. These mutations interrupt developmental processes and make it possible to identify

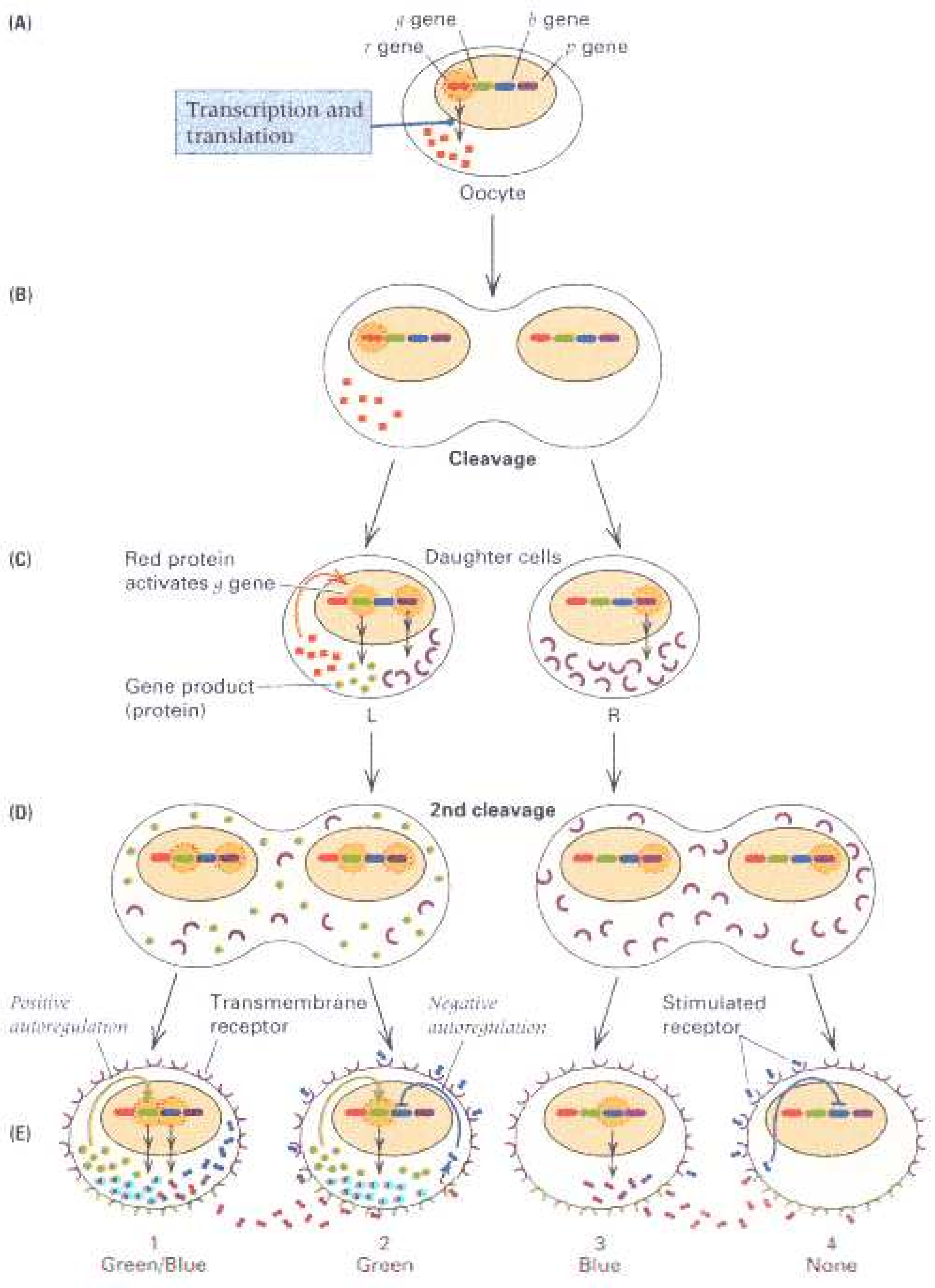


Figure 12.1
Hypothetical example illustrating some regulatory mechanisms that result in differences in gene expression in development. (A) The *r* gene is expressed in the oocyte. (B) Polarized presence of the r gene product in the egg. (C) Presence of the *r* gene product in cell L stimulates transcription of the *g* gene. The *p* gene codes for a transmembrane receptor expressed in both L and R cells. (D) The *g* gene has positive autoregulation and so continues to be expressed in cells 1 and 2. (E) Expression of the *b* gene in one cell represses its expression in the neighboring cell by stimulation of the *p* gene transmembrane receptor. The result of these mechanisms is that cells 1 through 4 differ in their combinations of *b* gene and *g* gene activity.

factors that control development and to study the interactions among them. This chapter demonstrates how genetics is used in the study of development. Many of the examples come from the nematode *Caenorhabditis elegans* and the fruit fly *Drosophila melanogaster*, because these organisms have been studied intensively from the standpoint of developmental genetics.

The key process in development is **pattern formation,** which means the emergence, from cell division and differentiation of the fertilized egg, of spatially organized and specialized cells that form the embryo. One might make a analogy between pattern formation in development and a pattern formed by fitting the pieces of a jigsaw puzzle together, but the analogy is not quite right. In a jigsaw puzzle, the pattern exists independently of the shape and position of the pieces; the cuts in the pattern are superimposed on a preexisting picture that is merely reassembled. In biological development, the emergence of the image on each piece is *caused by* the shape and position of the piece. It is remarkable also that the picture changes throughout life as the organism undergoes growth and aging.

12.1— Genetic Determinants of Development

Conceptually, the relationship between genotype and development is straightforward:

> The genotype contains a developmental program that unfolds and results in the expression of different sets of genes in different types of cells.

In other words, development consists of a program that results in the specific expression of some genes in one cell type and not in another. As development unfolds, cell types progressively appear that differ qualitatively in the genes that are expressed. Whether a particular genes expressed may depend on the presence or absence of a particular transcription factor, a change in chromatin structure, or the synthesis of a particular receptor molecule. These and other molecular mechanisms define the pattern of interactions by which one gene controls another. Collectively, these regulatory interactions ultimately determine the fate of the cell. Moreover, many interactions that are important in development make use of more general regulatory elements. For example, developmental events often depend on regional differences in the concentration of molecules within a cell or within an embryo, and the activity of enhancers may be modulated by the local concentration of these substances.

The formation of an embryo is also affected by the environment in which development takes place. Although genotype and environment work together in development, the genotype determines the developmental potential of the organism. Genes determine whether a developing embryo will become a nematode, a fruit fly, a chicken, or a mouse. However, the expression of the genetically determined developmental potential is also influenced by the environment—in some cases, very dramatically. Fetal alcohol syndrome is one example in which an environmental agent, chronic alcohol poisoning, affects various aspects of fetal growth and development.

Development includes many examples in which genes are selectively turned on or off by the action of regulatory proteins that respond to environmental signals. The identification of genetic regulatory interactions that operate during development is an important theme in developmental genetics. The implementation of different regulatory interactions means that genetically identical cells can become qualitatively different in the genes that are expressed.

Several common mechanisms by which genes become activated or repressed at particular stages in development or in particular tissues are illustrated in the hypothetical example in Figure 12.1. In this example, the developmentally regulated genes are denoted *r* (red), *g* (green), *b* (blue), and *p* (purple). The initial event is transcription of the *r* gene in the oocyte (Figure 12.1A). Unequal partitioning of the *r* gene product into one region of the egg establishes a polarity or regionalization of the egg with respect to presence of the *r* gene product (Figure 12.1B). When cleavage takes place (Figure 12.1C), the polarized cell produces daughter cells either with (L, left) or without (R, right) the *r* gene product. The *r* gene product is a transcriptional activator of the *g* gene, so the *g* gene product is expressed in cell L but not in cell R; hence, even at the two-cell stage, the daughter cells may differ in gene expression as a result of the initial polarization of the egg. Also in the two-cell stage, the *p* gene is expressed in both daughter cells. This gene codes for a **transmembrane receptor** containing amino acid sequences that span the cell membrane.

The presence of receptor molecules in the cell membrane provides an important mechanism for signaling between cells in development, as illustrated in this example with the regulation of the *b* gene in Figure 12.1D. In the absence of the transmembrane receptor, the *b* gene would be expressed in all four cells. However, expression of the *b* gene in one cell inhibits its expression in the neighboring cell because of the transmembrane receptor. The mechanism is that the product of the *b* gene stimulates the receptor of neighboring cells, and stimulation of the receptor represses transcription of *b.* The process in which gene expression in one cell inhibits expression of the same or differing genes in neighboring cells is known as *lateral inhibition.*

At the same time that *b*-gene expression is regulated by lateral inhibition, the initial activation of the *g* gene in cell L is retained in daughter cells 1 and 2 because the *g* gene product has a positive autoregulatory activity and stimulates its own transcription. Positive autoregulation is one way in which cells can amplify weak or transient regulatory signals into permanent changes in gene expression. The result of lateral inhibition of *b* expression and *g* autoregulation is that cells 1 through 4 in Figure 12.1E, though genetically identical, are developmentally different because they express different combinations of the *g* and *b* genes. Specifically, cell 1 has *g* and *b* activity, cell 2 has *g* activity only, cell 3 has *b* activity only, and cell 4 has neither.

Among the most intensively studied developmental genes are those that act early in development, before tissue differentiation, because these genes establish the overall pattern of development.

12.2—
Early Embryonic Development in Animals

The early development of the animal embryo establishes the basic developmental plan for the whole organism. Fertilization initiates a series of mitotic **cleavage** divisions in which the embryo becomes multicellular. There is little or no increase in overall size compared with the egg, because the cleavage divisions are accompanied by little growth and merely partition the fertilized egg into progressively smaller cells. The cleavage divisions form the **blastula,** which is essentially a ball of about 10^4 cells containing a cavity (Figure 12.2). Completion of the cleavage divisions is followed by the formation of the **gastrula** through an infolding of part of the blastula wall and extensive cellular migration. In the reorganization of cells in the gastrula, the cells become arranged in several distinct layers. These layers establish the basic body plan of an animal. In higher plants, as we shall see later, the developmental processes differ substantially

from those in animals.

A fertilized animal egg has full developmental potential because it contains the genetic program in its nucleus and macro-molecules present in the oocyte cytoplasm that are necessary for giving rise to an entire organism. Full developmental potential is maintained in cells produced by the early cleavage divisions. However, the developmental potential is progressively channeled and limited in early embryonic development. Cells within the blastula usually become committed to particular developmental outcomes, or **fates,** that limit the differentiated states possible among descendant cells.

Autonomous Development and Intercellular Signaling

Two principal mechanisms progressively restrict the developmental potential of cells within a lineage.

• Developmental restriction may be **autonomous,** which means that it is determined by genetically programmed changes in the cells themselves.

• Cells may respond to **positional information,** which means that developmental restrictions are imposed by the position of cells within the embryo. Positional information may be mediated by signaling interactions between neighboring cells or by gradients in concentration of morphogens. A **morphogen** is a molecule that participates directly in the control of growth and form during development.

The distinction between autonomous development and development that depends on positional information is illustrated in Figure 12.3. In the normal embryo (Figure 12.3A), cells 1 and 2 have different fates either because of autonomous developmental programs or because they respond to positional information near the anterior and posterior near the anterior and posterior ends. When the cells are transplanted (Figure 12.3B), autonomous development is indicated if the developmental fate of cells 1 and 2 is unchanged in spite of their new locations; in this case, the embryo has anterior and posterior reversed. When development depends on positional information, the transplanted cells respond to their new locations, and the embryo is normal.

Restriction of developmental fate is usually studied by transplanting cells of the embryo to new locations to determine whether they can substitute for the cells that they displace. Alternatively, individual cells are isolated from early embryos and cultured in laboratory dishes to study their developmental potential. In some eukaryotes, such as the soil nematode *Caenorhabditis elegans*, many lineages develop autonomously according to genetic programs that are induced by interactions with neighboring cells very early in embryonic development. Subsequent cell interactions also are important, and each stage in development is set in motion by the successful completion of the preceding stage. Figure 12.4 illustrates the first three cell

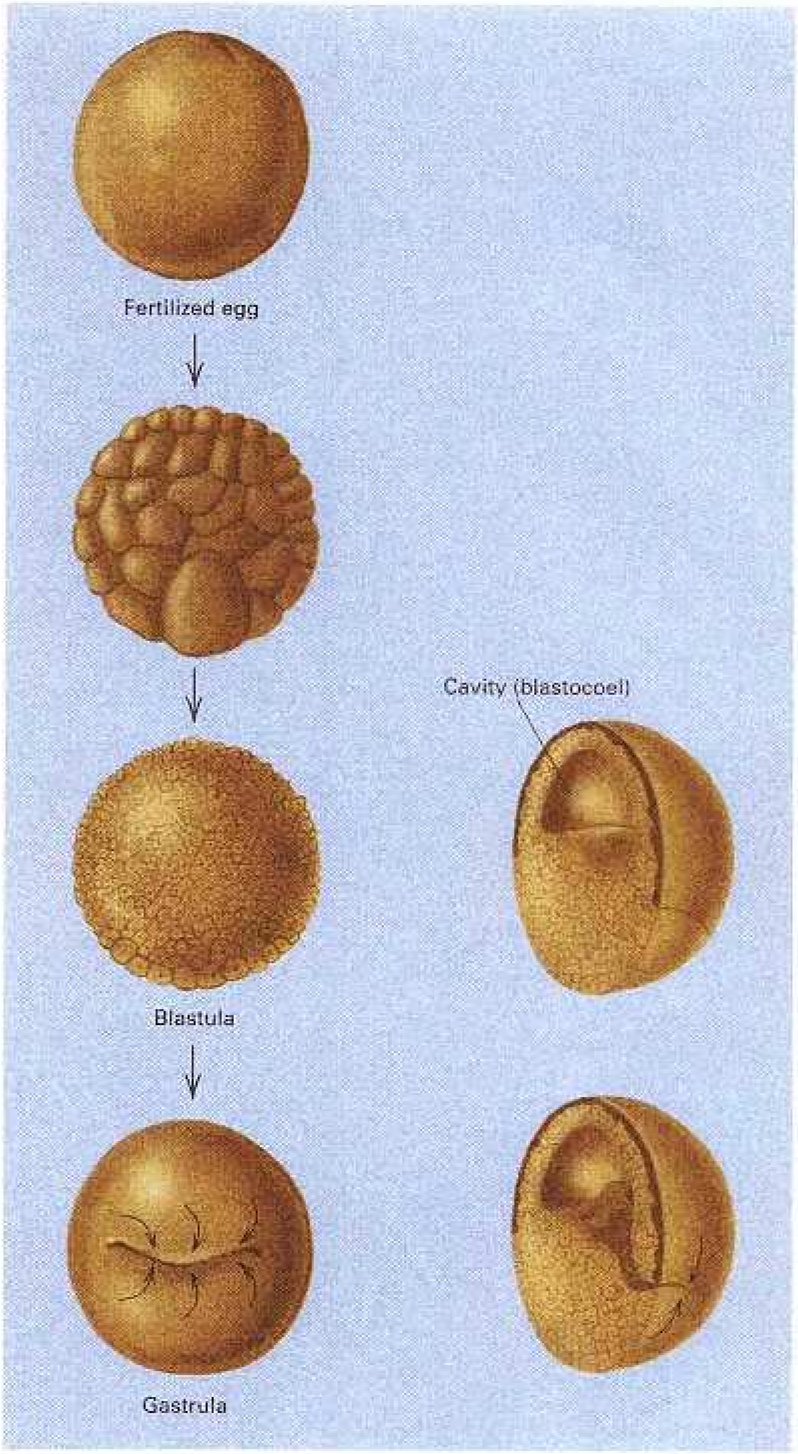


Figure 12.2
Early development of the animal embryo. The cleavage divisions of the fertilized egg result in first a clump of cells and then a hollow ball of cells the blastula). Extensive cell migrations form the gastrula and establish the basic body plan of the embryo.

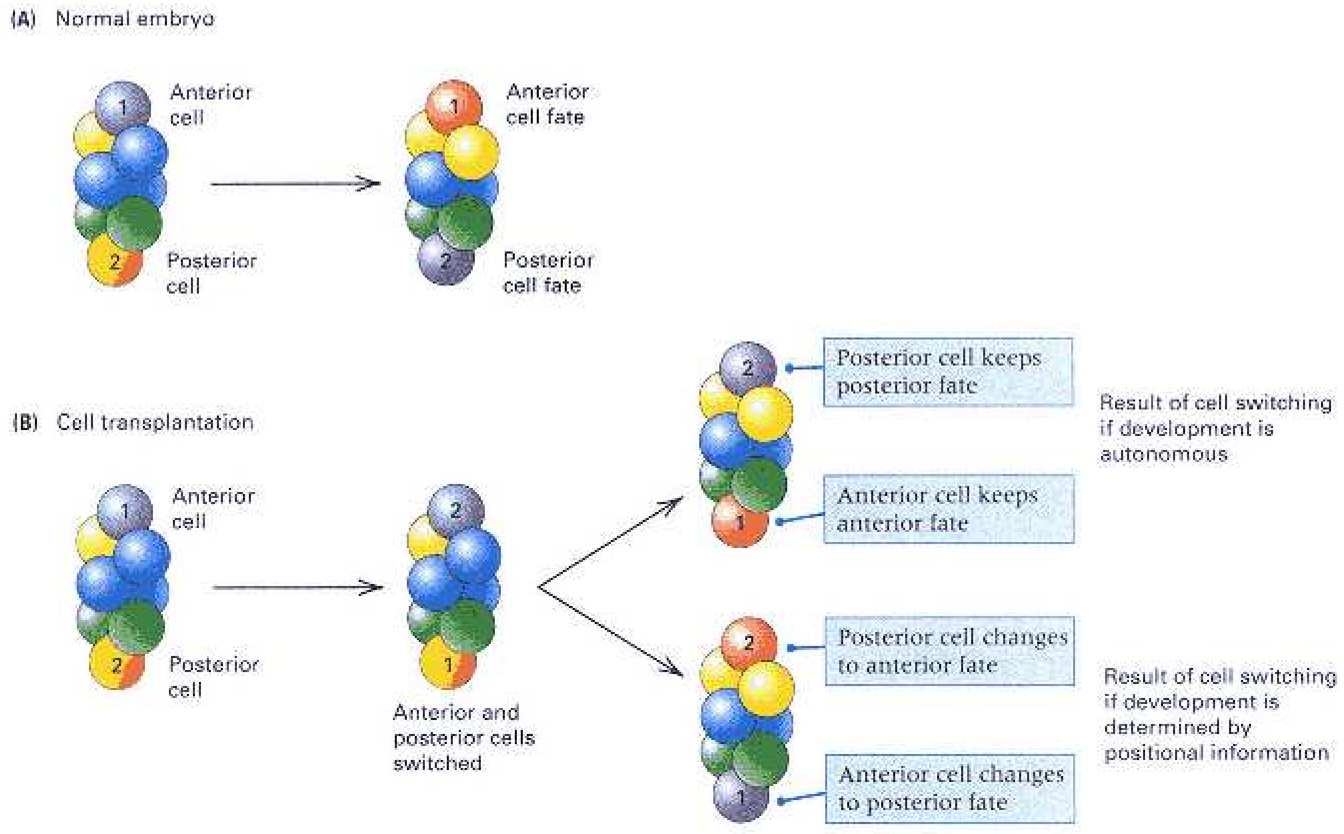


Figure 12.3
Distinction between autonomous determination and positional information (A) Cells 1 and 2 differentiate normally as shown.(B) Transplantation of the cells to reciprocal locations. If the transplanted cells differentiate autonomously, then they differentiate asthey would in the normal embryo, but in their new locations. If they differentiate in response to positional information (signals fromneighboring cells), then their new positions determine their fate.

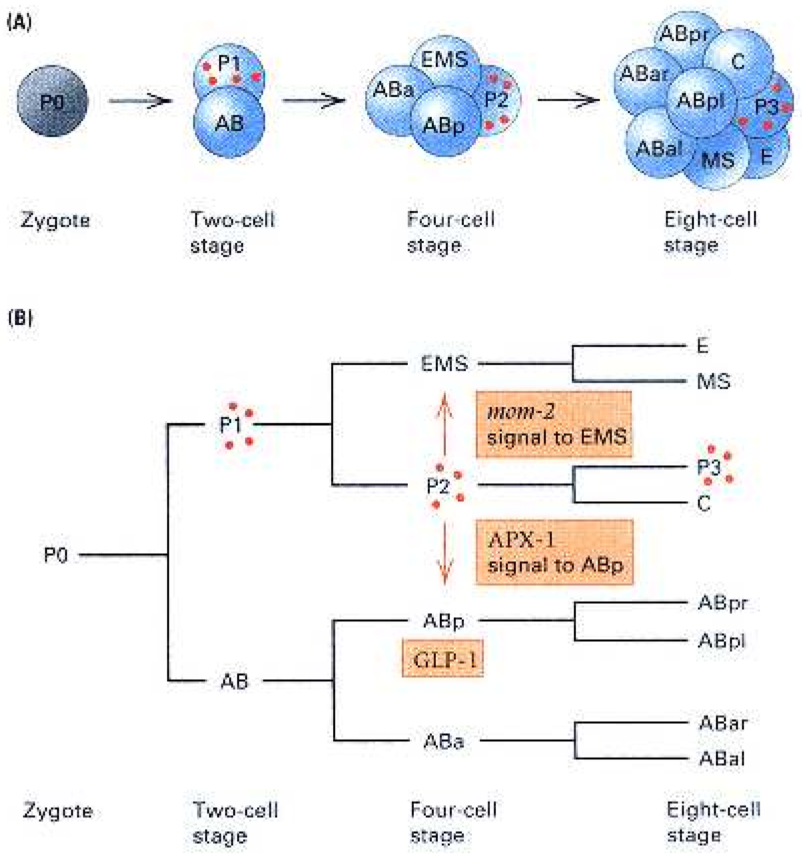


Figure 12.4
Early cell divisions in *C. elegans* development. (A) Spatial organization of cells. (B) Lineage relationships of the cells. The transmission of the polar granules illustrates cellautonomous development. The arrows denote cell-to-cell

signaling mechanisms that determine developmental fate.
(From Wade Roush 1996. *Science* 272: 1871.)

divisions in the development of *C. elegans,* which result in eight embryonic cells that differ in their genetic activity and developmental fate. The determination of cell fate in these early divisions is in part autonomous and in part a result of interactions between cells. Figure 12.4B shows the lineage relationships between the cells. Cell-autonomous mechanisms are illustrated by the transmission of cytoplasmic particles called polar granules from the cells P0 to P1 to P2 to P3. Normal segregation of the polar granules is a function of microfilaments in the cytoskeleton. Cell-signaling mechanisms are illustrated by the effects of P2 on EMS and on ABp. The EMS fate is determined by the activity of the *mom-2* gene in P2. The P2 cell also produces a signaling molecule, APX-1, which determines the fate of ABp through the cell-surface receptor GLP-1. (The specific mechanisms that determine early cell fate in *C. elegans* strongly resemble some of the general mechanisms outlined in Figure 12.1.) In contrast to *C. elegans*, in which many developmental decisions are cell-autonomous, in *Drosophila* and *Mus* (the mouse), regulation by cell-to-cell signaling is more the rule than the exception. The use of cell signaling to regulate development provides a sort of insurance that helps to overcome the death of individual cells that might happen by accident during development.

One special case of cell-to-cell signaling is **embryonic induction,** in which the development of a major embryonic structure is determined by a signal sent from neighboring cells. An example is found in early development of the sea urchin, *Strongylocentrotus purpuratus,* in which, after the fourth cleavage division, specialized cells called micromeres are produced at the ventral pole of the embryo (Figure 12.5A and B). After additional cell proliferation, the region of the embryo immediately above the micromeres folds inward to form a tube, the archenteron, that eventually develops into the stomach, intestine, and related structures. If micromere cells from a 16-cell embryo are removed and transplanted into the dorsal region of another embryo at the 8-cell stage (Figure 12.5C), then the next cleavage results in an embryo with two sets of micromeres (Figure 12.5D), the normal set being at the ventral end and the transplanted set at the dorsal end. In these embryos, as development proceeds, an archenteron forms in the normal position from cells lying above the ventral micromeres. However, cells lying beneath the upper, transplanted micromeres also form an archenteron (Figure 12.5E), which indicates that the transplanted micromeres are capable of inducing the development of archenteron in the cells with which they come into contact.

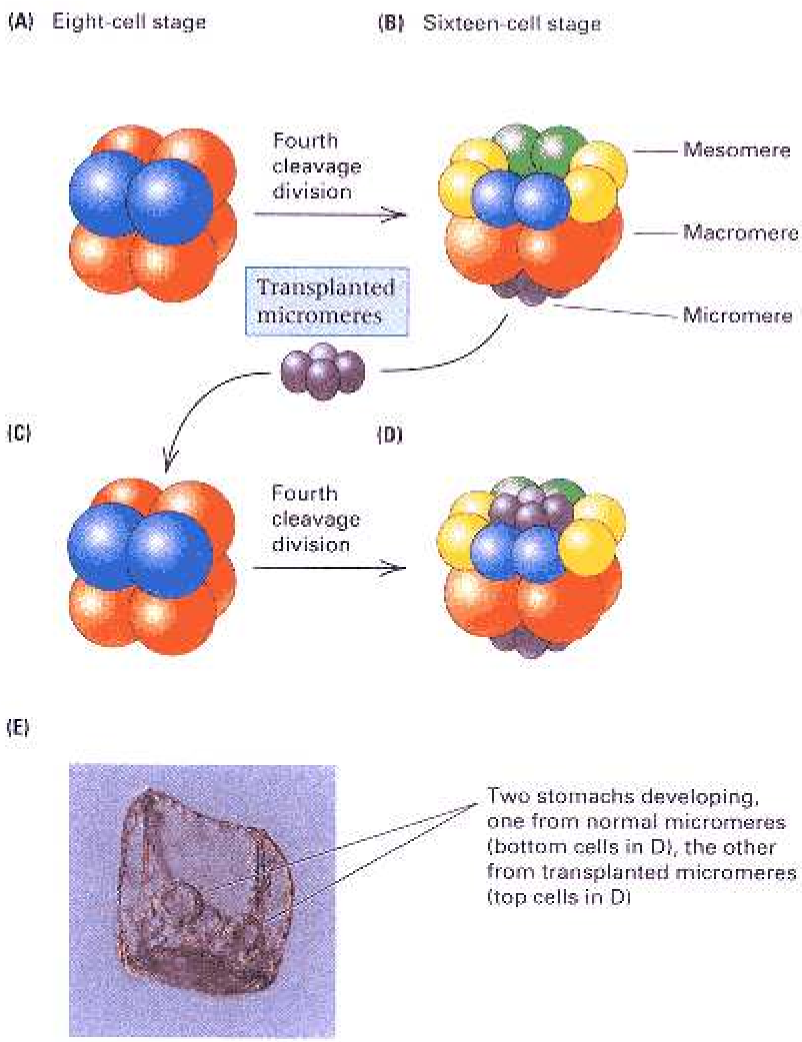


Figure 12.5
Embryonic induction of the archenteron by micromere cells in the sea

urchin. (A) Normal embryo at the 8-cell stage. (B) Normal embryo at the 16-cell stage showing micromeres on ventral side. (C) Transplanted 8-cell embryo with micromeres from another embryo placed at the dorsal end. (D) Transplanted embryo after the fourth cleavage division with two sets of micromeres. (E) The result of the transplant is an embryo with two archenterons. The doughnut-shaped objects in this embryo are the developing stomachs.
[Photomicrograph, courtesy of Eric H. Davidson and Andrew Ransick. From A. Ransick and E. H. Davidson. 1993. *Science* 259: 1134.]

Early Development and Activation of the Zygote Genome

In most animals, the earliest events in embryonic development do not depend on genetic information in the cell nucleus of the zygote. For example, fertilized frog eggs with the nucleus removed are still able to carry out the cleavage divisions and form rudimentary blastulas. Similarly, when gene transcription in sea urchin or amphibitian embryos is blocked shortly after fertilization, there is no effect on the cleavage divisions or on blastula formation, but gastrulation does not take place. The reason why the genes in the zygote are not needed in the early stages of development is that the oocyte cytoplasm produced by the mother contains all the necessary macromolecules.

Following the period early in development in which the genes in the zygote nucleus are not needed, the embryo becomes dependent on the activity of its own genes. In mice, and probably in all mammals, the zygotic genes are needed much earlier than in lower vertebrates. The shift from control by the maternal genome to control by the zygote genome begins in the two-cell stage of development, when proteins coded by the zygote nucleus are first detectable. Inhibitors of transcription stop development of the mouse embryo when they are applied at any time after the first cleavage division. However, even in mammals, the earliest stages of development are greatly influenced by the cytoplasm of the oocyte, which determines the planes of the initial cleavage divisions and other events that ultimately affect cell fate.

Early activation of the zygote nucleus in mammals may be necessary because, in gamete formation, certain genes undergo a process of *imprinting* that prevents their expression in either the egg or the sperm nuclei that unite to form the zygote nucleus. Imprinting of a gene is thought to be associated with methylation of the DNA in the gene (Chapter 11). Very few genes—probably fewer than ten in the mouse—are subject to imprinting. Among these are the gene for an insulin-like growth factor (*Igf2*), which is imprinted during oogenesis, and the gene for the *Igf2* transmembrane receptor (*Igf2-r*), which is imprinted during spermatogenesis. Therefore, expression of both *lgf2* and *lgf2-r* in the embryo requires activation of the sperm nucleus for *lgf2* and of the egg nucleus for *lgf2-r.* Imprinting also affects some genes in the human genome and has been implicated in some unusual aspects of the genetic transmission of the fragile-X syndrome of mental retardation (Chapter 7).

Composition and Organization of Oocytes

The oocyte is a diploid cell during most of oogenesis. The cytoplasm of the oocyte is extensively organized and regionally differentiated (Figure 12.6). This spatial differentiation ultimately determines the different developmental fates of cells in the blastula. The cytoplasm of the animal egg has two essential functions:

1. Storage of the molecules needed to support the cleavage divisions and the rapid RNA and protein synthesis that take place in early embryo genesis.

2. Organization of the molecules in the cytoplasm to provide the positional

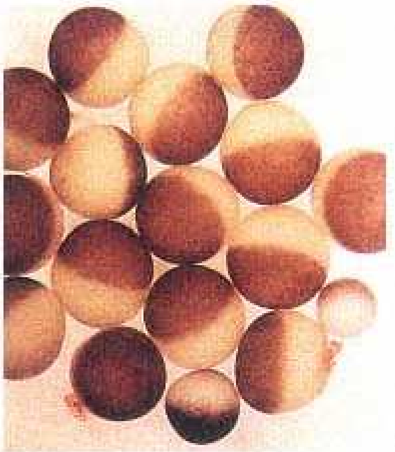


Figure 12.6
The animal oocyte is highly organized internally, which is revealed by the visible differences between the dorsal (dark) and ventral (light) parts of these Xenopus eggs. The regional organization of the oocyte determines many of the critical events in early development.

[Courtesy of Michael W.Klymkowsky.]

information that results in differences between cells in the early embryo.

To establish the proper composition and organization of the oocyte requires the participation of many genes within the oocyte itself and gene products supplied by adjacent helper cells of various types (Figure 12.7). Numerous maternal genes are transcribed in oocyte development, and the mature oocyte typically contains an abundance of transcripts that code for proteins needed in the early embryo. Some of the maternal mRNA transcripts are stored complexed with proteins in special ribonucleoprotein particles that cannot be translated, and release of the mRNA, enabling it to be translated, does not happen until after fertilization.

Developmental instructions in oocytes are determined in part by the presence of distinct types of molecules at different positions within the cell and in part by gradients of morphogens that differ in concentration from one position to the next. Although the oocyte contains the products of many genes, only a small number of genes are expressed exclusively in the formation of the oocyte. Most genes expressed in oocyte formation are also important in the development of other tissues or at later times in development. Therefore, it is not only gene products but also the spatial organization of the gene products within the cell that give the oocyte its unique developmental potential.

12.3— Genetic Control of Cell Lineages

The mechanisms that control early development can be studied genetically by isolating mutations with early developmental abnormalities and altered cell fates. In most organisms, it is difficult to trace the lineage of individual cells in development because the embryo is not transparent, the cells are small and numerous, and cell migrations are extensive. The **lineage** of a cell refers to the ancestor-descendant relationships among a group of cells. A cell lineage can be illustrated with a **lineage diagram,** the sort of cell pedigree in Figure 12.8 that

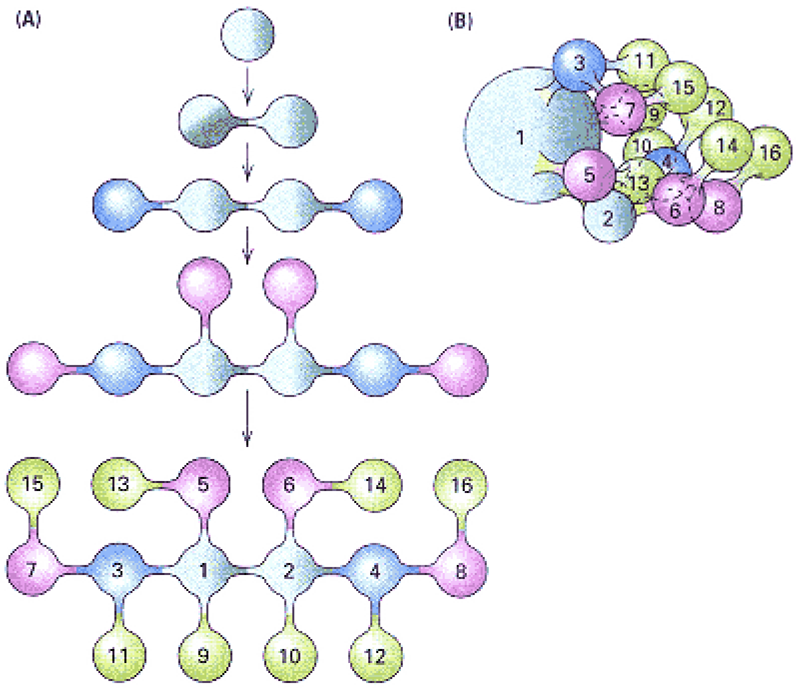


Figure 12.7
Pattern of cell divisions in the development of the Drosophila oocyte. (A) The cells surrounding the oocyte are connected to it and to each other by cytoplasmic bridges. These cells synthesize products transported into the oocyte and contribute to its regional organization. (B) Geometrical organization of the cells.
[From R. C. King. 1965. Genetics, Oxford University Press.]

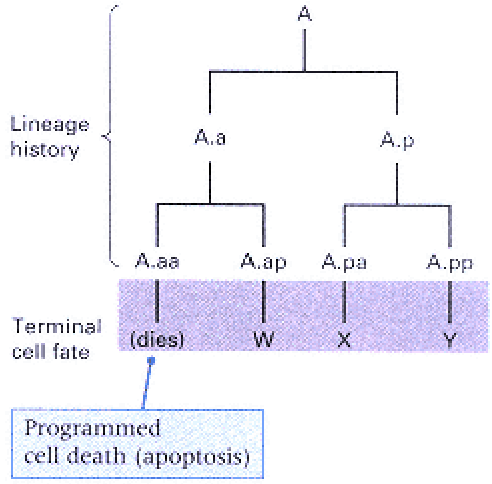


Figure 12.8
Hypothetical cell-lineage diagram. Different terminally differentiated cell fates are denoted W, X, and Y. One cell in the lineage (A.aa) undergoes programmed cell death. The lowercase letters a and p denote anterior and posterior daughter cells. For example, cell A.ap is the posterior daughter of the anterior daughter of cell A.

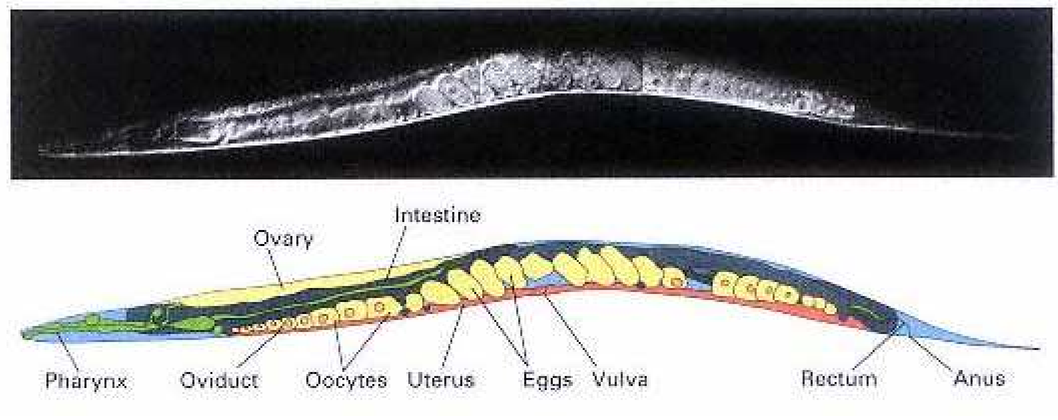


Figure 12.9
The soil nematode Caenorhabditis elegans. This organism offers several advantages for the genetic analysis of development, including the fact that all individuals of both sexes exhibit an identical pattern of cell lineages in the development of the somatic cells. DNA sequencing of the 100-megabase genome is nearing completion.
[Photograph courtesy of Tim Schedl.]

shows each cell division and that indicates the terminal differentiated state of each cell. Figure 12.8 is another lineage diagram of a hypothetical cell A in which the cell fate is either programmed cell death or one of the terminally differentiated cell types designated W, X, and Y. The letter symbols are the kind normally used for cells in nematodes, in which the name denotes the cell lineage according to ancestry and position in the embryo. For example, the cells A.a and A.p are the anterior and posterior daughters of cell A, respectively; and A.aa and A.ap are the anterior and posterior daughters of cell A.a.

Genetic Analysis of Development in the Nematode

The soil nematode *Caenorhabditis elegans* (Figure 12.9) is popular for genetic studies because it is small, is easy to culture, and has a short generation time with a large number of offspring. The worms are grown on agar surfaces in petri dishes and feed on bacterial cells such as *E. coli*. Because they are microscopic in size, as many as 10^5 animals can be contained in a single petri dish. Sexually mature adults of *C. elegans* are capable of laying more than 300 eggs within a few days. At 20°C, it requires about 60 hours for the eggs to hatch, undergo four larval molts, and become sexually mature adults.

Nematodes are diploid organisms with two sexes. In *C. elegans,* the two sexes are the hermaphrodite and the male. The hermaphrodite contains two X chromosomes (XX), produces both functional eggs and functional sperm, and is capable of self-fertilization. The male produces only sperm and fertilizes the hermaphrodites. The sex-chromosome constitution of *C. elegans* consists of a single X chromosome; there is no Y chromosome, and the male karyotype is XO.

The transparent body wall of the worm has made possible the study of the division, migration, and death or differentiation of all cells present in the course of development. Nematode development is unusual in that the pattern of cell division and differentiation is virtually identical from one individual to the next; that is, cell division and differntiation are highly stereotyped. The result is that both sexes show the same geometry in the number and arrangement of somatic cells. The hermaphrodite contains exactly 959 somatic cells, and the male contains exactly 1031 somatic cells. The complete developmental history of each somatic cell is known at the cellular and ultrastructural level, including the *wiring diagram,* which describes all the

CONNECTION Distinguished Lineages

John E. Sulston,[1] E. Schierenberg,[2] J. G. White,[1] and J. N. Thomson[1] 1983

[1]Medical Research Council Laboratory for Molecular Biology, Cambridge, England
[2]Max-Planck Institute for Experimental Medicine, Gottingen, Germany
The Embryonic Cell Lineage of the Nematode Caenorhabditis elegans

The data produced in this landmark study form the basis for interpreting developmental mutants in the nematode worm. This long paper offers voluminous data, and it is presently available through the Internet. During embryogenesis, 671 cells are generated; 113 of these, or 17%, undergo programmed cell death. What is the reason for such a high proportion of programmed cell deaths? The embryonic lineage is highly invariant—the same from one organism to the next. Why is there not more developmental flexibility, as is found in most other organisms? These issues are addressed in this excerpt, in which the emphasis is on the historical background and motivation of the study, the big picture of development, and interpretation of the lineage in terms of the evolution of the nematode. The technique of Nomarski microscopy is a modern invention also called differential interference contrast microscopy. When light passes through living material, it changes phase according to the refractive index of the material. Adjacent parts of a cell or organism that differ in refractive index cause different changes in phase. When two sets of waves combine after passing through an object, the difference in phase creates an interference pattern that yields an image of the object. The major advantage of Nomarski microscopy is that it can be used to observe living tissue.

This report marks the completion of a project begun over one hundred years ago—namely the determination of the entire cell lineage of a nematode. Nematode embryos were attractive to nineteenth-century biologists because of their simplicity and the reproducibility of their development, and considerable progress was made in determining their lineages by the use of fixed specimens. By the technique of Nomarski microscopy, which is nondestructive and yet produces high resolution, cells can be followed in living larvae. The use of living material lends a previously unattainable continuity and certainty to the observations, and has permitted the origin and fate of every cell in one nematode species [*Caenorhabditis elegans*] to be determined. Thus, not only are the broad relationships between tissues now known unambiguously but also the detailed pattern of cell fates is clearly revealed. . . . The lineage is of

The nematode belongs to an ancient phylum, and its cell lineage is a piece of frozen evolution. In the course of time, new cell types were generated from precursors selected not so much for their intrinsic properties as for the accident of their position in the embryo

significance both for what it can tell us immediately about relationships between cells and also as a framework into which future observations can be fitted. . . . The embryonic cell lineage is essentially invariant. The patterns of division, programmed cell death, and terminal differentiation are constant from one individual to another, and no great differences are seen in timing. . . . We shall use the term "sublineage" as an abbreviation for the more descriptive, but cumbersome, phrase "intrinsically determined sublineage"—namely, a fragment of the lineage which is thought to be generated by a programme within its precursor cell. . . . Two of the available criteria for postulating the existence of a sublineage are: (1) the generation of the same lineage, giving rise to the same cell fates, from a series of precursors of diverse origin and position; (2) evidence for cell autonomy within the lineage, obtained from laser ablation experiments or the study of mutants. . . . The large number of programmed cell deaths, and their reproducibility, is evident from the lineage. The most likely reason for the occurrence of most of them is that, because of the existence of sublineages, unneeded cells are frequently generated along with needed ones. . . . Perhaps the most striking findings are firstly the complexity and secondly the cell autonomy of the lineages. . . . With hindsight, we can rationalize both this complexity and this rigidity. The nematode belongs to an ancient phylum, and its cell lineage is a piece of frozen evolution. In the course of time, new cell types were generated from precursors selected not so much for their intrinsic properties as for the accident of their position in the embryo. . . . Cell-cell interactions that were initially necessary for developmental decisions may have been gradually supplanted by autonomous programmes that were fast, economical, and reliable, the loss of flexibility being outweighed by the gain in efficiency. On this view, the perverse assignments, the cell deaths, the long-range

migration—all the features which could, it seems, be eliminated from a more efficient design—are so many developmental fossils. These are the places to look for clues both to the course of evolution and to the mechanisms by which the lineage is controlled today.
Source: Developmental Biology 100: 64–119

interconnections among cells in the nervous system.

Nematode development is largely autonomous, which means that in most cells, the developmental program unfolds automatically with no need for interactions with other cells. However, in the early embryo, some of the developmental fates are established by interactions among the cells. In later stages of development of these cells, the fates established early are reinforced by still other interactions between cells. Worm development also provides important examples of the effects of intercellular signaling on determination (for example, those shown in Figure 12.4).

Mutations That Affect Cell Lineages

Many mutations that affect cell lineages have been studied in nematodes, and they reveal several general features by which genes control development.

• The division pattern and fate of a cell are generally affected by more than one gene and can be disrupted by mutations in any of them.

• Most genes that affect development are active in more than one type of cell.

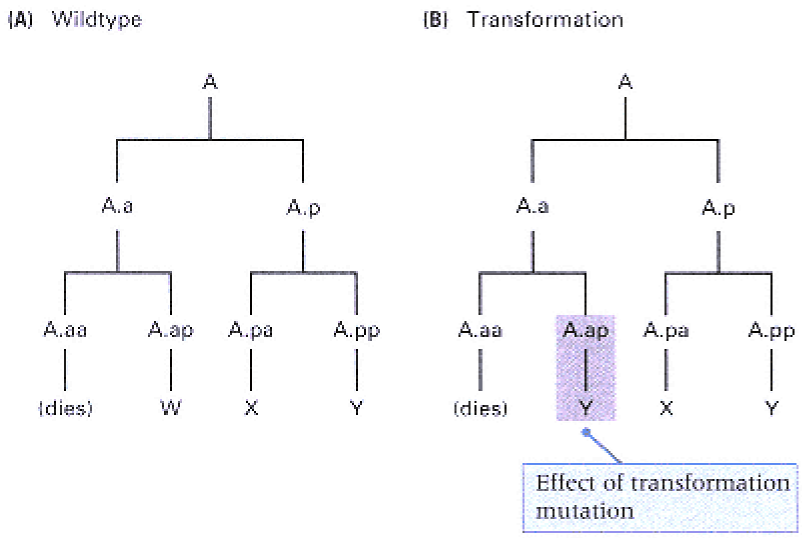


Figure 12.10
Transformation mutations cause cells to undergo abnormal terminal differentiation. (A) The wildtype lineage. (B) A mutant lineage in which the cell A.ap differentiates into a Y-type cell rather than the normal W-type cell.

• Complex cell lineages often include simpler, genetically determined lineages within them; these components are called *sublineages* because they are expressed as an intergrated pattern of cell division and terminal differentiation.

• The lineage of a cell may be triggered autonomously within the cell itself or by signaling interactions with other cells.

• Regulation of development is controlled by genes that determine the different sublineages that cells can undergo and the individual steps within each sublineage.

The next section deals with some of the types of mutations that affect cell lineages and development.

Types of Lineage Mutations

Mutations can affect cell lineages in two major ways. One is through nonspecific metabolic blocks—for example, in DNA replication—that prevent the cells from undergoing division or differentiation. The other is through specific molecular defects that result in patterns of division or differentiation that are characteristic of cells normally found elsewhere in the embryo or at a different time in development. From the standpoint of genetic analysis of development, the latter class of mutants is the more informative, because the mutant genes must be involved in developmental processes rather than in general "housekeeping" functions found in all cells. *C. elegans* provides several examples of each of the following general types of lineage mutations.

Mutations that cause cells to undergo developmental fates characteristic of other types of cells are called **transformation mutations.** For example, in the wildtype cell lineage shown in Figure 12.10A, the cell designated A.ap differentiates into cell-type W. In the transformation mutant (Figure 12.10B), cell A.ap differentiates into cell-type Y. Transformation mutations have the consequence that one or more differentiated cell types are missing from the embryo and are replaced with extra, supernumerary copies of other cell types. In this example, the missing cell type is W, and the extra cell type is Y. In the nematode, the mutation *unc-86* (*unc* = *uncoordinated*) provides an example in which the supernumerary cells become neurons. Supernumerary cells can impair development by interfering with the normal developmental process in nearby cells. In Section 12.4, we will see that transformation mutations in nematodes are analogous to a class of mutations in *Drosophila* called *homeotic* mutations, in which certain cells undergo developmental fates that are normally characteristic of other cells. Indeed, the wildtype *unc-86* gene codes for a transcription factor that contains a DNA-binding domain resembling that found in the proteins of the *Drosophila* homeotic genes.

Developmental programs often require sister cells or parent and offspring cells to adopt different fates. Mutations that cause sister cells or parent-offspring cells to fail to become differentiated from each other are called **segregation mutations** because the factors that govern the daughter cells fates fail to segregate (become separated) in the daughter cells. Two examples based on the wildtype lineage in Figure 12.8 are illustrated in Figure 12.11. In the mutation in Figure 12.11A, the sister cells A.a and A.p give rise to sublineages in which the anterior derivative undergoes programmed cell death and the posterior derivative adopts fate W. This pattern contrasts with the wildtype situation in Figure 12.8, in which A.a and A.p give rise to different sublineages. In Figure 12.11B, the daughter cell A.pp adopts the fate of its parent cell, A.p, in that A.pp divides, the anterior daughter A.ppa adopts fate X, and the posterior daughter A.ppp divides again. Continuation of this pattern results in a group of supernumerary cells of type X.

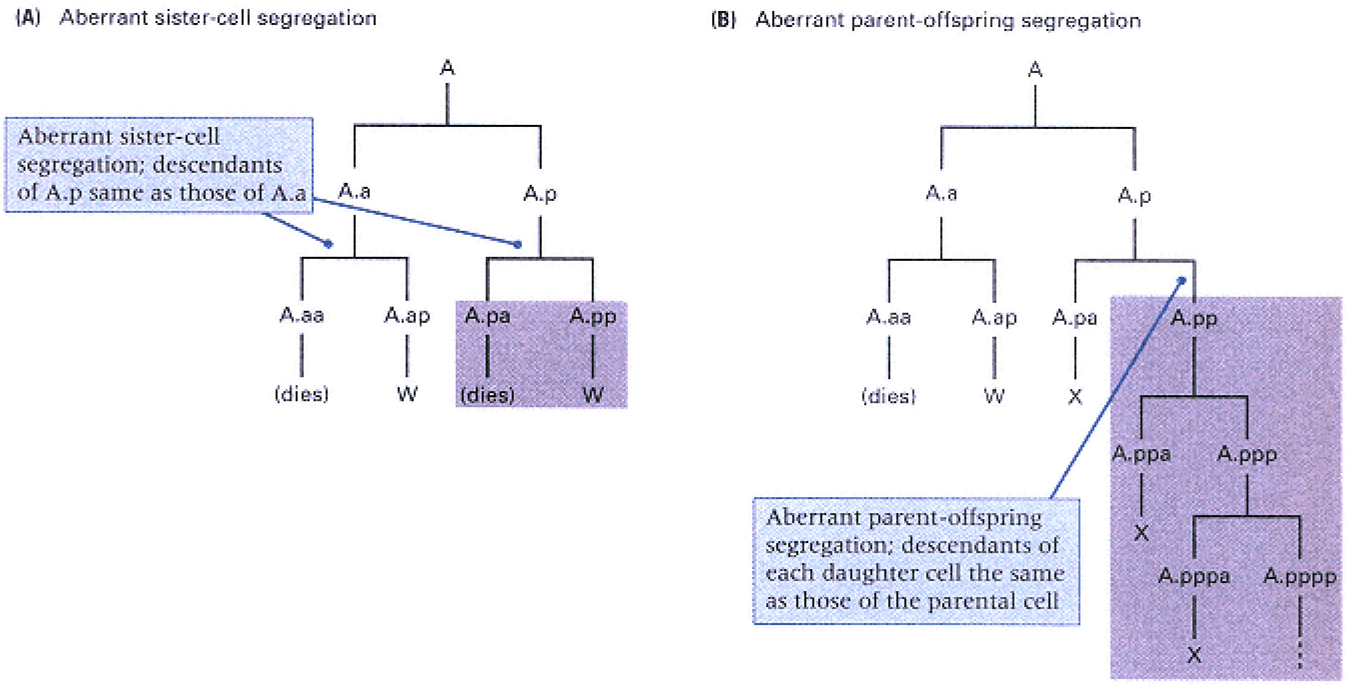


Figure 12.11
Two types of segregation mutations, which cause related cells to fail to become differentiated from each other (see Figure 12.8 for the wildtype lineage). (A) Aberrant sister-cell segregation, in which the mutant cell A.p adopts the same lineage as its sister cell A.a instead of undergoing its normal fate. (B) Aberrant parent-offspring segregation, in which the mutant cell A.pp, like its parent A.p, undergoes cell division, the anterior daughter of which differentiates into an X-type cell. In both of these types of segregation mutations, the result of abnormal segregation is the absence of certain cell types and the presence of supernumerary copies of other cell types.

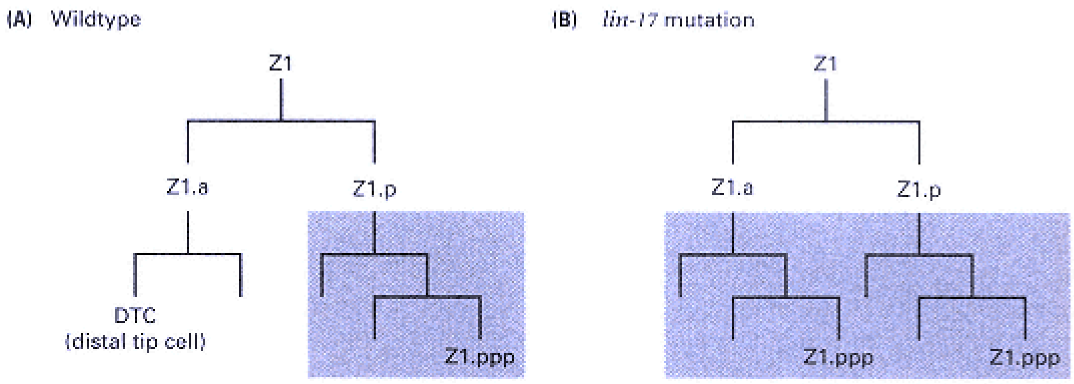


Figure 12.12
Aberrant sister-cell segregation caused by the *lin-17* mutation in *C. elegans.* (A) Z1.a and Z1.p normally undergo different fates, and the Z1.a lineage includes a distal tip cell (DTC). (B) In the *lin-17* segregation mutant, Z1.a and Z1.p have the same fate, and the distal tip cell is absent.

Mutations in the *lin-17* gene in *C. elegans* result in sister-cell segregation defects. In males, a gonadal precursor cell, Z1, produces daughter cells that are different in that the sister cells Z1.a and Z1.p have different lineages and fates (Figure 12.12A). In *lin-17* mutants, the developmental determinants do not segregate, and the sister cells have the same lineage and fate as the normal Z1.p (Figure 12.12B).

When mutant cells are unable to execute their normal developmental fates, the mutation is an **execution mutation.** The *lin-11* gene in *C. elegans* provides an example of an execution mutation. In the normal development of the vulva in the hermaphrodite, a lineage designated the 2° lineage gives rise to four cells in the spatial pattern N-T-L-L (Figure] 12.13A), in which N indicates a cell that does not divide, T a cell that divides in a transverse plane relative to the orientation of the larva, and L a cell that divides in a longitudinal plane. In *lin-11* mutants, the 2° lineage is not executed, and the four cells divide in the abnormal pattern L-L-L-L (Figure 12.13B).

The process of *programmed cell death*, technically known as **apoptosis,** is an

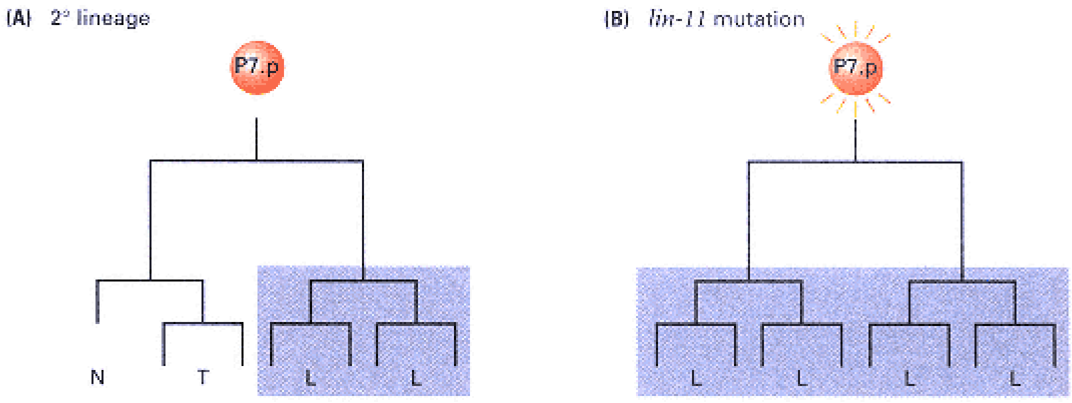


Figure 12.13
Example of an execution mutation. (A) Wildtype development of the 2° lineage in the vulva of C. *elegans*. The letter N denotes no further cell division, and T and L denote cell division in either a transverse (T) or a longitudinal (L) plane with respect to the embryo. (B) The *lin-11* mutation results in failure to execute the N and T sublineage. All cells divide in a longitudinal plane.

important feature of normal development in many organisms. Apoptosis is a completely normal process in which, at the appropriate time in development, a cell commits suicide. In many cases, the signaling molecules that determine this fate have been identified (a number are known to be transcription factors). Failure of programmed cell death often results in specific developmental abnormalities. For example, compared with the wildtype lineage in Figure 12.14A, the lineage in Figure 12.14B is abnormal in that cell A.aa fails to undergo apoptosis and, instead, differentiates into the cell-type V. Phenotypically, when apoptosis fails and the surviving cells differentiate into recognizable cell types, the result is the presence of supernumerary cells of that type. For example, with mutations in the *ced-3* gene (*ced* = *cell death abnormal*) in *C. elegans*, a particular cell that normally undergoes programmed cell death survives and often differentiates into a supernumerary neuron. There are exactly 113 programmed cell deaths in the development of the *C. elegans* hermaphrodite. None of these deaths is essential. Mutants that cannot execute programmed cell death are viable and fertile but are slightly impaired in development and in some sensory capabilities. On the other hand, mutants in *Drosophila* that fail to execute apoptosis are lethal, and in mammals, including human beings, failure of programmed cell death results in severe developmental abnormalities or, in some instances, leukemia or other forms of cancer.

The events of development are coordinated in time, so mutations that affect the timing of developmental events are of great interest. Mutations that affect timing are called **heterochronic mutations.** An example is shown in Figure 12.14C. In comparison with the wildtype situation in Figure 12.14A, the heterochronic mutant cell A.a delays cell division until the daughter cells A.aa and A.ap become contemporaneous with X and Y. Therefore, the heterochronic mutant in Figure 12.14C is *retarded* in that developmental events are normal but delayed. Heterochronic mutations can also be *precocious* in the expression of developmental events at times earlier than normal. For example, in certain heterochronic mutants in *C. elegans,* specific sublineages that normally develop in males

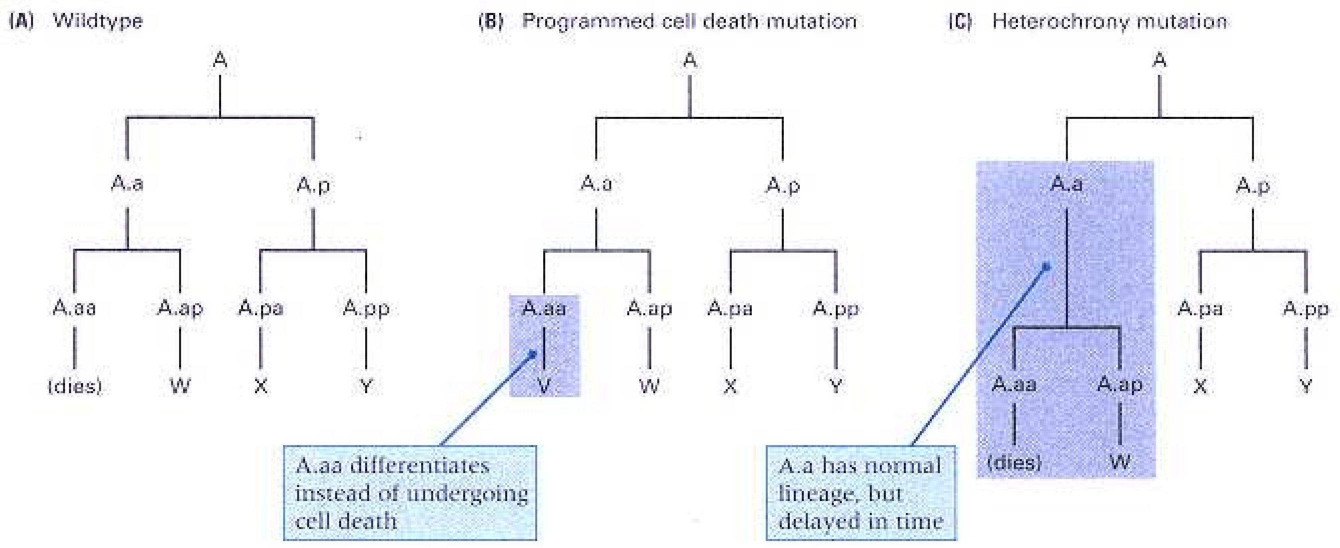


Figure 12.14
Apoptosis (programmed cell death) and heterochrony. (A) A wildtype lineage. (B) Failure of apoptosis, in which cell A.aa does not die but instead differentiates into a V-type cell. In some cases in which programmed cell death fails, the surviving cells differentiate into identifiable types. (C) Heterochronic mutants are abnormal in the timing of events in development. Retarded mutants undergo normal events too late, and precocious mutants undergo normal events too early. The example shown here is a retarded mutant in which cell A.a delays division until its products (A.aa and A.ap) are contemporaneous with the X and Y cells derived from A.p.

only at the fourth larval molt develop precociously in the mutant in the second or third larval molt.

The lin-12 Developmental-Control Gene

Control genes that cause cells to diverge in developmental fate are not always easy to recognize. For example, an execution mutation may identify a gene that is necessary for the expression of a particular developmental fate, but the gene may not actually control or determine the developmental fate of the cells in which it is expressed. This possibility complicates the search for genes that control major developmental decisions.

Genes that control decisions about cell fate can sometimes be identified by the unusual characteristic that dominant or recessive mutations have opposite effects;

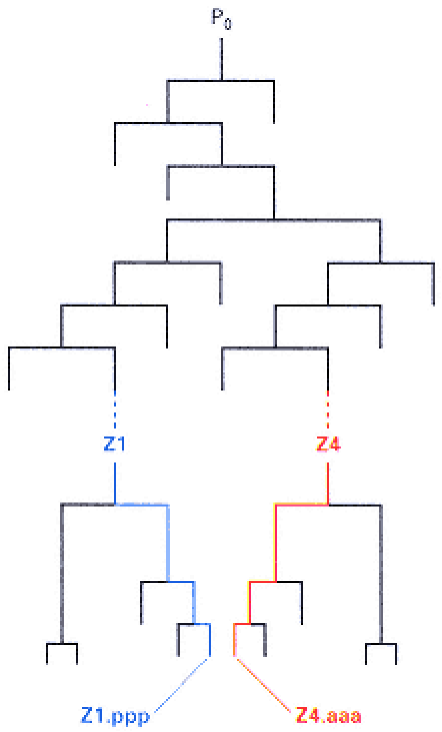


Figure 12.15
Complete lineage of Z1.ppp and Z4.aaa in *C. elegans*. P_0 represents the zygote, and the dashed lines indicate three cell divisions not shown. In the normal development of the vulva, Z1.ppp and Z4. aaa are equally likely to differentiate into the anchor cell. Whichever cell remains differentiates into a ventral uterine precursor cell.

that is, if alternative alleles of a gene result in opposite cell fates, then the product of the gene must be both necessary and sufficient for expression of the fate. Identification of possible regulatory genes in this way excludes the large number of genes whose functions are merely necessary, but not sufficient, for the expression of cell fate. Recessive mutations in genes controlling development often result from **loss of function** in that the mRNA or the protein is not produced; loss-of-function mutations are exemplified by nonsense mutations that cause polypeptide chain termination in translation (Chapter 10). Dominant mutations in developmental-control genes often result from **gain of function** in that the gene is overexpressed or is expressed at the wrong time.

In *C. elegans*, only a small number of genes have dominant and recessive alleles that affect the same cells in opposite ways. Among them is the *lin-12* gene, which controls developmental decisions in a number of cells. One example concerns the cells denoted Z1.ppp and Z4.aaa in Figure 12.15. These cells lie side by side in the embryo, but they have quite different lineages (cell P_0 is the zygote). Normally, one of the cells differentiates into an *anchor cell* (AC), which participates in development of the vulva, and the other differentiates into a *ventral uterine precursor cell* (VU) Figure 12.16A). Either Z1.ppp or Z4.aaa may become the anchor cell with equal likelihood.

Direct cell-cell interaction between Z1.ppp and Z4.aaa controls the AC-VU decision. If either cell is burned away (ablated) by a laser microbeam, then the remaining cell differentiates into an anchor cell (Figure 12.16B). This result implies that the preprogrammed fate of both Z1.ppp and Z4.aaa is that of an anchor cell. When either cell becomes committed to the anchor-cell fate, its contact with the other cell elicits the latter's fate as the ventral uterine precursor cell. As noted, recessive and dominant mutations of *lin-12* have opposite effects. Mutations in which *lin-12* activity is lacking or is greatly reduced are denoted *lin-12(0).* These mutations are recessive, and in the mutants, both Z1.aaa and Z4.aaa become anchor cells (Figure 12.16C). In contrast, *lin-12(d)* mutations are those in which *lin-12* activity is overex-

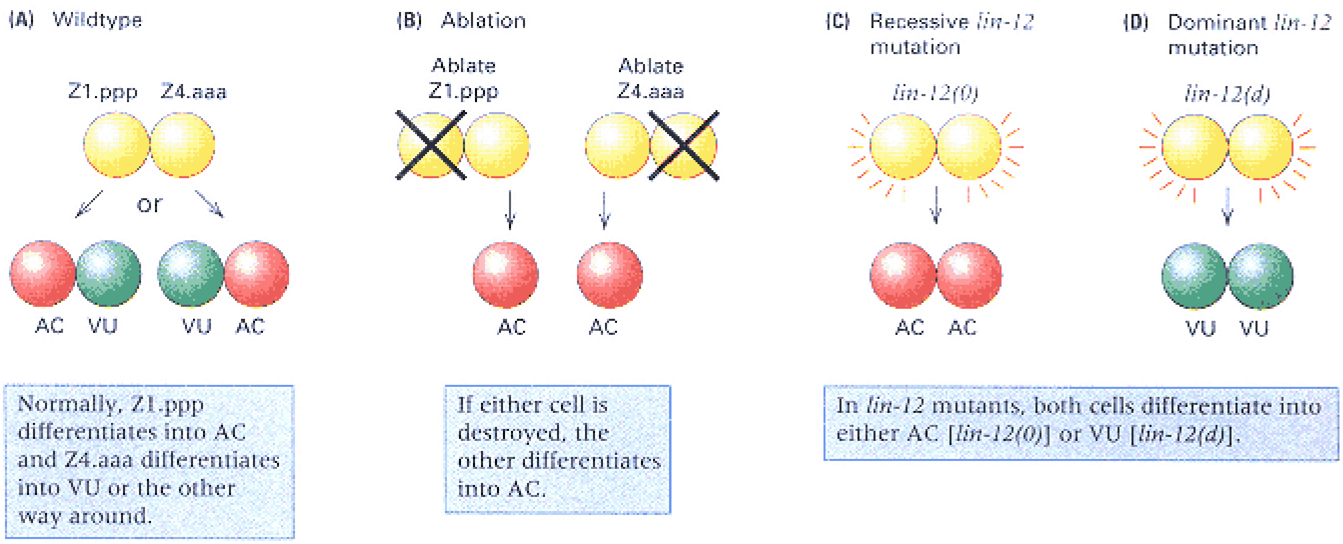


Figure 12.16
Control of the fates of Z1.ppp and Z4.aaa in vulval development. For the complete lineage of these cells, see Figure 12.15. (A) In wildtype cells, each cell has an equal chance of becoming the anchor cell (AC); the other becomes a ventral uterine precursor cell (VU). (B) If either cell is destroyed (ablated) by a laser beam, then the other differentiates into the anchor cell. (C) Genetic control of cell fate by the *lin-12* gene. With recessive loss-of-function mutations [*lin-12(0)*], both cells become anchor cells. (D) With dominant gain-of-function mutations [*lin-12(d)*], both cells become ventral uterine precursor cells.

pressed. These mutations are dominant or partly dominant, and in the mutants, both Z1.aaa and Z4.ppp become ventral uterine precursor cells (Figure 12.16D).

The effects of *lin-12* mutations suggest that the wildtype gene product is a receptor of a developmental signal. The molecular structure of the *lin-12* gene product is typical of a transmembrane receptor protein, and it shares domains with other proteins that are important in developmental control (Figure 12.17). The transmembrane region separates the LIN-12 protein into an extracellular part (the amino end) and an intracellular part (the carboxyl end). The extracellular part contains 13 repeats of a domain found in a mammalian peptide hormone, epidermal growth factor (EGF), as well as in the product of the *Notch* gene in *Drosophila*, which controls the decision between epidermal and neural cell fates. Nearer the transmembrane region, the amino end contains three repeats of a cysteine-rich domain also found in the *Notch* gene product. Inside the cell, the carboxyl part of the LIN-12 protein contains six repeats of a domain also found in the genes *cdc10* and *SW16*, which control cell division in two species of yeast.

The anchor cell expresses a signaling gene, called *lin-3*, that illustrates another case in which either loss-of-function or gain-of-function alleles have opposite effects on phenotype. In the anchor cell, the gene *lin-3* controls the fate of certain cells in the development of the vulva. Figure 12.18 illustrates five precursor cells, P4.p through P8.p, that participate in the development of the vulva. Each precursor cell has the capability of differentiating into one of three fates, called the 1°, 2°, and 3° lineages, which differ in whether descendant cells remain in a syncytium (S) or divide longitudinally (L), transversely (T), or not at all (N). The precursor cells normally differentiate as shown in Figure 12.18, giving five lineages in the order 3°-2°-1°-2°-3°. The vulva itself is formed from the 1° and 2° cell lineages. The spatial arrangement of some of the key cells is shown in the photograph in Figure 12.19. The black arrow indicates the anchor cell,

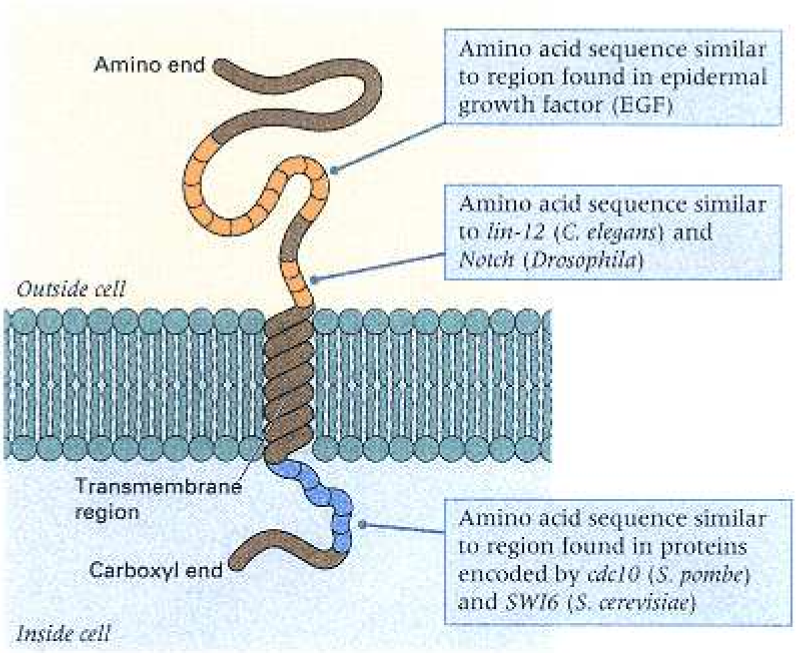


Figure 12.17
The structure of the LIN-12 protein is that of a receptor protein containing a transmembrane region and various types of repeated units that resemble those in epidermal growth factor (EGF) and other developmental control genes.

and the white lines show the pedigrees of 12 cells. The four cells in the middle derive from P6.p and the four on each side derive from P5.p and P7.p.

The important role of the *lin-3* gene product (LIN-3) is suggested by the opposite phenotypes of loss-of-function and gain-of-function alleles. Loss of LIN-3 results in the complete absence of vulval development. Conversely, overexpression of LIN-3 results in excess vulval induction. LIN-3 is a typical example of an interacting molecule, or **ligand,** that interacts with an EGF-type transmembrane receptor. In this case, the receptor is located in cell P6.p and is the product of the gene *let-23.* The LET-23 protein is a tyrosine-kinase receptor that, when stimulated by the LIN-3 ligand, stimulates a series of intracellular signaling events that ultimately results in the synthesis of transcription factors that determine the 1° fate. Among the genes that are induced is a gene for yet another ligand, which stimulates receptors on the cells P5.p and P7.p, causing these cells to adopt the 2° fate (blue horizontal arrows in Figure 12.18). The evidence for horizontal signaling is found in genetic mosaic worms in which the LET-23 receptor is missing in some or all of P5.p through P7.p. If the receptor is missing in all three cell types, none of the cells adopts its normal fate.

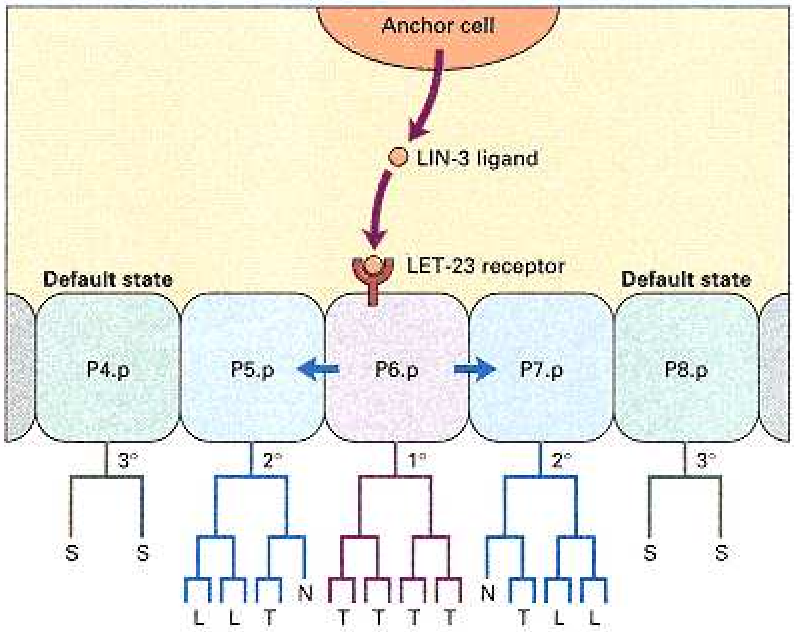


Figure 12.18

Determination of vulval differentiation by means of intercellular signaling. Cells P4.p through P8.p in the hermaphrodite give rise to lineages in the development of the vulva. The three types of lineages are designated 1°, 2°, and 3°. The 1° lineage is induced in P6.p by the ligand LIN-3 produced in the anchor cell (AC), which stimulates the LET-23 receptor tyrosine kinase in P6.p. The P6.p cell, in turn, produces a ligand that stimulates receptors in P5.p and P7.p to induce the 2° fate. On the other hand, the 3° fate is the default or baseline condition which P4.p and P8.p adopt normally and all cells adopt in the absence of AC.

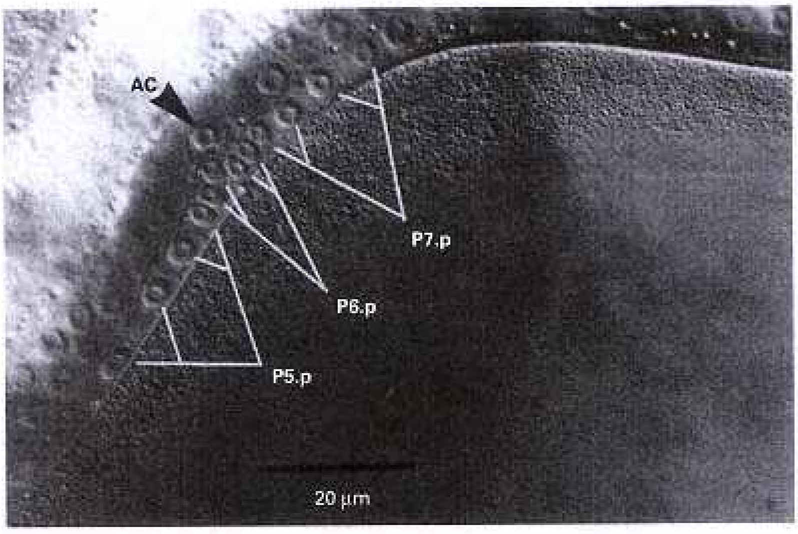


Figure 12.19
Spatial organization of cells in the vulva, including the anchor cell (black arrowhead) and the daughter cells produced by the first two divisions of P5.p through P7.p (white tree diagrams). The length of the scale bar equals 20 micrometers.
[Courtesy of G. D. Jongeward, T. R. Clandinin, and P. W. Sternberg. 1995. *Genetics* 139: 1553.]

However, if the receptor is present in P6.p but absent in P5.p and P7.p, all three cell types differentiate as they should, which implies that receipt of the LIN-3 signal is necessary for 1° determination and that a stimulated P6.p is necessary for 2° determination.

In vulval development, the adoption of the 3° lineages by the P4.p and P8.p cells is determined not by a positive signal, but by the lack of a signal, because in the absence of the anchor cell, all of the cells P4.p through P8.p express the 3° lineage. Thus development of the 3° lineage is the uninduced or *default* state, which means that the 3° fate is preprogrammed into the cell and must be overridden by another signal if the cell's fate is to be altered.

12.4— Development in Drosophila

Many important insights into developmental processes have been gained from genetic analysis in *Drosophila.* In 1995, the pioneering work of Christiane Nüsslein-Volhard, Eric Wieschaus, and Edward B. Lewis was recognized with the awarding of the Nobel Prize in Physiology or Medicine.

The developmental cycle of *D. melanogaster,* summarized in Figure 12.20, includes egg, larval, pupal, and adult stages. Early development includes a series of cell divisions, migrations, and infoldings that result in the gastrula. About 24 hours after fertilization, the first-stage larva, composed of about 10^4 cells, emerges from the egg. Each larval stage is called an *instar.* Two successive larval molts that give rise to the second and third instar larvae are followed by pupation and a complex metamorphosis that gives rise to the adult fly composed of more than 10^6 cells. In wildtype strains reared at 25°C, development requires from 10 to 12 days.

Early development in *Drosophila* takes place within the egg case (Figure 12.21A). The first nine mitotic divisions occur in rapid succession without division of the cytoplasm and produce a cluster of nuclei within the egg (Figure 12.21B). The nuclei migrate to the periphery, and the germ line is formed from about 10 **pole cells** set off at the posterior end (Figure 12.21C); the pole cells undergo two additional divisions and are reincorporated into the embryo by invagination. The nuclei within the embryo undergo four more mitotic divisions without division of the cytoplasm, forming the **syncytial blastoderm,** which contains about 6000 nuclei (Figure 12.21D). Cellularization of the blastoderm takes place from about 150 to 180 minutes after fertilization by the synthesis of membranes that separate the nuclei. The **blastoderm** formed by cellularization (Figure 12.21E) is a flattened hollow ball of cells that corresponds to the blastula in other animals.

The experimental destruction of patches of cells within a *Drosophila* blastoderm results in localized defects in the larva and adult. The spatial correlation between the position of the cells destroyed and the type

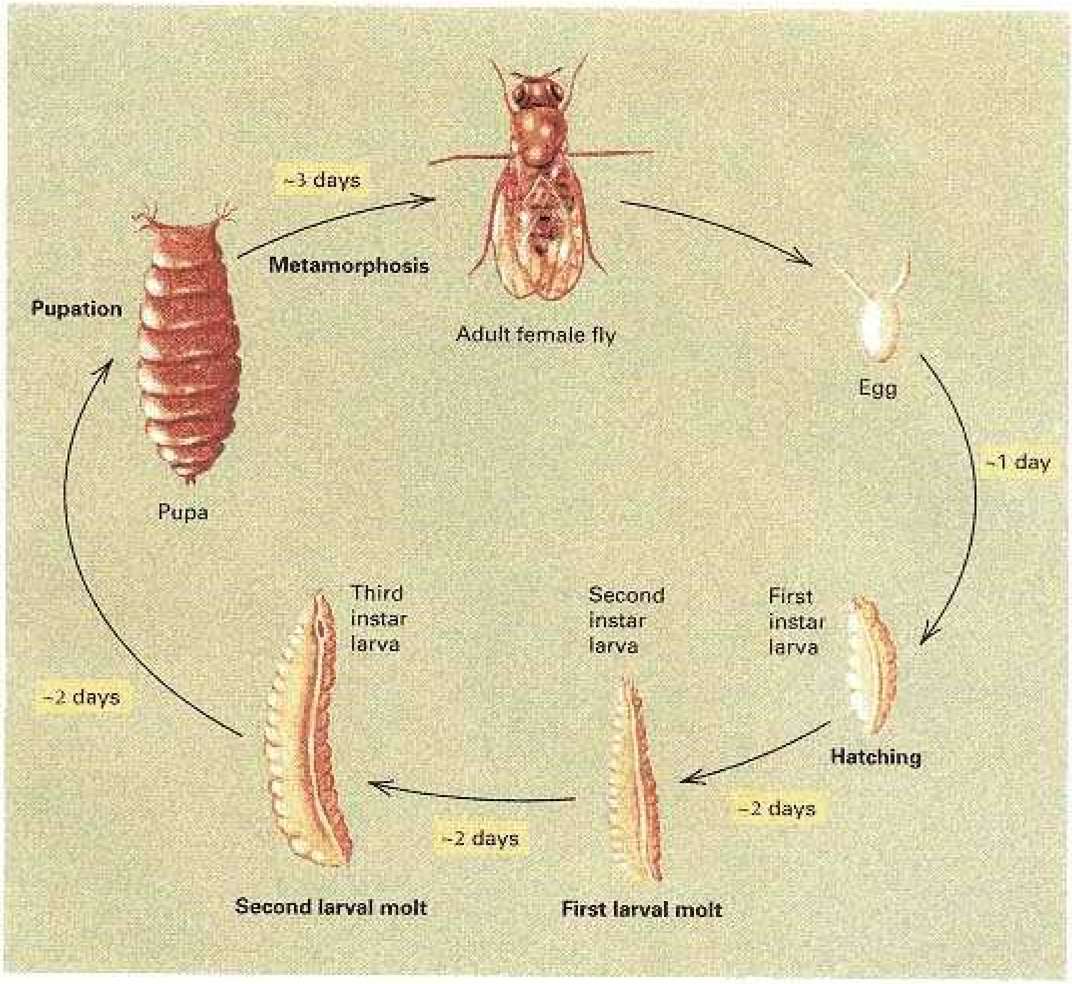


Figure 12.20
Developmental program of *Drosophila melanogaster*. The durations of the stages are at 25°C.

of defects results in a **fate map** of the blastoderm, which specifies the cells in the blastoderm that give rise to the various larval and adult structures (Figure 12.22). Use of genetic markers in the balstoderm has made possible further refinement of the fate map. Cell lineages can be genetically marked during development by induicing recombination between homologous chromosomes in mitosis, resulting in genetically different daughter cells (Chapter 4). Much like the cells in the early blastula of *Caenorhabditis,* cells in the blastoderm of *Drosophila* have predetermined developmental fates, with little ability to substitute in development for other, sometimes even adjacent, cells.

Evidence that blastoderm cells in *Drosophila* have predetermined fates comes from experiments in which cells from a genetically marked blastoderm are implanted into host blastoderms. Blastoderm cells implanted into the equivalent regions of the host become part of the normal adult structures. However, blastoderm cells implanted into different regions develop autonomously and are not integrated into host structures.

Because of the relatively high degree of determination in the blastoderm, genetic analysis of *Drosophila* development has tended to focus on the early stages of development when the basic body plan of the embryo is established and key regulatory

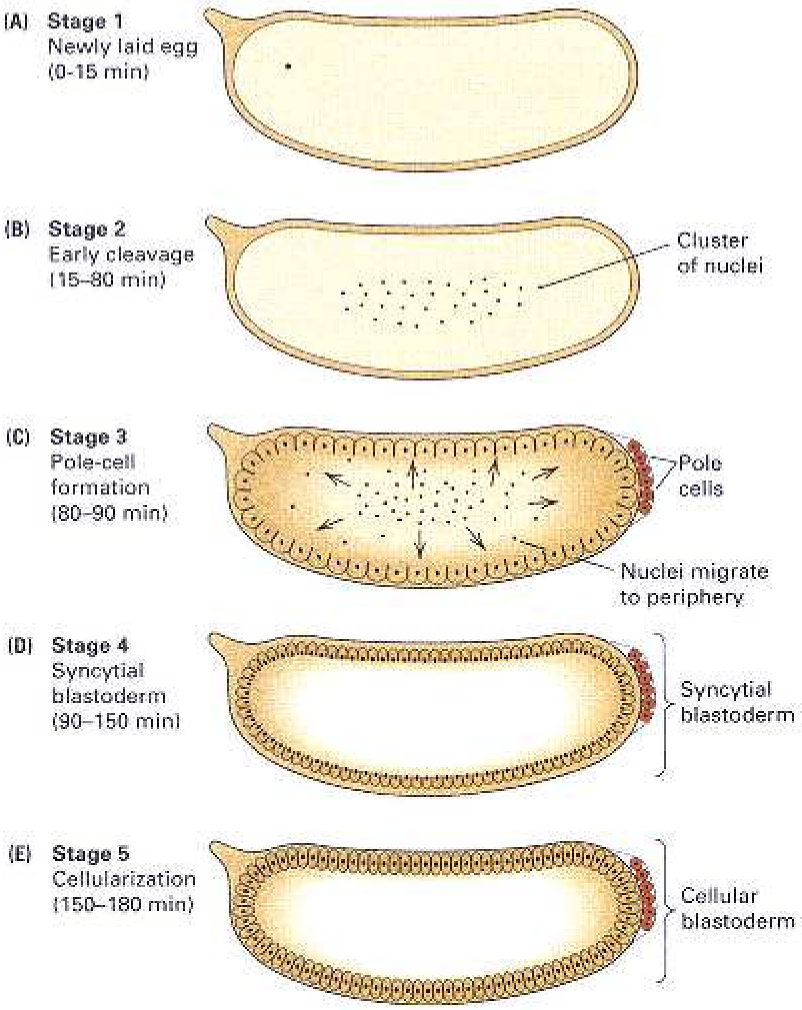
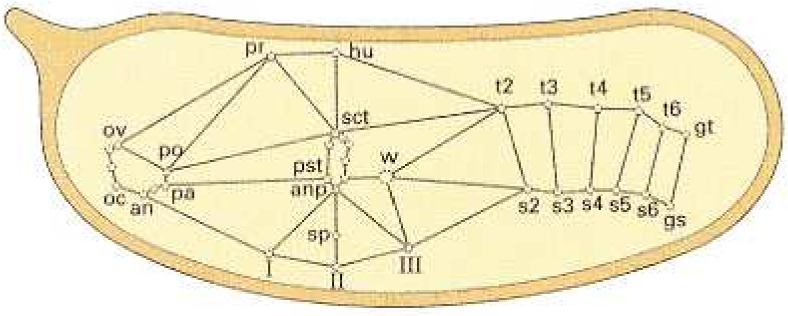


Figure 12.21
Early development in *Drosophila.* (A) The nucleus in the fertilized egg. (B) Mitotic divisions take place synchronously within a syncytium. (C) Some nuclei migrate to the periphery of the embryo, and at the posterior end, the pole cells (which form the germ line) become cellularized. (D) Additional mitotic divisions occur within the syncytial blastoderm. (E) Membranes are formed around the nuclei, giving rise to the cellular blastoderm.



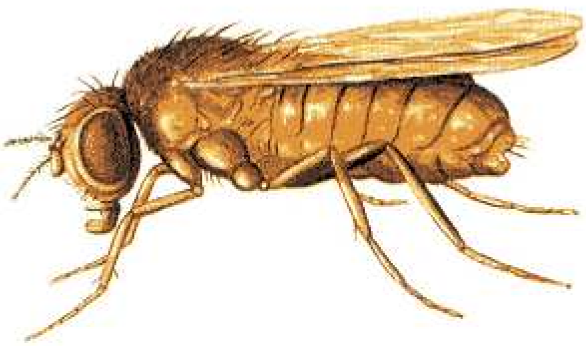


Figure 12.22
Fate map of the *Drosophila* blastoderm, which shows the adult structures that derive from various parts of the blastoderm. The map was determined by correlating the expression of genetic markers in different adult structures in genetic mosaics. The abbreviations stand for various body parts in the adult fly. For example, ov and oc are head structures; w is the wing; I, II, and III are the first, second, and third legs; and gs and gt are genital structures.
[After J. C. Hall, W. M. Gelbart, and D. R. Kankel. 1976. In *The Genetics and Biology of Drosophila*, vol. la, M. Ashburner and E. Novitski, eds. New York: Academic Press, pp. 265–314.]

processes become activated. The following sections summarize the genetic control of these early events.

Maternal-Effect Genes and Zygotic Genes

Early development in *Drosophila* requires translation of maternal mRNA molecules present in the oocyte. Blockage of protein synthesis during this period arrests the early cleavage divisions. Expression of the zygote genome is also required, but the timing varies. Blockage of transcription of the zygote genome at any time after the ninth cleavage division prevents formation of the blastoderm.

Because the earliest stages of *Drosophila* development are programmed in the oocyte, mutations that affect oocyte composition or structure can upset development of the embryo. Genes that function in the mother and are needed for development of the embryo are called **maternal-effect genes,** and developmental genes that function in the embryo are called **zygotic genes.** The interplay between the two types of genes is as follows:

> The zygotic genes interpret and respond to the positional information laid out in the egg by the maternal-effect genes.

Mutations in maternal-effect genes are easy to identify because homozygous females produce eggs that are unable to support normal embryonic development, whereas homozygous males produce normal sperm. Therefore, reciprocal crosses give dramatically different results. For example, a recessive maternal-effect mutation, *m,* will yield the following results in crosses:

$$m/m \,♀ \times +/+ \,♂ \rightarrow +/m \text{ progeny}$$
$$\text{(abnormal development)}$$

$$+/+ \,♀ \times m/m \,♂ \rightarrow +/m \text{ progeny}$$
$$\text{(normal development)}$$

The +/*m* progeny of the reciprocal crosses are genetically identical, but development is upset when the mother is homozygous *m/m.*

The reason why maternal-effect genes are needed in the mother is that the maternal-effect genes establish the polarity of the *Drosophila* oocyte even before fertilization takes place. They are active during the earliest stages of embryonic development, and they determine the basic body plan of the embryo. Maternal-effect mutations provide a valuable tool for investigating the genetic control of pattern formation and for identifying the molecules that are important in morphogenesis.

Genetic Basis of Pattern Formation in Early Development

The *Drosophila* embryo features 14 superficially similar repeating units visible as a pattern of stripes along the main trunk (Figure 12.23). The stripes can be recognized externally by the bands of *denticles,* which are tiny, pigmented, tooth-like projections from the surface of the larva. The 14 stripes in the embryo correspond to the segments in the larva that forms from the embryo. Each **segment** is defined morphologically as the region between successive indentations formed by the sites of muscle attachment in the larval cuticle. The designations of the segments are indicated in Figure 12.23. There are three head segments (C1-C3), three thoracic segments (T1-T3), and eight abdominal segments (A1-A8.). In addition to the segments, another type of repeating unit is also important in development. These repeating units are called **parasegments;** each parasegment consists of the posterior region of one segment and the anterior region of the adjacent segment. Para-segments have a transient existence in embryonic development. Although they are not visible morphologically, they are important in gene expression because the patterns of expression of many genes coincide with the boundaries of the parasegments rather than with the boundaries of the segments.

The early stages of pattern formation are determined by genes that are often called **segmentation genes** because they determine the origin and fate of the segments and parasegments. There are four classes of segmentation genes, which differ in their times and patterns of expression in the embryo.

1. The *coordinate genes* determine the principal coordinate axes of the embryo: the anterior-posterior axis, which defines

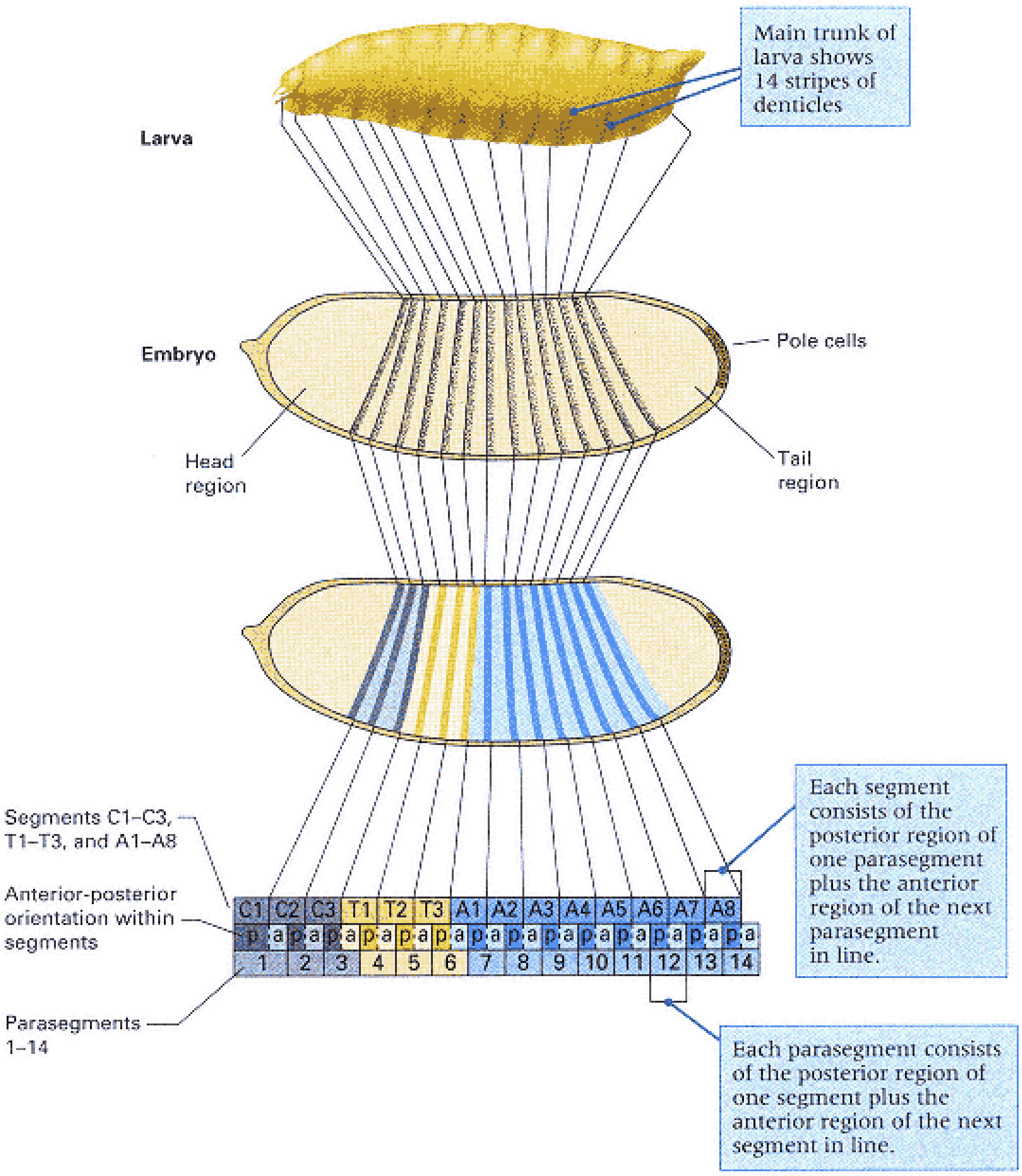


Figure 12.23
Segmental organization of the *Drosophila* embryo and larva. The segments are defined by successive indentations formed by the sites of muscle attachment in the larval cuticle. The parasegments are not apparent morphologically but include the anterior and posterior regions of adjacent segments. The distinction is important because the patterns of expression of segmentation genes are more often correlated with the parasegment boundaries than with the segment boundaries.

the front and rear, and the dorsalventral axis, which defines the top and bottom.

2. The *gap genes* are expressed in contiguous groups of segments along the embryo (Figure 12.24A), and they establish the next level of spatial organization. Mutations in gap genes result in the absence of contiguous body segments, so gaps appear in the normal pattern of structures in the embryo.

3. The *pair-rule genes* determine the separation of the embryo into discrete segments (Figure 12.24B). Mutations in pair-rule genes result in missing pattern elements in alternate segments. The reason for the two-segment periodicity of pair-rule genes is that the genes are expressed in zebra-stripe pattern along the embryo.

4. The *segment-polarity genes* determine the pattern of anterior-posterior development within each segment of the embryo (Figure 12.24C). Mutations in segment-polarity genes affect all segments or parasegments in which the normal gene is active. Many segment- polarity mutations have the normal number of segments, but part of each

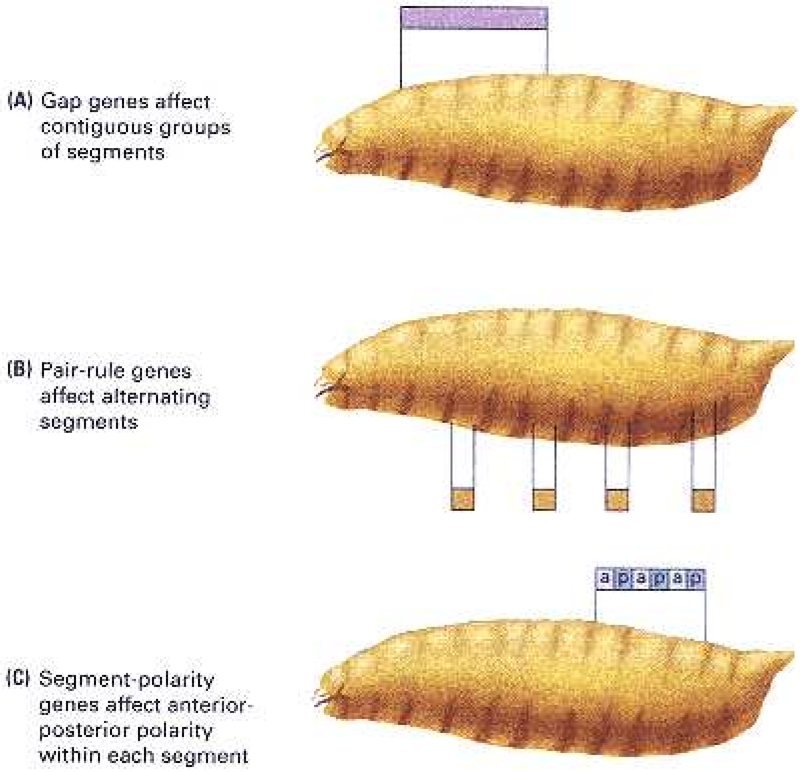


Figure 12.24
Patterns of expression of different types of segmentation genes. (A) The gap genes are expressed in a set of contiguous segments (B) The pair-rule genes are expressed in alternating segments. (C) The segment-polarity genes are expressed in each segment and determine the anterior-posterior pattern of differentiation within each parasegment.

segment is deleted and the remainder is duplicated in mirror-image symmetry.

Evidence for the existence of the four classes of segmentation genes—coordinate genes, gap genes, pair-rule genes, and segment-polarity genes—is presented in the following sections.

Coordinate Genes

The **coordinate genes** are maternal-effect genes that establish early polarity through the presence of their products at defined positions within the oocyte or through gradients of concentration of their products. The genes that determine the anterior-posterior axis can be classified into three groups according to the effects of mutations in them, as illustrated in Figure 12.25.

1. The first group of coordinate genes includes the *anterior genes,* which affect the head and thorax. The key gene in this class is *bicoid.* Mutations in *bicoid* produce embryos that lack the head and thorax and occasionally have abdominal segments in reverse polarity duplicated at the anterior end. The *bicoid* phenotype resembles that produced by certain kinds of surgical manipulations. For example, when *Drosophila* eggs are punctured and small amounts of cytoplasm allowed to escape, loss of cytoplasm from the anterior end results in embryos in which some posterior structures develop in place of the head. Similarly, replacement of anterior cytoplasm with posterior cytoplasm by injection yields embryos with two mirror-image abdomens and no head.

The *bicoid* gene product is a transcription factor for genes determining anterior structures. Because the *bicoid* mRNA is localized in the anterior part of the early-cleavage embryo, these genes are activated primarily in the anterior region. The *bicoid* mRNA is produced in nurse cells (the cells surrounding the oocyte in Figure 12.7) and exported to a localized region at the anterior pole of the oocyte. The protein product is less localized and, during the syncytial cleavages, forms an anterior-posterior concentration gradient with the maximum at

Connection Embryo Genesis

Christiane Nüsslein-Volhard and Eric Wieschaus 1980

European Molecular Biology
Laboratory, Heidelberg, Germany
Mutations Affecting Segment Number and Polarity in Drosophila

Nüsslein-Volhard and Wieschaus were exceptionally bold in supposing that the molecular mechanisms governing a process as complex as early embryonic development could be understood by the genetic and molecular analysis of mutations. The phenotype of such mutants is superficially identical: The embryo dies. The Drosophila genetic map was already littered with mutations classified collectively as "recessive lethals." These were generally considered as not amenable to further analysis because, in any particular case, the search for the specific defect was regarded as a needle-in-a-haystack problem. Nüsslein-Volhard and Wieschaus ignored most of the existing mutants. They set out to acquire systematically a new set of recessive-lethal mutants, each showing a specific and characteristic type of defect in the formation of organized patterns in the early embryo. Their first efforts, reported in this paper, yielded a number of mutations in each of three major classes of genes concerned with development. The paper sparked an enormous interest in Drosophila *developmental genetics. Today, a typical Annual Drosophila Research Conference includes approximately 500 presentations (mainly posters) dealing with aspects of* Drosophila *development. Nüsslein-Volhard and Wieschaus were awarded a Nobel Prize in 1995. They shared it with Edward B. Lewis for his pioneering genetic studies of the homeotic genes.*

The construction of complex form from similar repeating units is a basic feature of spatial organization in all higher animals. Very little is known for any organism about the genes involved in this process. In *Drosophila,* the metameric [repeating] nature of the pattern is most obvious in the thoracic and abdominal segments of the larval epidermis and we are attempting to

In Drosophila, it would seem feasible to identify all genetic components involved in the complex process of embryonic pattern formation

identify all loci required for the establishment of this pattern. . . . We have undertaken a systematic search for mutations that affect the segmental pattern. We describe here mutations at 15 loci which show one of three novel types of pattern alteration: pattern duplication (segment polarity mutants; six loci), pattern deletion in alternating segments (pair-rule mutants; six loci) and deletion of a group of adjacent segments (gap mutants; three loci). . . . Segment polarity mutants have the normal number of segments. However, in each segment a defined fraction of the normal pattern is deleted and the remainder is present as a mirror-image duplication. The duplicated part is posterior to the 'normal' part and has reversed polarity. . . . In pairrule mutants homologous parts of the pattern are deleted in every other segment. Each of the six loci is characterized by its own pattern of deletions. . . . One of the striking features of the [segment polarity and pair-rule] classes is that the alteration in the pattern is repeated at specific interval along the antero-posterior axis of the embryo. No such repeated pattern is found in mutants of the gap class and instead a group of up to eight adjacent segments is deleted. . . . The lack of a repeated pattern suggests that the loci are involved in processes in which position along the antero-posterior axis of the embryo is define by unique values. . . . The majority of mutants described here have been isolated in systematic searches for mutations affecting the segmentation pattern. These experiments are still incomplete. . . . In *Drosophila,* it would seem feasible to identify all genetic components involved in the complex process of embryonic pattern formation.

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the anterior tip of the embryo (Figure 12.26). The bicoid protein is a principl morphogen in determining the blastoderm fate map. The protein is a transcriptional activator that contains a helix-turn-helix motif for DNA binding (Chapter 11). Genes affected by the bicoid protein contain multiple upstream binding domains that consist of nine nucleotides resembling the consensus sequence 5'-TCTAATCCC-3'. Binding sites that differ by as many as two base pairs from the consensus sequence can bind the bicoid protein with high affinity, and sites that contain four mismatches bind with low affinity. The combination of high- and low-affinity binding sites determines the concentration of bicoid protein needed for gene activation; genes with many highaffinity binding sites can be activated at low concentrations, but those with many low-affinity bindings sites need higher concentrations. Such differences in binding affinity mean that the level of gene expression can differ from one regulated gene to